

VR3511F Data Book

VRender™ 3D

Data Book

MDVR3511FDB

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VR3511F

Data Book for VR3511F 3D Graphic Application Processor

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1. Overview

VRender3D 32bit RISC microprocessor is designed to provide a cost-effective, low power, small die size and high performance micro-controller for hand-held devices and game applications. To reduce total system cost, VRender3D also provides the following : separate 16KB instruction and 16KB Data Cache, MMU to handle virtual memory management, Digital LCD Interface, 2-channel UART with handshake, 12-channel DMA, System Manager, 4-channel Timers, 2-channel PWM, 10bit ADC for touch-screen interface, IIC Controller, AC97/I2S codec Interface, USB device, Multimedia card interface,.

The VRender3D was developed using an ARM920T core, 0.18um CMOS standard cells and a memory compiler. Its low-power, simple, elegant and fully static design is particularly suitable for cost-sensitive and power sensitive applications.

An outstanding features of the VRender3D is 3D graphic accelerator, GTE, Wavetable Synthesizer.

By Providing complete set of common system peripherals, the VRender3D minimizes overall system costs and eliminates the need to configure additional components. The integrated on-chip functions that are described in this document include;

CPU

- ARM920T@200Mhz
- Data Cache : 16KByte
- Instruction Cache : 16KByte

MEMORY

- DDR-SDRAM
- x16, x32bit Interface
- 64/128/256/512Mbit SDRAM
- Static Memory
- x8, x16, Interface
- PCMCIA/CF/IDE Controller

Enhanced 3D Features

GTE (Geometric Transform Engine)

- Vector Processing Unit (24bit Floating point: s15e8)
- Embedded Projection, View frustum Clipping, Viewport scaling
- Programmable Feature
- add/mul/dp3/dp4/dp3clamp/dp4clamp/rsqrt/min/max...

TSE (Triangle Setup Engine)

- 24bit Floating Point Engine
- Various Primitive
- Triangle strip, Triangle fan, Triangle list, Rectangle list

Rasterizer

- Sub-Pixel Correction
- Max resolution: 1024x768
- Texture Mapping
- 2-stage multi-texturing
- Perspective Correction
- Sub Texel Accuracy
- Bilinear Filtering
- Internal Texture Cache : 8KByte
- Enhanced Prefetch Texture Algorithm
- MIPMAP
- Use MPEG Video Texture
- Z-Buffer, Alpha Blending, Dithering
- W-based pixel fog
- Render to texture

MPEG Video

MPEG4 Video Decoder

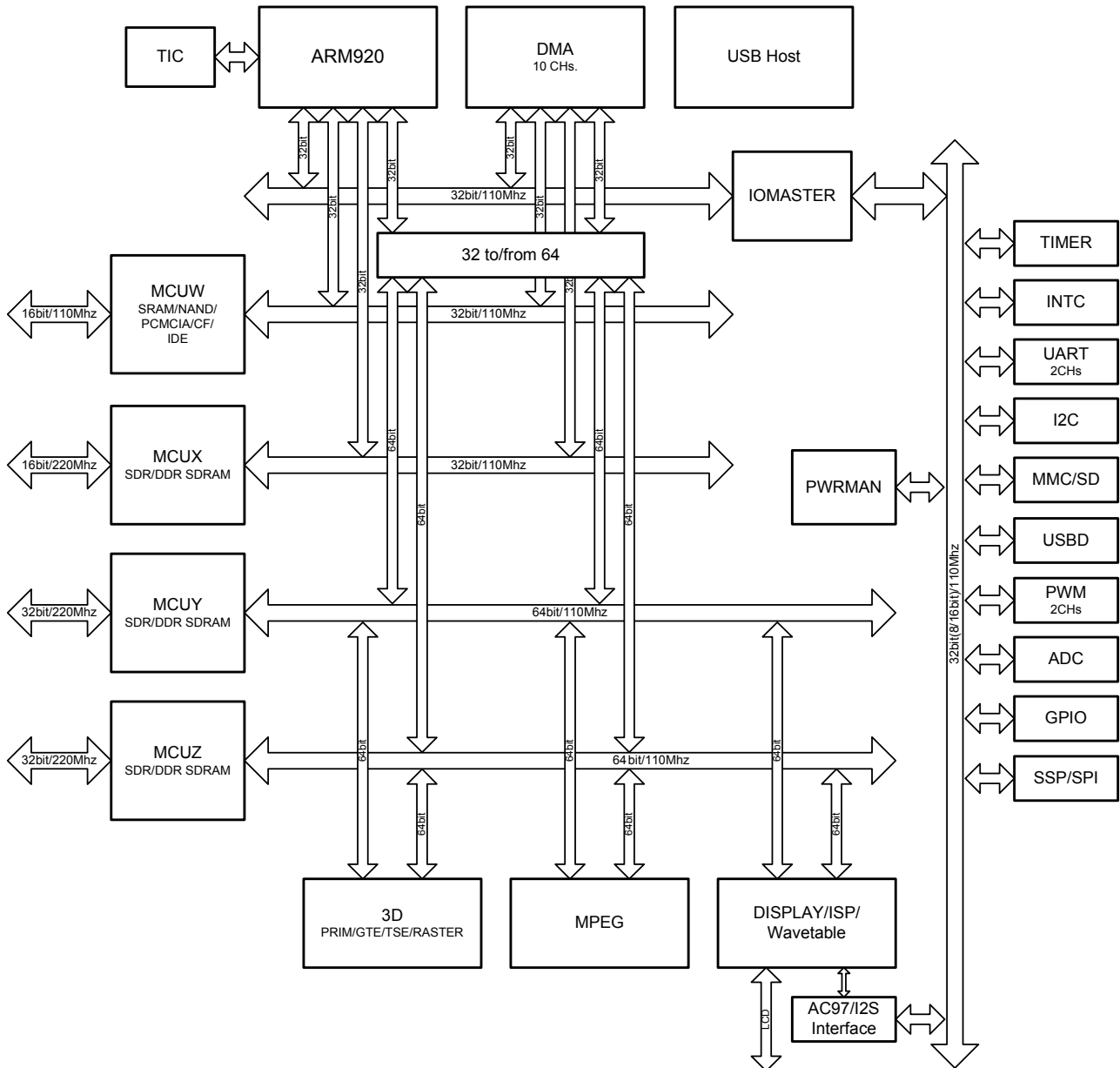
- Simple Profile, Level2 Decoder
- Up to 1024 x 1024 Resolution
- Supports MPEG Video Texture for 2D/3D
- 720x480 @30Hz real time decoding

Wavetable Synthesizer

- Output Format
- 16bit Stereo, ~44.1Khz
- Interface : AC97 Codec, I2S
- Up to 64 output channels
- Input Format
- 8/16bit PCM, 8bit u-law
- Effects
- individual volume/panning control
- individual 6 stage envelope control

Peripheral

- 10 CHs DMA
- 4 CHs Timer (Watchdog Timer)
- 2 CHs PWM (Pulse Width Modulation)
- Interrupt Controller
- USB 1.2 Host/Slave, I2C
- 2 CHs UART
- MMC / SD / NAND Flash
- AC97, I2S Codec Interface
- Camera interface (ISP)



2.

3. PAD Description

3.1 PAD List

BALL	PIN NAME	I/O	Description	Function #1	Function #2	Reset	Type
C9	TCK	I	TCK (TAP Controller Clock) provides the clock input for the JTAG logic. A 10K pull-up resistor must be connected to TCK pin.	TCK		F1	PI
B9	TMS	I	TMS (TAP Controller Mode Select) controls sequence of the TAP controller's states. A 10K pull-up resistor has to be connected to TMS pin.	TMS		F1	PI
D8	TDI	I	TDI (TAP Controller Data Input) is the serial input for test instructions and data. A 10K pull-up resistor has to be connected to TDI pin.	TDI		F1	PI
A7	TDO	O	TDO (TAP Controller Data Output) is the serial output for test instructions and data	TDO		F1	P8
E10	nTRST	I	nTRST (TAP Controller Reset) resets the TAP controller at start. If debugger is used. A 10K pull-up resistor has to be connected. If debugger is not used, nTRST pin must be issued by a low active pulse (Typically connected to nRESET)	nTRST		F1	PI
A8	TESTEN	I	Test Mode Enable 0: Normal Operation, 1: Test Mode	TESTEN		F1	PI
D9	TESTSCAN	I	For test purpose only 0: Normal Operation	TESTSCAN		F1	PI
C1	XTALIN	AI	Crystal Input for internal PLL. 10.000Mhz	XTALIN		F1	M26
B1	XTALOUT	AO	Crystal Output for Internal PLL	XTALOUT		F1	M26
D1	EXTCLK	I	External Clock Source for test purpose only. It has to be low (GND)	EXTCLK		F1	PI
A9	PWRXTALIN	AI	Crystal Input for Power Management Recommend : 32.768Khz	PWRXTALIN		F1	GP
A10	PWRXTALOUT	AO	Crystal Output for Power Management	PWRXTALOUT		F1	GP
E9	nBATF	I	Battery Fault, Probe for battery state (Does not wake up at power-off mode in case of low battery state). If it isn't used, it has to be High(3.3v). In VR3Dr2, Because It isn't supported, it must be connected to High.	nBATF		F1	PI
B8	nVDDF	I	VDD Fault, Probe for VDD state (Does not wake up at power-off mode in case of unstable power state). If it isn't used, it has to be High(3.3v) Because The Power-off mode (SLEEP, DEEP-SLEEP Mode) is not supported in VR3Dr2, This pin must be connected to High.	nVDDF		F1	PI
D7	nPORST	I	nPORST places VR3D into a known reset state. For a reset, nPORST must be hold to Low level for at least 4 PWRXTAL after power has been stabilized.	nPORST		F1	PI
E8	nRSTOUT	O	For external device reset control, It is activated by nPORST, Watchdog Reset, SW Reset.	nRSTOUT		F1	P8
C8	PWRON	O	1.8V core power on-off control signal. If Power-off mode is not used, PWRON pad is "don't care"	PWRON		F1	P8
E14	XA0/GPIOA0	IO	X-BUS DDR-DRAM Address : XA[12:0] (DDR DRAM Address bus for X-Bank) outputs the memory address of the DDR for x-bank / GPIO	XA0		GI	SB
D14	XA1/GPIOA1	IO	X-BUS DDR-DRAM Address / GPIO	XA1		GI	SB
C14	XA2/GPIOA2	IO	X-BUS DDR-DRAM Address / GPIO	XA2		GI	SB
C13	XA3/GPIOA3	IO	X-BUS DDR-DRAM Address / GPIO	XA3		GI	SB

E13	XA4/GPIOA4	IO	X-BUS DDR-DRAM Address / GPIO	XA4		GI	SB
D13	XA5/GPIOA5	IO	X-BUS DDR-DRAM Address / GPIO	XA5		GI	SB
F13	XA6/GPIOA6	IO	X-BUS DDR-DRAM Address / GPIO	XA6		GI	SB
C12	XA7/GPIOA7	IO	X-BUS DDR-DRAM Address / GPIO	XA7		GI	SB
E12	XA8/GPIOA8	IO	X-BUS DDR-DRAM Address / GPIO	XA8		GI	SB
A12	XA9/GPIOA9	IO	X-BUS DDR-DRAM Address / GPIO	XA9		GI	SB
A17	XA10/GPIOA10	IO	X-BUS DDR-DRAM Address / GPIO	XA10		GI	SB
F12	XA11/GPIOA11	IO	X-BUS DDR-DRAM Address / GPIO	XA11		GI	SB
D12	XA12/GPIOA12	IO	X-BUS DDR-DRAM Address / GPIO	XA12		GI	SB
B17	XBA0/GPIOA13	IO	X-BUS DDR-DRAM Bank Address : XBA[1:0] (DDR DRAM Bank Address for X-Bank) output bank address for DDR-DRAM / GPIO	XBA0		GI	SB
F14	XBA1/GPIOA14	IO	X-BUS DDR-DRAM Bank Address / GPIO	XBA1		GI	SB
C10	XCK	O	X-BUS DDR-DRAM Clock (DDR DRAM Clock for X-Bank)	XCK		F1	SD
B10	XnCK	O	X-BUS DDR-DRAM Inverting Clock (DDR DRAM Inverted Clock for X-Bank)	XnCK		F1	SD
C11	XCKE	O	X-BUS DDR-DRAM Clock Enable	XCKE		F1	SO
D15	XnCS/GPIOB0	IO	X-BUS DDR-DRAM Chip Select / GPIO	XnCS0		GI	SB
D16	XnRAS/GPIOB1	IO	X-BUS DDR-DRAM RAS (Row address strobe) / GPIO	XnRAS		GI	SB
E15	XnCAS/GPIOB2	IO	X-BUS DDR-DRAM CAS (Column address strobe) / GPIO	XnCAS		GI	SB
C16	XnWE/GPIOB3	IO	X-BUS DDR-DRAM Write Enable / GPIO	XnWE		GI	SB
E16	XLDM/GPIOB4	IO	X-BUS DDR-DRAM Data Mask : Lower Byte Data mask / GPIO	XLDM		GI	SB
D11	XUDM/GPIOB5	IO	X-BUS DDR-DRAM Data Mask : Higher Byte Data mask / GPIO	XUDM		GI	SB
C17	XLDQS	IO	X-BUS DDR-DRAM Data Strobe : Lower Byte Data strobe	XLDQS		F1	SB
E11	XUDQS	IO	X-BUS DDR-DRAM Data Strobe : High Byte Data strobe	XUDQS		F1	SB
A21	XD0	IO	X-BUS DDR-DRAM Data : XD[15:0], Data bus	XD0		Hi-Z	SB
B20	XD1	IO	X-BUS DDR-DRAM Data	XD1		Hi-Z	SB
A20	XD2	IO	X-BUS DDR-DRAM Data	XD2		Hi-Z	SB
B19	XD3	IO	X-BUS DDR-DRAM Data	XD3		Hi-Z	SB
A19	XD4	IO	X-BUS DDR-DRAM Data	XD4		Hi-Z	SB
B18	XD5	IO	X-BUS DDR-DRAM Data	XD5		Hi-Z	SB
A18	XD6	IO	X-BUS DDR-DRAM Data	XD6		Hi-Z	SB
B16	XD7	IO	X-BUS DDR-DRAM Data	XD7		Hi-Z	SB
A16	XD8	IO	X-BUS DDR-DRAM Data	XD8		Hi-Z	SB
B15	XD9	IO	X-BUS DDR-DRAM Data	XD9		Hi-Z	SB
A15	XD10	IO	X-BUS DDR-DRAM Data	XD10		Hi-Z	SB
B14	XD11	IO	X-BUS DDR-DRAM Data	XD11		Hi-Z	SB
A14	XD12	IO	X-BUS DDR-DRAM Data	XD12		Hi-Z	SB
B13	XD13	IO	X-BUS DDR-DRAM Data	XD13		Hi-Z	SB
A13	XD14	IO	X-BUS DDR-DRAM Data	XD14		Hi-Z	SB
B11	XD15	IO	X-BUS DDR-DRAM Data	XD15		Hi-Z	SB
P18	YA0/GPIOC0	IO	Y-BUS DDR-DRAM Address : YA[12:0] (DDR DRAM Address bus for Y-Bank) outputs the memory address of the DDR for Y-bank / GPIO	YA0		GI	SB
U22	YA1/GPIOC1	IO	Y-BUS DDR-DRAM Address / GPIO	YA1		GI	SB
T20	YA2/GPIOC2	IO	Y-BUS DDR-DRAM Address / GPIO	YA2		GI	SB
T19	YA3/GPIOC3	IO	Y-BUS DDR-DRAM Address / GPIO	YA3		GI	SB
F20	YA4/GPIOC4	IO	Y-BUS DDR-DRAM Address / GPIO	YA4		GI	SB
H19	YA5/GPIOC5	IO	Y-BUS DDR-DRAM Address / GPIO	YA5		GI	SB

E22	YA6/GPIOC6	IO	Y-BUS DDR-DRAM Address / GPIO	YA6		GI	SB
J20	YA7/GPIOC7	IO	Y-BUS DDR-DRAM Address / GPIO	YA7		GI	SB
H20	YA8/GPIOC8	IO	Y-BUS DDR-DRAM Address / GPIO	YA8		GI	SB
J19	YA9/GPIOC9	IO	Y-BUS DDR-DRAM Address / GPIO	YA9		GI	SB
R20	YA10/GPIOC10	IO	Y-BUS DDR-DRAM Address / GPIO	YA10		GI	SB
K17	YA11/GPIOC11	IO	Y-BUS DDR-DRAM Address / GPIO	YA11		GI	SB
K18	YA12/GPIOC12	IO	Y-BUS DDR-DRAM Address / GPIO	YA12		GI	SB
N18	YBA0/GPIOC13	IO	Y-BUS DDR-DRAM Bank Address : YBA[1:0] (DDR DRAM Bank Address for Y-Bank) output bank address for DDR-DRAM / GPIO	YBA0		GI	SB
R19	YBA1/GPIOC14	IO	Y-BUS Bank Address / GPIO	YBA1		GI	SB
K20	YCK	O	Y-BUS DDR-DRAM Clock (DDR DRAM Clock for Y-Bank)	YCK		F1	SD
L19	YnCK	O	Y-BUS DDR-DRAM Inverting Clock (DDR DRAM Inverted Clock for Y-Bank)	YnCK		F1	SD
L17	YCKE	O	Y-BUS DDR-DRAM Clock Enable	YCKE		F1	SO
P20	YnCS/GPIOB6	IO	Y-BUS DDR-DRAM Chip Select / GPIO	YnCS		GI	SB
M18	YnRAS/GPIOB7	IO	Y-BUS DDR-DRAM RAS (Row address strobe) / GPIO	YnRAS		GI	SB
N20	YnCAS/GPIOB8	IO	Y-BUS DDR-DRAM CAS (Column address strobe) / GPIO	YnCAS		GI	SB
M19	YnWE/GPIOB9	IO	Y-BUS DDR-DRAM Write Enable / GPIO	YnWE		GI	SB
U20	YLDM0/GPIOB10	IO	Y-BUS DDR-DRAM Data Mask / GPIO	YLDM0		GI	SB
M17	YUDM0/GPIOB11	IO	Y-BUS DDR-DRAM Data Mask / GPIO	YUDM0		GI	SB
N19	YLDM1/GPIOB12	IO	Y-BUS DDR-DRAM Data Mask / GPIO	YLDM1		GI	SB
G20	YUDM1/GPIOB13	IO	Y-BUS DDR-DRAM Data Mask / GPIO	YUDM1		GI	SB
U21	YLDQS0	IO	Y-BUS DDR-DRAM Data Strobe	YLDQS0		F1	SB
L18	YUDQS0	IO	Y-BUS DDR-DRAM Data Strobe	YUDQS0		F1	SB
P19	YLDQS1	IO	Y-BUS DDR-DRAM Data Strobe	YLDQS1		F1	SB
E21	YUDQS1	IO	Y-BUS DDR-DRAM Data Strobe	YUDQS1		F1	SB
Y22	YD0	IO	Y-BUS DDR-DRAM Data	YD0		Hi-Z	SB
W21	YD1	IO	Y-BUS DDR-DRAM Data	YD1		Hi-Z	SB
W22	YD2	IO	Y-BUS DDR-DRAM Data	YD2		Hi-Z	SB
V21	YD3	IO	Y-BUS DDR-DRAM Data	YD3		Hi-Z	SB
V22	YD4	IO	Y-BUS DDR-DRAM Data	YD4		Hi-Z	SB
T21	YD5	IO	Y-BUS DDR-DRAM Data	YD5		Hi-Z	SB
T22	YD6	IO	Y-BUS DDR-DRAM Data	YD6		Hi-Z	SB
R21	YD7	IO	Y-BUS DDR-DRAM Data	YD7		Hi-Z	SB
R22	YD8	IO	Y-BUS DDR-DRAM Data	YD8		Hi-Z	SB
P21	YD9	IO	Y-BUS DDR-DRAM Data	YD9		Hi-Z	SB
P22	YD10	IO	Y-BUS DDR-DRAM Data	YD10		Hi-Z	SB
N21	YD11	IO	Y-BUS DDR-DRAM Data	YD11		Hi-Z	SB
N22	YD12	IO	Y-BUS DDR-DRAM Data	YD12		Hi-Z	SB
M21	YD13	IO	Y-BUS DDR-DRAM Data	YD13		Hi-Z	SB
M22	YD14	IO	Y-BUS DDR-DRAM Data	YD14		Hi-Z	SB
L21	YD15	IO	Y-BUS DDR-DRAM Data	YD15		Hi-Z	SB
L22	YD16	IO	Y-BUS DDR-DRAM Data	YD16		Hi-Z	SB
K21	YD17	IO	Y-BUS DDR-DRAM Data	YD17		Hi-Z	SB
K22	YD18	IO	Y-BUS DDR-DRAM Data	YD18		Hi-Z	SB
J21	YD19	IO	Y-BUS DDR-DRAM Data	YD19		Hi-Z	SB

J22	YD20	IO	Y-BUS DDR-DRAM Data	YD20		Hi-Z	SB
H21	YD21	IO	Y-BUS DDR-DRAM Data	YD21		Hi-Z	SB
H22	YD22	IO	Y-BUS DDR-DRAM Data	YD22		Hi-Z	SB
G21	YD23	IO	Y-BUS DDR-DRAM Data	YD23		Hi-Z	SB
G22	YD24	IO	Y-BUS DDR-DRAM Data	YD24		Hi-Z	SB
F22	YD25	IO	Y-BUS DDR-DRAM Data	YD25		Hi-Z	SB
F21	YD26	IO	Y-BUS DDR-DRAM Data	YD26		Hi-Z	SB
D22	YD27	IO	Y-BUS DDR-DRAM Data	YD27		Hi-Z	SB
D21	YD28	IO	Y-BUS DDR-DRAM Data	YD28		Hi-Z	SB
C22	YD29	IO	Y-BUS DDR-DRAM Data	YD29		Hi-Z	SB
C21	YD30	IO	Y-BUS DDR-DRAM Data	YD30		Hi-Z	SB
B22	YD31	IO	Y-BUS DDR-DRAM Data	YD31		Hi-Z	SB
Y8	ZA0	O	Z-BUS DDR-DRAM Address	ZA0		F1	SO
V8	ZA1	O	Z-BUS DDR-DRAM Address	ZA1		F1	SO
W8	ZA2	O	Z-BUS DDR-DRAM Address	ZA2		F1	SO
AB3	ZA3	O	Z-BUS DDR-DRAM Address	ZA3		F1	SO
W16	ZA4	O	Z-BUS DDR-DRAM Address	ZA4		F1	SO
U15	ZA5	O	Z-BUS DDR-DRAM Address	ZA5		F1	SO
W15	ZA6	O	Z-BUS DDR-DRAM Address	ZA6		F1	SO
W14	ZA7	O	Z-BUS DDR-DRAM Address	ZA7		F1	SO
U14	ZA8	O	Z-BUS DDR-DRAM Address	ZA8		F1	SO
U13	ZA9	O	Z-BUS DDR-DRAM Address	ZA9		F1	SO
V9	ZA10	O	Z-BUS DDR-DRAM Address	ZA10		F1	SO
Y14	ZA11	O	Z-BUS DDR-DRAM Address	ZA11		F1	SO
V13	ZA12	O	Z-BUS DDR-DRAM Address	ZA12		F1	SO
Y9	ZBA0	O	Z-BUS DDR-DRAM Bank Address	ZBA0		F1	SO
U9	ZBA1	O	Z-BUS DDR-DRAM Bank Address	ZBA1		F1	SO
U12	ZCK	O	Z-BUS DDR-DRAM Clock	ZCK		F1	SD
W13	ZnCK	O	Z-BUS DDR-DRAM Inverting Clock	ZnCK		F1	SD
V12	ZCKE	O	Z-BUS DDR-DRAM Clock Enable	ZCKE		F1	SO
Y10	ZnCS0	O	Z-BUS DDR-DRAM Chip Select	ZnCS0		F1	SO
V10	ZnCS1	O	Z-BUS DDR-DRAM Chip Select	ZnCS1		F1	SO
W10	ZnRAS	O	Z-BUS DDR-DRAM RAS	ZnRAS		F1	SO
W11	ZnCAS	O	Z-BUS DDR-DRAM CAS	ZnCAS		F1	SO
U11	ZnWE	O	Z-BUS DDR-DRAM Write Enable	ZnWE		F1	SO
Y7	ZLDM0	O	Z-BUS Data Mask for ZD[7:0]	ZLDM0		F1	SO
Y12	ZUDM0	O	Z-BUS Data Mask for ZD[15:8]	ZUDM0		F1	SO
Y11	ZLDM1	O	Z-BUS Data Mask for ZD[23:16]	ZLDM1		F1	SO
V15	ZUDM1	O	Z-BUS Data Mask for ZD[31:24]	ZUDM1		F1	SO
Y6	ZLDQS0	IO	Z-BUS Data Strobe for ZD[7:0]	ZLDQS0		F1	SB
V11	ZUDQS0	IO	Z-BUS Data Strobe for ZD[15:8]	ZUDQS0		F1	SB
T11	ZLDQS1	IO	Z-BUS Data Strobe for ZD[23:16]	ZLDQS1		F1	SB
Y16	ZUDQS1	IO	Z-BUS Data Strobe for ZD[31:24]	ZUDQS1		F1	SB
AA4	ZD0	IO	Z-BUS DDR-DRAM Data	ZD0		F1	SB
AB4	ZD1	IO	Z-BUS DDR-DRAM Data	ZD1		F1	SB
AA5	ZD2	IO	Z-BUS DDR-DRAM Data	ZD2		F1	SB

AB5	ZD3	IO	Z-BUS DDR-DRAM Data	ZD3		F1	SB
AA6	ZD4	IO	Z-BUS DDR-DRAM Data	ZD4		F1	SB
AB6	ZD5	IO	Z-BUS DDR-DRAM Data	ZD5		F1	SB
AA7	ZD6	IO	Z-BUS DDR-DRAM Data	ZD6		F1	SB
AB7	ZD7	IO	Z-BUS DDR-DRAM Data	ZD7		F1	SB
AB8	ZD8	IO	Z-BUS DDR-DRAM Data	ZD8		F1	SB
AA8	ZD9	IO	Z-BUS DDR-DRAM Data	ZD9		F1	SB
AB9	ZD10	IO	Z-BUS DDR-DRAM Data	ZD10		F1	SB
AA9	ZD11	IO	Z-BUS DDR-DRAM Data	ZD11		F1	SB
AB10	ZD12	IO	Z-BUS DDR-DRAM Data	ZD12		F1	SB
AA10	ZD13	IO	Z-BUS DDR-DRAM Data	ZD13		F1	SB
AB11	ZD14	IO	Z-BUS DDR-DRAM Data	ZD14		F1	SB
AA11	ZD15	IO	Z-BUS DDR-DRAM Data	ZD15		F1	SB
AB12	ZD16	IO	Z-BUS DDR-DRAM Data	ZD16		F1	SB
AA12	ZD17	IO	Z-BUS DDR-DRAM Data	ZD17		F1	SB
AB13	ZD18	IO	Z-BUS DDR-DRAM Data	ZD18		F1	SB
AA13	ZD19	IO	Z-BUS DDR-DRAM Data	ZD19		F1	SB
AB14	ZD20	IO	Z-BUS DDR-DRAM Data	ZD20		F1	SB
AA14	ZD21	IO	Z-BUS DDR-DRAM Data	ZD21		F1	SB
AB15	ZD22	IO	Z-BUS DDR-DRAM Data	ZD22		F1	SB
AA15	ZD23	IO	Z-BUS DDR-DRAM Data	ZD23		F1	SB
AB16	ZD24	IO	Z-BUS DDR-DRAM Data	ZD24		F1	SB
AA16	ZD25	IO	Z-BUS DDR-DRAM Data	ZD25		F1	SB
AB17	ZD26	IO	Z-BUS DDR-DRAM Data	ZD26		F1	SB
AA17	ZD27	IO	Z-BUS DDR-DRAM Data	ZD27		F1	SB
AB18	ZD28	IO	Z-BUS DDR-DRAM Data	ZD28		F1	SB
AA18	ZD29	IO	Z-BUS DDR-DRAM Data	ZD29		F1	SB
AB19	ZD30	IO	Z-BUS DDR-DRAM Data	ZD30		F1	SB
AA19	ZD31	IO	Z-BUS DDR-DRAM Data	ZD31		F1	SB
Y1	WA0_WnDQM0 /GPIOE0	IO	W-BUS Address & ByteEnb[0] /GPIO : WA[25:0] outputs the memory address of the corresponding static banks. If the bus width of static bank is 16bit. Especially WA0 is used to be Lower byte enable.	WA0_WnDQM0		Note1	PB12
V2	WA1/GPIOE1	IO	W-BUS Address / GPIO	WA1		Note1	PB12
W1	WA2/GPIOE2	IO	W-BUS Address / GPIO	WA2		Note1	PB12
R5	WA3/GPIOE3	IO	W-BUS Address / GPIO	WA3		Note1	PB12
R2	WA4/GPIOE4	IO	W-BUS Address / GPIO	WA4		Note1	PB12
P5	WA5/GPIOE5	IO	W-BUS Address / GPIO	WA5		Note1	PB12
R1	WA6/GPIOE6	IO	W-BUS Address / GPIO	WA6		Note1	PB12
P2	WA7/GPIOE7	IO	W-BUS Address / GPIO	WA7		Note1	PB12
N3	WA8/GPIOE8	IO	W-BUS Address / GPIO	WA8		Note1	PB12
N2	WA9/GPIOE9	IO	W-BUS Address / GPIO	WA9		Note1	PB12
M4	WA10/GPIOE10	IO	W-BUS Address / GPIO	WA10		Note1	PB12
L1	WA11/GPIOE11	IO	W-BUS Address / GPIO	WA11		Note1	PB12
K1	WA12/GPIOE12	IO	W-BUS Address / GPIO	WA12		Note1	PB12
K2	WA13/WDREQ1 /GPIOE13	IO	W-BUS Address / DMA Request #1 / GPIO WA[13] can be used to be external DMA Request. Refer to memory controller part.	WA13	WDREQ1	Note1	PB12

H1	WA14/WDACK1 /GPIOE14	IO	W-BUS Address / DMA Acknowledge #1 / GPIO WA[14] can be used to be external DMA acknowledge	WA14	WDACK1	Note1	PB12
J2	WA15/IVSYNC /GPIOE15	IO	W-BUS Address / ISP Vertical Sync Input / GPIO WA[25:15] can be used for capturing external video.	WA15	IVSYNC	Note1	PB12
H2	WA16/IHSYNC/GPIOF0	IO	W-BUS Address / ISP Horizontal Sync Input / GPIO	WA16	IHSYNC	Note1	PB12
G2	WA17/ICLK/GPIOF1	O	W-BUS Address / ISP Video Input,Output Clock / GPIO If WA[17] is used to be video capture clock, ICLK can be video input clock and video output clock	WA17	ICLK	Note1	PB12
J4	WA18/ID0/GPIOF2	O	W-BUS Address / ISP Video Input Data / GPIO ID[7:0], WA[25:18] can be used to be video input data bus.	WA18	ID0	Note1	PB12
H3	WA19/ID1/GPIOF3	O	W-BUS Address / ISP Video Input Data / GPIO	WA19	ID1	Note1	PB12
T1	WA20/ID2/GPIOF4	IO	W-BUS Address / ISP Video Input Data / GPIO	WA20	ID2	Note1	PB12
P4	WA21/ID3/GPIOF5	IO	W-BUS Address / ISP Video Input Data / GPIO	WA21	ID3	Note1	PB12
N5	WA22/ID4/GPIOF6	IO	W-BUS Address / ISP Video Input Data / GPIO	WA22	ID4	Note1	PB12
N4	WA23/ID5/GPIOF7	IO	W-BUS Address / ISP Video Input Data / GPIO	WA23	ID5	Note1	PB12
M3	WA24/ID6/GPIOF8	IO	W-BUS Address / ISP Video Input Data / GPIO	WA24	ID6	Note1	PB12
M1	WA25/ID7/GPIOF9	IO	W-BUS Address / ISP Video Input Data / GPIO	WA25	ID7	Note1	PB12
V5	WD0/GPIOG0	IO	W-BUS Data / GPIO	WD0		Note1	PB12
W4	WD1/GPIOG1	IO	W-BUS Data / GPIO	WD1		Note1	PB12
V4	WD2/GPIOG2	IO	W-BUS Data / GPIO	WD2		Note1	PB12
U5	WD3/GPIOG3	IO	W-BUS Data / GPIO	WD3		Note1	PB12
Y2	WD4/GPIOG4	IO	W-BUS Data / GPIO	WD4		Note1	PB12
AA1	WD5/GPIOG5	IO	W-BUS Data / GPIO	WD5		Note1	PB12
T5	WD6/GPIOG6	IO	W-BUS Data / GPIO	WD6		Note1	PB12
W2	WD7/GPIOG7	IO	W-BUS Data / GPIO	WD7		Note1	PB12
Y3	WD8/VD16/GPIOG8	IO	W-BUS Data / Video Output Data /GPIO If 24bit digital video output is required. WD[15:8] can be used to be Upper byte (VD[23:16]).	WD8	VD16	Note1	PB12
W3	WD9/VD17/GPIOG9	IO	W-BUS Data / Video Output Data /GPIO	WD9	VD17	Note1	PB12
V3	WD10/VD18/GPIOG10	IO	W-BUS Data / Video Output Data /GPIO	WD10	VD18	Note1	PB12
U4	WD11/VD19/GPIOG11	IO	W-BUS Data / Video Output Data /GPIO	WD11	VD19	Note1	PB12
U3	WD12/VD20/GPIOG12	IO	W-BUS Data / Video Output Data /GPIO	WD12	VD20	Note1	PB12
T4	WD13/VD21/GPIOG13	IO	W-BUS Data / Video Output Data /GPIO	WD13	VD21	Note1	PB12
U2	WD14/VD22/GPIOG14	IO	W-BUS Data / Video Output Data /GPIO	WD14	VD22	Note1	PB12
T2	WD15/VD23/GPIOG15	IO	W-BUS Data / Video Output Data /GPIO	WD15	VD23	Note1	PB12
C5	WnCS0/WnNCS6	O	Static Memory Bank0 Chip Select / NAND Bank6 It is used to be Static memory bank0 chip select or chip select for NAND bank6. its function is selected by cpu, Reset default function is WnCS0 If ROM/NOR flash booting is required. The chip selection of boot memory is connected to it.	WnCS0_WnNCS6		F1	P8
B4	WnCS1/WnNCS7	O	Static Memory Bank1 Chip Select / NAND Bank7	WnCS1_WnNCS7		F1	P8
B3	WnNCS0/WnCS2	O	NAND Bank0 / Static Memory Bank2 Chip Select If NAND Boot is required. The Chip selection of NAND Boot memory must to be connected to it.	WnNCS0_WnCS2		F1	P8
A2	WnNCS1/WnCS3	O	NAND Bank1 / Static Memory Bank3 Chip Select	WnNCS1_WnCS3		F1	P8
E6	WnPCS0/WnCS4/WnNCS2	O	PCMCIA Chip Select 0 / Static Bank4 / NAND Bank2	WnPCS0_WnCS4_WnNCS2		F1	P8
E7	WnPCS1/WnCS5/WnNCS3	O	PCMCIA Chip Select 1 / Static Bank5 / NAND Bank3	WnPCS1_WnCS5_WnNCS3		F1	P8
B7	WnICS0/WnNCS4	O	IDE Controller Chip select /NAND Chip Select Bank4	WnICS0_WnNCS4		F1	P8

			In IDE Chip select, It is used to select IDE card, and enable the High byte lane.	4			
C7	WnICS1/WnNCS5	O	IDE Controller Chip select /NAND Chip Select Bank5 In IDE Chip select, It is used to select IDE card, and enable the Low byte land.	WnICS1_WnNCS5		F1	P8
A3	WnWE	O	Static/PCMCIA/IDE Write Enable	WnWE		F1	P8
C6	WnOE	O	Static/PCMCIA/IDE Read Enable	WnOE		F1	P8
F6	WnDQM1	O	High Byte Enable, ByteEnb[1]	WnDQM1		F1	P8
D6	WnNOE	O	NAND Read Enable	WnNOE		F1	P8
A4	WnNWE	O	NAND Write Enable	WnNWE		F1	P8
B6	WRDnWR	O	Buffer Direction Control if static memory is connected through buffer, It is used to be buffer controller signal of external buffer.	WRDnWR		F1	P8
P3	WnPior_CLE/GPIOH0	IO	PCMCIA,IDE I/O Read Enable/ NAND CLE (Command Latch Enable) / GPIO	WnPior_CLE		F1	PB12
P1	WnPiow_ALE/GPIOH1	IO	PCMCIA,IDE I/O Read Enable / NAND ALE (Address Latch Enable) / GPIO	WnPiow_ALE		F1	PB12
M5	WSKTSEL/GPIOH2	IO	PCMCIA Socket Select / GPIO It is an output and is used by external steering logic to route control, address and data signals to one of the two PCMCIA sockets	WSKTSEL		F1	PB12
N1	WnPREG/GPIOH3	IO	PCMCIA nPREG / GPIO This signal is an output and indicates that, on a memory transaction, the target address is attribute space. This signal has same timing as address	WnPREG		F1	PB12
U1	WnIOIS16/GPIOH4	IO	PCMCIA IOIS16 Signal / GPIO Is is an input and is an acknowledgement from the PCMCIA card that the current address is a valid 16bit wide I/O address	WnIOIS16		F1	PB12
R3	WRnB/GPIOH5	IO	Static/NAND/IDE/PCMCIA Wait Control / GPIO Is is an input and is driven low by the PCMCIA/CF/IDE/Static Memory to extend the length of the transfers to/from application processor	WRnB		F1	PB12
L5	WDREQ0/GPIOH6	IO	External DMA Request / GPIO	WDREQ0		F1	PB12
K5	WDACK0/GPIOH7	IO	External DMA Acknowledge / GPIO	WDACK0		F1	PB12
B5	EIRQ0	IO	External Interrupt 0	EIRQ0		F1	PI
A6	EIRQ1	IO	External Interrupt 1	EIRQ1		F1	PI
F1	EIRQ2/GPIOK12	IO	External Interrupt 2 / GPIO	EIRQ2		GI	PB12
H5	EIRQ3/CTS1/GPIOK13	IO	External Interrupt 3 / UART1 CTS / GPIO	EIRQ3	CTS1	GI	PB12
W5	USBD0+	IO	USB0 Port : D+ : USB Host Data port	USBD+		F1	US
AB2	USBD0-	IO	USB0 Port : D- : USB Host Data port	USBD-		F1	US
AB1	USBD1+	IO	USB1 Port : D+ : USB Host Data port / USB Device Data port It can be used to be USB Host data or USB Device data, Its function is decided by cpu, refer to GPIO Misc. register.	USBD+		F1	US
AA2	USBD1-	IO	USB1 Port : D- : USB Host Data port / USB Device Data port	USBD-		F1	US
U6	SCL	IO	I2C Clock	SCL		F1	OD
T3	SDA	IO	I2C Data	SDA		F1	OD
K3	SDDAT0/ GPIOH8	IO	MMC-SD Data bit 0 / GPIO	SDDAT0		GI	PB12
L3	SDDAT1/ GPIOH9	IO	MMC-SD Data bit 1 / GPIO	SDDAT1		GI	PB12
J1	SDDAT2/GPIOH10	IO	MMC-SD Data bit 2 / GPIO	SDDAT2		GI	PB12
L4	SDDAT3/GPIOH11	IO	MMC-SD Data bit 3 / GPIO	SDDAT3		GI	PB12
L2	SDCLK/GPIOH12	O	MMC-SD Clock / GPIO	SDCLK		GI	PB12
M2	SDCMD/ GPIOH13	IO	MMC-SD Command / GPIO	SDCMD		GI	PB12
V6	PWM0/RTS1/GPIOH14	IO	PWM0 Output / UART1 RTS / GPIO	PWM0	RTS1	GI	PB12
Y4	PWM1/GPIOH15	IO	PWM1 Output / GPIO	PWM1		GI	PB12
C3	ADCIN0	AI	ADC Channel 0 Input	ADCIN0		F1	GP

D4	ADCIN1	AI	ADC Channel 1 linput	ADCIN1		F1	GP
E5	ADCIN2	AI	ADC Channel 2 linput	ADCIN2		F1	GP
F4	ADCIN3	AI	ADC Channel 3 linput	ADCIN3		F1	GP
F5	AREF	AI	ADC Voltage Reference	AREF		F1	GP
E20	NGPIO0/DTR1/ID0 /GPIOI0	IO	NGPIO / UART1 DTR / Video In / GPIO NGPIO[15:0] : When interfacing with external CPU, Its bus is used to get the data which external cpu send to VR3D. ID[7:0] : When capturing the video, Its bus is used as Video input data bus.	DTR1	ID0	GI	PB12
G19	NGPIO1/DSR1/ID1 /GPIOI1	IO	NGPIO / UART1 DSR / Video In / GPIO	DSR1	ID1	GI	PB12
A22	NGPIO2/DCD1/ID2 /GPIOI2	IO	NGPIO / UART1 DCD / Video In / GPIO	DCD1	ID2	GI	PB12
F19	NGPIO3/RTS1/ID3 /GPIOI3	IO	NGPIO / UART1 RTS / Video In / GPIO	RTS1	ID3	GI	PB12
D20	NGPIO4/CTS1/ID4 /GPIOI4	IO	NGPIO / UART1 CTS / Video In / GPIO	CTS1	ID4	GI	PB12
E18	NGPIO5/RI1/ID5 /GPIOI5	IO	NGPIO / UART1 RI / Video In / GPIO	RI1	ID5	GI	PB12
E19	NGPIO6/ID6/GPIOI6	IO	NGPIO / Video In / GPIO		ID6	GI	PB12
D19	NGPIO7/ID7/GPIOI7	IO	NGPIO / Video In / GPIO		ID7	GI	PB12
C20	NGPIO8/DTR0/IVSYNC /GPIOI8	IO	NGPIO / UART0 DTR / Video In Vertical Sync / GPIO	DTR0	IVSYNC	GI	PB12
C19	NGPIO9/DSR0/IHSYNC /GPIOI9	IO	NGPIO / UART0 DSR / Video In Horizontal Sync / GPIO	DSR0	IHSYNC	GI	PB12
D17	NGPIO10/DCD0/ICLK /GPIOI10	IO	NGPIO / UART0 DCD / Video In Clock / GPIO	DCD0	ICLK	GI	PB12
B21	NGPIO11/RTS0 /SSPRXD/GPIOI11	IO	NGPIO / UART0 RTS / SSP Receiver Port / GPIO	RTS0	SSPRXD	GI	PB12
D18	NGPIO12/CTS0 /SSPCLK/GPIOI12	IO	NGPIO / UART0 CTS / SSP Clock / GPIO	CTS0	SSPCLK	GI	PB12
G3	NGPIO13/RI0 /SSPFRM/GPIOI13	IO	NGPIO / UART0 RI / SSP FRM / GPIO	RI0	SSPFRM	GI	PB12
E1	NGPIO14/RX1 /GPIOI14	IO	NGPIO / UART1 RX / GPIO	RX1		GI	PB12
F2	NGPIO15/TX1/GPIOI15	IO	NGPIO / UART1 TX / / GPIO	TX1		GI	PB12
V1	NGPIOLAT/SSPTXD /GPIOK0	IO	GPIO Latch Enable / SSP Transmitter port / GPIO	NGPIOLAT	SSPTXD	GI	PB12
H4	ABCK/BCK/GPIOK1	IO	AC97 BCK / I2S BCK / GPIO	ABCK	BCK	GI	PB12
G1	ASYNCR/LRCK/GPIOK2	IO	AC97 Sync / I2S LRCK / GPIO	ASYNCR	LRCK	GI	PB12
J3	ADATIN/SDAT2 /GPIOK3	IO	AC97 Data In / I2S Data Channel 2 / GPIO	ADATIN	SDAT2	GI	PB12
G4	nARST/SDAT1/GPIOK4	IO	AC97 Reset / I2S Data Channel 1 / GPIO	nARST	SDAT1	GI	PB12
K4	ADATOUT/SDAT0/GPIOK5	IO	AC97 Data Output / I2S Data Channel 2 / GPIO	ADATOUT	SDAT0	GI	PB12
V16	VD0/GPIOJ0/CFG0	IO	Digital Video Data Output / GPIO / CfgShadow VD[15:0] is Digital video output bus, It is used as reset configuration parameter to determine operation mode. It must be tied to GND to VDD(3.3v) through pull-up/pull-down resistor. CfgShadow : It determine to enable/disable memory shadow. 0 : Non Shadow (Base 0 : Bank-W), Refer to memory controller. 1 : Shadow (Base 0 : Bank-z)		VD0	GI	PB12
Y18	VD1/GPIOJ1/CFG1	IO	Digital Video Data Output / GPIO / CfgBootType[0] CfgBootType[1:0] : It determines the booting method. 00 : ROM/NOR Flash 10 : NAND Boot		VD1	GI	PB12

W18	VD2/GPIOJ2/CFG2	IO	Digital Video Data Output / GPIO / CfgBootType[1]		VD2	GI	PB12
AA20	VD3/GPIOJ3/CFG3	IO	Digital Video Data Output / GPIO / CfgNFBSIZE, CfgSDRLATx CfgNFBSIZE : Reserved, Its name and function is changed to CfgSDRLATx in VR3Dr2 CfgSDRLATx : Its determines the CAS latency of DDR-DRAM 0 : 2 Cycle 1 : 3 Cycle		VD3	GI	PB12
AB21	VD4/GPIOJ4/CFG4	IO	Digital Video Data Output / GPIO / CfgNFBUSWidth CfgNFBUSWidth : It determines the bus width of NAND Flash which is used as NAND booting memory. 0 : 8bit NAND Flash 1 : 16bit NAND Flash		VD4	GI	PB12
Y19	VD5/GPIOJ5/CFG5	IO	Digital Video Data Output / GPIO / CfgNFTYPE CfgNFTYPE : It determines the NAND Address Type, 0 : 3 address type 1 : 4 address type		VD5	GI	PB12
V17	VD6/GPIOJ6/CFG6	IO	Digital Video Data Output / GPIO / CfgSR0BUSWidth CfgSR0BUSWidth : It determines the bus width of Static memory (ROM/NOR flash) which is controlled by nSCS0 0 : 8bit 1 : 16bit		VD6	GI	PB12
AB22	VD7/GPIOJ7/CFG7	IO	Digital Video Data Output / GPIO / CfgSDRTYPE CfgSDRTYPE : Reserved : It must be tied to VDD through pull-up resistor		VD7	GI	PB12
V18	VD8/GPIOJ8/CFG8	IO	Digital Video Data Output/ GPIO / CfgSDRBUSBW[0] CfgSDRBUSBW[1:0] : It determines the bus width of Bank-Z 10 : 16bit 11 : 32bit If CfgShadow is 1, The Reset Configuration be related to DDR-DRAM must be set.		VD8	GI	PB12
AA21	VD9/GPIOJ9/CFG9	IO	Digital Video Data Output / GPIO / CfgSDRBUSBW[1]		VD9	GI	PB12
W19	VD10/GPIOJ10/CFG10	IO	Digital Video Data Output / GPIO / CfgSDRBW[0] CfgSDRBW[1:0] : It determines the data bit width of a DDR-DRAM component. 00 : Reserved 01 : 8bit 10 : 16bit 11 : 32bit		VD10	GI	PB12
Y20	VD11/GPIOJ11/CFG11	IO	Digital Video Data Output / GPIO / CfgSDRBW[1]		VD11	GI	PB12
Y21	VD12/GPIOJ12/CFG12	IO	Digital Video Data Output / GPIO / CfgSDRCAP[0] CfgSDRCAP[1:0] : It determines the capacity of a DDR-DRAM component. 00 : 64Mbit 01 : 128Mbit 10 : 256Mbit 11 : 512Mbit		VD12	GI	PB12
W20	VD13/GPIOJ13/CFG13	IO	Digital Video Data Output / GPIO / CfgSDRCAP[1]		VD13	GI	PB12
AA22	VD14/GPIOJ14	IO	Digital Video Data Output / GPIO		VD14	GI	PB12
V19	VD15/GPIOJ15	IO	Digital Video Data Output / GPIO		VD15	GI	PB12
W17	VSYNCGPIOK6	IO	Vertical Sync Output / GPIO	VSYNCGPIOK6		GI	PB12
AB20	HSYNCGPIOK7	IO	Horizontal Sync Output / GPIO	HSYNCGPIOK7		GI	PB12
W6	CSYNCGPIOK8	IO	Composite Sync Output / GPIO	CSYNCGPIOK8		GI	PB12
V20	VCLKGPIOK9	IO	Video Clock Input/Output / GPIO	VCLKGPIOK9		GI	PB12
F3	RX0GPIOK10	IO	UART0 RX / GPIO	RX0GPIOK10		F1	PB12
G5	TX0GPIOK11	IO	UART0 TX / GPIO	TX0GPIOK11		F1	PB12
C4	ADCVDD3D	P	3.3V for ADC Digital Block				
E4	ADCVDDA	P	3.3V for ADC Analog Block				
D5	ADCVSS3D	P	Ground for ADC Digital Block				
E3	ADCVSSA	P	Ground for ADC Analog Block				
B2	PLLVDDA	P	1.8V for A-PLL				
D2	PLLVDDF	P	1.8V for F-PLL				
D3	PLLVDDU	P	1.8V for U-PLL				
C2	PLLVSSA	P	Ground for A-PLL				
E2	PLLVSSF	P	Ground for F-PLL				
A1	PLLVSU	P	Ground for U-PLL				

A5	VDD3A	P	3.3V for I/O PAD		
F9, F10, F16, G18, J6, L6, P6, T18, U17, Y5, Y17, D10	VDD1	P	1.8V for Core		
AA3, F18, J5, M6, R4, U19	VDD3	P	3.3V for I/O PAD		
E17, K6, U18	VDD1A	P	1.8V for Core		
A11	XREF	P	Bank X Voltage Reference, DDR:1.25V		
L20	YREF	P	Bank Y Voltage Reference, DDR:1.25V		
W12	ZREF	P	Bank Z Voltage Reference, DDR:1.25V		
C18	VDD1X	P	Bank X 1.8V		
J17	VDD1Y	P	Bank X 1.8V		
W9	VDD1Z	P	Bank X 1.8V		
B12, C15, F11	VDD2X	P	Bank X PAD Power, DDR:2.5V		
H18, J18, K19, M20, P17, R18	VDD2Y	P	Bank Y PAD Power, DDR:2.5V		
U10, V7, V14, W7, Y13, Y15	VDD2Z	P	Bank Z PAD Power, DDR:2.5V		
F7, F8, F15, F17G6, G10, G11, G12, G13, G17, H6, H17, K7, K16, L7, L16, M7, M16, N6, N7, N16, N17, R6, R17, T6, T10, T12, T13, T17, U7, U8, U16	GND	P	Ground		

I/O Type	Description
PB8	Bi-Directional pad, LVCMOS Schmitt-trigger, 100Kohm pull-up resistor with control, tri-state, Io=8mA
PB12	Bi-Directional pad, LVCMOS Schmitt-trigger, 100Kohm pull-up resistor with control, tri-state, Io=12mA
PI	Input pad, LVCMOS Schmitt-trigger
M26	Oscillator cell with enable and feedback resistor
OD	Open Drain PAD
GP	Analog PAD
SO	SSTL output pad, Io=24mA
SD	SSTL Differential pad, Io=24mA
SB	SSTL Bi-Directional pad, Io=24mA
US	USB Pad

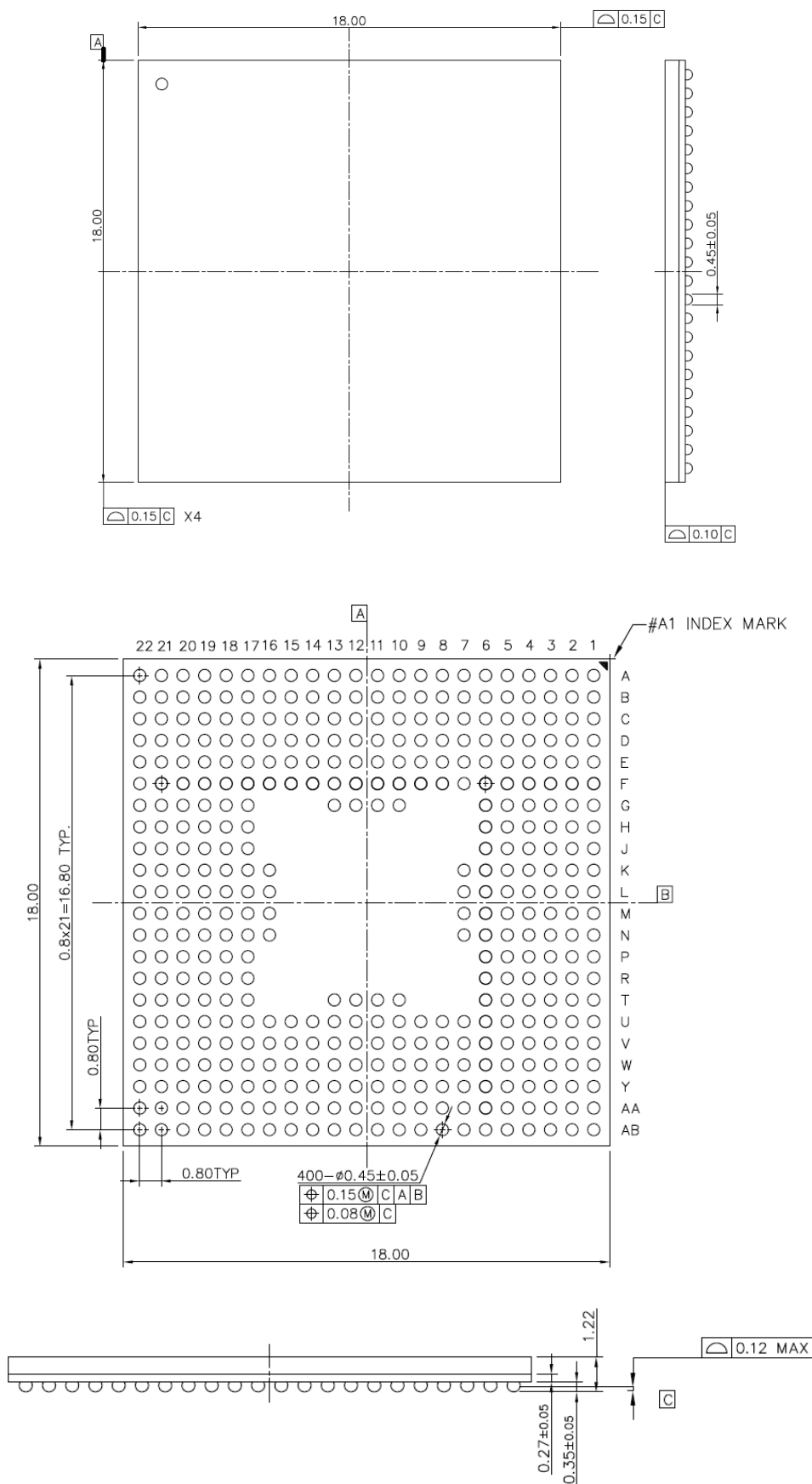
Note1 : It is determined by reset configuration, Refer to memory controller part

3.2 Ball Location

	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	
A	NGPIO2	XD0	XD2	XD4	XD6	XA10	XD8	XD10	XD12	XD14	XA9	XREF	PWRXTO	PWRXTI	TESTEN	TDO	EIRQ1	VDD3	WnNWE	WnWE	WnNCS1	PLLVS	A
B	YD31	NGPIO11	XD1	XD3	XD5	XBA0	XD7	XD9	XD11	XD13	VDD2x	XD15	XnCK	TMS	nVDDF	WnICS0	WRDnWR	EIRQ0	WnCS1	WnNCS0	PLLVD	XTALO	B
C	YD29	YD30	NGPIO8	NGPIO9	VDD1	XLDQS	XnWE	VDD2x	XA2	XA3	XA7	XCKE	XCK	TCK	PWRON	WnICS1	WnOE	WnCS0	ADC VDD3D	ADCIN0	PLLVS	XTALI	C
D	YD27	YD28	NGPIO4	NGPIO7	NGPIO12	NGPIO10	XnRAS	XnCS	XA1	XA5	XA12	XUDM	VDD1	TEST SCAN	TDI	nPORST	WnNOE	ADC VSS3D	ADCIN1	PLLVD	PLLVD	EXTCLK	D
E	YA6	YUDQS1	NGPIO0	NGPIO6	NGPIO5	VDD1	XLDM	XnCAS	XA0	XA4	XA8	XUDQS	nTRST	nBATF	nRSTOUT	WnPCS1	WnPCS0	ADCIN2	ADC VDDA	ADC VSSA	PLLVS	NGPIO14	E
F	YD25	YD26	YA4	NGPIO3	VDD3	GND	VDD1	GND	XBA1	XA6	XA11	VDD2x	VDD1	VDD1	GND	GND	WnDQM1	AREF	ADCIN3	RX0	NGPIO15	EIRQ2	F
G	YD24	YD23	YUDM1	NGPIO1	VDD1	GND				GND	GND	GND	GND				GND	TX0	nARST	NGPIO13	WA17	ASYN	G
H	YD22	YD21	YA8	YA5	VDD2y	GND											GND	EIRQ3	ABCK	WA19	WA16	WA14	H
J	YD20	YD19	YA7	YA9	VDD2y	VDD1											VDD1	VDD3	WA18	ADATIN	WA15	SDDAT2	J
K	YD18	YD17	YCK	VDD2y	YA12	YA11	GND									GND	VDD1	WDACK0	ADAT0	SDDAT0	WA13	WA12	K
L	YD16	YD15	YREF	YnCK	YUDQS0	YCKE	GND									GND	VDD1	WDREQ0	SDDAT3	SDDAT1	SDCLK	WA11	L
M	YD14	YD13	VDD2y	YnWE	YnRAS	YUDM0	GND									GND	VDD3	WSKTSEL	WA10	WA24	SDCMD	WA25	M
N	YD12	YD11	YnCAS	YLDM1	YBA0	GND	GND									GND	GND	WA22	WA23	WA8	WA9	WnPREG	N
P	YD10	YD9	YnCS	YLDQS1	YA0	VDD2y											VDD1	WA5	WA21	WnPIOR	WA7	WnPIOW	P
R	YD8	YD7	YA10	YBA1	VDD2y	GND											GND	WA3	VDD3	WRnB	WA4	WA6	R
T	YD6	YD5	YA2	YA3	VDD1	GND				GND	GND	ZLDQS1	GND				GND	WD6	WD13	SDA	WD15	WA20	T
U	YA1	YLDQS0	YLDM0	VDD3	VDD1	VDD1	GND	ZA5	ZA8	ZA9	ZCK	ZnWE	VDD2z	ZBA1	GND	GND	SCL	WD3	WD11	WD12	WD14	WnIOIS16	U
V	YD4	YD3	VCLK	VD15	VD8	VD6	VD0	ZUDM1	VDD2z	ZA12	ZCKE	ZUDQS0	ZnCS1	ZA10	ZA1	VDD2z	PWM0	WD0	WD2	WD10	WA1	NGPIO1AT	V
W	YD2	YD1	VD13	VD10	VD2	VSYN	ZA4	ZA6	ZA7	ZnCK	ZREF	ZnCAS	ZnRAS	VDD1	ZA2	VDD2z	CSYN	USB0+	WD1	WD9	WD7	WA2	W
Y	YD0	VD12	VD11	VD5	VD1	VDD1	ZUDQS1	VDD2z	ZA11	VDD2z	ZUDM0	ZLDM1	ZnCS0	ZBA0	ZA0	ZLDM0	ZLDQS0	VDD1	PWM1	WD8	WD4	WA0	Y
AA	VD14	VD9	VD3	ZD31	ZD29	ZD27	ZD25	ZD23	ZD21	ZD19	ZD17	ZD15	ZD13	ZD11	ZD9	ZD6	ZD4	ZD2	ZD0	VDD3	USB01-	WD5	AA
AB	VD7	VD4	HSYN	ZD30	ZD28	ZD26	ZD24	ZD22	ZD20	ZD18	ZD16	ZD14	ZD12	ZD10	ZD8	ZD7	ZD5	ZD3	ZD1	ZA3	USB00-	USB01+	AB
	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	

Bottom
View

3.3 Package Dimension



4. Power Management Controller

4.1 Overview

Power Management executes the below functions.

- Power Up Sequence
- Reset Generation
- Power Management
- Change PLL Value

PWRXTI clock (32.768Khz) is used as clock by power management.

The main functionality is Power up sequence control which make the VR3D to be stable state after power on, and latch the reset configuration value which determines the initial operation boot mode, etc in reset state.

This block makes a several reset signals such as External Reset Output (nRSTOUT), GPIO Reset, Soft Reset according to nPORST (Power On Reset PAD) and watchdog reset.

And VR3D has various power management schemes to keep optimal power consumption for a given task. The power management block consists of three power mode as below.

- Normal Mode
- IDLE Mode
- STOP Mode

IDLE mode : In this mode, All clocks are supplied to each block except for CPU (FCLK). The functionality of this mode is same as WaitForIRQ on ARM9. The IDLE mode results in reduced power consumption due to CPU core. Any interrupt request to CPU can be woken up from the IDLE mode. The cpu core define the interrupt source by setting interrupt enable bit in Interrupt controller. In addition to interrupt request generated by interrupt controller, The USB, Touch PAD are used as wakeup source.

STOP mode : In this mode, The Clock controller does not supply the clock for all block including ARM. The VR3D changes the state of DRAM to the self refresh mode before stopping the clock. In stop mode, The PLL, XTAL pad are changed to power down mode in order to achieve the low power consumption.

In STOP mode, the limited wakeup source (such as external interrupt, USB, TouchPAD) can be woken up from the STOP mode, because of The clock controller doest not supply the clock for all block except Power management.

4.2 Reset Generation

4.2.1 Power On Reset Sequence

Power management block has the reset generation block. The reset generation block uses the nPORST which is sampled at PWRXTI clock (32.768KHz). and the PWRXTI clock is used as main clock for power management. So You must supply the PWRXTO clock.

Figure 3.1 shows the clock and reset behavior during the power-on reset sequence.

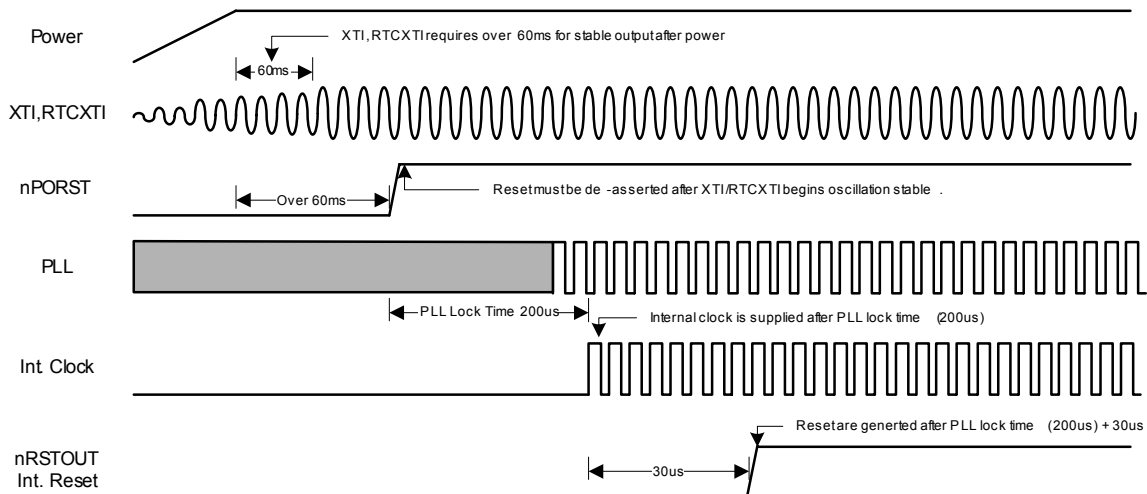


Figure 3.1 Power On Reset Sequence

nPORST(Power on reset) PAD must be hold to Low(GND) while the crystal oscillator begins stable oscillation after power on. It take a least 60ms to stabilize oscillations. When nPORST is released after the stabilization of OSC pad (XTl, PWRXTl), the internal PLL start to operate according to the default PLL configuration. But PLL needs the locking time before the stable clock is output. While locking time, power manage doesn't supply the clock for internal logic, After PLL clock is stable, PLL clock output is supplied for each block, then The internal reset and external reset are released after 30us.

Figure 3.2 shows the state diagram of Power Management Block.

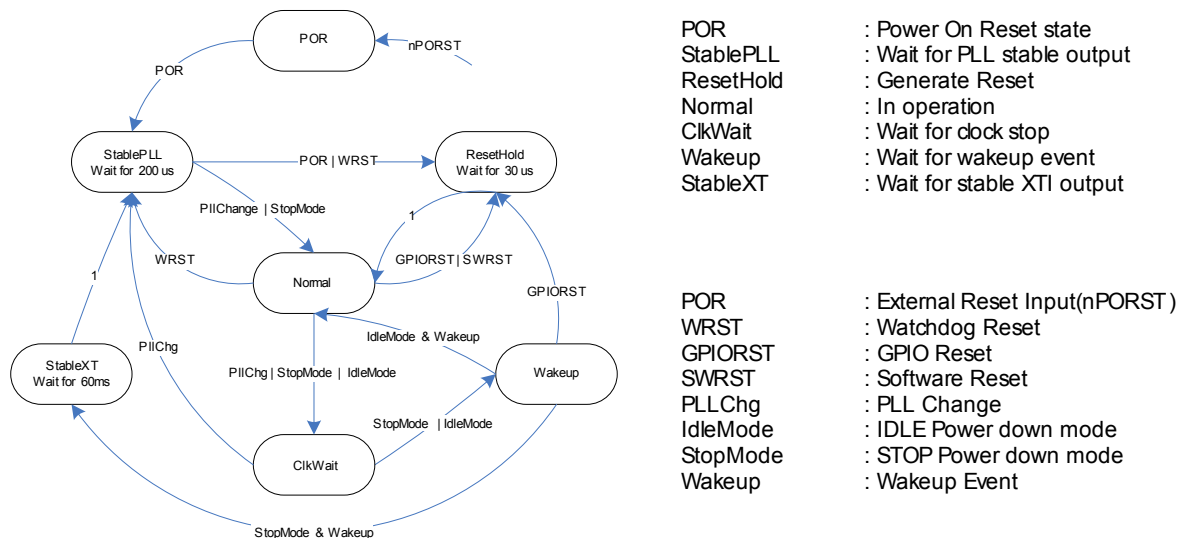


Figure 3.2 Power Management Sequence

4.2.2 nPORST & Software Reset & Watchdog Reset & GPIO Reset

VR3D has four reset state (Power On Reset, Watchdog Reset, Software Reset, GPIO Reset) as below.

Reset Mode	Condition	nRSTOUT	GPIO Alternate	Clock Controller	Power Manage	Memory Controller	Others
Power On Reset	nPORST	Reset	Reset	Reset	Reset	Reset	Reset
Watchdog Reset	Watchdog	Reset	Reset	Reset	Reset	Reset	Reset
Software Reset	CPU	Reset	None	None	Reset	None	Reset
GPIO Reset	GPIO	Reset	None	None	Reset	None	Reset

4.2.2.1 Software Reset & GPIO Reset

VR3D supports that CPU can reset itself with Software Reset.

To generate Software Reset, SRSTENB bit of PWRMODE Register must be set to 1 before setting SWRST bit of PWRMODE register.

Software Reset mode does not need the time for stabilization clock because the software reset is requested in stable state differently from power on reset.

VR3D supports user defined GPIO Reset. The Power management block generates reset when the GPIO pad defined as GPIO Reset source is asserted or de-asserted. The GPIO pad is used as GPIO Reset source are defined at Wakeup Source Register. The GPIORESET bit of PWRSTSTATUS register set to 1 enables the GPIO Reset source feature.

The reset which is generated from Software Reset and GPIO Reset does not initialize the Clock Controller, Memory Controller, GPIO Alternate Function Selection Register. Therefore There are no problem about accessing DRAM memory without re-setting the parameters to memory controller by cpu, and Clock Controller doesn't need the Initial clock parameter.

4.2.2.2 Watchdog Reset

The Watchdog timer block is used to resume the controller operation whenever it is disturbed by malfunctions such as system error, etc. When power management block detects the event from watchdog timer, It generates exactly the same reset as power on reset because the watchdog reset event occur in malfunctions and unknown state.

4.3 Power Down Mode

The power management has two power down mode as below to reduce power consumption.

To enter into each power down mode, CURPWRMODE bits of Power Mode Control Register set to power down mode (IDLE Mode, STOP Mode) changes NORMAL Power mode to each power down mode. Especially CPU must check that the blocks except for MCU, DMA, Interrupt controller in in IDLE state (No operation) before changing to STOP mode. If it is changed to STOP mode without respect of the operation state of other blocks, VR3D would enter into unknown state.

Before entering into STOP mode, Power management block make the DRAM to enter into Self refresh mode to prevent the contents of memory.

VR3D can wakeup from IDLE mode by the interrupt supplied to CPU, So The Interrupt Enable bit be used as wakeup source must be set before entering into IDLE mode.

In Stop Mode, The wakeup sources are determined by Wakeup Source Register in STOP mode.

Power Down Mode	CPU	ETC	SDRAM	Wakeup Condition
IDLE Mode	CLK OFF	CLK ON	NORMAL	Wakeup Source Register All Interrupt, nBATF, nVDDF,
STOP Mode	CLK OFF	CLK OFF	Self Refresh	Wakeup Source Register nBATF, nVDDF

4.3.1 IDLE Mode

In Idle mode, The power management does not supply the clock for CPU only. The wakeup source form IDLE mode are the interrupt supplied to CPU and nBATF, nVDDF PAD, USB PAD, Touch Panel.

After wakeup from IDLE mode, CPU will be back to previous state immediately.

4.3.2 Stop Mode

In STOP Mode, The power management does not supply the all clock to all block except for Power management for optimal low power consumption.

Before stopping the all clocks, It make DRAM enter into Self-refresh mode to prevent the content of DRAM. So, After wakeup, the contents of DRAM is not broken.

The wakeup source in STOP mode are differ from it of IDLE mode because of stopping the almost clocks, therefore The function defined in GPIO Wakeup Control Register are used as wakeup source in addition to nBATF, nVDDF, USB, Touch Panel.

To reduce more power consumption, The Power management block forces Three internal PLL and XTI PAD to enter into power down mode. Therefore It takes more time (about 70ms) than IDLE mode to wakeup from STOP mode.

4.3.3 Wakeup Source

The wakeup sources are such as below

- GPIOI[15:3]
- USB
- External Interrupt
- Touch Panel

Above Wakeup sources are defined in GPIO Wakeup Control Register (**GPIOWAKEUP0**)

Power management block is able to detect the rising edge and falling edge from event source. And, to use USB, Touch panel as wakeup source, GPIO Misc. Register in GPIO Controller must be set to appropriate value.

4.4 Change PLL Value

When CPU want to change the PLL divider value. The PLL Change Bit (**PWRMODE**) must be set to 1 after setting the PLL Setting Reset(**FPLLSETREG**, **APLLSETREG**, **UPLLSETREG**) to appropriate value.

Power management and Clock Controller blocks up the clock supplied to internal controllers because PLLs are to unstable when PLL divider value is chaged. After locking time, These block supply clock, CPU must check whether the blocks run or stop such as STOP mode.

4.5 GPIO Alternate Function Selection

The GPIO PAD has several functions.

GPIO Alternate Function Register determines what function the GPIO pad has.

Almost GPIO PAD has one of four functions as below table. The GPIO Alternate Function Selection Register (**GPIOALT**) must be set to proper value according to system configuration. Normally GPIO PADs are in GPIO Input state in power on Reset. And GPIOALT Register is not initialized by Software Reset and GPIO Reset except of power on reset and watchdog reset.

Value	Function
00b	GPIO In : Default
01b	GPIO Output
10b	Alternate Function #1 : Refer to PAD Description Chapter
11b	Alternate Function #2 : Refer to PAD Description Chapter

To update **GPIOALT** register, The GPIO Index Register (**GPIOINDEX**) must be set to Index number of GPIO group before setting

GPIOALT register.

To get the information what function the GPIO pads has, refer to PAD Description chapter.

To get the information how to control GPIO output/input, pull-up control, GPIO Interrupt control, refer to GPIO Controller.

4.6 XTAL Connection

The Figure 3-3 shows a circuit of the Power management unit oscillation at 32.768KHz

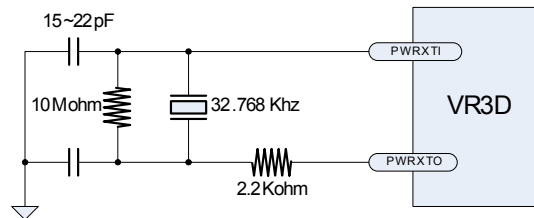


Figure 3-3, Power Management Oscillator Circuit Example

The PWR Clock must be connected to VR3D because the PWR clock is used as the clock for power management operation.

The load capacitance value has to be adjusted to the optimal value for X-tal. The Capacitance should be changed depend on the maker of x-tal.

4.7 Power Management Controller Registers

4.7.1 GPIO Wakeup Control Register (PWRWAKEUP0)

Address : 0xC000 0900

[illegible]

4.7.2 RTC Wakeup Control Register (Reserved)

Address : 0xC000 0902 (Reserved)

4.7.3 Rising Edge Detect Control Register (PWRRISEENB)

Address : 0xC000 0904

[illegible]

4.7.4 Falling Edge Detect Control Register (PWRFALLENB)

Address : 0xC000 0906

Bit	R/W	Description	Default Value
15	R/W	FALLWKSRC15 : Wakeup Source 15 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
14	R/W	FALLWKSRC14 : Wakeup Source 14 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
13	R/W	FALLWKSRC13 : Wakeup Source 13 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
12	R/W	FALLWKSRC12 : Wakeup Source 12 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
11	R/W	FALLWKSRC11 : Wakeup Source 11 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
10	R/W	FALLWKSRC10 : Wakeup Source 10 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
9	R/W	FALLWKSRC9 : Wakeup Source 9 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
8	R/W	FALLWKSRC8 : Wakeup Source 8 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
7	R/W	FALLWKSRC7 : Wakeup Source 7 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
6	R/W	FALLWKSRC6 : Wakeup Source 6 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
5	R/W	FALLWKSRC5 : Wakeup Source 5 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
4	R/W	FALLWKSRC4 : Wakeup Source 4 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
3	R/W	FALLWKSRC3 : Wakeup Source 3 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
2	R/W	FALLWKSRC2 : Wakeup Source 2 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	0
1	R/W	FALLWKSRC1 : Wakeup Source 1 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	1
0	R/W	FALLWKSRC0 : Wakeup Source 0 Falling Edge Detect Enable Control 0 : Disable 1 : Enable	1

4.7.5 Wakeup Source Detect Register (PWRWAKEUPDET)

Address : 0xC000 0908

Bit	R/W	Description	Default Value
15	R/W	WKDET15 : Wakeup Source 15 Detection Set 1 to clear this bit 0 : None 1 : Detected	0
14	R/W	WKDET14 : Wakeup Source 14 Detection 0 : None 1 : Detected	0
13	R/W	WKDET13 : Wakeup Source 13 Detection 0 : None 1 : Detected	0
12	R/W	WKDET12 : Wakeup Source 12 Detection 0 : None 1 : Detected	0
11	R/W	WKDET11 : Wakeup Source 11 Detection 0 : None 1 : Detected	0
10	R/W	WKDET10 : Wakeup Source 10 Detection 0 : None 1 : Detected	0
9	R/W	WKDET9 : Wakeup Source 9 Detection 0 : None 1 : Detected	0
8	R/W	WKDET8 : Wakeup Source 8 Detection 0 : None 1 : Detected	0
7	R/W	WKDET7 : Wakeup Source 7 Detection 0 : None 1 : Detected	0
6	R/W	WKDET6 : Wakeup Source 6 Detection 0 : None 1 : Detected	0
5	R/W	WKDET5 : Wakeup Source 5 Detection 0 : None 1 : Detected	0
4	R/W	WKDET4 : Wakeup Source 4 Detection 0 : None 1 : Detected	0
3	R/W	WKDET3 : Wakeup Source 3 Detection 0 : None 1 : Detected	0
2	R/W	WKDET2 : Wakeup Source 2 Detection 0 : None 1 : Detected	0
1	R/W	WKDET1 : Wakeup Source 1 Detection 0 : None 1 : Detected	1
0	R/W	WKDET0 : Wakeup Source 0 Detection 0 : None 1 : Detected	1

4.7.6 Scratch PAD Register (PWRSCRPAD)

Address : 0xC000 090C

Bit	R/W	Description	Default Value
15 : 0	R/W	SCRPAD : General Purpose	0

It is used for general purpose.

It is not effect on other blocks.

4.7.7 Reset Status Register (PWRRSTSTATUS)

Address : 0xC000 090E

Bit	R/W	Description	Default Value
15 : 6	R	Reserved	0

Bit	R/W	Description	Default Value
5	R/W	VDDF : Voltage Fault(nVDDF) Detection, Set 1 to clear it.	0
4	R/W	BATF : Battery Fault(nBATF) Detection, Set 1 to clear it.	0
3	R/W	GPIORESET : When the event defined in GPIO Wakeup Source Register occurs, It determines whether reset is generated or is not. 0 : Disable 1 : Generate GPIO Reset when detecting.	0
2	R/W	Reserved	0
1	R/W	WATCHDOGRST : Inform Watchdog Reset Booting Set 0 to clear it.	0
0	R/W	POWERONRST : Inform Power- On Reset Booting Set 0 to clear it	1

4.7.8 GPIO Index Register (GPIOINDEX)

Address : C000 091Eh

Bit	R/W	Description	Default Value
15 : 4	R	Reserved	-
0	R/W	GPIOIDX : Set GPIO Group to access GPIOA : 0, GPIOB : 1 GPIOC : 2 Reserved : 3 GPIOE : 4, GPIOF : 5 GPIOG : 6 GPIOH : 7 GPIOI : 8 GPIOJ : 9 GPIOK : 10	1

4.7.9 GPIO Alternate Function Selection Register (GPIOALT)

Address : C000 0910h

Bit	R/W	Description	Default Value
31 : 30	W	ALTFN15 : Set Alternate Function of GPIO*15 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
29 : 28	W	ALTFN14 : Set Alternate Function of GPIO*14 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
27 : 26	W	ALTFN13 : Set Alternate Function of GPIO*13 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
25 : 24	W	ALTFN12 : Set Alternate Function of GPIO*12 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
23 : 22	W	ALTFN11 : Set Alternate Function of GPIO*11 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
21 : 20	W	ALTFN10 : Set Alternate Function of GPIO*10 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
19 : 18	W	ALTFN9 : Set Alternate Function of GPIO*9 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
17 : 16	W	ALTFN8 : Set Alternate Function of GPIO*8 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
15 : 14	W	ALTFN7 : Set Alternate Function of GPIO*7 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
13 : 12	W	ALTFN6 : Set Alternate Function of GPIO*6 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0

Bit	R/W	Description	Default Value
11 : 10	W	ALTFN5 : Set Alternate Function of GPIO*5 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
9 : 8	W	ALTFN4 : Set Alternate Function of GPIO*4 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
7 : 6	W	ALTFN3 : Set Alternate Function of GPIO*3 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
5 : 4	W	ALTFN2 : Set Alternate Function of GPIO*2 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
3 : 2	W	ALTFN1 : Set Alternate Function of GPIO*1 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0
1 : 0	W	ALTFN0 : Set Alternate Function of GPIO*0 00 : GPIO Input 01 : GPIO Output 10 : Alternate Function 0 11 : Alternate Function 1	0

4.7.10 SLEEP GPIO Output Register (PWRSLPOUT)

Address : C000 0914h - Reserved

4.7.11 SLEEP GPIO Direction Register (PWRSLPDIR)

Address : C000 0916h - Reserved

4.7.12 SLEEP GPIO Pull-Up Control Register (PWRSLPPU)

Address : C000 0916h - Reserved

4.7.13 Power Mode Control Register (PWRMODE)

Address : C000 0952h

Bit	R/W	Description	Default Value
15	R/W	CHGPLL : Change PLL value with new value defined in PLL Setting Register (FPLLSET, UPLLSET, APLLSET) in Clock Controller 0 : Idle 1 : PLL Value Change	0
14	R/W	Reserved	0
13	R/W	SRSTENB : Status for GPIORESET function 0 : Disable 1 : Enable	0
12	R/W	SWRST : Software Reset, SRSTENB should be 1 for reflection 0 : Idle 1 : RESET	0
11 : 2	R/W	Reserved	0
1 : 0	R/W	CURPWRMODE : Set new Power Mode 00 : NORMAL mode 01 : IDLE mode 10 : STOP mode 11 : Reserved	0

5. Clock Controller

5.1 Overview

The Clock controller in VR3D can generate the required clock signals including FCLK for CPU, BCLK for Bus Controller, PCLK for peripheral interface, and various clock for peripheral operation. The VR3D has three internal PLL (FPLL, UPLL, APLL).

The Clock controller support dynamic power control to reduce power consumption in normal operation state. If the dynamic power control function is enabled, Each block enables or disables the its clock itself.

Figure 5.1 shows the internal clock path diagram of Clock controller.

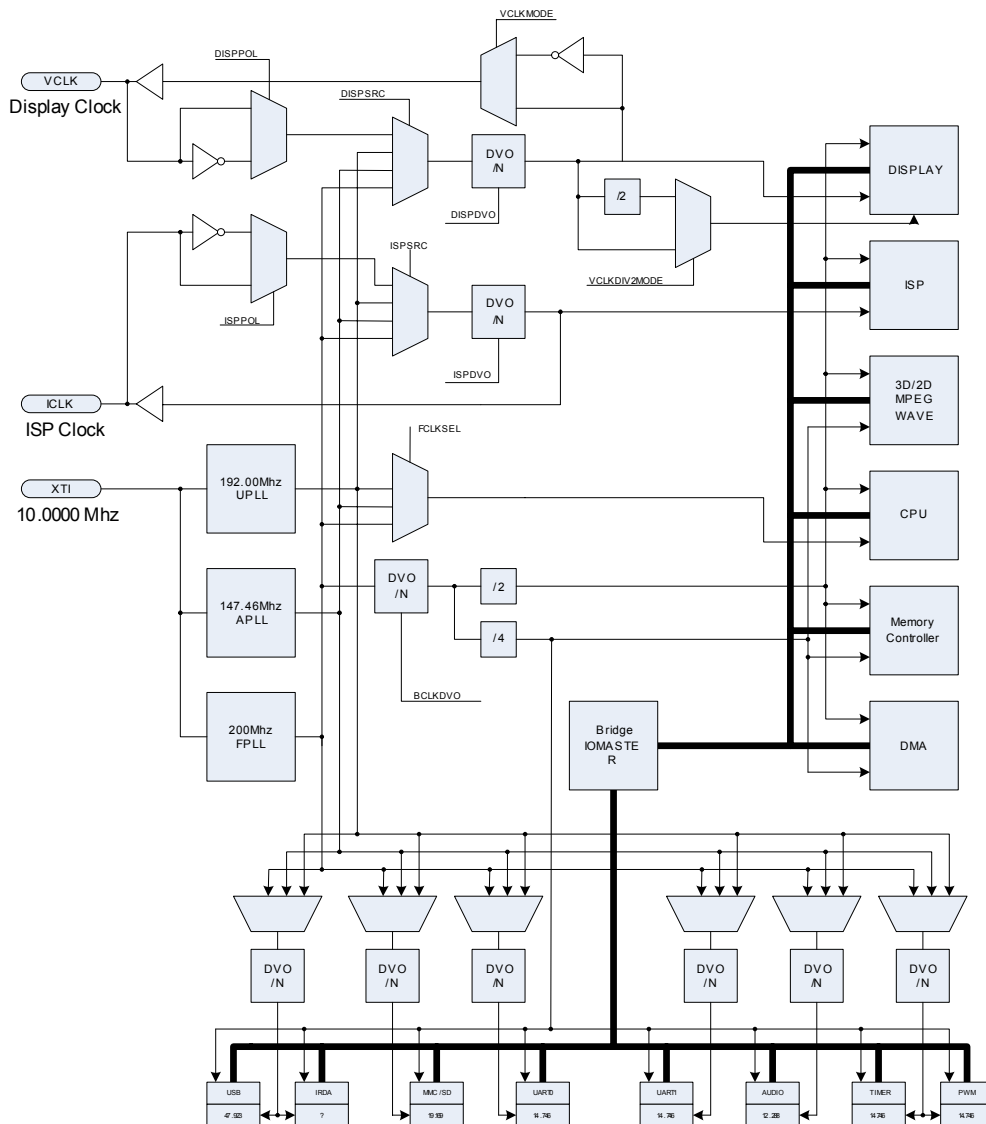


Figure 4.1 Clock Path diagram

The main clock source comes from an external crystal(XTI). The internal PLL generate the high frequency clock required in VR3D.

VR3D recommends that Input frequency of XTI is slower than 10.0000Mhz. If the input frequency is faster than 10.0000Mhz, VR3D would operate unstable because the BCLK (for Bus Controller, System Controller) is faster than 100Mhz in booting stage.

5.2 PLL

5.2.1 Change PLL Divider

VR3D has three same PLL. Each PLL is programmed to generate various clocks. To change the PLL output frequency, The procedure to change is required.

Whether All blocks except of CPU, Memory Controllers run or stop must be checked before changing the PLL output frequency. Then The PLL Change Bit (**PWRMODE**) must be set to 1 after setting the PLL Setting Reset(**FPLLSETREG**, **APLLSETREG**, **UPLLSETREG**) to appropriate value.

5.2.2 PMS

Figure 5.2.2 shows PLL Internal block diagram.

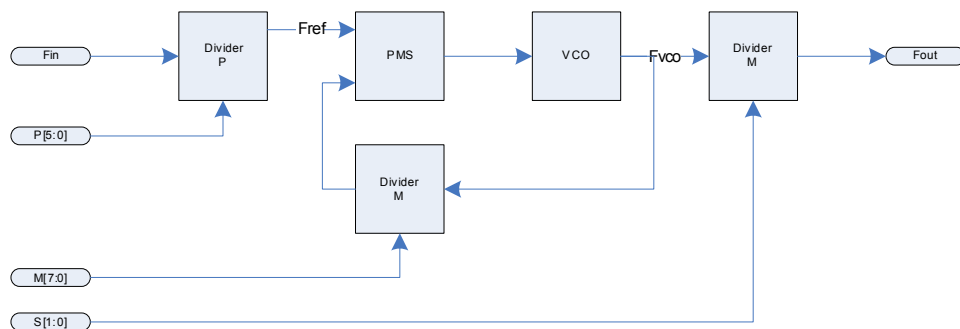


Figure 4.2.2 Block diagram of PLL

The voltage controlled oscillator(VCO) to generate the output frequency proportional to input DC voltage, the divider P to divide the input frequency (Fin) by p, the divider M to divide the VCO output frequency by m which is input to Phase Frequency Detector(PFD), the divider S to divide the VCO output frequency by s which PLL. The output clock frequency PLL is related to the reference input clock frequency Fin by the following equation

$$F_{PLL} = (m * F_{in}) / (p * 2^s)$$

$$m = (MDIV + 8), \quad p = (PDIV + 2), \quad s = SDIV$$

the program which calculate the PMS value according to the required output frequency is supplied. (VR3DPLLCalc.exe)

PLL	Initial Frequency	PMS		
		P	M	S
FPLL	200.000Mhz	1	52	0
UPLL	130.000Mhz	1	70	1
APLL	200.000Mhz	1	52	0

FCLK is used by ARM920T

BCLK is used by Core Logic (3D, MPEG, Wavetable, Memory Controller, DMA, etc.)

PCLK is used for peripheral Interface and IOMASTER

5.2.3 PLL Power Down

When each ALL, UPLL is not used, CPU can force for APLL, UPPL to enter into power down mode. If the FCLK for CPU is generated by APLL or UPLL, CPU would stop operation when related PLL enter into power down mode,.

5.3 Generate Clock

5.3.1 CPU Clock Selection

It is important that the duty of FCLK for CPU is 50%, The duty of VCO output can be changed up to 45~55% according to external condition and each chip, therefore, If the clock of VCO which is not divided by divider S is used for FCLK, The max operation frequency would not be guaranteed.

The clock path of FCLK has another divider except divider of PLL. When CPU operates in max operation frequency, The programmable divider not used is recommended for 50% duty.

5.3.2 Dynamic Clock Control

3D, MPEG, Wavetable Synthesizer, DMA block supports Dynamic power control for reducing power consumption in operation.

5.3.3 Display Clock Control

The clock for display is generated by dividing the PLL clock (FPLL or UPLL or APLL) or External Video Clock PAD by DISPDVO value.

The polarity of external Video input clock or external Video output clock can be changed according to DISPPOL and VCLKMODE. The direction (Input or output) of VCLK pad is determined by DISPSRC bit, If DISPSRC is set to 00 (External Video Clock PAD), the direction is input. When External Device want to use the clock generated by VR3D, DISPSRC is set to PLL set except 00b. then VCLK pad outputs clocks.

5.3.4 ISP Clock Control

The clock for ISP is generated by dividing the PLL clock (FPLL or UPLL or APLL) or External ISP Clock PAD by ISPDVO value as same as Display Clock generation.

The polarity of external ISP output clock can be changed according to ISPPOL.

The direction (Input or output) of ICLK pad is determined by ISPSRC bit, If ISPSRC is set to 00 (External ISP Clock PAD), the direction is input. When External Device want to use the clock generated by VR3D, ISPSRC is set to PLL set except 00b. then ICLK pad outputs clocks

5.3.5 XTAL Connection

The Figure 6-1 shows a circuit of the Main unit oscillation at 10Mhz

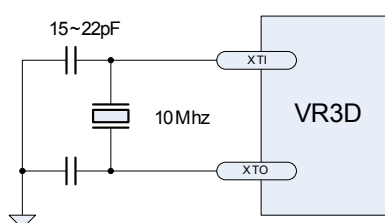


Figure 4.3.5, Main Oscillator Circuit Example

5.4 Clock Control Register

5.4.1 PLL Power Down Control Register (PLLPWRDN)

Address : C000 5100h

Bit	R/W	Description	Default Value
15	R/W	UPLLPWDN : UPLL Power Down 0 : Normal 1 : Power Down	0
14	R/W	APLLPWDN : APLL Power Down 0 : Normal 1 : Power Down	0
13	R/W	PWDDTENB : Power Down Request Control of Peripheral, for Test purpose	0
12 : 0	R	Reserved	-

5.4.2 FPLL Setting Register (FPLLSETREG)

Address : C000 5102h

Bit	R/W	Description	Default Value
15 : 8	R/W	MDIV of PLL	34h
7 : 2	R/W	PDIV of PLL	1
1 : 0	R/W	SDIV of PLL	0

PLL CONTROL REGISTER (FPLLSETVREG / UPLLSETVREG / APLLSETVREG)

$$M_{pII} = (m * F_{in}) / (p * 2^s)$$

$$m = (MDIV + 8), \quad p = (PDIV + 2), \quad s = SDIV$$

Example

If $F_{in} = 14.31818\text{MHz}$ and $F_{CLK} = 60\text{MHz}$, the calculated value is as folloes:

MDIV=59, PDIV=6 and SDIV=1 (This value may be calculated using PLLSET.EXE utility, provided by MagicEyes)

PLL VALUE SELECTION GUIDE

1. $M_{pll} * 2^A$ has to be less than 400Mhz
2. S should be as great as possible
3. (Fin / p) is recommended to be 2Mhz~1Mhz

5.4.3 UPLL Setting Register (UPLLSETREG)

Address : C000 5104h

Bit	R/W	Description	Default Value
15 : 8	R/W	MDIV of PLL	46
7 : 2	R/W	PDIV of PLL	1
1 : 0	R/W	SDIV of PLL	1

5.4.4 APLL Setting Register (APLLSETREG)

Address : C000 5106h

Bit	R/W	Description	Default Value
15 : 8	R/W	MDIV of PLL	34h
7 : 2	R/W	PDIV of PLL	1

Bit	R/W	Description	Default Value
1 : 0	R/W	SDIV of PLL	1

5.4.5 BCLK Control Register0 (CLKBCLKCTRL0)

Address : C000 5108h

Bit	R/W	Description	Default Value
15	R/W	BCLKENBISP : BCLK Enable of ISP (Image Signal Processor) 0 : Disable 1 : Enable	0
14	R/W	BCLKENBDISP : BCLK Enable of Display Controller 0 : Disable 1 : Enable	0
13	R/W	BCLKENBMCUW : BCLK Enable of MCU W Bank Controller 0 : Disable 1 : Enable	1
12	R/W	BCLKENBMCUZ : BCLK Enable of MCU Z Bank Controller 0 : Disable 1 : Enable	1
11	R/W	BCLKENBMCUY : BCLK Enable of MCU Y Bank Controller 0 : Disable 1 : Enable	1
10	R/W	BCLKENBMCUX : BCLK Enable of MCU X Bank Controller 0 : Disable 1 : Enable	1
9 : 8	R/W	BCLKENBDMA : BCLK Control Mode of DMA 00 : Disable 01 : Reserved 10 : Reserved 11 : Enable	11
7 : 6	R/W	BCLKENB2D : BCLK Control Mode of 2D Graphic Controller 00 : Disable 01 : Reserved 10 : Reserved 11 : Enable	10
5 : 4	R/W	BCLKENBWAV : BCLK Control Mode of Wavetable Synthesizer 00 : Disable 01 : Reserved 10 : Dynamic Power Control 11 : Enable	10
3 : 2	R/W	BCLKENBMPEG : BCLK Control Mode of MPEG 00 : Disable 01 : Reserved 10 : Dynamic Power Control 11 : Enable	10
1 : 0	R/W	BCLKENB3D : BCLK Control Mode of 3D Controller 00 : Disable 01 : Reserved 10 : Dynamic Power Control 11 : Always	10

5.4.6 BCLK Control Register1 (CLKBCLKCTRL1)

Address : C000 510ah

Bit	R/W	Description	Default Value
15	R/W	PCLKMODE : Set PCLK Mode 0 : Output only when CPU Accesses 1 : Always output	1
14	R	Reserved	-
13 : 8	R/W	BCLKDVO : Generation PLL Divide Set Value (N-1) for BCLK	0
7 : 3	R	Reserved	-
2 : 1	R/W	FCLKSEL : Select FCLK Source 00 : FPLL 01 : FPLL 10 : APLL 11 : UPLL	0
0	R/W	BCLKENBHUFVLD : BCLK Enable of VLD 0 : Disable 1 : Enable	1

5.4.7 MISC Control Register (CLKMISCCTRL)

Address : C000 510Ch

[illegible]

5.4.8 MMC/SD, USB Clock Divisor Register (MSMUDIVCTRL)

Address : C000 510Eh

Bit	R/W	Description	Default Value
15 : 14	R/W	MSMSRC : Set Clock Source for MMC/SD 00 : Disable 01 : FPLL 10 : UPLL 11 : APLL	00
13 : 8	R/W	MSMDVO : Set division value of MMC/SD Clock (N-1)division	3Fh
7 : 6	R/W	USBIRDASRC : Set Clock Source for USB 00 : Disable 01 : FPLL 10 : UPLL 11 : APLL	00
5 : 0	R/W	USBIRDADVO : Set division value of USB Clock (N-1)division	3Fh

5.4.9 AUDIO Clock Divisor Register (AUDDIVCTRL)

Address : C000 5110h

Bit	R/W	Description	Default Value
15 : 8	R	Reserved	-
7 : 6	R/W	AUDSRC : Set Clock Source for AC97/I2S 00 : Disable 01 : FPLL 10 : UPLL 11 : APLL	00
5 : 0	R/W	AUDDVO : Set division value of AC97/I2S Clock (N-1)division	3Fh

5.4.10 Display Clock Divisor Register (DISPDIVCTRL)

Address : C000 5112h

Bit	R/W	Description	Default Value
15 : 14	R/W	DISPSRC : Set Clock Source for Display Controller 00 : External Video Clock PAD 01 : FPLL 10 : UPLL 11 : APLL	00
13 : 8	R/W	DISPDVO : Set division value of Display Clock (N-1)division	3Fh
7	R/W	DISPPOL : Set Polarity of External Video Clock. 0 : Inverted 1 : Normal	1
6	R/W	VCLKDIV2MODE : Set Clock VCLKDIV2. 0 : VCLKDIV2= VCLK 1 : VCLKDIV2= VCLK/2	0
5	R/W	VCLKMODE : Set Polarity of Display Clock. 0 : Normal 1 : Inverted	0
4 : 0	R	Reserved	0

5.4.11 ISP Clock Divisor Register (ISPDIVCTRL)

Address : C000 5114h

Bit	R/W	Description	Default Value
15 : 14	R/W	ISPSRC : Set Clock Source for ISP Controller. 00 : External ISP Clock PAD 01 : FPLL 10 : UPLL 11 : APLL	00
13 : 8	R/W	ISPDVO : Set division value of Image Clock. (N-1)division	3Fh
7	R/W	ISPPOL : Set Polarity of External ISP Video Clock. 0 : Inverted 1 : Normal	0
6 : 0	R	Reserved	-

5.4.12 UART Clock Divisor Register (UARTDIVCTRL)

Address : C000 5116h

Bit	R/W	Description	Default Value
15 : 14	R/W	UART1SRC : Set Clock Source for UART1. 00 : Disable 01 : FPLL 10 : UPLL 11 : APLL	00
13 : 8	R/W	UART1DVO : Set division value of UART1 Clock. (N-1)division	3Fh
7 : 6	R/W	UART0SRC : Set Clock Source for UART0. 00 : Disable 01 : FPLL 10 : UPLL 11 : APLL	01
5 : 0	R/W	UART0DVO : Set division value of UART0 Clock. (N-1)division.	03h

5.4.13 Timer Clock Divisor Register (TIMERDIVCTRL)

Address : C000 5118h

Bit	R/W	Description	Default Value
15 : 14	R/W	TIMERSRC : Set Clock Source for Timer. 00 : Disable 01 : FPLL 10 : UPLL 11 : APLL	00
13 : 8	R/W	TIMERDVO : Set division value of Timer Clock. (N-1)division	3Fh
7 : 0	R	Reserved	-

6. GPIO Controller

6.1 Overview

The VR3D has the ten GPIO Group, GPIOA~GPIOK except GPIOD. Each GPIO group is composed of several GPIO bit. Each GPIO bit can be GPIO output or GPIO In, In GPIO Input mode, GPIO Controller can detect the rising edge or falling edge or High or low level according to GPIO Detect Mode register (**GPIODET**). And The detected event is accessed by CPU or can activate the GPIO Interrupt to CPU. And Input signal through GPIO pad are sampled in PCLK(50Mhz) because of reducing the external noise on the signal.

Figure 5.1 shows the data path for a GPIO pad.

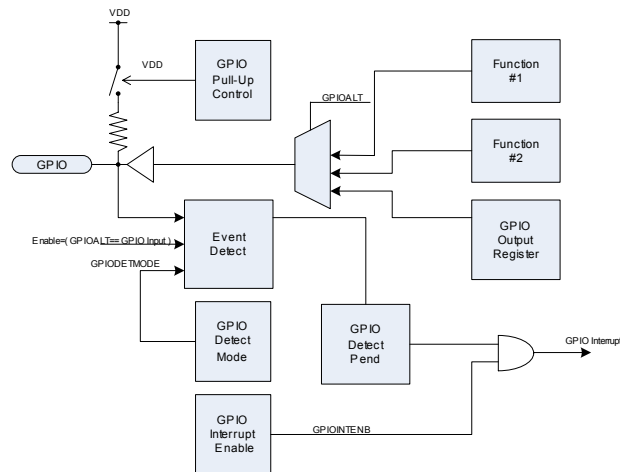


Figure 5.1 GPIO Data path block diagram

The GPIOA~GPIOC Pad are SSTL pad for DDR Memory, When these pad is used as input pad, These pad dissipates more power than normal pad and output mode. So, These pad has power down control when these pad are not used. The GPIOD~GPIOE has controllable Pull-Up Resistor .

GPIO Misc Register (**GPIONISC**) determines the wakeup enable of Touch PAD, USB, and the pad status of USB for detecting the usb event in power down mode, and the USB function selection (USB Host or USB device), determines the event detection mode for external Interrupt pad.

The USBD1+/- Pads can be used as USB device pad or USB Host pad, The UDCSEL bit of GPIO Misc. Register(**GPIONISC**) determines what functionality.

6.2 GPIO Controller Register

6.2.1 GPIO Interrupt Enable Register (GPIODETENB)

Address :	GPIOA : C000 1000h,	GPIOB : C000 1020h,	GPIOC : C000 1040h,
	GPIOD : Reserved,	GPIOE : C000 1080h,	GPIOF : C000 10A0h,
	GPIOG : C000 10C0h,	GPIOH : C000 10E0h,	GPIOI : C000 0F00h,
	GPIOJ : C000 0F20h,	GPIOK : C000 0F40h	

Bit	R/W	Description	Default Value
15	R/W	GPIO*INTENB[15] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
14	R/W	GPIO*INTENB[14] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
13	R/W	GPIO*INTENB[13] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
12	R/W	GPIO*INTENB[12] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
11	R/W	GPIO*INTENB[11] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
10	R/W	GPIO*INTENB[10] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
9	R/W	GPIO*INTENB[9] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
8	R/W	GPIO*INTENB[8] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
7	R/W	GPIO*INTENB[7] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
6	R/W	GPIO*INTENB[6] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
5	R/W	GPIO*INTENB[5] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
4	R/W	GPIO*INTENB[4] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
3	R/W	GPIO*INTENB[3] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
2	R/W	GPIO*INTENB[2] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
1	R/W	GPIO*INTENB[1] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0
0	R/W	GPIO*INTENB[0] : Set GPIO detection Interrupt according to GPIO Detect Mode 0 : Disable 1 : Enable	0

6.2.2 GPIO Detect0 Mode Register (GPIODETMODE)

Address :	GPIOA : C000 1002h,	GPIOB : C000 1022h,	GPIOC : C000 1042h,
	GPIOD : Reserved,	GPIOE : C000 1082h,	GPIOF : C000 10A2h,
	GPIOG : C000 10C2h,	GPIOH : C000 10E2h,	GPIOI : C000 0F02h,
	GPIOJ : C000 0F22h,	GPIOK : C000 0F42h	

Bit	R/W	Description	Default Value
15 : 14	R/W	GPIO*DETMODE[7] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
13 : 12	R/W	GPIO*DETMODE[6] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
11 : 10	R/W	GPIO*DETMODE[5] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
9 : 8	R/W	GPIO*DETMODE[4] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
7 : 6	R/W	GPIO*DETMODE[3] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
5 : 4	R/W	GPIO*DETMODE[2] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
3 : 2	R/W	GPIO*DETMODE[1] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
1 : 0	R/W	GPIO*DETMODE[0] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0

6.2.3 GPIO Detect1 Mode Register (GPIODETMODE)

Address : **GPIOA : C000 1004h,** **GPIOB : C000 1024h,** **GPIOC : C000 1044h,**
 GPIOD : Reserved, **GPIOE : C000 1084h,** **GPIOF : C000 10A4h,**
 GPIOG : C000 10C4h, **GPIOH : C000 10E4h,** **GPIOI : C000 0F04h,**
 GPIOJ : C000 0F24h, **GPIOK : C000 0F44h**

Bit	R/W	Description	Default Value
15 : 14	R/W	GPIO*DETMODE[15] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
13 : 12	R/W	GPIO*DETMODE[14] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
11 : 10	R/W	GPIO*DETMODE[13] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
9 : 8	R/W	GPIO*DETMODE[12] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
7 : 6	R/W	GPIO*DETMODE[11] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
5 : 4	R/W	GPIO*DETMODE[10] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
3 : 2	R/W	GPIO*DETMODE[9] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0
1 : 0	R/W	GPIO*DETMODE[8] : Set GPIO Detect Mode 00 : Low Level 01 : High Level 10 : Falling Edge 11 : Rising Edge	0

6.2.4 GPIO Output Register (GPIOOUT)

Address :	GPIOA : C000 1006h,	GPIOB : C000 1026h,	GPIOC : C000 1046h,
	GPIOD : Reserved,	GPIOE : C000 1086h,	GPIOF : C000 10A6h,
	GPIOG : C000 10C6h,	GPIOH : C000 10E6h,	GPIOI : C000 0F06h,
	GPIOJ : C000 0F26h,	GPIOK : C000 0F46h	

Bit	R/W	Description	Default Value
15	R/W	GPIO*OUT[15] : Set GPIO Output value 0 : Low Level 1 : High Level	0
14	R/W	GPIO*OUT[14] : Set GPIO Output value 0 : Low Level 1 : High Level	0
13	R/W	GPIO*OUT[13] : Set GPIO Output value 0 : Low Level 1 : High Level	0
12	R/W	GPIO*OUT[12] : Set GPIO Output value 0 : Low Level 1 : High Level	0
11	R/W	GPIO*OUT[11] : Set GPIO Output value 0 : Low Level 1 : High Level	0
10	R/W	GPIO*OUT[10] : Set GPIO Output value 0 : Low Level 1 : High Level	0
9	R/W	GPIO*OUT[9] : Set GPIO Output value 0 : Low Level 1 : High Level	0
8	R/W	GPIO*OUT[8] : Set GPIO Output value 0 : Low Level 1 : High Level	0
7	R/W	GPIO*OUT[7] : Set GPIO Output value 0 : Low Level 1 : High Level	0
6	R/W	GPIO*OUT[6] : Set GPIO Output value 0 : Low Level 1 : High Level	0
5	R/W	GPIO*OUT[5] : Set GPIO Output value 0 : Low Level 1 : High Level	0
4	R/W	GPIO*OUT[4] : Set GPIO Output value 0 : Low Level 1 : High Level	0
3	R/W	GPIO*OUT[3] : Set GPIO Output value 0 : Low Level 1 : High Level	0
2	R/W	GPIO*OUT[2] : Set GPIO Output value 0 : Low Level 1 : High Level	0
1	R/W	GPIO*OUT[1] : Set GPIO Output value 0 : Low Level 1 : High Level	0
0	R/W	GPIO*OUT[0] : Set GPIO Output value 0 : Low Level 1 : High Level	0

6.2.5 GPIO Pull-Up Control Register (GPIOPULLUP)

Address :	GPIOA : C000 100Ch,	GPIOB : C000 102Ch,	GPIOC : C000 104Ch,
	GPIOD : Reserved,	GPIOE : C000 108Ch,	GPIOF : C000 10ACh,
	GPIOG : C000 10CCh,	GPIOH : C000 10ECh,	GPIOI : C000 0F0Ch,
	GPIOJ : C000 0F2Ch,	GPIOK : C000 0F4Ch	

Bit	R/W	Description	Default Value
15	R/W	GPIO*PULLUP[15] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up If GPIO PADs are DDR-PAD and DDR-PAD are set to GPIO-IN, GPIOxPULLUP Control register control the power down mode of PAD. 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
14	R/W	GPIO*PULLUP[14] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
13	R/W	GPIO*PULLUP[13] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
12	R/W	GPIO*PULLUP[12] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
11	R/W	GPIO*PULLUP[11] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
10	R/W	GPIO*PULLUP[10] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
9	R/W	GPIO*PULLUP[9] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
8	R/W	GPIO*PULLUP[8] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
7	R/W	GPIO*PULLUP[7] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
6	R/W	GPIO*PULLUP[6] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
5	R/W	GPIO*PULLUP[5] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
4	R/W	GPIO*PULLUP[4] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
3	R/W	GPIO*PULLUP[3] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
2	R/W	GPIO*PULLUP[2] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
1	R/W	GPIO*PULLUP[1] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0
0	R/W	GPIO*PULLUP[0] : Set GPIO Pull-Up Controller 1 : Normal 0 : Pull-up 1 : Power-Down (DDR-PAD GPIO-In) 0 : Normal	0

6.2.6 GPIO Detect Clear Register (GPIODETCLR)

Address :	GPIOA : C000 100Eh,	GPIOB : C000 102Eh,	GPIOC : C000 104Eh,
	GPIOD : Reserved,	GPIOE : C000 108Eh,	GPIOF : C000 10AEh,
	GPIOG : C000 10CEh,	GPIOH : C000 10EEh,	GPIOI : C000 0F0Eh,
	GPIOJ : C000 0F2Eh,	GPIOK : C000 0F4Eh	

Bit	R/W	Description	Default Value
15	R/W	GPIO*DET[15] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
14	R/W	GPIO*DET[14] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
13	R/W	GPIO*DET[13] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
12	R/W	GPIO*DET[12] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
11	R/W	GPIO*DET[11] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
10	R/W	GPIO*DET[10] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
9	R/W	GPIO*DET[9] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
8	R/W	GPIO*DET[8] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
7	R/W	GPIO*DET[7] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
6	R/W	GPIO*DET[6] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
5	R/W	GPIO*DET[5] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
4	R/W	GPIO*DET[4] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
3	R/W	GPIO*DET[3] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
2	R/W	GPIO*DET[2] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
1	R/W	GPIO*DET[1] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0
0	R/W	GPIO*DET[0] : Read GPIO Detect status Set 1 to clear this bit. 0 : No Event 1 : Detect Event	0

6.2.7 GPIO Misc. Low Register (GPIOMISCL)

Address : C000_0F12h

Bit	R/W	Description	Default Value
15 : 14	R/W	EXTIRQ3DETMODE : Set External Interrupt Detect Mode 00 : Low Level 11 : High Level 01 : Rising Edge 10 : Falling Edge	0
13 : 12	R/W	EXTIRQ2DETMODE : Set External Interrupt Detect Mode 00 : Low Level 11 : High Level 01 : Rising Edge 10 : Falling Edge	0
11 : 10	R/W	EXTIRQ1DETMODE : Set External Interrupt Detect Mode 00 : Low Level 11 : High Level 01 : Rising Edge 10 : Falling Edge	0
9 : 8	R/W	EXTIRQ0DETMODE : Set External Interrupt Detect Mode 00 : Low Level 11 : High Level 01 : Rising Edge 10 : Falling Edge	0
7 : 1	R/W	Reserved	0
0	R/W	UDCSEL : Set USB Port1 Type 0 : USB Host 1 : USB Device	0

6.2.8 GPIO Misc. High Register (GPIOMISCH)

Address : C000_0F14h

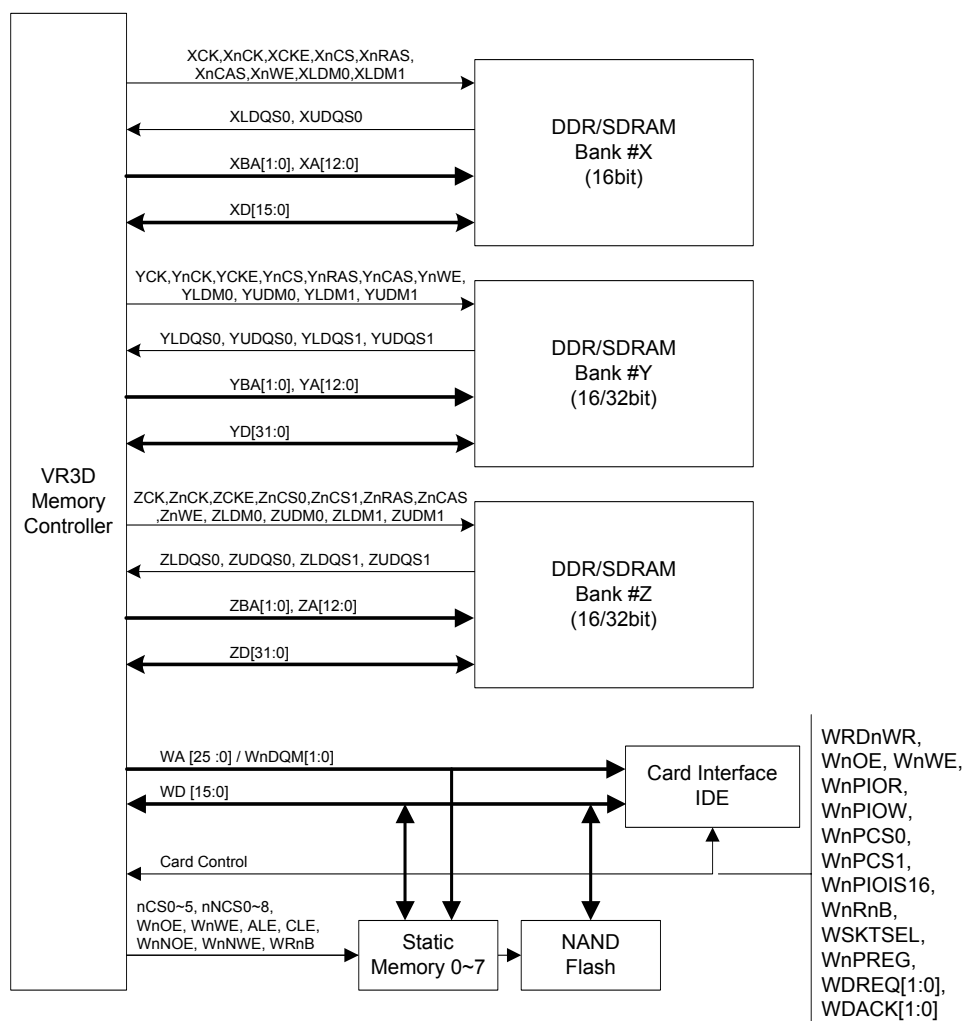
Bit	R/W	Description	Default Value
15	R/W	EXTWAKEUP : Set Wake option usage of USB, Touch at Idle, Stop Power mode 0 : Disable 1 : Enable	0
14 : 12	R/W	WAKEPAT : Set PAD status of RXD, RXDP, RXDN for USB Wakeup [30] : RXD, [29] : RXDP [28] : RXDN	0
11 : 0	R	Reserved	0

7. Memory Controller

7.1 Overview

The VRender3D memory controller is based on a Unified Memory Architecture (UMA). This controller consists of three main control unit : dynamic memory, static memory, PCMCIA/CF interface. Each unit has dedicated control pins

- Unified Memory Architecture (UMA)
 - Three Banks of DDR-SDRAM (16 or 32bit data supported)
 - ◆ Support Active Power down mode automatically.
 - ◆ Support Self Refresh mode
 - ◆ Support three CAS Latency : 2, 2.5, 3 Cycles
 - Six Static Memory Chip Selects
 - Card interface supporting PCMCIA or Compact Flash
 - Supports UDMA(Ultra DMA), MDMA(Multiword DMA) for IDE
 - Four Separate Memory Controller (Bank#W, Bank #X , Bank #Y, Bank #Z)



7.2 Operations

7.2.1 Boot Mode

VR3D provide various Boot mode below.

CfgShadow	CfgNFBoot	CfgNFBUSWidth	CfgSR0BUSWidth	Boot Mode & GPIO
0	0	X	0	8bit nSCS0 (STATIC#0) WD[15:8] is GPIO-IN. WA[25:16] is GPIO-IN
0	0	X	1	16bit nSCS0 (STATIC#0) WA[25:16] is GPIO-IN
0	1	0	0	8bit NAND Boot, WD[15:8] is GPIO-IN WA[25 :16] is GPIO-IN Copy 8bit NAND to 8bit STATIC#0
0	1	0	1	8bit NAND Boot, WA[25 :16] is GPIO-IN Copy 8bit NAND to 16bit STATIC#0
0	1	1	0	16bit NAND Boot, WA[25 :16] is GPIO-IN Copy 16bit NAND to 8bit STATIC#0
0	1	1	1	16bit NAND Boot, WA[25 :16] is GPIO-IN Copy 16bit NAND to 16bit STATIC#0
1	0	0	X	No Boot
1	0	1	X	No Boot
1	1	0	X	8bit NAND Boot WD[15:8] is GPIO-IN WA[25:0] is GPIO-IN Copy 8bit NAND to DDR (MCU-Z)
1	1	1	X	16bit NAND Boot WA[25:0] is GPIO-IN Copy 16bit NAND to DDR (MCU-Z)

If NAND boot and Shadow on, MCU-W Bus goes GPIO status when initial Reset because it is not used when boot. GPIO status of MCU-W Address and Data bus are defined from boot mode. It can be used in various way to use this function.

MCU-X, MCU-Y address and data bus go GPIO-Input because they are not used in initial reset. They should be treat carefully because they have more static leakage current than normal pad because they are SSTL pad.

Only small block NAND Flash can be used in NAND Boot. However, both small block and large block NAND Flash can be use for date NAND Flash.

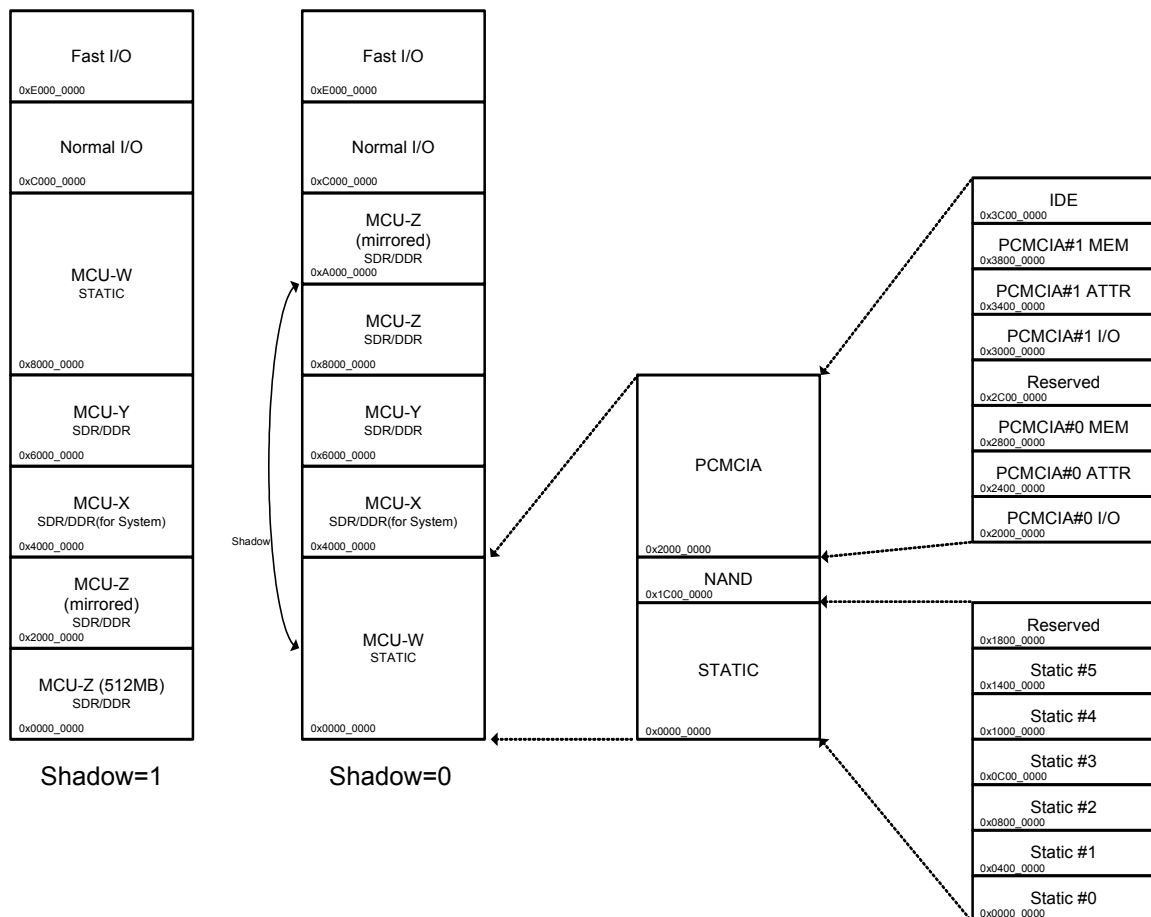
7.2.2 Memory Map

DDR-SDRAM Bank and Static Memory Bank could be exchanged by CfgShadow Pin.

VR3D has total 6 memory region. It consists of 3 DRAM bank and 1 static bank. MCU-W bank as static bank has 2 PCMCIA/CF, 1 NAND Flash memory bank, 1 IDE bank, 6 static memory bank for SRAM and NOR-Flash.

When shadow off, static#0 region of MCU-W should have NOR-Flash type memory for storing code and CfgSR0BUSWidth Pin should be Pull-up or Pull-down to define initial Static#0 bus width.

RAM should exist at address 0 when NAND Boot. If CfgShadow pin set to 1, DDR-memory exist at address 0. Type and data width of DDR-memory are defined by CfgSDRTYPE, CfgSDRBUSBW, CfgSDRBW, CfgSDRTYPE Pin when reset.



7.2.3 MCU-W Bank (Static Memory Controller)

Memory Master : CPU, DMA, USB Host

Data Bus Width : 8/16bit Interface

Address Region : If Shadow=0 then 0x0000_0000 ~ 0x3FFF_FFFF. If Shadow=1 then 0x8000_0000 ~ 0xBFFF_FFFF

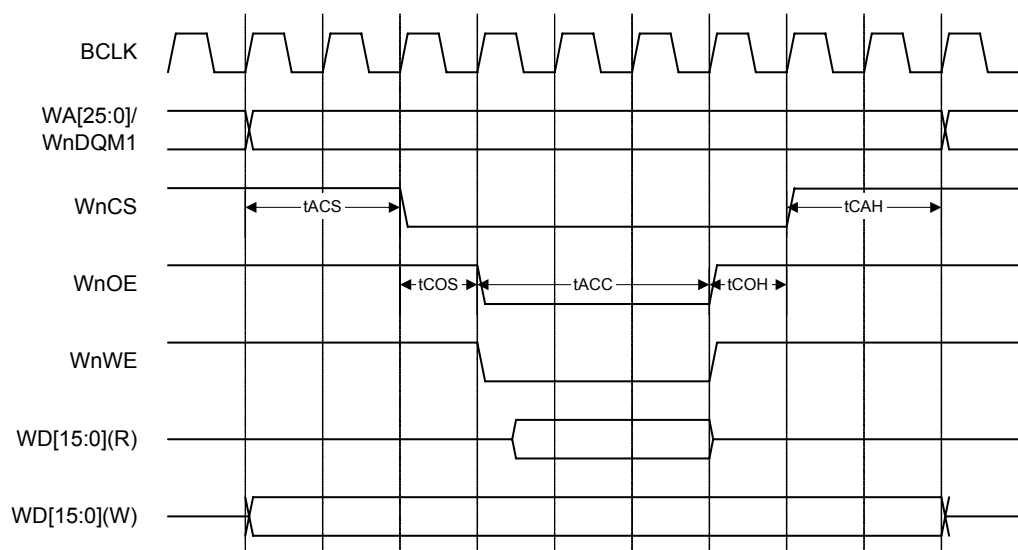
Supported Memory Type :

Static Memory : Asynchronous SRAM, ROM, NOR-Flash, NAND-Flash

Card Interface : PCMCIA/CF

IDE Interface : Support MDMA, UDMA mode for HDD, CD-ROM

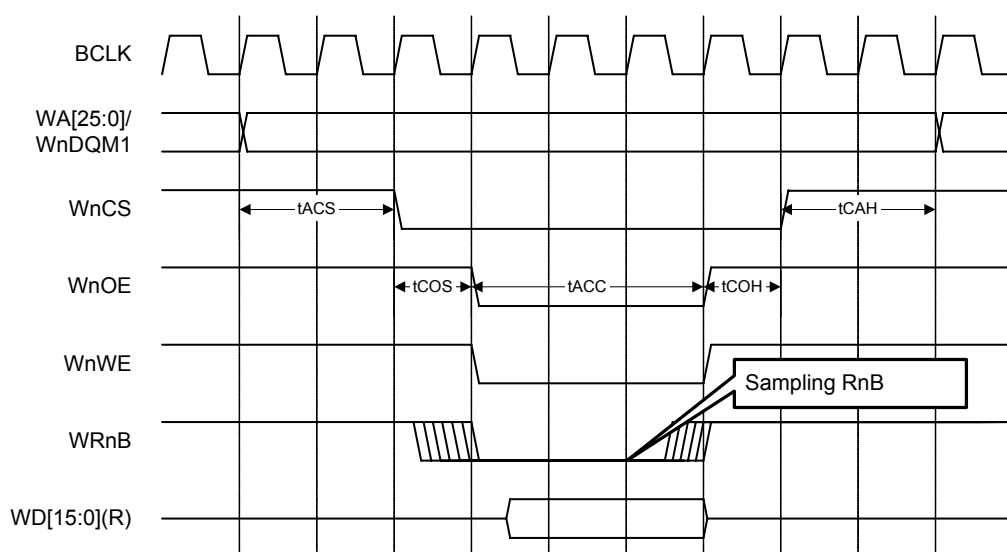
7.2.3.1 NoWait Static Memory Timing Diagram



No Wait Timing Diagram

7.2.3.2 Wait Static Memory Timing Diagram

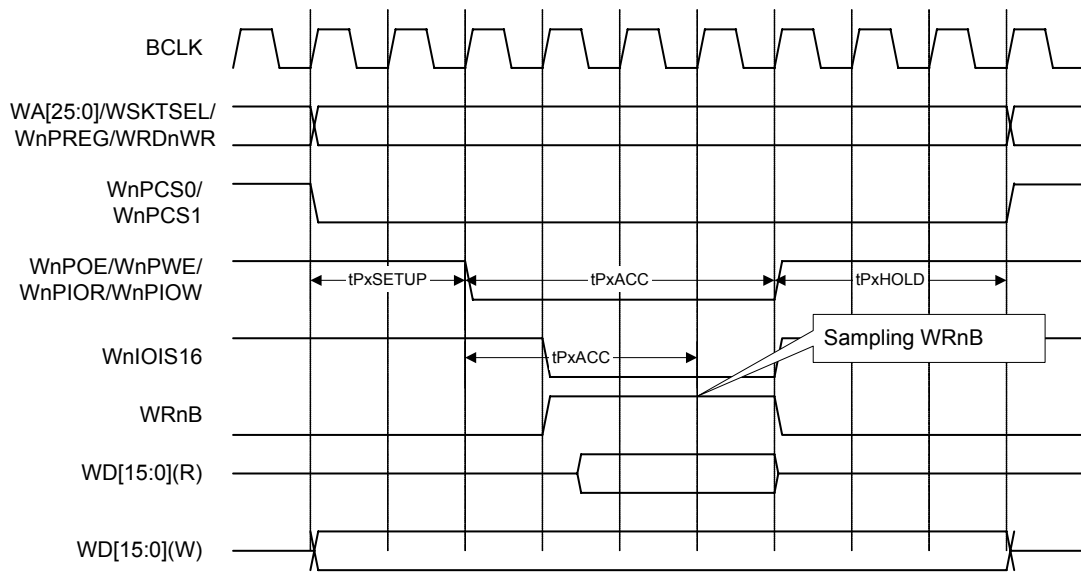
If Wait Control corresponding to each static memory bank is enabled. The Access duration should be prolonged by the external RnB (Ready&Busy) pin while the memory bank is active.



Wait Timing Diagram

7.2.3.3 PCMCIA/CF Memory or I/O 16bit Access Timing Diagram

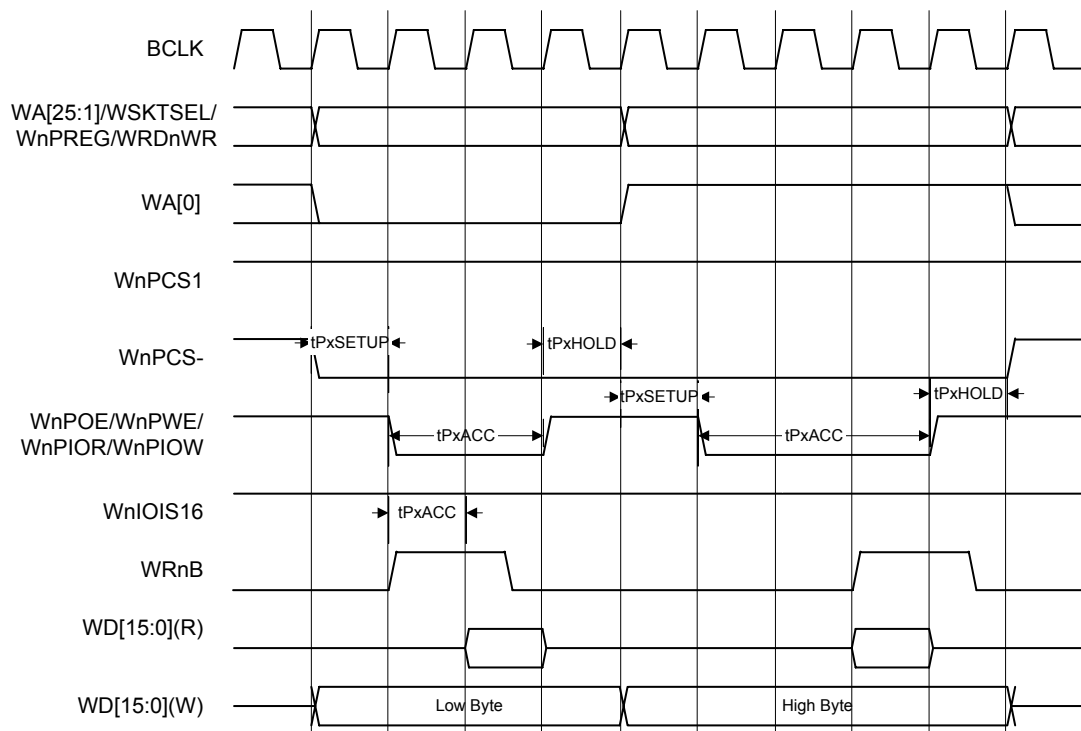
If Wait Control corresponding to each static memory bank is enabled. The Access duration should be prolonged by the external RnB



PCMCIA/CF Memory or I/O 16bit Access Timing Diagram

7.2.3.4 PCMCIA/CF I/O 16bit Access to 8bit Device Timing Diagram

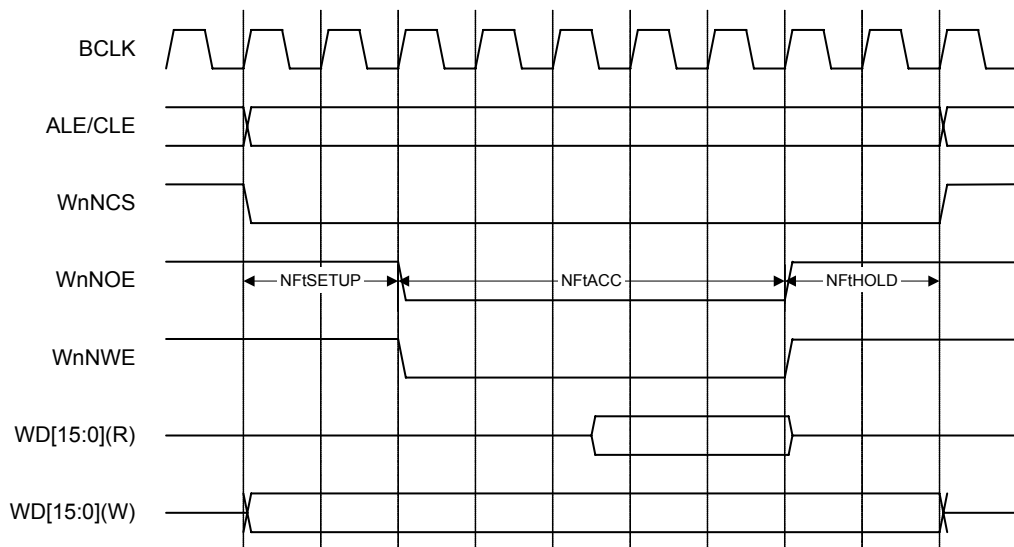
If Wait Control corresponding to each static memory bank is enabled. The Access duration should be prolonged by the external RnB



PCMCIA/CF 16bit Access to 8bit Device Timing Diagram

7.2.3.5 NAND Flash Timing Diagram

If Wait Control corresponding to each static memory bank is enabled. The Access duration should be prolonged by the external RnB

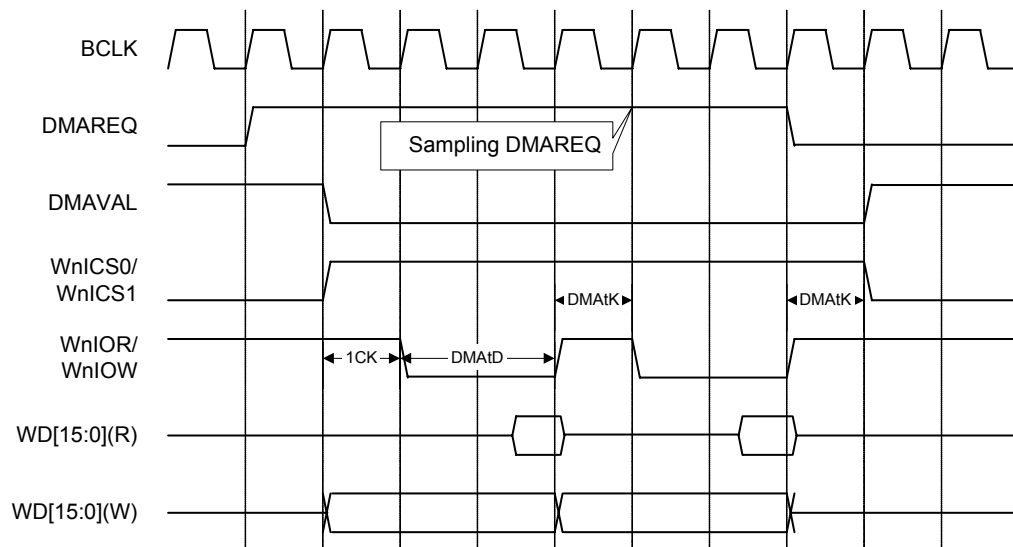


NAND Flash Timing Diagram

CAUTION : The Small Block NAND Flash needs to be hold nNCS as '0' during detecting RnB. So, nNCS signal of VR3D hold low level even if CPU does not access nand flash.

7.2.3.6 MDMA(Multi-Word DMA) Timing Diagram

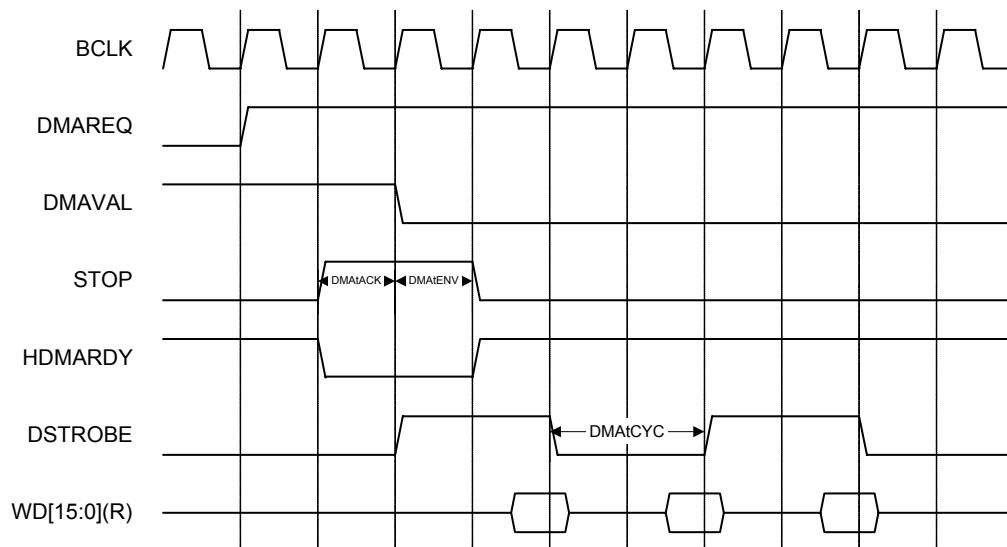
If Wait Control corresponding to each static memory bank is enabled. The Access duration should be prolonged by the external RnB



MDMA (MultiWord DMA) Timing Diagram

7.2.3.7 IDE UDMA (Ultra DMA) Timing Diagram

If Wait Control corresponding to each static memory bank is enabled. The Access duration should be prolonged by the external RnB



IDE UDMA Timing Diagram

7.2.4 MCU-X Bank (DDR SDRAM Controller)

Memory Master : CPU, DMA, USB Host

Data Bus Width : 16bit Interface

Address Region : 0x4000_0000 ~ 0x5FFF_FFFF

Supported Memory Type :

DDR-SDRAM :

64Mbit : x8, x16, x32bit,

128Mbit : x8, x16, x32bit

256Mbit : x8, x16, x32bit,

512Mbit : x8, x16, x32bit

7.2.4.1 DDR-SDRAM Memory Timing Diagram

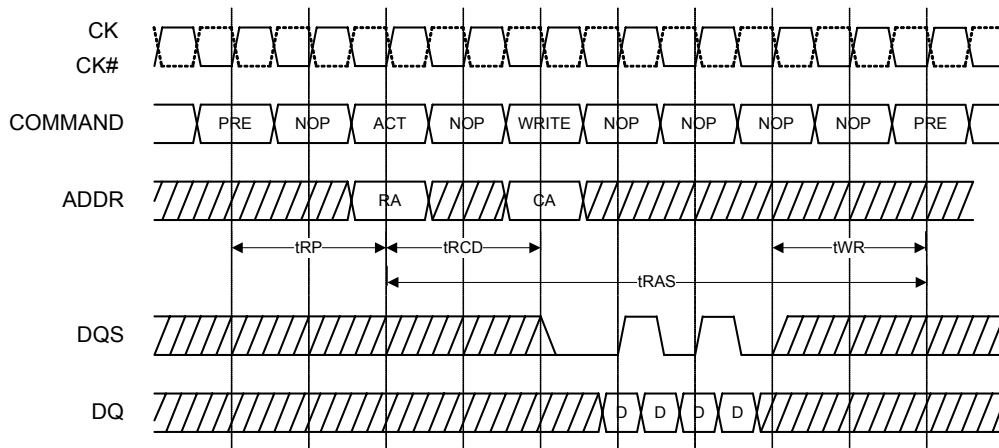


Figure 7.2.4.1 DDR-SDRAM Write Timing

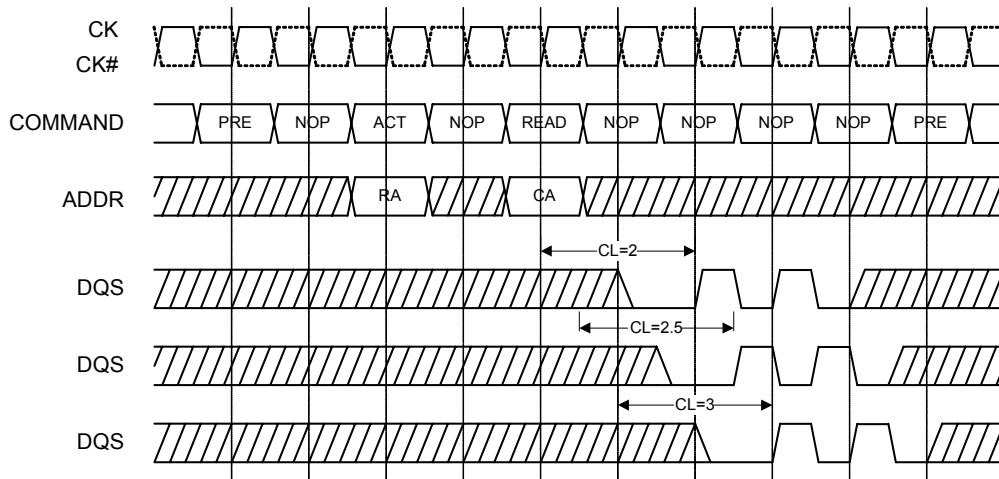


Figure 7.2.4.2 DDR-SDRAM Read Timing

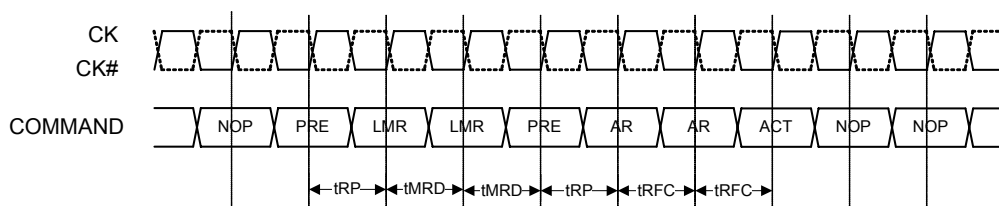


Figure 7.2.4.3 DDR-SDRAM Intialize and Load Mode Timing

7.2.5 MCU-Y Bank (DDR SDRAM Controller)

Memory Master : CPU, DMA, USB Host, 2D/3D Controller, Wavetable Synthesizer, ISP, Display, MPEG

Data Bus Width : 32bit Interface

Address Region : 0x6000_0000 ~ 0x7FFF_FFFF

Supported Memory Type :

DDR-SDRAM :

64Mbit : x8, x16, x32bit,

128Mbit : x8, x16, x32bit

256Mbit : x8, x16, x32bit,

512Mbit : x8, x16, x32bit

7.2.6 MCU-Z Bank (DDR SDRAM Controller)

Memory Master : CPU, DMA, USB Host, 2D/3D Controller, Wavetable Synthesizer, ISP, Display, MPEG

Data Bus Width : 32bit Interface

Address Region : If Shadow=0 then 0x8000_0000 ~ 0x9FFF_FFFF. If Shadow=1 then 0x0000_0000 ~ 0x1FFF_FFFF

Supported Memory Type :

DDR-SDRAM :

64Mbit : x8, x16, x32bit,

128Mbit : x8, x16, x32bit

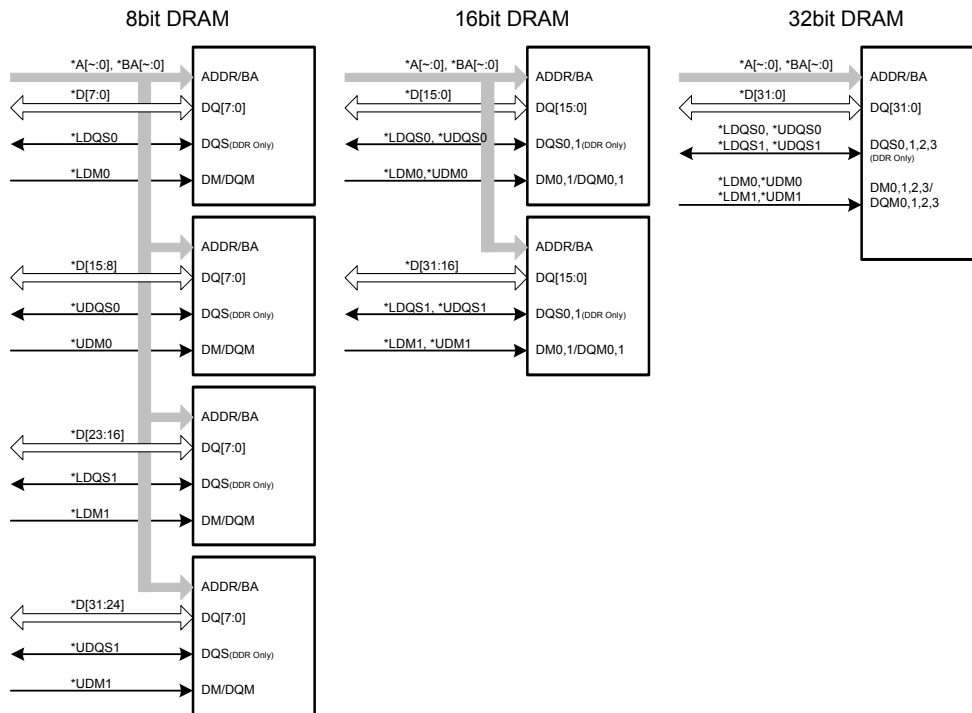
256Mbit : x8, x16, x32bit,

512Mbit : x8, x16, x32bit

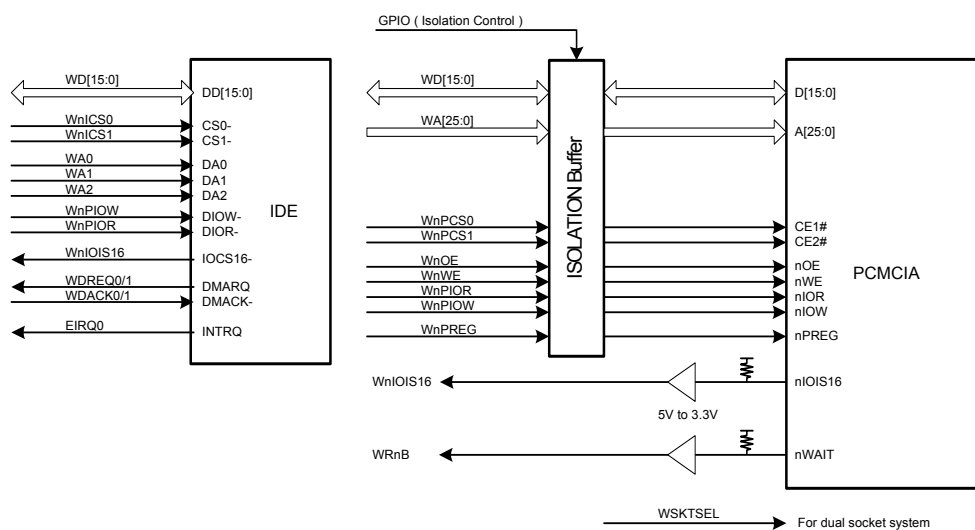
7.2.7 Interface

7.2.8 DRAM Connection

Connect VR3D address pin0 to DRAM address pin0 regardless of DRAM size or data bus width. Connect bank address at the same way.



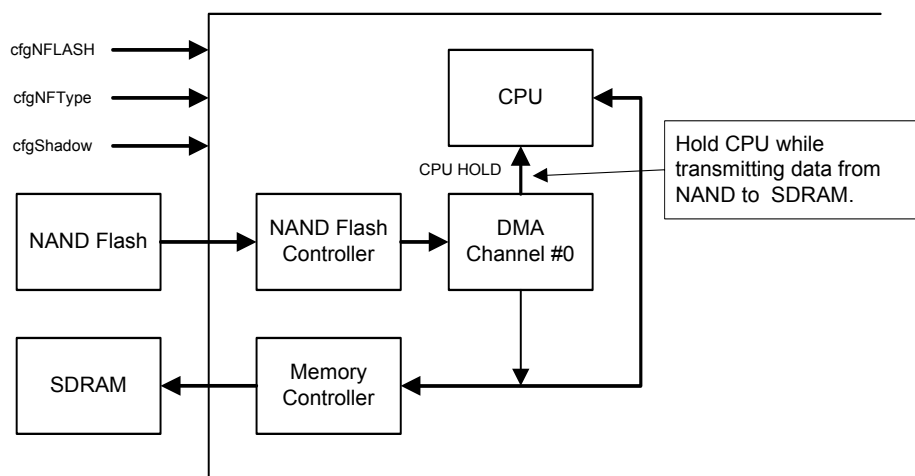
7.2.9 IDE/PCMCIA Interface



When using PCMCIA, additional GPIOs are needed for PCMCIA Interface reset, PCMCIA Power Control. Some signals needed buffer because of different power requirement as above.

7.3 NAND Boot Overview

VRender3D supports the booting function with NAND Flash.
The following figure shows the data path of booting with NAND.



FEATURE

1. Support 8 NAND Flash Banks. : x8, x16bit Bus Width
2. Contain Hardware ECC Block
3. Support NAND Flash Booting

NAND BOOT SEQUENCE

1. In case cfgNFBOOT is high, DMA prohibits main memory access as DMA holds CPU.
2. NAND Flash Controller which reads the data of external NAND flash saves at internal buffer.
3. NAND Flash requests DMA transmission to DMA Channel#0.
4. DMA transmits to main memory.
5. Execute 16/32KB repeatedly according to cfgNFSize.
6. If the transmission is completed, make CPU become normal operation as lifting CPU hold.

Caution when NAND Boot

- When NAND Boot, there should be Read/Writable Memory of 2/4KB in physical address 0 according to cfgNFSize.
- Because ECC is not executed when NAND boot, there should be no bit fail in the early 2/4Kbyte of flash.

NORMAL ACCESS SEQUENCE

READ Cycle

1. Write NAND FLASH Command to NFCMD. (Refer to NAND flash data book)
2. Access address of memory which tries to access to NFADDR register by 3~4 times according to the sort of NAND flash. (Refer to NAND flash data book)
3. Check whether NAND flash is ready or not by reading PendIRQ Bit of MEMNANDCTRLW register.
4. Read data through NFDATA register.
5. Read the result of ECC through NFECC01 and NFECC2 register.

Write Cycle

1. Write NAND FLASH Command to NFCMD. (Refer to NAND flash data book).

2. Access address of memory which tries to access to NFADDR register by 3~4 times according to the sort of NAND flash. (Refer to NAND flash data book)
3. Write data through NFDATA register.
4. Read the result of ECC through NFECC01 and NFECC2 register.
5. Write NAND FLASH Command to NFCMD. (Refer to NAND flash data book)
6. Check whether NAND flash is ready or not by reading PendIRQ Bit of MEMNANDCTRLW register.. For the exact sequence of NAND flash, please refer to NAND Flash data book)

512Byte ECC Parity Code Assignment Table

	I/O 7	I/O 6	I/O 5	I/O 4	I/O 3	I/O 2	I/O 1	I/O 0
ECC0	P64	P64'	P32	P32'	P16	P16'	P8	P8'
ECC1	P1024	P1024'	P512	P512'	P256	P256'	P128	P128'
ECC2	P4	P4'	P2	P2'	P1	P1'	P2048	P2048'

Generate only 512Byte ECC Parity Code during Write / Read operating

ECC Parity Code consists of 3Byte per 512 Byte

24bit ECC Parity Code = 18bit line parity + 6bit Column Parity

ECC generator block executes the followings

1. When MCU writes data to NAND, ECC generator block generates ECC code
2. When MCU reads data from NAND, ECC generator block generates ECC code and compare it with pre-written ECC code

7.4 BankW Memory Control Register

7.4.1 BankW Config Register (MEMCFGW)

Address : C000 5700h

Bit	R/W	Description	Default Value
15 : 8	R	Reserved	-
7	R/W	NANDBW : Set Data Bit Width of NAND Flash 0 : Byte 1 : Half-Word	CfgNFBUSWidth
6	R	Reserved	1
5	R/W	SR5BW : Set Data Bit Width of STATIC#5 0 : Byte 1 : Half-Word	1
4	R/W	SR4BW : Set Data Bit Width of STATIC#4 0 : Byte 1 : Half-Word	1
3	R/W	SR3BW : Set Data Bit Width of STATIC#3 0 : Byte 1 : Half-Word	1
2	R/W	SR2BW : Set Data Bit Width of STATIC#2 0 : Byte 1 : Half-Word	1
1	R/W	SR1BW : Set Data Bit Width of STATIC#1 0 : Byte 1 : Half-Word	1
0	RW	SR0BW : Set Data Bit Width of STATIC#0 0 : Byte 1 : Half-Word	CfgSR0BUSWidth

Above register sets data width of static bank. STATIC#0 points booting code memory when shadow off and data bit width can be defined by CfgSR0BUSWidth when reset.

NANDBW sets NAND Flash memory data bit width. NAND Flash data bit width should be set by CfgNFBUSWidth because NAND Flash contents must be copied to address 0 when CfgNFBoot on.

7.4.2 BankW Timming Register #0 (MEMTIMEW0)

Address : C000 5702h

Bit	R/W	Description	Default Value
15 : 12	R	Reserved	-
11 : 10	R/W	tACS5 : tACS (Address setup-up before nCS) of STATIC#5	11
9 : 8	R/W	tACS4 : tACS of STATIC#4	11
7 : 6	R/W	tACS3 : tACS of STATIC#3	11
5 : 4	R/W	tACS2 : tACS of STATIC#2	11
3 : 2	R/W	tACS1 : tACS of STATIC#1	01
1 : 0	RW	tACS0 : tACS of STATIC#0	11

7.4.3 BankW Timming Register #1 (MEMTIMEW1)

Address : C000 5704h

Bit	R/W	Description	Default Value
15 : 12	R	Reserved	-
11 : 10	R/W	tCOS5 : tCOS (Chip select set-up nOE) of STATIC#5	11
9 : 8	R/W	tCOS4 : tCOS of STATIC#4	11

Bit	R/W	Description	Default Value
7 : 6	R/W	tCOS3 : tCOS of STATIC#3	11
5 : 4	R/W	tCOS2 : tCOS of STATIC#2	11
3 : 2	R/W	tCOS1 : tCOS of STATIC#1	01
1 : 0	RW	tCOS0 : tCOS of STATIC#0	11

7.4.4 BankW Timming Register #2 (MEMTIMEW2)

Address : C000 5706h

Bit	R/W	Description	Default Value
15 : 12	R	Reserved	-
11 : 10	R/W	tOCH5 : tOCH (Chip selection hold on nOE) of STATIC#5	11
9 : 8	R/W	tOCH4 : tOCH of STATIC#4	11
7 : 6	R/W	tOCH3 : tOCH of STATIC#3	11
5 : 4	R/W	tOCH2 : tOCH of STATIC#2	11
3 : 2	R/W	tOCH1 : tOCH of STATIC#1	01
1 : 0	RW	tOCH0 : tOCH of STATIC#0	11

7.4.5 BankW Timming Register #3 (MEMTIMEW3)

Address : C000 5708h

Bit	R/W	Description	Default Value
15 : 12	R	Reserved	-
11 : 10	R/W	tCAH5 : tCAH (Address Holding time after nCS) of STATIC#5	11
9 : 8	R/W	tCAH4 : tCAH of STATIC#4	11
7 : 6	R/W	tCAH3 : tCAH of STATIC#3	11
5 : 4	R/W	tCAH2 : tCAH of STATIC#2	11
3 : 2	R/W	tCAH1 : tCAH of STATIC#1	01
1 : 0	RW	tCAH0 : tCAH of STATIC#0	11

7.4.6 BankW Timming Register #4 (MEMTIMEW4)

Address : C000 570Ah

Bit	R/W	Description	Default Value
15 : 12	R/W	tACC3 : tACC of STATIC#3	1111b
11 : 8	R/W	tACC2 : tACC of STATIC#2	1111b
7 : 4	R/W	tACC1 : tACC of STATIC#1	1000b
3 : 0	RW	tACC0 : tACC of STATIC#0	1111b

7.4.7 BankW Timming Register #5 (MEMTIMEW5)

Address : C000 570Ch

Bit	R/W	Description	Default Value
15 : 8	R	Reserved	0
7 : 4	R/W	tACC5 : tACC of STATIC#5	1111b

Bit	R/W	Description	Default Value
3 : 0	RW	tACC4 : tACC of STATIC#4	1111b

7.4.8 BankW Wait Control Register (MEMWAITCTRLW)

Address : C000 570Eh

Bit	R/W	Description	Default Value
15 : 14	R	Reserved	-
11 : 10	R/W	WAITENB5 : Wait Control Enable of STATIC#5 Bit[1] Enable Contrtol 0 : Disable 1 : Enable Bit[0] Polarity of WRnB 0 : Inversion 1 : Normal	01
9 : 8	R/W	WAITENB4 : Wait Control Enable of STATIC #4	01
7 : 6	R/W	WAITENB3 : Wait Control Enable of STATIC #3	01
5 : 4	R/W	WAITENB2 : Wait Control Enable of STATIC #2	01
3 : 2	R/W	WAITENB1 : Wait Control Enable of STATIC #1	01
1 : 0	RW	WAITENB0 : Wait Control Enable of STATIC #0	01

7.4.9 BankW Page Control Register (MEMPAGEW)

Address : C000 5710h

Bit	R/W	Description	Default Value
15 : 14	R	Reserved	0
13	R/W	SRPAGE5 : SRPAGE (Page Mode) of STATIC#5 0 : Disable 1 : Enable	0
12	R/W	SRPAGE4 : SRPAGE of STATIC#4	0
11	R/W	SRPAGE3 : SRPAGE of STATIC#3	0
10	R/W	SRPAGE2 : SRPAGE of STATIC#2	0
9	R/W	SRPAGE1 : SRPAGE of STATIC#1	0
8	R/W	SRPAGE0 : SRPAGE of STATIC#0	0
7	R	Reserved	0
6	R/W	IOIS16ENB : Enable IOIS16 Control of PCMCIA Region	1
5 : 0	RW	Reserved	0

7.4.10 BankW IDE Timing Control Register (MEMIDETIMEW)

Address : C000 5712h

Bit	R/W	Description	Default Value
15	R	Reserved	-
14 : 10	R/W	tIHOLD : Data Hold Time of IDE BUS	2
9 : 5	R/W	tISETUP : Data Setup Time of IDE BUS	2
4 : 0	RW	tIACC : Data Access Time of IDE BUS	2

7.4.11 BankW PCMCIA Memory Timing Control Register (MEMPCMCIAMW)

Address : C000 5714h

Bit	R/W	Description	Default Value
-----	-----	-------------	---------------

Bit	R/W	Description	Default Value
15	R	Reserved	-
14 : 10	R/W	tPMHOLD : Data Hold Time of PCMCIA Memory Region	3
9 : 5	R/W	tPMSETUP : Data Setup Time of PCMCIA Memory Region	3
4 : 0	RW	tPMACC : Data Access Time of PCMCIA Memory Region	3

7.4.12 BankW PCMCIA Attribute Timing Control Register (MEMPCMCIAAW)

Address : C000 5716h

Bit	R/W	Description	Default Value
15	R	Reserved	-
14 : 10	R/W	tPAHOLD : Data Hold Time of PCMCIA Attribute Region	4
9 : 5	R/W	tPASETUP : Data Setup Time of PCMCIA Attribute Region	4
4 : 0	RW	tPAACC : Data Access Time of PCMCIA Attribute Region	4

7.4.13 BankW PCMCIA I/O Timing Register (MEMPCMCIAIW)

Address : C000 5718h

Bit	R/W	Description	Default Value
15	R	Reserved	-
14 : 10	R/W	tPIHOLD : Data Hold Time of PCMCIA I/O Region	5
9 : 5	R/W	tPISETUP : Data Setup Time of PCMCIA I/O Region	5
4 : 0	RW	tPIACC : Data Access Time of PCMCIA I/O Region	5

7.4.14 BankW PCMCIA Wait Timing Register #0 (MEMPCMCIAWAITW)

Address : C000 571Ah

Bit	R/W	Description	Default Value
15	R	Reserved	-
14 : 10	R/W	PCMWAITM : PCMCIA Wait Control for Memory Region	5
9 : 5	R/W	PCMWAITA : PCMCIA Wait Control for Attribute Region	5
4 : 0	RW	PCMWAITI : PCMCIA Wait Control for I/O Region	5

7.4.15 BankW IDE Wait Timing Register #1 (MEMIDEWAITW)

Address : C000 571Ch

Bit	R/W	Description	Default Value
15 : 5	R	Reserved	-
4 : 0	RW	IDEWAIT : IDE Wait Control	2

7.4.16 BankW DMA Timeout Register (MEMDTIMEOUTW)

Address : C000 5720h

Bit	R/W	Description	Default Value
15	R	Reserved	-

Bit	R/W	Description	Default Value
14 : 10	R/W	DMATIMEOUT :Automatically performs Host termination when there is no response from external device during assigned TIMEOUT. Unit is 8Cycle (BCLK).	4
9 : 0	RW	DMALIMIT : The maximum number of data transfer for occupying IDE BUS.	8

7.4.17 BankW DMA Control Register (MEMDMACTRLW)

Address : C000 5722h

Bit	R/W	Description	Default Value
15 : 14	R	Reserved	-
13	R/W	DMAENB1 : DMA Enable of Channel 1 0 : Disable 1 : Enable	0
12	R/W	DMAENB0 : DMA Enable of Channel 0	1
11 : 10	R	Reserved	-
9	R/W	DMAWIDTH1 : Set DMA of Transfer Width of Channel 1 0 : 8bit 1 : 16bit	1
8	R/W	DMAWIDTH0 : Set DMA Transfer Width of Channel 0	1
7 : 4	R	Reserved	-
3 : 2	R/W	DMAMODE1 : Set DMA Transfer Width of Channel 1 00 : MultiWord DMA Read 01 : MultiWord DMA Write 10 : UltraDMA Read 11 : UltraDMA Write	01
1 : 0	RW	DMAMODE0 : Set DMA mode of Channel 0.	11

7.4.18 BankW DMA Polarity Register (MEMDMAPOLW)

Address : C000 5724h

Bit	R/W	Description	Default Value
15 : 14	R	Reserved	-
13	R/W	DMAREQPOL1 : Set WDREQ1 Polarity of Channel 1 0 : Low Active 1 : High Active	1
12	R/W	DMAREQPOL0 : Set WDREQ1 Polarity of Channel 0	1
11 : 10	R	Reserved	-
9	R/W	DMAACKPOL1 : Set WDACK1 Polarity of Channel 1 0 : Low Active 1 : High Active	0
8	R/W	DMAACKPOL0 : Set WDACK1 Polarity of Channel 0	0
7 : 4	R	Reserved	-
3 : 2	R/W	DMAPATH1 : Set DMA Path of Channel 1 00 : Static Memory 01 : IDE 10 : PCMCIA #0 11 : PCMCIA #1	00
1 : 0	RW	DMAPATH0 : Set DMA Path of Channel 0	01

7.4.19 BankW DMA Timing Register #0 (MEMDMATIMEW0)

Address : C000 5726h

Bit	R/W	Description	Default Value
15 : 10	R	Reserved	-
9 : 5	R/W	DMAtD1 : Set tD(nIOR/nIOW asserted pulse width) Timing of Channel 1	4
4 : 0	RW	DMAtD0 : Set tD Timing of Channel 0	4

7.4.20 BankW DMA Timing Register #1 (MEMDMATIMEW1)

Address : C000 5728h

Bit	R/W	Description	Default Value
15 : 10	R	Reserved	-
9 : 5	R/W	DMAtK1 : Set tK(nIOR/nLOW negated pulse width) Timing of Channel 1	4
4 : 0	RW	DMAtK0 : Set tK Timing of Channel 0	4

7.4.21 BankW DMA Timing Register #2 (MEMDMATIMEW2)

Address : C000 572Ah

Bit	R/W	Description	Default Value
15 : 10	R	Reserved	-
9 : 5	R/W	DMAtACK1 : Set tACK(Setup Time before nDMACK) Timing of Channel 1	4
4 : 0	RW	DMAtACK0 : Set tACK Timing of Channel 0	4

7.4.22 BankW DMA Timing Register #3 (MEMDMATIMEW3)

Address : C000 572Ch

Bit	R/W	Description	Default Value
15 : 10	R	Reserved	-
9 : 5	R/W	DMAtENV1 : Set tENV (Envelope time) Timing of Channel 1	4
4 : 0	RW	DMAtENV0 : Set tENV Timing of Channel 0	4

7.4.23 BankW DMA Timing Register #4 (MEMDMATIMEW4)

Address : C000 572Eh

Bit	R/W	Description	Default Value
15 : 10	R	Reserved	-
9 : 5	R/W	DMAtRP1 : Set tRP (Ready to pause time) Timing of Channel 1	4
4 : 0	RW	DMAtRP0 : Set tRP Timing of Channel 0	4

7.4.24 BankW DMA Timing Register #5 (MEMDMATIMEW5)

Address : C000 5730h

Bit	R/W	Description	Default Value
15 : 10	R	Reserved	-
9 : 5	R/W	DMAtSS1 : Set tSS (Time from DSTROVE edge to negation of DMARQ) Timing of Channel 1	4
4 : 0	RW	DMAtSS0 : Set tSS Timing of Channel	4

7.4.25 BankW DMA Timing Register #6 (MEMDMATIMEW6)

Address : C000 5732h

Bit	R/W	Description	Default Value
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7.4.30 BankW NAND Timing Register (MEMNANDTIMEW)

Address : C000 573Ch

Bit	R/W	Description	Default Value
15 : 12	R	Reserved	-
11 : 8	R/W	NFtSETUP : Set Data Setup Time of NAND Flash	4
7 : 4	R/W	NFtACC : Set Data Access Time of NAND Flash	15
3 : 0	RW	NFtHOLD : Set Data Hold Time of NAND Flash	4

7.4.31 BankW NAND ECC0 Register (MEMNANDECC0W)

Address : C000 573Eh

Bit	R/W	Description	Default Value
15 : 8	R	ECC0UPPER : Upper 8bit Data out of 16bit will be output in case of NAND Flash It will be disregarded in case of 8bit	-
7 : 0	R	ECC0LOWER : Lower 8bit Data out of 16bit will be output in case of NAND Flash. 8bit ECC data will be output in case of 8bit NAND Flash.	-

7.4.32 BankW NAND ECC1 Register (MEMNANDECC1W)

Address : C000 5740h

Bit	R/W	Description	Default Value
15 : 8	R	ECC1UPPER : Upper 8bit Data out of 16bit will be output in case of NAND Flash It will be disregarded in case of 8bit	-
7 : 0	R	ECC1LOWER : Lower 8bit Data out of 16bit will be output in case of NAND Flash. 8bit ECC data will be output in case of 8bit NAND Flash.	-

7.4.33 BankW NAND ECC2 Register (MEMNANDECC2W)

Address : C000 5742h

Bit	R/W	Description	Default Value
15 : 8	R	ECC1UPPER : Upper 8bit Data out of 16bit will be output in case of NAND Flash It will be disregarded in case of 8bit	-
7 : 0	R	ECC1LOWER : Lower 8bit Data out of 16bit will be output in case of NAND Flash. 8bit ECC data will be output in case of 8bit NAND Flash.	-

7.4.34 BankW NAND Data Counter Register (MEMNANDCNTW)

Address : C000 5744h

Bit	R/W	Description	Default Value
15 : 10	R	Reserved	-
9 : 0	R	DATAcnt : Read transferred number of Data	-

7.4.35 BankW Chip Selection Control Register (MEMCHIPSELTW)

Address : C000 5746h

Bit	R/W	Description	Default Value
-----	-----	-------------	---------------

Bit	R/W	Description	Default Value
15 : 14	R	Reserved	-
13	R/W	SELNCS0 : Set function of WnNCS0/WnCS2 0 : WnCS2 (Static Memory 2) 1 : WnNCS0 (NAND Bank 0)	1
12	R/W	SELNCS1 : Set function of WnNCS1/WnCS3 0 : WnCS3 (Static Memory 3) 1 : WnNCS1 (NAND Bank 1)	1
11	R/W	SELPCS0 : Set function of WnNCS0/WnCS4/WnNCS2 0 : WnCS4 or WnNCS2 1 : WnPCS0 (PCMCIA CS0)	1
10	R/W	SELPCS1 : Set function of WnNCS1/WnCS5/WnNCS3 0 : WnCS5 or WnNCS3 1 : WnPCS1 (PCMCIA CS1)	1
9	R/W	SELICS0 : Set function of WnICS0/WnNCS4 0 : WnNCS4 1 : WnICS0 (IDE CS0)	1
8	R/W	SELICS1 : Set function of WnICS1/WnNCS5 0 : WnNCS5 1 : WnICS1 (IDE CS1)	1
7	R/W	SELCS0 : Set function of WnCS0/WnNCS6 0 : WnNCS6 1 : WnCS0 (Static Memory 0)	1
6	R/W	SELCS1 : Set function of WnCS1/WnNCS7 0 : WnNCS7 1 : WnCS1 (Static Memory 1)	1
5	R/W	SELCS4 : Set function of WnNCS0/WnCS4/WnNCS2 0 : WnNCS2 1 : WnCS4 (Static Memory 4)	0
4	R/W	SELCS5 : Set function of WnNCS1/WnCS5/WnNCS3 0 : WnNCS3 1 : WnCS4 (Static Memory 5)	0
3	R/W	SELPCS0 : Write same value of Bit[11]	1
2	R/W	SELPCS1 : Write same value of Bit[10]	1
1	R/W	SELICS0 : Write same value of Bit[9]	1
0	RW	SELICS1 : Write same value of Bit[8]	1

7.4.36 NAND Flash Data Register (NFDATA)

Address : 1C00 0000h (CfgShadow=0)

Address : 9C00 0000h (CfgShadow=1)

Bit	R/W	Description	Default Value
15 : 0	R/W	NFDATA : Data Register	00000000h

This will be connected to data bus of NAND flash.

In case of 16bit access, NAND flash will access twice by 8bit automatically.

7.4.37 NAND Flash Command Register (NFCMD)

Address : 1C00 0010h (CfgShadow=0)

Address : 9C00 0010h (CfgShadow=1)

Bit	R/W	Description	Default Value
15:8	R	Reserved	0
7:0	R	NFCMD : Command Register	0

This will be connected to data bud of NAND flash. As CLE becomes high, notify command cycle.

This register will be accessed by byte different from NFDATA register.

7.4.38 NAND Flash Address Register (NFADDR)

Address : 1C00 0018h (CfgShadow=0)

Address : 9C00 0018h (CfgShadow=1)

Bit	R/W	Description	Default Value
15:8	R	Reserved	0
7:0	W	NFADDR : Address Register	0

This will be connected to data bus of NAND flash. As ALE becomes high, notify address cycle.
This register will be accessed by byte different from NFDATA register.

7.5 BankX Memory Control Register

7.5.1 BankX SDRAM Configuration Register (MEMCFGX)

Address : C000 5400h

Bit	R/W	Description	Default Value
15 : 7	R	Reserved	-
6	R/W	SDRTYPE : Set DRAM Type. 0 : Reserved 1 : DDR-DRAM	CfgSDRTYPE
5 : 4	R/W	SDRBUSBW : Set data bit width of Bank. 00 : Reserved 01 : Reserved 10 : 32bit 11 : 64bit (DDR)	CfgSDRBUSBW
3 : 2	R/W	SDRBW : Set SDRAM bit width of DRAM which is used in Bank 00 : 4bit 01 : 8bit 10 : 16bit 11 : Reserved	CfgSDRBW
1 : 0	RW	SDRCAP : Set SDRAM Capacity of DRAM which is used in Bank 00 : 64Mbit 01 : 128Mbit 10 : 256Mbit 11 : 512Mbit	CfgSDRType

7.5.2 BankX TIMMING Register #0 (MEMTIMEX0)

Address : C000 5402h

Bit	R/W	Description	Default Value
15 : 12	R/W	tMRD : LOAD MODE REGISTER command cycle time	2
11 : 8	R/W	tRFC : AUTO REFRESH command period	Fh
7 : 4	R/W	tRP : PRECHARGE command period	4
3 : 0	RW	tRCD : ACTIVE to READ or WRITE delay	4

7.5.3 BankX TIMMING Register #1 (MEMTIMEX1)

Address : C000 5404h

Bit	R/W	Description	Default Value
15	R/W	ModeSet (Write) : Set 1 to set new SDRAM mode parameter. ModeStatus (Read) : Mode Setting Status 0 : Complete 1 : Busy	0
14	R	Reserved	
13:12	R/W	LAT : CAS Latency 00 : 2 Cycle 01 : 2.5 Cycle 1x : 3 Cycle	00
11 : 8	R/W	tRC : ACTIVE to ACTIVE/AUTO REFRESH command period 0 : 1Cycle	Ah
7 : 4	R/W	tRAS : ACTIVE to PRECHARGE command	7
3 : 0	RW	tWR : Write recovery time	2

7.5.4 BankX Active Power Down Control Register (MEMACTPWDX)

Address : C000 5406h

[illegible]

Bit	R/W	Description	Default Value
7 : 0	RW	ACTPERD : Active Power Down Register	FFh

If **ACTPWDEN** is set to 1 and Memory Controller is IDLE, move to Active Power Down Mode. The value of **ACTPERD** decides to move to Active Power down mode after how long clock the IDLE is maintained.

7.5.5 BankX Refresh Control Register (MEMREFX)

Address : C000 5408h

Bit	R/W	Description	Default Value
15 : 0	RW	REFPERD : Refresh Period (Unit BCLK/2)	0100h

Bit	R/W	Description	Default Value
15 : 9	R	Reserved	0
8	R/W	ACTPDWEN : Active Power Down Mode Enable 0 : Disable 1 : Enable	0
7 : 0	RW	ACTPERD : Active Power Down Register	FFh

If **ACTPWDEN** is set to 1 and Memory Controller is IDLE, move to Active Power Down Mode. The value of **ACTPERD** decides to move to Active Power down mode after how long clock the IDLE is maintained.

7.6.5 BankY Refresh Control Register (MEMREFY)

Address : C000 5508h

Bit	R/W	Description	Default Value
15 : 0	RW	REFPERD : Refresh Period, Unit PCLK	0100h

7.7 BankZ Memory Control Register

7.7.1 BankZ SDRAM Configuration Register (MEMCFGZ)

Address : C000 5600h

Bit	R/W	Description	Default Value
15 : 7	R	Reserved	-
6	R/W	SDRTYPE Set DRAM Type. 0 : Reserved 1 : DDR-DRAM	CfgSDRTYPE
5 : 4	R/W	SDRBUSBW Set data bit width of Bank. 00 : Reserved 01 : Reserved 10 : 32bit 11 : 64bit (DDR)	CfgSDRBUSBW
3 : 2	R/W	SDRBW Set SDRAM bit width of DRAM which is used in Bank 00 : 4bit 01 : 8bit 10 : 16bit 11 : Reserved	CfgSDRBW
1 : 0	RW	SDRCAP Set SDRAM Capacity of DRAM which is used in Bank 00 : 64Mbit 01 : 128Mbit 10 : 256Mbit 11 : 512Mbit	CfgSDRType

7.7.2 BankZ TIMMING Register #0 (MEMTIMEZ0)

Address : C000 5602h

Bit	R/W	Description	Default Value
15 : 12	R/W	tMRD : LOAD MODE REGISTER command cycle time	2
11 : 8	R/W	tRFC : AUTO REFRESH command period	Fh
7 : 4	R/W	tRP : PRECHARGE command period	4
3 : 0	RW	tRCD : ACTIVE to READ or WRITE delay	4

7.7.3 BankZ TIMMING Register #1 (MEMTIMEZ1)

Address : C000 5604h

Bit	R/W	Description	Default Value
15	R/W	ModeSet (Write) : Set 1 to set new SDRAM mode parameter. ModeStatus (Read) : Mode Setting Status 0 : Complete 1 : Busy	0
14	R	Reserved	
13 : 12	R/W	LAT : CAS Latency 00 : 2 Cycle (CfgSDRLATx='0') 01 : 2.5 Cycle 1x : 3 Cycle (CfgSDRLATx='1')	00
11 : 8	R/W	tRC : ACTIVE to ACTIVE/AUTO REFRESH command period 0 : 1Cycle	Ah
7 : 4	R/W	tRAS : ACTIVE to PRECHARGE command	7
3 : 0	RW	tWR : Write recovery time	2

7.7.4 BankZ Active Power Down Control Register (MEMACTPWDZ)

Address : C000 5606h

Bit	R/W	Description	Default Value
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Bit	R/W	Description	Default Value
15 : 9	R	Reserved	0
8	R/W	ACTPW DEN : Active Power Down Mode Enable 0 : Disable 1 : Enable	0
7 : 0	RW	ACTPERD : Active Power Down Register	FFh

If **ACTPWDEN** is set to 1 and Memory Controller is IDLE, move to Active Power Down Mode. The value of **ACTPERD** decides to move to Active Power down mode after how long clock the IDLE is maintained.

7.7.5 BankZ Refresh Control Register (MEMREFZ)

Address : C000 5608h

Bit	R/W	Description	Default Value
15 : 0	RW	REFPERD : Refresh Period, Unit PCLK	0100h

8. Interrupt Controller

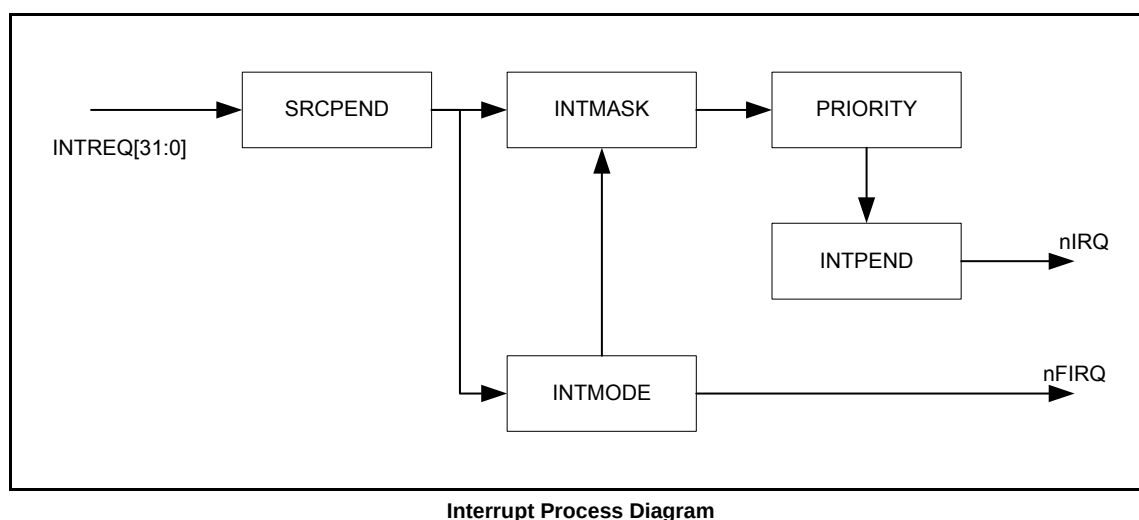
8.1 Overview

ARM920T in 28 interrupt sources from internal peripherals, including the DMA controller, UART, I2C and GPIO, etc. supply requests to an Interrupt Controller.

An FIQ or an IRQ (interrupt requests) are signaled by the Interrupt Controller to ARM920T or ARM940T. After arbitrating multiple requests from internal peripherals and GPIO, the Controller requests an interrupt.

The hardware arbitration logic decides the interrupt arbitration process, and the results are recorded in the interrupt pending registers.

8.1.1 Block Diagram



8.2 Interrupt controller Operation

The interrupt controller consists of five control registers including the interrupt pending register, priority register, mask register, interrupt mode register and source pending register.

The first registration of all the interrupt requests from the interrupt source is done in the source pending register. Based on the interrupt mode register, interrupt requests are divided into two groups: FIQ requests and IRQ requests. Arbitration is performed for multiple IRQ requests according to the priority register.

F-bit and I-bit of PSR (program status register)

A FIQ (fast interrupt request) from the Interrupt Controller is not accepted by the CPU when the F-bit of PSR (program status register in ARM920T CPU) is set to 1. The IRQ (interrupt request) from the interrupt controller is not accepted by the CPU when the I-bit of PSR (program status register in ARM920T CPU) is set to 1. Therefore, the PSR F-bit or I-bit must be cleared to 0 and the corresponding bit of INTMSK must be set to 0 to enable the interrupt reception.

Interrupt Mode

ARM920T has 2 types of interrupt mode, FIQ or IRQ. All the interrupt sources determine the mode of interrupt to be used at interrupt request.

Interrupt Pending Register

There are two interrupt pending registers in VR3D - the source pending register (SRCPND) and the interrupt pending register (INTPND). The interrupt pending status is indicated by these pending registers. One bit of the INTPND register is set to 1 automatically after arbitration, and when the interrupt sources request interrupt service the corresponding bit of the SRCPND register is set to 1. The INTPND register bit is not changed when the corresponding bits of the SRCPND register are set to 1 and the interrupts are masked. The interrupt service routine starts whenever the I-flag or F-flag is set to 0 when a pending bit of the INTPND register is set. You can

read and write to the SRCPND and the INTPND registers and the service routine has to clear the pending status by first writing a 1 to the corresponding bit in the SRCPND register and then clearing the pending status in the INTPND register the same way.

Interrupt Mask Register

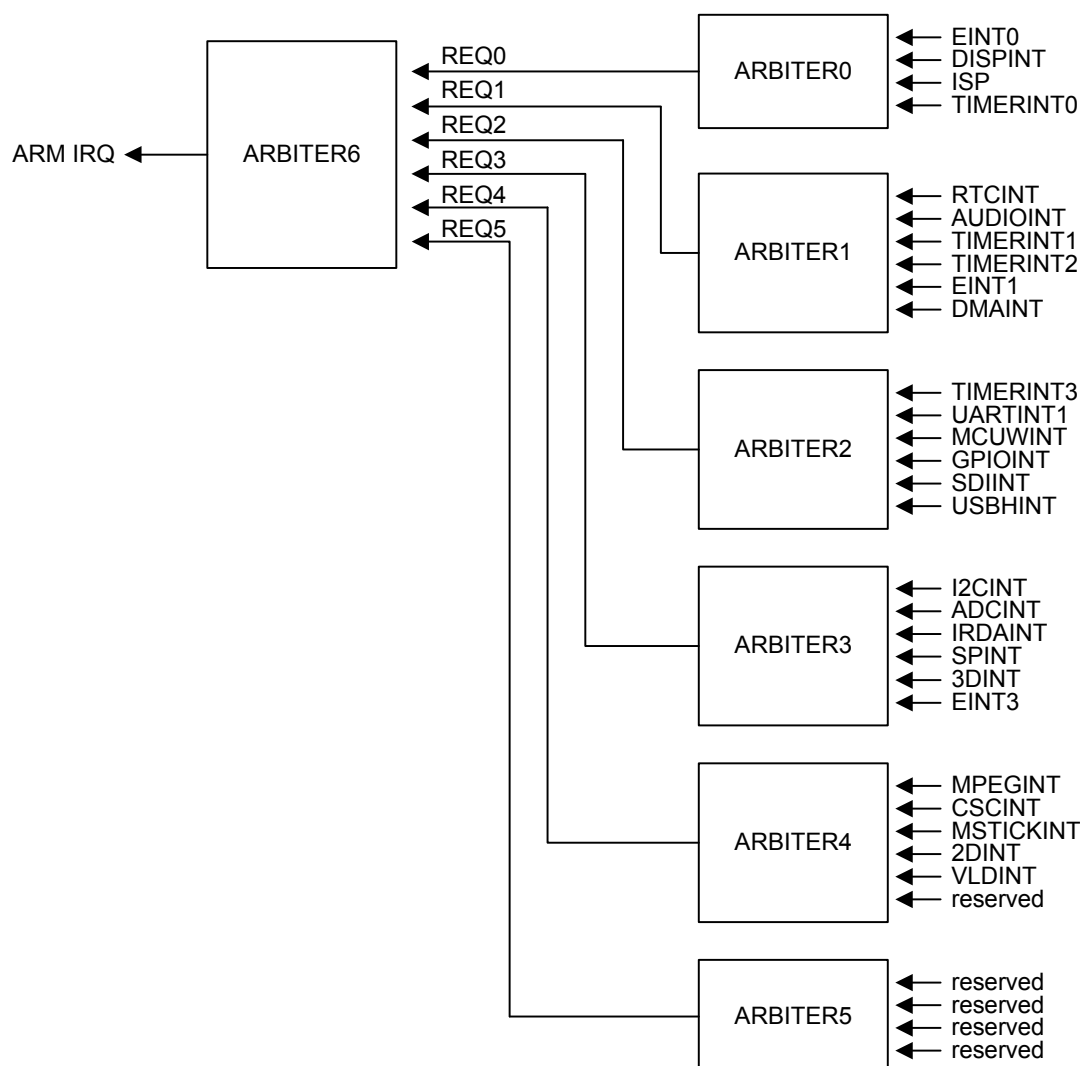
Indicates that an interrupt has been disabled if the corresponding mask bit is 1. If an interrupt mask bit of INTMSK is 0, the interrupt will be serviced normally. If the corresponding mask bit is 1 and the interrupt is generated, the source pending bit will be set.

8.2.1 Interrupt Sources

Interrupt Num	Source	Description
0	EINT0	External Interrupt 0
1	DISPINT	Multi Layer Controller
2	ISP	ISP
3	TIMERINT0	Timer 0
4	Reserved	Reserved
5	AUDIOINT	Audio
6	TIMERINT1	Timer 1
7	TIMERINT2	Timer 2
8	EINT1	External Interrupt 1
9	DMAINT	DMA
10	TIMERINT3	Timer 3
11	UARTINT	UART
12	MCUWINT	
13	GPIOINT	
14	EINT2	External Interrupt 2
15	SDIINT	MMC/SD
16	USBHINT	USB Host
17	PWRMANINT	
18	USBDMINT	USB Device
19	I2CINT	I2C
20	ADCINT	ADC
21	Reserved	
22	SSPINT	SSP
23	3DINT	3D graphic core
24	EINT3	External Interrupt 3
25	MPEGINT	
26	CSCINT	
27	Reserved	
28	2DINT	2D graphic core
29	VLDINT	
30~31	Reserved	Reserved

8.2.2 Interrupt Priority Generating Block

The priority logic for 23 interrupt requests is composed of seven rotation based arbiters: six first-level arbiters and one second-level arbiter as shown in the following figure.



Each arbiter can handle six interrupt requests based on the one bit arbiter mode control (ARB_MODE) and two bits of selection control signals (ARB_SEL) as follows:

If ARB_SEL bits are 00b, the priority order is REQ0, REQ1, REQ2, REQ3, REQ4, and REQ5.

If ARB_SEL bits are 01b, the priority order is REQ0, REQ2, REQ3, REQ4, REQ1, and REQ5.

If ARB_SEL bits are 10b, the priority order is REQ0, REQ3, REQ4, REQ1, REQ2, and REQ5.

If ARB_SEL bits are 11b, the priority order is REQ0, REQ4, REQ1, REQ2, REQ3, and REQ5.

Note that REQ0 of an arbiter is always the highest priority, and REQ5 is the lowest one. In addition, by changing the ARB_SEL bits, we can rotate the priority of REQ1 - REQ4.

Here, if ARB_MODE bit is set to 0, ARB_SEL bits are not automatically changed, thus the arbiter operates in the fixed priority mode. (Note that even in this mode, we can change the priority by manually changing the ARB_SEL bits.). On the other hand, if ARB_MODE bit is 1, ARB_SEL bits are changed in rotation fashion, e.g., if REQ1 is serviced, ARB_SEL bits are changed to 01b automatically so as to make REQ1 the lowest priority one. The detailed rule of ARB_SEL change is as follows.

If REQ0 or REQ5 is serviced, ARB_SEL bits are not changed at all.

If REQ1 is serviced, ARB_SEL bits are changed to 01b.

If REQ2 is serviced, ARB_SEL bits are changed to 10b.

If REQ3 is serviced, ARB_SEL bits are changed to 11b.

If REQ4 is serviced, ARB_SEL bits are changed to 00b.

8.3 Interrupt Controller Registers

8.3.1 Source Pending Register (SRCPENDREG)

Address : C000 0800h

Bit	R/W	Description	Default Value
31 : 0	RW	Read : Returns the sources that are waiting for the interrupt to be serviced. Write : Clears only the bit positions that is set in the data. The bit positions that are not set in the data remain with no change. In IRQ service routine, the content of INTPENDREG is appropriate to be used as data to clean the SRCPENDREG. In FIQ service routine, the content of INTMODEREG can be used as data.	00000000h

8.3.2 Interrupt Mode Register (INTMODEREG)

Address : C000 0804h

Bit	R/W	Description	Default Value
31 : 0	RW	Write : Sets the interrupt mode for each interrupt source ; 0 is for IRQ and 1 is for FIQ. At most, only one source can be serviced via FIQ.	00000000h

8.3.3 Interrupt Mask Register (MASKREG)

Address : C000 0808h

Bit	R/W	Description	Default Value
31 : 0	RW	Read : Shows the contents of this SFR. Write : Sets the mask bit of each interrupt source ; 1 is masked off and 0 is non-masked(serviced). We could not find any advantage of using global masks yet. So that feature is not supported yet. But it is really easy to support global masking feature.	00000000h

8.3.4 Interrupt Priority Register (PRIORITYREG)

Address : C000 080Ch

Bit	R/W	Description	Default Value
20:19	R/W	ARB_SEL6 : Arbiter 6 group priority order set 00 : REQ 0-1-2-3-4-5 01 : REQ 0-2-3-4-1-5 10 : REQ 0-3-4-1-2-5 11 : REQ 0-4-1-2-3-5	0
18:17	R/W	ARB_SEL5 : Arbiter 5 group priority order set 00 : REQ 0-1-2-3-4-5 01 : REQ 0-2-3-4-1-5 10 : REQ 0-3-4-1-2-5 11 : REQ 0-4-1-2-3-5	0
16:15	R/W	ARB_SEL4 : Arbiter 4 group priority order set	0

Bit	R/W	Description	Default Value
		00 : REQ 0-1-2-3-4-5 01 : REQ 0-2-3-4-1-5 10 : REQ 0-3-4-1-2-5 11 : REQ 0-4-1-2-3-5	
14:13	R/W	ARB_SEL3 : Arbiter 3 group priority order set 00 : REQ 0-1-2-3-4-5 01 : REQ 0-2-3-4-1-5 10 : REQ 0-3-4-1-2-5 11 : REQ 0-4-1-2-3-5	0
12:11	R/W	ARB_SEL2 : Arbiter 2 group priority order set 00 : REQ 0-1-2-3-4-5 01 : REQ 0-2-3-4-1-5 10 : REQ 0-3-4-1-2-5 11 : REQ 0-4-1-2-3-5	0
10:9	R/W	ARB_SEL1 : Arbiter 1 group priority order set 00 : REQ 0-1-2-3-4-5 01 : REQ 0-2-3-4-1-5 10 : REQ 0-3-4-1-2-5 11 : REQ 0-4-1-2-3-5	0
8:7	R/W	ARB_SEL0 : Arbiter 0 group priority order set 00 : REQ 0-1-2-3-4-5 01 : REQ 0-2-3-4-1-5 10 : REQ 0-3-4-1-2-5 11 : REQ 0-4-1-2-3-5	0
6	R/W	ARB_MODE6 : Arbiter6 group rotate enable 0 : Priority does not rotate 1 : Priority rotate enable	1
5	R/W	ARB_MODE5 : Arbiter5 group rotate enable 0 : Priority does not rotate 1 : Priority rotate enable	1
4	R/W	ARB_MODE4 : Arbiter4 group rotate enable 0 : Priority does not rotate 1 : Priority rotate enable	1
3	R/W	ARB_MODE3 : Arbiter3 group rotate enable 0 : Priority does not rotate 1 : Priority rotate enable	1
2	R/W	ARB_MODE2 : Arbiter2 group rotate enable 0 : Priority does not rotate 1 : Priority rotate enable	1
1	R/W	ARB_MODE1 : Arbiter1 group rotate enable 0 : Priority does not rotate 1 : Priority rotate enable	1
0	RW	ARB_MODE0 : Arbiter0 group rotate enable 0 : Priority does not rotate 1 : Priority rotate enable	1

8.3.5 Interrupt Pending Register (INTPENDREG)

Address : C000 0810h

Bit	R/W	Description	Default Value
31 : 0	RW	Read : Shows the highest priority interrupt that is waiting to be serviced or already under being serviced. Only one bit should be 1. Write : Clears only the bit positions that is set in the data. The bit positions that are not set in the data remain with no change. In IRQ service routine, the content of INTPENDREG is appropriate to be used as data to clean. Note : This register is only for the IRQ sources. FIQ request under being serviced is not shown in this register. Thus, in FIQ service routine, we don't need to clean up this register.	00000000h

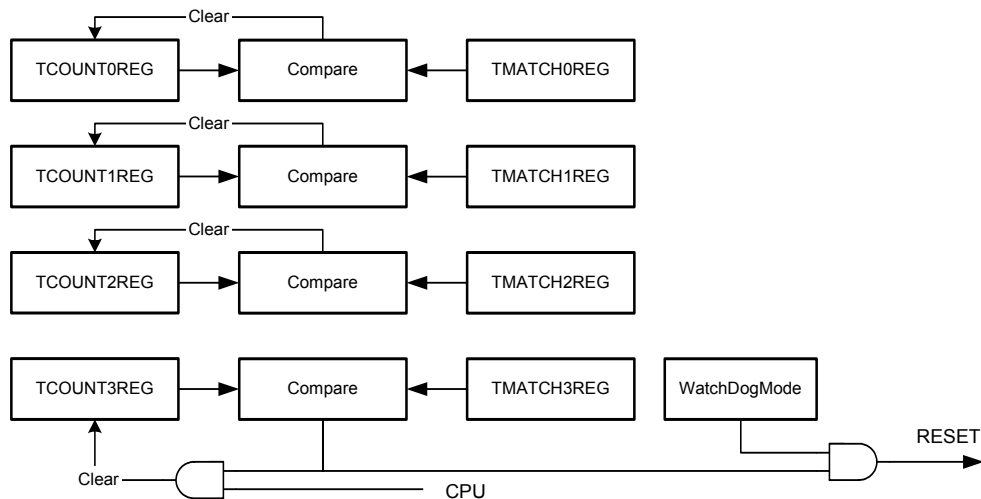
Because of Interrupt Controller Bug. The bits of INTPENDREG are should set in simultaneously. So. The interrupt handler must handle it. Refer to Interrupt handler application note.

9. Timer

9.1 Overview

There are 4 Timers in VR3D and each Timer operates by divided clock from clock of Clock Generator. Each Timer performs Free running up-counting and is cleared when has same value of each TMATCH*REG(Timer Match Counter).

Timer3 is used as Watchdog Timer. Timer3 Clock should be not divided clock but original clock when it used for Watchdog Timer. Writing 0 in Interrupt Pending Register of Timer3 will clear Watchdog Timer.



9.2 Timer Controller Registers

9.2.1 Timer Counter0 Register (TCOUNT0REG)

Address : C000 0A00h

Bit	R/W	Description	Default Value
15 : 0	R	Timer counter0 register When TMATCH0REG equal TCOUNT0REG, This Value will be to 0.	0000h

9.2.2 Timer Counter1 Register (TCOUNT1REG)

Address : C000 0A02h

Bit	R/W	Description	Default Value
15 : 0	R	Timer counter1 register	0000h

9.2.3 Timer Counter2 Register (TCOUNT2REG)

Address : C000 0A04h

Bit	R/W	Description	Default Value
15 : 0	R	Timer counter2 register	0000h

9.2.4 Timer Counter3 Register (TCOUNT3REG)

Address : C000 0A06h

Bit	R/W	Description	Default Value
15 : 0	R	Timer counter3 register	0000h

9.2.5 Timer Match 0 Register (TMATCH0REG)

Address : C000 0A08h

Bit	R/W	Description	Default Value
15 : 0	R	Timer match 0 register	0000h

9.2.6 Timer Match 1 Register (TMATCH1REG)

Address : C000 0A0Ah

Bit	R/W	Description	Default Value
15 : 0	R	Timer match 1 register least significant half word	0000h

9.2.7 Timer Match 2 Register (TMATCH2REG)

Address : C000 0A0Ch

Bit	R/W	Description	Default Value
15 : 0	R	Timer match 2 register least significant half word	0000h

9.2.8 Timer Match 3 / Watchdog Register (TMATCH3REG)

Address : C000 0A0Eh

Bit	R/W	Description	Default Value
15 : 0	R	Timer match 3 register least significant half word	0000h

9.2.9 Watchdog Timer Control Register (TCONTROLREG)

Address : C000 0A10h

Bit	R/W	Description	Default Value
15:14	RW	Select Timer0 Clock, 00:TCLK/2, 01:TCLK/4, 10:TCLK/8, 11:TCLK	11b
13:12	RW	Select Timer1 Clock, 00:TCLK/2, 01:TCLK/4, 10:TCLK/8, 11:TCLK	11b
11:10	RW	Select Timer2 Clock, 00:TCLK/2, 01:TCLK/4, 10:TCLK/8, 11:TCLK	11b
9:8	RW	Select Timer3 Clock, 00:TCLK/2, 01:TCLK/4, 10:TCLK/8, 11:TCLK 'TCLK' should be used in Watchdog Mode	11b
7	RW	Timer3 mode select 0: Normal timer mode 1: Watchdog timer mode	0
6	RW	Timer Operation Enable 0: Disable 1: Enable	0
5:4		Reserved	Xx
3	RW	Timer3 Interrupt Enable	0
2	RW	Timer2 Interrupt Enable	0
1	RW	Timer1 Interrupt Enable	0
0	RW	Timer0 Interrupt Enable. 0 : Disable , 1: Enable	0b

9.2.10 Timer Interrupt Pending Register (TINTPENDREG)

Address : C000 0A12h

Bit	R/W	Description	Default Value
3	R	Timer Match3 interrupt Pending / TCOUNT3REG clear [R] 0: no interrupt is generated 1: interrupt caused [W]	0b

Bit	R/W	Description	Default Value
		0: Clear TCOUNT3REG in Watchdog Mode Does nothing in Normal Mode 1: interrupt pending bit clear	
2	R	Timer Match2 interrupt Pending	0b
1	R	Timer Match1 interrupt Pending	0b
0	R	Timer Match0 interrupt Pending	0b

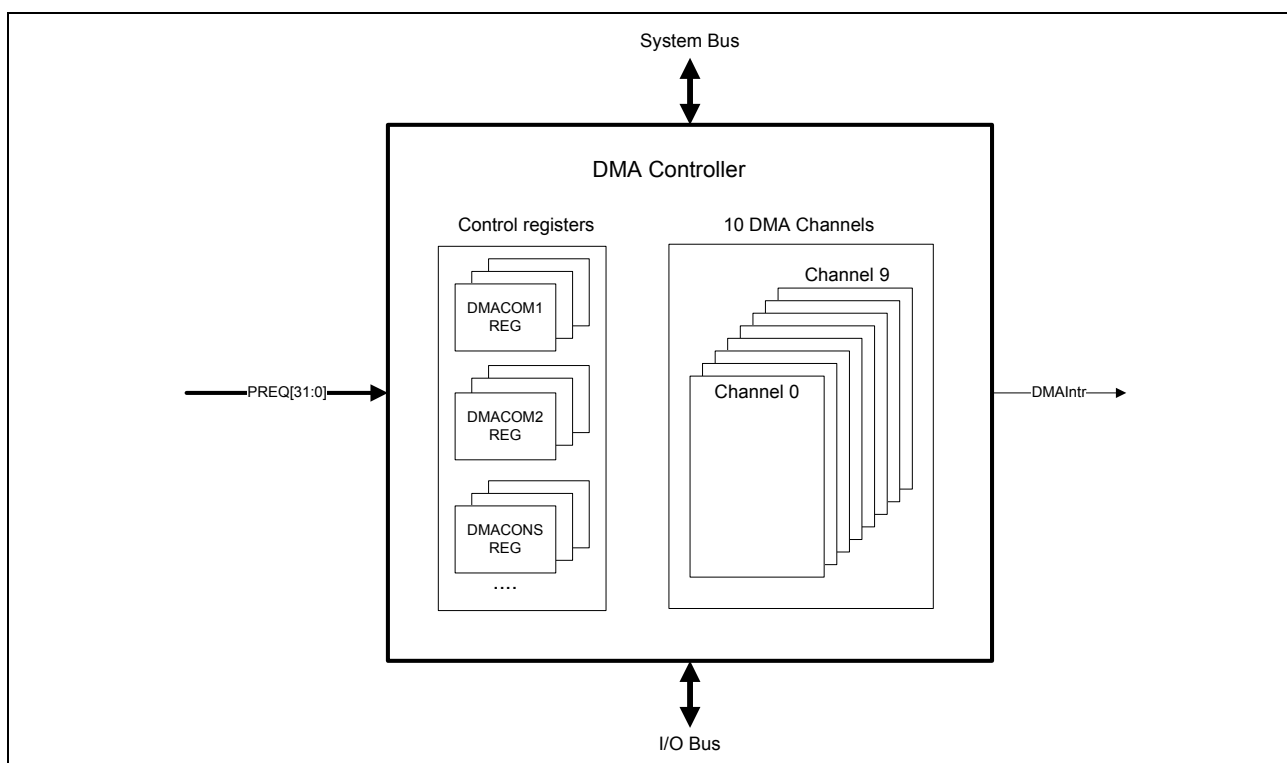
10. DMA

10.1 Overview

The DMA controller has 10 Channels and 4 words or 8 words of burst transmission. In addition, each channel has 8 EA of 32-bit buffers. VR3D supports the 10-channel DMA controller between system buses and peripheral buses. Each channel of the DMA controller can unrestrictedly transmit Data to each device connected to a system bus or a peripheral bus. Therefore, each channel allows devices to transfer data under the following conditions: 1) when the source and the destination are system buses, 2) when the source is a system bus and the destination is a peripheral bus, 3) when the source is a peripheral bus and the destination is a system bus, 4) when the source and the destination are peripheral buses.

The best advantage of the DMA controller is that it can perform the transfer of Data without CPU intervention. The DMA controller performs DMA operation by software, an internal peripheral request or external request.

10.1.1 Block Diagram



DMA Controller Block Diagram

10.2 DMA Sources

DMA Index	Source	Description
0	UART0_TX	UART Channel 0, TX
1	UART0_RX	UART Channel 0, RX
2	UART1_TX	UART Channel 1, TX
3	UART1_RX	UART Channel 1, Rx
4~5	Reserved	Reserved
6	UDC_EP1	USB Device Endpoint 1
7	UDC_EP2	USB Device Endpoint 2
8	UDC_EP3	USB Device Endpoint 3
9	UDC_EP4	USB Device Endpoint 4
10	SDI	MMC/SD DMA Request
11	Reserved	Reserved
12	ADC	ADC
13	Reserved	Reserved
14	AUDIO_PLAY	Audio Play DMA Request
15	AUDIO_REC	Audio Record DMA Request
16	SSP_WRITE	SSP Write Request
17	SSP_READ	SSP Read Request
18~49	Reserved	Reserved
50	EDMA0	External DMA Request 0
51	EDMA1	External DMA Request 1
52~54	Reserved	Reserved
55	SP	MPEG Stream Processor
56~59	Reserved	Reserved
60	HUFFMAN	MPEG HUFFMAN
61	RVLD	MPEG VLD

10.3 DMA Control Registers

DMA_CH_BASE :

CH0 : 0200h
CH4 : 0240h
CH8 : 0280h

CH1 : 0210h
CH5 : 0250h
CH9 : 0290h

CH2 : 0220h
CH6 : 0260h

CH3 : 0230h
CH7 : 0270h

10.3.1 DMA Command Register_0 (DMACOM0REG)

Address : C000 0000h + DMA_CH_BASE

Bit	R/W	Description	Default Value
[15:14]	R/W	BurstOperSize : Burst Operation Request Size "01" : 4 Words, "10" - 8 Words, Other - 1 Word	0b
[13]	R/W	IncSrcAddr : Source Address Increment setting 0: Disable 1: Enable Increment/Decrement	0b
[12]	R/W	FlowSrc : '1' Wait for a request signal before data transaction	0b
[11:10]	R/W	SrcMemFormat : Memory Data Format "00"-WORD, "01"-LS HWORD, "10"-MS HWORD	1b
[9:8]	R/W	SrcPeriWidth : IO Peripheral Device Data Bit Width e "00" : BYTE, "01" : HWORD	00b
[7:6]	R/W	Reserved	0b
[5]	R/W	IncTrgAddr : Target Address Increment/Decrement setting 0: Disable 1: Enable Increment/Decrement	0b
[4]	R/W	FlowTrg : '1' Wait for a request signal before data transaction	0b
[3:2]	R/W	TrgMemFormat : Memory Data Format "00"-WORD, "01"-LS HWORD, "10"-MS HWORD	0b
[1:0]	R/W	TrgPeriWidth : IO Peripheral Device Data Bit Width "00" : BYTE, "01" : HWORD	00b

10.3.2 DMA Command Register_1 (DMACOM1REG)

Address : C000 0002h+ DMA_CH_BASE

Bit	R/W	Description	Default Value
[15:8]	R/W	SrcAddrStep : Address Increment/Decrement in Word Step : -128 ~ 127	0b
[7:0]	R/W	TrgAddrStep : Address Increment/Decrement in Word Step : -128 ~ 127	0b

10.3.3 DMA Command Register_2 (DMACOM2REG)

Address : C000 0004h+ DMA_CH_BASE

Bit	R/W	Description	Default Value
[15:0]	R/W	Length : Request Length in Byte	0000h

10.3.4 DMA Control/Status Register (DMACONSREG)

Address : C000 0006h+ DMA_CH_BASE

Bit	R/W	Description	Default Value
[10]	R/W	DMARun : '1' : DMA Start '0' : DMA End	0b

Bit	R/W	Description	Default Value
[9]	R/W	EndIrqEn : enable the interrupt if the channel enters end state	0b
[8]	R/W	Must be 0 : Reserved.	0b
[7:5]	R/W	Reserved.	1b
[4]	R/W	FlyByS : '0' Flowthrough, '1' Fly-By for Memory Read Operation	00b
[3]	R/W	FlyByT : '0' Flowthrough, '1' Fly-By for Memory Write Operation	0b
[2]	R/W	ReqPend : channel has a pending request	0b
[1]	R/W	EndIntr : due to successful of the current transaction	0b
[0]	R/W	Reserved	0b

10.3.5 DMA Source Low Address Register (SRCLADDRREG)

Address : C000 0008h+ DMA_CH_BASE

Bit	R/W	Description	Default Value
[15:0]	R/W	DMA Source Low Address Register : 4bytes word address alignment	0000h

10.3.6 DMA Source Address High Register (SRCHADDRREG)

Address : C000 000Ah+ DMA_CH_BASE

Bit	R/W	Description	Default Value
[15:0]	R/W	DMA Source High Address Register	0000h

10.3.7 DMA Target Address Low Register (TRGLADDRREG)

Address : C000 000Ch+ DMA_CH_BASE

Bit	R/W	Description	Default Value
[15:0]	R/W	DMA Target Low Address Register : 4bytes word address alignment	0000h

10.3.8 DMA Target Address High Register (TRGHADDRREG)

Address : C000 000Eh+ DMA_CH_BASE

Bit	R/W	Description	Default Value
[15:0]	R/W	DMA Target High Address Register : 4bytes word address alignment	0000h

10.3.9 8 x 16bit Circular Buffer (DMADATAALBUF)

Address : C000 0020h, C000 0024h, C000 0028h, C000 002Ch, C000 0030h, C000 0034h, C000 0038h, C000 003Ch

Bit	R/W	Description	Default Value
[15:0]	R/W	DMA DATA LSB Buffer Register	0000h

10.3.10 8 x 16bit Circular Buffer (DMADATAAHBUF)

Address : C000 0022h, C000 0026h, C000 002Ah, C000 002Eh, C000 0032h, C000 0036h, C000 003Ah, C000 003Eh

Bit	R/W	Description	Default Value
[15:0]	R/W	DMA DATA MSB Buffer Register	0000h

10.3.11 DMA Interrupt Status Register (DMAINTREG)

Address : C000 00C0h

Bit	R/W	Description	Default Value
[7:0]	R	DMA Interrupt Status Register	0b

10.3.12 DMA Channel0 Source I/O Request Map Register (DCH0SRMREG)

Address : C000 00D0h

Bit	R/W	Description	Default Value
[5]	R/W	DMA Source I/O Map Valid bit	0b
[4:0]	R/W	DMA Source I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.13 DMA Channel0 Target I/O Request Map Register (DCH0TRMREG)

Address : C000 00D2h

Bit	R/W	Description	Default Value
[5]	R/W	DMA Target I/O Map Valid bit	0b
[4:0]	R/W	DMA Target I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.14 DMA Channel1 Source I/O Request Map Register (DCH1SRMREG)

Address : C000 00D4h

Bit	R/W	Description	Default Value
[5]	R/W	DMA Source I/O Map Valid bit	0b
[4:0]	R/W	DMA Source I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.15 DMA Channel1 Target I/O Request Map Register (DCH1TRMREG)

Address : C000 00D6h

Bit	R/W	Description	Default Value
[5]	R/W	DMA Target I/O Map Valid bit	0b
[4:0]	R/W	DMA Target I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.16 DMA Channel2 Source I/O Request Map Register (DCH2SRMREG)

Address : C000 00D8h

Bit	R/W	Description	Default Value
[5]	R/W	DMA Source I/O Map Valid bit	0b
[4:0]	R/W	DMA Source I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.17 DMA Channel2 Target I/O Request Map Register (DCH2TRMREG)

Address : C000 00DAh

Bit	R/W	Description	Default Value
[5]	R/W	DMA Target I/O Map Valid bit	0b
[4:0]	R/W	DMA Target I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.18 DMA Channel3 Source I/O Request Map Register (DCH3SRMREG)

Address : C000 00DCh

Bit	R/W	Description	Default Value
[5]	R/W	DMA Source I/O Map Valid bit	0b
[4:0]	R/W	DMA Source I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.19 DMA Channel3 Target I/O Request Map Register (DCH3TRMREG)

Address : C000 00DEh

Bit	R/W	Description	Default Value
[5]	R/W	DMA Target I/O Map Valid bit	0b
[4:0]	R/W	DMA Target I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.20 DMA Channel4 Source I/O Request Map Register (DCH4SRMREG)

Address : C000 00E0h

Bit	R/W	Description	Default Value
[5]	R/W	DMA Source I/O Map Valid bit	0b
[4:0]	R/W	DMA Source I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.21 DMA Channel4 Target I/O Request Map Register (DCH4TRMREG)

Address : C000 00E2h

Bit	R/W	Description	Default Value
[5]	R/W	DMA Target I/O Map Valid bit	0b
[4:0]	R/W	DMA Target I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.22 DMA Channel5 Source I/O Request Map Register (DCH5SRMREG)**Address : C000 00E4h**

Bit	R/W	Description	Default Value
[5]	R/W	DMA Source I/O Map Valid bit	0b
[4:0]	R/W	DMA Source I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.23 DMA Channel5 Target I/O Request Map Register (DCH5TRMREG)**Address : C000 00E6h**

Bit	R/W	Description	Default Value
[5]	R/W	DMA Target I/O Map Valid bit	0b
[4:0]	R/W	DMA Target I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.24 DMA Channel6 Source I/O Request Map Register (DCH6SRMREG)**Address : C000 00E8h**

Bit	R/W	Description	Default Value
[5]	R/W	DMA Source I/O Map Valid bit	0b
[4:0]	R/W	DMA Source I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.25 DMA Channel6 Target I/O Request Map Register (DCH6TRMREG)**Address : C000 00EAh**

Bit	R/W	Description	Default Value
[5]	R/W	DMA Target I/O Map Valid bit	0b
[4:0]	R/W	DMA Target I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.26 DMA Channel7 Source I/O Request Map Register (DCH7SRMREG)**Address : C000 00ECh**

Bit	R/W	Description	Default Value
[5]	R/W	DMA Source I/O Map Valid bit	0b
[4:0]	R/W	DMA Source I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.27 DMA Channel7 Target I/O Request Map Register (DCH7TRMREG)**Address : C000 00EEh**

Bit	R/W	Description	Default Value
[5]	R/W	DMA Target I/O Map Valid bit	0b
[4:0]	R/W	DMA Target I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.28 DMA Channel8 Source I/O Request Map Register (DCH8SRMREG)**Address : C000 00F0h**

Bit	R/W	Description	Default Value
[5]	R/W	DMA Source I/O Map Valid bit	0b
[4:0]	R/W	DMA Source I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.29 DMA Channel8 Target I/O Request Map Register (DCH8TRMREG)**Address : C000 00F2h**

Bit	R/W	Description	Default Value
[5]	R/W	DMA Target I/O Map Valid bit	0b
[4:0]	R/W	DMA Target I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.30 DMA Channel9 Source I/O Request Map Register (DCH9SRMREG)**Address : C000 00F4h**

Bit	R/W	Description	Default Value
[5]	R/W	DMA Source I/O Map Valid bit	0b
[4:0]	R/W	DMA Source I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

10.3.31 DMA Channel9 Target I/O Request Map Register (DCH9TRMREG)**Address : C000 00F6h**

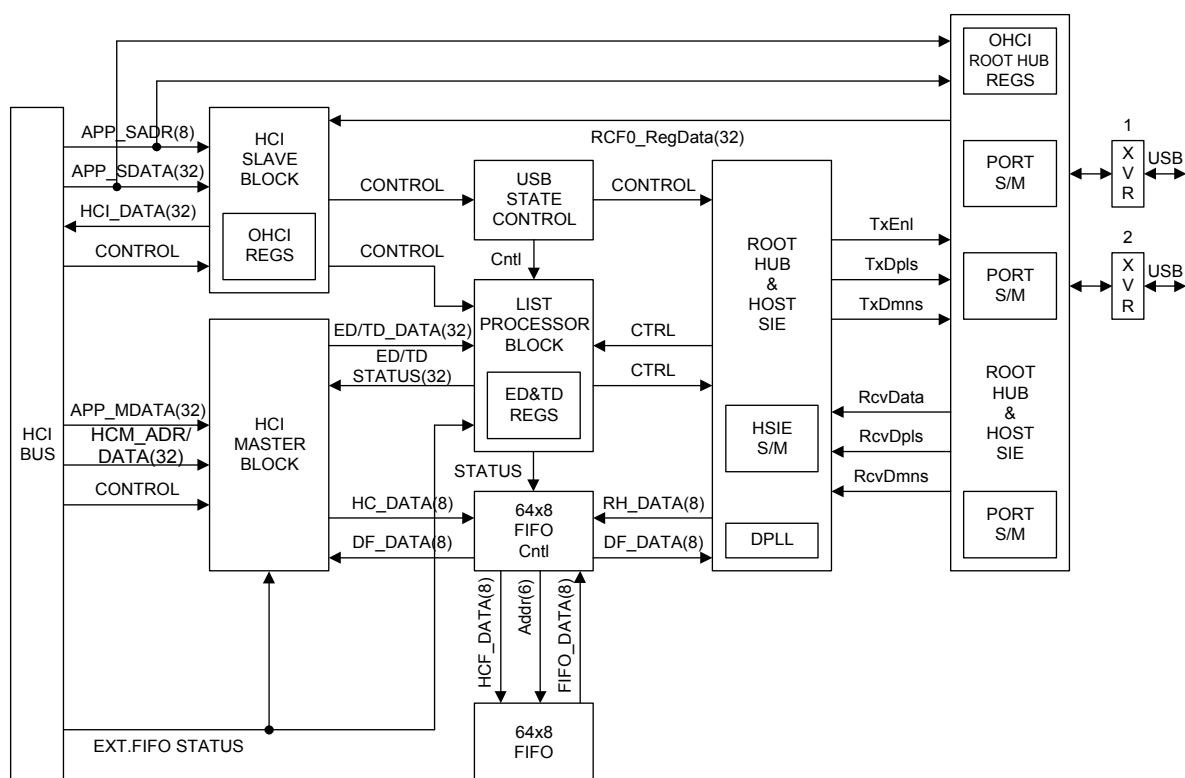
Bit	R/W	Description	Default Value
[5]	R/W	DMA Target I/O Map Valid bit	0b
[4:0]	R/W	DMA Target I/O Peri Device Number, total 32 Peripheral Devices can access each DMA Channel	

11. USB HOST CONTROLLER (UHC)

11.1 Overview

VR3D supports 2 port USB Host interface as follows

- OpenHCI Rev 1.0 compatible
- USB Rev1.1 compatible
- 2 down stream ports
- Support for both LowSpeed and HighSpeed USB Devices



11.2 USB Host Controller Register

VR3D USB Host Controller compiles with OPEN HCI Rev 1.0, Please refer to Open Host Controller Interface Rev1.0 specification for detail information.

Register	Address	Description
HcRevision	0xC000_4800	Control and Status Group
HcControl	0xC000_4804	
HcCommonStatus	0xC000_4808	
HcInterruptStatus	0xC000_480C	
HcInterruptEnable	0xC000_4810	
HcInterruptDisable	0xC000_4814	
HcHCCA	0xC000_4818	Memory Pointer Group
HcPeriodCuttentED	0xC000_481C	
HcControlHeadED	0xC000_4820	
HcControlCurrentED	0xC000_4824	
HcBulkHeadED	0xC000_4828	
HcBulkCurrentED	0xC000_482C	
HcDoneHead	0xC000_4830	
HcRmInterval	0xC000_4834	Frame Counter Group
HcFmRemaining	0xC000_4838	
HcFmNumber	0xC000_483C	
HcPeriodicStart	0xC000_4840	
HcLSThreshold	0xC000_4844	
HcRhDescriptorA	0xC000_4848	Root Hub Group
HcRhDescriptorB	0xC000_484C	
HcRhStatus	0xC000_4850	
HcRhPortStatus1	0xC000_4854	
HcRhPortStatus2	0xC000_4858	

12. USB DEVICE CONTROLLER (UDC)

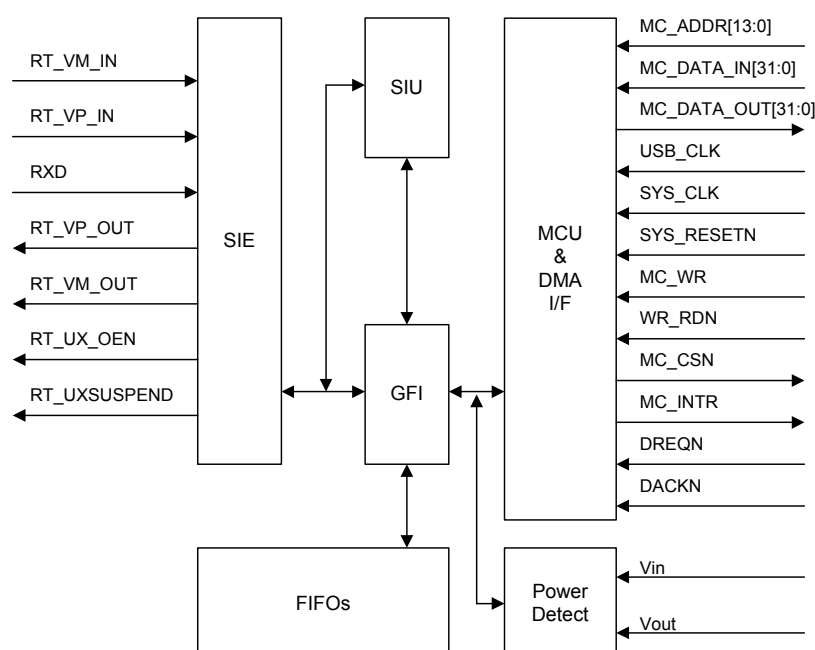
12.1 Overview

USB Function controller is designed to provide a high performance full speed function controller solution with DMA I/F. USB function controller allows bulk transfer with DMA, interrupt transfer and control transfer.

The functions are as follows:

- Full Speed USB Function Controller compatible with the USB Specification Version 1.1
- DMA Interface for Bulk Transfer
- 5 Endpoint with FIFO
 - EP0: 16byte (Register)
 - EP1: 64byte IN/OUT FIFO (dual port asynchronous RAM): interrupt or DMA
 - EP2: 64byte IN/OUT FIFO (dual port asynchronous RAM): interrupt or DMA
 - EP3: 64byte IN/OUT FIFO (dual port asynchronous RAM): interrupt or DMA
 - EP4: 64byte IN/OUT FIFO (dual port asynchronous RAM): interrupt or DMA
- Integrated USB Transceiver

12.1.1 Block Diagram



12.1.2 Features

Fully compliant to USB Specification Version 1.1

- Full Speed (12Mbps) Device
- Integrated USB Transceiver
- Supports Control, Interrupt and Bulk transfer
- 5 Endpoint with FIFO:
One Bi-directional Control Endpoint with 16 byte FIFO (EP0)
Four bi-directional bulk endpoint with 64-byte FIFO (EP1, EP2, EP3, EP4)
- Supports DMA interface for receive and transmit bulk endpoints. (EP1, EP2, EP3, EP4)
- Independent 64byte receive and transmit FIFO to maximize throughput
- Support suspend and remote wake-up function
- USB device operating frequency: 48MHz

12.2 OPERATION

It is recommended that you refer to Universal Serial Bus Specification. Rev. 11.1 for further information.

12.2.1 USB DEVICE SPECIAL REGISTERS.

Detail functionality of the register set of the USB device controller is described in this section. All special function registers are byte-accessible or word accessible. All reserved bits are zero. Common indexed registers depend on the INDEX register (INDEX_REG) (offset address; 0X141c) value. EX: Before writing to the IN_CSR1 register, you must write '0x00' to the INDEX_REG if you want to write to the EPO_CSR register.

NOTE. All registers should be reconfigured after Host Reset Signaling.

Register Name	Description	Offset Address
NON INDEXED REGISTERS		
FUNC_ADDR_REG	Function address register	0x4000
PWR_REG	Power management register	0x4002
EP_INT_REG (EP0–EP4)	Endpoint interrupt register	0x4004
USB_INT_REG	USB interrupt register	0x400C
EP_INT_EN_REG (EP0–EP4)	Endpoint interrupt enable register	0x400e
USB_INT_EN_REG	USB interrupt enable register	0x4016
FRAME_NUM1_REG	Frame number 1 register	0x4018
FRAME_NUM2_REG	Frame number 2 register	0x401a
INDEX_REG	Index register	0x401c
EP0_FIFO_REG	Endpoint0 FIFO register	0x4040
EP1_FIFO_REG	Endpoint1 FIFO register	0x4042
EP2_FIFO_REG	Endpoint2 FIFO register	0x4044
EP3_FIFO_REG	Endpoint3 FIFO register	0x4046
EP4_FIFO_REG	Endpoint4 FIFO register	0x4048
EP1_DMA_CON	Endpoint1 DMA control register	0x4060
-	reserved	0x4062
EP1_DMA_FIFO	Endpoint1 DMA FIFO counter register	0x4064
EP1_DMA_TTC_L	Endpoint1 DMA transfer counter low-byte register	0x4066
EP1_DMA_TTC_M	Endpoint1 DMA transfer counter middle-byte register	0x4068
EP1_DMA_TTC_H	Endpoint1 DMA transfer counter high-byte register	0x406a

EP2_DMA_CON	Endpoint2 DMA control register	0x406c
-	reserved	0x406e
EP2_DMA_FIFO	Endpoint2 DMA FIFO counter register	0x4070
EP2_DMA_TTC_L	Endpoint2 DMA transfer counter low-byte register	0x4072
EP2_DMA_TTC_M	Endpoint2 DMA transfer counter middle-byte register	0x4074
EP2_DMA_TTC_H	Endpoint2 DMA transfer counter high-byte register	0x4076
EP3_DMA_CON	Endpoint3 DMA control register	0x4080
-	reserved	0x4082
EP3_DMA_FIFO	Endpoint3 DMA FIFO counter register	0x4084
EP3_DMA_TTC_L	Endpoint3 DMA transfer counter low-byte register	0x4086
EP3_DMA_TTC_M	Endpoint3 DMA transfer counter middle-byte register	0x4088
EP3_DMA_TTC_H	Endpoint3 DMA transfer counter high-byte register	0x408a
EP4_DMA_CON	Endpoint4 DMA control register	0x408c
-	reserved	0x408e
EP4_DMA_FIFO	Endpoint4 DMA FIFO counter register	0x4090
EP4_DMA_TTC_L	Endpoint4 DMA transfer counter low-byte register	0x4092
EP4_DMA_TTC_M	Endpoint4 DMA transfer counter middle-byte register	0x4094
EP4_DMA_TTC_H	Endpoint4 DMA transfer counter high-byte register	0x4096
COMMON INDEXED REGISTERS		
MAXP_REG	Endpoint MAX packet register	0x4020
OUT_MAXP_REG	OUT endpoint MAX packet register	0x4026
IN INDEXED REGISTERS		
IN_CSR1_REG	EP In control status register 1/EP0 control status register	0x4022
IN_CSR2_REG	EP In control status register 2	0x4024
OUT INDEXED REGISTERS		
OUT_CSR1_REG	EP Out control status register 1	0x4028
OUT_CSR2_REG	EP Out control status register 2	0x402a
OUT_FIFO_CNT1_REG	EP Out write count register 1	0x402c
OUT_FIFO_CNT2_REG	EP Out write count register 2	0x402e

12.3 UDC Registers

This section describes the detail functionality about register set USB Device.

All special function register is byte access or word access. If you access byte mode offset-address is different in little-endian and big-endian. All reserved bit is zero.

Common indexed registers depend on INDEX_REG(offset address : 0x178) value. For example if you want to write EP0 CSR register, you must write '0x00' on INDEX_REG before writing IN_CSR1 register.

All register must be reconfigured after Host Reset Signaling

12.3.1 USB Function Address Register (FUNC_ADDR_REG)

Address : C000 4000h

Bit	R/W	Description	Default Value
7	R/W	ADDR_UPDATE : The MCU sets this bit whenever it updates the FUNCTION_ADDR field in this register. This bit will be cleared by USB when DATA_END bit in EP0_CSR register.	0
6 : 0	R/W	FUNC_ADDR : The MCU write the unique address, assigned by host, to this field.	0

This register maintains the USB Device Address assigned by the host. The MCU writes the value received through a SET_ADDRESS descriptor to this register. This address is used for the next token

12.3.2 USB Power Management Register (PWR_REG)

Address : C000 4002h

Bit	R/W	Description	Default Value
7	R/W	ISO_UPDATE : If set, GFI waits for a SOF token to set IN_PKT_RDY even though a packet to send is already loaded by MCU. If an IN token is received before a SOF token, then a zero length data packet will be sent	0
6 : 4	R	Reserved	0
3	R	USB_RESET : The USB sets this bit if reset signaling is received from the host. This bit remains set as long as reset signaling persists on the bus	0
2	R/W	MCU_RESUME : The MCU sets this bit for MCU resume. The USB generates the resume signaling depending RESUME CON Register, while this bit is set in suspend mode	0
1	R/W	SUSPEND_MODE : This bit can be set by USB, automatically when the device enter into suspend mode. It is cleared under the following conditions 1) The MCU clears the MCU_RESUME bit by writing '0', to end remote resume signaling. 2) The resume signal form host is received	0
0	R/W	SUSPEND_EN : Suspend mode enable control bit 0 = Disable(default). The device will not enter suspend mode. 1 = Enable suspend mode	0

12.3.3 USB EndPoint Interrupt Register (EP_INT_REG)

Address : C000 4004h

Bit	R/W	Description	Default Value
15:5	R	Reserved	-
4	R/W	EP4 Interrupt : For BULK/INTERRUPT IN endpoints: The USB sets this bit under the following conditions: 1. IN_PKT_RDY bit is cleared. 2. FIFO is flushed 3. SENT_STALL set. For BULK/INTERRUPT OUT endpoints: USB sets this bit under the following conditions: 1. Sets OUT_PKT_RDY bit	0

Bit	R/W	Description	Default Value
		2. Sets SENT_STALL bit For ISO IN endpoints: the USB sets this bit under the following conditions: 1. UNDER_RUN bit is set 2. IN_PKT_RDY bit is cleared. 3. FIFO is flushed Note: conditions 1 and 2 are mutually exclusive For ISO OUT endpoints: USB sets this bit under the following conditions: 1. OUT_PKT_RDY bit is set 2. OVER RUN bit is set. Note: Conditions 1 and 2 are mutually exclusive.	
3	R/W	EP3 Interrupt	0
2	R/W	EP2 Interrupt	0
1	R/W	EP1 Interrupt	0
0	R/W	EP0 Interrupt : This bit corresponds to endpoint 0 interrupt The USB sets this bit under the following conditions: 1. OUT_PKT_RDY bit is set. 2. IN_PKT_RDY bit is cleared. 3. SENT_STALL bit is set 4. SETUP_END bit is set 5. DATA_END bit is cleared(Indicates end of control transfer)	0

The USB core has two interrupt registers.

These registers act as status registers for the MCU when it is interrupted. The bits are cleared by writing a '1'(not '0') to each bit that was set.

Once the MCU is interrupted, MCU should read the contents of interrupt-related registers and write back to clear the contents if it is necessary

12.3.4 USB Interrupt Register (USB_INT_REG)

Address : C000 400Ch

Bit	R/W	Description	Default Value
15 : 3	R/W	Reserved	0
2	R/W	USB Reset Interrupt : The USB set this bit, when it receives reset signaling	0
1	R/W	USB Resume Interrupt : The USB sets this bit, when it receives resume signaling, <i>while in suspend mode.</i> If the resume is due to a USB reset, then the MCU is first interrupted with a RESUME interrupt. Once the clocks resume and the SE0 condition persists for 3ms, USB RESET interrupt will be asserted	0
0	R/W	Suspend Interrupt : The USB sets this bit when it receives suspend signaling. This bit is set whenever there is no activity for 3ms on the bus. Thus, if the MCU does not stop the clock after the first suspend interrupt, it will be continue to be interrupted every 3ms as long as there is no activity on the USB bus. By default this interrupt is disabled.	0

The USB core has two interrupt registers.

These registers act as status registers for the MCU when it is interrupted. The bits are cleared by writing a '1'(not '0') to each bit that was set.

Once the MCU is interrupted, MCU should read the contents of interrupt-related registers and write back to clear the contents if it is necessary

12.3.5 USB Endpoint Interrupt Enable Register (EP_INT_EN_REG)

The USB device controller also has two interrupt enable registers (not resume interrupt enable) that correspond to each interrupt register. The default is USB reset interrupt enabled.

Address : C000 400Eh

Bit	R/W	Description	Default Value
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Bit	R/W	Description	Default Value
15:5	R	Reserved	0
4	R/W	EP4 Interrupt Enable 0 : Disable 1 : Enable	1
3	R/W	EP3 Interrupt Enable	1
2	R/W	EP2 Interrupt Enable	1
1	R/W	EP1 Interrupt Enable	1
0	R/W	EP0 Interrupt Enable	1

12.3.6 USB Interrupt Enable Register (USB_INT_EN_REG)

Address : C000 4016h

Bit	R/W	Description	Default Value
15 : 3	R	Reserved	0
2	R/W	RESET_INT_EN 0 : Disable 1 : Enable	1
1	R	Reserved	0
0	R/W	SUSPEND_INT_EN : Suspend Interrupt enable bit 0 : Disable 1 : Enable	0

12.3.7 USB Frame Number Register (FRAME_NUM1_REG, FRAME_NUM2_REG)

Address : C000 4018h

Bit	R/W	Description	Default Value
7 : 0	R/W	FRAME_NUM1_REG :	0

Address : C000 401Ah

Bit	R/W	Description	Default Value
7 : 0	R/W	FRAME_NUM2_REG :	0

When host transfer USB packet, there is frame number in SOF(Start Of Frame). The USB catch this frame number and load it into this register, automatically.

A frame number is included in each Start Of Frame (SOF) if the host transfers a USB packet. The USB device controller catches this frame number and loads it into this register

12.3.8 USB Index Register (INDEX_REG)

Address : C000 401Ch

Bit	R/W	Description	Default Value
7 : 0	R/W	It indicates a certain endpoint.	0

This INDEX register is used to indicate certain endpoint registers effectively. MCU can access the endpoint registers(MAXP_REG,IN_CSR1_REG,IN_CSR2_REG,OUT_CSR1_REG,OUT_CSR2_REG,OUT_FIFO_CNT1_REG,OUT_FIFO_CNT2_REG) for an endpoint inside the core using the INDEX register

12.3.9 USB EP0 Cont & Status Reg(CSR) (EP0_CSR)

For Endpoint0, this register has the control and status bits. As a control transaction is involved with both IN and OUT tokens, only one CSR register is mapped to the IN CSR1 register. (Shares IN1_CSR and can be accessed by writing the index register"0" and read/write IN1_CSR)

Address : C000 4022h

Bit	R/W	Description	Default Value
15 : 8		Reserved	-
7	W	SERVICED_SETUP_END : The MCU should write a "1" to this bit to clear SETUP_END	0
6	W	SERVICED_OUT_PKT_RDY : The MCU should write a "1" to this bit to clear OUT_PKT_RDY	0
5	R/W	SEND_STALL : MCU should writes a "1" to this bit at the same time it clears OUT_PKT_RDY, if it decodes an invalid token. 0 = Finish the STALL condition 1 = The USB issues a STALL and shake to the current control transfer.	0
4	R	SETUP_END : The USB sets this bit when a control transfer ends before DATA_END is set. When the USB sets this bit, an interrupt is generated to the MCU. When such a condition occurs, the USB flushes the FIFO and invalidates MCU access to the FIFO.	0
3	SET	DATA_END : The MCU sets this bit below conditions: 1. After loading the last packet of data into the FIFO, at the same time IN_PKT_RDY is set. 2. While it clears OUT_PKT_RDY after unloading the last packet of data. 3. For a zero length data phase.	0
2	CLEAR	SEND_STALL : The USB sets this bit if a control transaction is stopped due to a protocol violation. An interrupt is generated when this bit is set. The MCU should write "0" to clear this bit.	0
1	SET	IN_PKT_RDY : The MCU sets this bit after writing a packet of data into EP0 FIFO. The USB clears this bit once the packet has been successfully sent to the host. An interrupt is generated when the USB clears this bit, so as the MCU to load the next packet. For a zero length data phase, the MCU sets DATA_END at the same time.	0
0	R	OUT_PKT_RDY : The USB sets this bit once a valid token is written to the FIFO. An interrupt is generated when the USB sets this bit. The MCU clears this bit by writing a "1" to the SERVICED_OUT_PKT_RDY bit.	0

This register has the control and status bits for Endpoint 0. Since a control transaction involves both IN and OUT tokens, there is only one CSR register, mapped to the IN_CSR1 register.

(share IN1_CSR and can access by writing index register "0" and read/write IN1_CSR)

12.3.10 End Point In Control/Status Register (IN_CSR1_REG)

Address : C000 4022h

Bit	R/W	Description	Default Value
15:7	R	Reserved	0
6	R/W	CLR_DT_TOG : Used in set-up procedure. 0 = There are alternations of DATA0 and DATA1. 1 = The data toggle bit is cleared and the PID in the packet will maintain DATA0	0
5	R/W	SNT_STALL : Set by the USB when an IN token issues a STALL handshake, and after the MCU sets the SEND_STALL bit to start STALL handshaking. When the USB issues a STALL handshake, IN_PKT_RDY is cleared	0
4	R/W	SND_STALL : 0 = The MCU clears this bit to finish the STALL condition. 1 = The MCU issues a STALL handshake to the USB	0
3	R/W	FF_FLUSH : Set by the MCU if it intends to flush the packet in Input-related FIFO. This bit is cleared by the USB when the FIFO is flushed. The MCU is interrupted when this happens. If a token is in process, the USB waits until the transmission is complete before FIFO flushing. If two packets are loaded into the FIFO, only the first packet (The packet that is intended to be sent to the host) is flushed, and the corresponding IN_PKT_RDY bit is cleared	0
2	R/W	UND_RUN : Valid only For ISO Mode Set by the USB when in ISO mode, an IN token is received, and the IN_PKT_RDY bit is not set. The USB sends a zero length data packet for such conditions, and the next packet that	0

Bit	R/W	Description	Default Value
		is loaded into the FIFO is flushed. This bit is cleared by writing 0	
1	R	Reserved	0
0	R/W	IN_PKT_RDY : Set by the MCU after writing a packet of data into the FIFO. The USB clears this bit once the packet has been successfully sent to the host. An interrupt is generated when the USB clears this bit, so the MCU can load the next packet. While this bit is set, the MCU will not be able to write to the FIFO. If the MCU sets the SND_STALL bit, this bit cannot be set.	0

12.3.11 End Point In Control/Status Register (IN_CSR2_REG)

Address : C000 4024h

Bit	R/W	Description	Default Value
15:8	R	Reserved	0
7	R/W	AUTO_SET : If set, whenever the MCU writes MAXP data, IN_PKT_RDY will automatically be set by the core without any intervention from the MCU. If the MCU writes less than MAXP data, then the IN_PKT_RDY bit has to be set by the MCU	0
6	R/W	ISO : Used only for endpoints whose transfer type is programmable. 0 = Configures endpoint to Bulk mode 1 = Configures endpoint to ISO mode	0
5	R/W	MODE_IN : Used only for endpoints whose direction is programmable. 0 = Configures Endpoint Direction as OUT 1 = Configures Endpoint Direction as IN	1
4	R/W	IN_DMA_INT_EN : Determines whether the interrupt should be issued or not when the EP1 IN_PKT_RDY condition happens. This is only useful for DMA mode. 0 = Interrupt Enable 1 = Interrupt Disable	0
3:0	R	Reserved	-

12.3.12 End Point Out Control/Status Register (OUT_CSR1_REG)

Address : C000 4028h

Bit	R/W	Description	Default Value
15:8	R	Reserved	0
7	R/W	CLR_DT_TOG : Used in set-up procedure. 0 = There are alternations of DATA0 and DATA1. 1 = The data toggle bit is cleared and the PID in the packet will maintain DATA0	0
6	R/W	SNT_STALL : Set by the USB when an IN token issues a STALL handshake, and after the MCU sets the SEND_STALL bit to start STALL handshaking. When the USB issues a STALL handshake, IN_PKT_RDY is cleared	0
5	R/W	SND_STALL : 0 = The MCU clears this bit to finish the STALL condition. 1 = The MCU issues a STALL handshake to the USB	0
4	R/W	FF_FLUSH : Set by the MCU if it intends to flush the packet in Input-related FIFO. This bit is cleared by the USB when the FIFO is flushed. The MCU is interrupted when this happens. If a token is in process, the USB waits until the transmission is complete before FIFO flushing. If two packets are loaded into the FIFO, only the first packet (The packet that is intended to be sent to the host) is flushed, and the corresponding IN_PKT_RDY bit is cleared	0
3	R/W	DATA_ERROR : Valid only in ISO mode. This bit should be sampled with OUT_PKT_RDY . When set, it indicates the data packet due to be unloaded by the MCU has an error (either bit stuffing or CRC). If two packets are loaded into the FIFO, and the second packet has an error, then this bit gets set only after the first packet is unloaded. This bit is automatically cleared when OUT_PKT_RDY gets cleared	0

Bit	R/W	Description	Default Value
2	R/W	OVER_RUN : Valid only in ISO mode. This bit is set if the core is not able to load an OUT ISO token into the FIFO. The MCU clears this bit by writing 0	0
1	R	Reserved	0
0	R/W	OUT_PKT_RDY : Set by the USB after it has loaded a packet of data into the FIFO. Once the MCU reads the packet from FIFO, this bit should be cleared by the MCU. (Write a "0")	0

12.3.13 End Point out control status Register (OUT_CSR2_REG)

Address : C000 402Ah

Bit	R/W	Description	Default Value
15:8	R/W	Reserved	0
7	R/W	AUTO_CLR : If set, whenever the MCU reads data from the OUT FIFO, OUT_PKT_RDY will be cleared automatically by the logic, without any intervention from the MCU	0
6	R/W	ISO : Determines the endpoint transfer type. 0 : Configures the endpoint to Bulk mode. 1 : Configures the endpoint to ISO mode	0
5	RW	OUT_DMA_INT_MASK : Determines whether the interrupt should be issued or not. The OUT_PKT_RDY condition happens. This is only useful for DMA mode. 0 = Interrupt Enable 1 = Interrupt Disable	0
4:0	R	Reserved	0

12.3.14 Endpoint FIFO Register (EPn_FIFO_REG)

EPn_FIFO_REG should be accessed by CPU to access EPn FIFO

Address : C000 4040h (EP0_FIFO_REG) / C000 4042h (EP1_FIFO_REG) / C000 4044h (EP2_FIFO_REG) / C000 4046h (EP3_FIFO_REG) / C000 4048h (EP4_FIFO_REG)

Bit	R/W	Description	Default Value
15:8	R	Reserved	0
7:0	R/W	FIFO_DATA	

12.3.15 IN MAX Packet Size Register (IN_MAXP)

Address : C000 4020h

Bit	R/W	Description	Default Value
15:4	R	Reserved	-
3:0	R/W	MAXP : 0000 : Reserved 0001 : MAXP = 8 Byte 0010 : MAXP = 16 Byte 0100 : MAXP = 32 Byte 1000 : MAXP = 64 Byte For EP0, MAXP=8 is recommended. For EP1~4, MAXP=32 or MAXP=64 is recommended. And, if MAXP=32, the dual packet mode will be enabled automatically.	0001b

12.3.16 OUT MAX Packet Size Register (OUT_MAXP)

Address : C000 4026h

Bit	R/W	Description	Default Value
15:4	R	Reserved	-
3:0	R/W	MAXP : 0000 : Reserved 0001 : MAXP = 8 Byte 0010 : MAXP = 16 Byte 0100 : MAXP = 32 Byte 1000 : MAXP = 64 Byte For EP0, MAXP=8 is recommended. For EP1~4, MAXP=32 or MAXP=64 is recommended. And, if MAXP=32, the dual packet mode will be enabled automatically.	0001b

12.3.17 END POINT Out Write Count Register (OUT_FIFO_CNT1)

The number of bytes in the packet due to be unloaded by the MCU is maintained by these registers.

Address : C000 402Ch

Bit	R/W	Description	Default Value
15:8	R	Reserved	-
7:0	R/W	OUT_CNT_LOW : Lower Byte of Write Count	0x00

Address : C000 402Eh

Bit	R/W	Description	Default Value
15:8	R	Reserved	-
7:0	R/W	OUT_CNT_HIGH: Higher Byte of Write Count The OUT_CNT_HIGH may be always 0 normally	0x00

12.3.18 DMA Interface Control Register (EPn_DMA_CON)

Address : C000 406Ch (EP1_DMA_CON) / C000 4080h (EP2_DMA_CON) / C000 408Ch (EP3_DMA_CON) / C000 4060h (EP4_DMA_CON)

Bit	R/W	Description	Default Value
15:8	R/W	Reserved	0
7	R/W	IN_RUN_OB Read : DMA Run Observation 0 : DMA is Stopped 1 : DMA is running Write : Ignore EPx_TTC_x register 0 : DMA requests will be ignored if EPn_TTC_x reaches 0 1 : DMA requests will be continued if EPn_TTC_x reaches 0	0
6:4	R	STATE : DMA State Monitoring	0
3	R	Reserved	-
2	RW	OUT_RUN_OB/OUT_DMA_RUN : Functionally separated into write and read operations Read : OUT DMA Run observation Write : 0 : Stop 1 : Run	0
1	RW	IN_DMA_RUN : Start DMA Run 0 : Stop 1 : Run	0
0	RW	DMA_MODE_EN : Set DMA Mode. If RUN_OB has been written as 0 and EPn_TTC_x reaches 0, the DMA_MODE_EN will be cleared by USB. 0 : Interrupt Mode 1 : DMA Mode	0

12.3.19 DMA FIFO Count Register (EP1_FIFO_CNT, EP2_FIFO_CNT, EP3_FIFO_CNT, EP4_FIFO_CNT)

The byte size in FIFO to be transferred by DMA is in this register. When OUT_DMA is enabled, the value in the OUT FIFO Write Count Register 1 will be loaded into this register. For IN DMA Mode, the MCU should set the proper value by R/W

Address : C000 4064h (EP1_FIFO_CNT) / C000 4070h / C000 4084h / C000 4090h

Bit	R/W	Description	Default Value
15:8	R	Reserved	0
7:0	R/W	EP1_FIFO_CNT: DMA total transfer count value (higher Byte)	0x00

12.3.20 DMA Total transfer count value Register (EP1_TTC_L/M/H)

The total number of bytes to be transferred using DMA is in this register. (Total 20 bit Counter)

Address : C000 4066h (EP1_TTC_L) / C000 4068h (EP1_TTC_M) / C000 406A (EP1_TTC_H)

Bit	R/W	Description	Default Value
15:8	R	Reserved	0
7:0	R/W	EP1_TTC_L : DMA total transfer count value (higher Byte)	0x00

12.3.21 DMA Total transfer count value Register (EP2_TTC_L/M/H)

Address : C000 4072h (EP2_TTC_L) / C000 4074h (EP2_TTC_M) / C000 4076h (EP2_TTC_H)

Bit	R/W	Description	Default Value
15:8	R	Reserved	0
7:0	R/W	EP2_TTC_L : DMA total transfer count value (higher Byte)	0x00

12.3.22 DMA Total transfer count value Register (EP3_TTC_L/M/H)

Address : C000 4086h (EP3_TTC_L) / C000 4088h (EP3_TTC_M) / C000 408Ah (EP3_TTC_H)

Bit	R/W	Description	Default Value
15:8	R	Reserved	0
7:0	R/W	EP3_TTC_L : DMA total transfer count value (higher Byte)	0x00

12.3.23 DMA Total transfer count value Register (EP4_TTC_L/M/H)

Address : C000 4092h (EP4_TTC_L) / C000 4094h (EP4_TTC_M) / C000 4096h (EP4_TTC_H)

Bit	R/W	Description	Default Value
15:8	R	Reserved	0
7:0	R/W	EP4_TTC_L : DMA total transfer count value (higher Byte)	0x00

13. UART

13.1 Overview

Four independent asynchronous serial I/O (SIO) ports, each of which can operate in interrupt-based or DMA-based mode, are provided by the VR3D UART (Universal Asynchronous Receiver and Transmitter) unit. That is, an interrupt or DMA request to transfer data between the CPU and UART can be generated by the UART. Bit rates of up to 3.1 Mbps can be supported by UART. Two 16-byte FIFOs are contained in each UART channel for receive and transmit.

Programmable baud-rates, one or two stop bit insertion, 5-bit, 6-bit, 7-bit or 8-bit data width and parity checking are included in the VR3D UART.

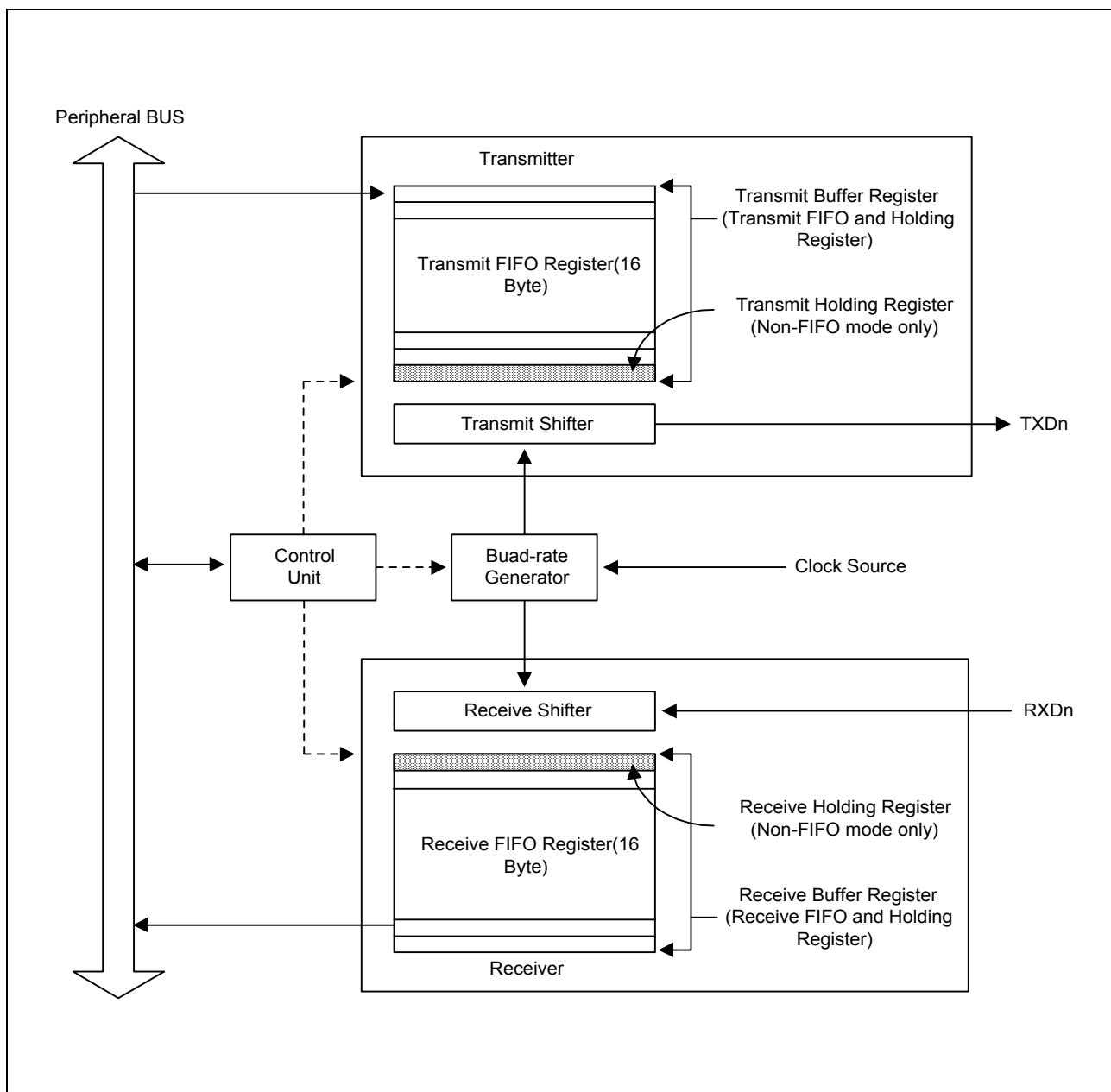
Fig UART Block Diagram (with FIFO) shows that baud-rate generator, transmitter, receiver and control unit are contained in each UART. PCLK can clock the baud-rate generator. The transmitter and the receiver contain 6-byte FIFOs and data shifters. Transmission data is written to FIFO and then copied to the transmit shifter. The transmit data pin (TXDn) then shifts out the data. The received data is shifted from the receive data pin (RXDn), and then copied to FIFO from the shifter.

13.1.1 Features

- RXD0, TXD0, RXD1, TXD1, RXD2, TXD2, RXD3, TXD3 with DMA-based or interrupt-based operation
- UART Ch 1 or Ch2 with full modem function
- UART Ch 2 with IrDA 1.0
- Supports handshake transmit / receive (UART Ch1 or Ch2)
- Ability to add or delete standard asynchronous communications bits (start, stop, and parity) in the serial data
- Independently controlled transmit, receive, line status, and data set interrupts
- Programmable baud rate
- Modem control pins that allow flow control through software.
- Fully programmable serial-interface:
 - 5-, 6-, 7-, or 8-bit characters
 - Even, odd, and no parity detection
 - 1, or 2 stop bit generation
 - Baud rate generation up to around 3.1Mbps
- 16-byte transmit FIFO
- 16-byte receive FIFO
- Complete status reporting capability
- Ability to generate and detect line breaks
- Internal diagnostic capabilities that include:
 - Loopback controls for communications link fault isolation
- Separate DMA requests for transmit and receive data services

Slow infrared asynchronous interface that conforms to the Infrared Data Association (IrDA) standard

13.1.2 Block Diagram



UART Block Diagram (with FIFO)

13.2 Pin Description

UART Pins		
TX0	O	UART0 Transmit Pin
RX0	I	UART0 Receive Pin
nRIO	I	UART1, UART2 Ring Indicator Pin
nDCD	I	UART1, UART2 Data-Carrier-Detect Pin
nDSR	I	UART1, UART2 Data-Set-Ready Pin
nDTR	O	UART1, UART2 Data-Terminal-Ready Pin
nCTS	I	UART1, UART2 Clear-To-Send Pin
nRTS	O	UART1, UART2 Ready-To-Send Pin
TX1	O	UART1 Transmit Pin
RX1	I	UART1 Receive Pin
RX2	I	UART2 Receive Pin
TX2	O	UART2 Transmit Pin
RX3	I	UART3 Receive Pin
TX3	O	UART3 Transmit Pin

13.3 UART Operation

The UART operations that include Auto Flow Control, infra-red mode, loopback mode, baud-rate generation, interrupt generation, data reception, and data transmission are described in the following sections.

13.3.1 Data Transmission

For transmission the data frame is programmable. A start bit, 5 to 8 data bits, an optional parity bit, and 1 to 2 stop bits are included in the data frame, which can be specified by the Line Control Register (LCONn). The break conditions can also be produced by the transmitter. The serial out put is forced by the break condition to a logic 0 state for one frame transmission time. The break signal is transmitted by this block after the present transmission word transmits perfectly. Data are continuously transmitted into the TX FIFO (TX holding register in this case of non-FIFO mode) after the break signal transmission.

13.3.2 Break Condition

The break is defined as a continuous low level signal for one frame transmission time on the transmit data output.

13.3.3 Data Reception

Similar to the transmission, the data frame for reception is also programmable. The data frame for reception consists of a start bit, 5 to 8 data bits, an optional parity bit, and 1 to 2 stop bits in the Line Control Register LCONn). Overrun error, parity error, frame error and break condition, each of which can set an error flag, can be detected by the receiver.

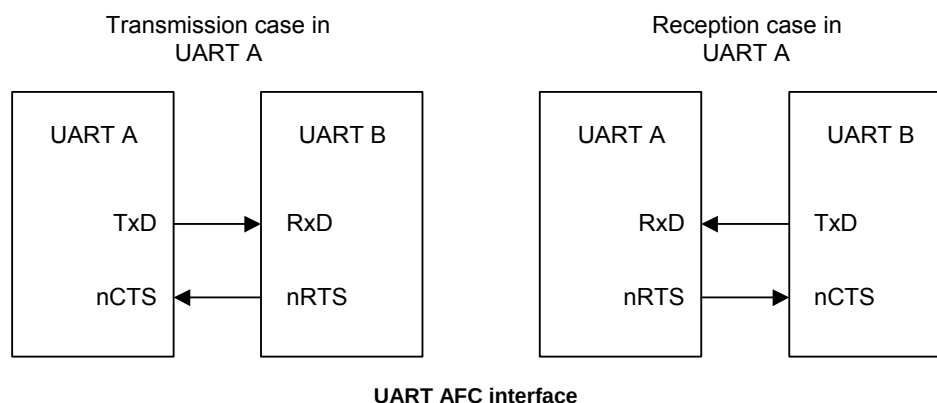
- The fact that new data has overwritten the old data before the old has been read is indicated by the overrun error.
- The fact that the receiver has detected an unexpected parity condition is indicated by the parity error.
- The fact that the received data does not have a valid stop bit is indicated by the frame error.
- The fact that the RXDn input is held in the logic 0 state for a duration longer than one frame transmission time is indicated by the break condition.

Receive timeout condition occurs when [the LCONn] does not receive data during the value of timeout register word time (The setting of the Word Length bit is followed by this interval) and the RX FIFO is not empty in the FIFO mode.

13.3.4 Auto Flow Control(AFC)

In case VR3D's UART would have to connect UART to UART, it supports Auto Flow Control with nRTS and nCTS signals. The AFC bit in the MCONn register is set to zero and control the signal of nRTS by software when users connect the UART to a modem. In AFC, the condition of the receiver controls nRTS, and nCTS signal controls the operation of the transmitter.. In FIFO, the data is transferred by the UART's transmitter only when the nCTS signal is active (In AFC, nCTS means that the other UART's FIFO is ready to receive data.). nRTS must be activated when its receive FIFO has a spare more than 2-byte before the UART receives data and must

be inactivated when its receive FIFO has a spare under 1-byte before the UART receives data (In AFC, nRTS means that its own receive FIFO is ready to receive data).



13.3.5 Non Auto-Flow control (Controlling nRTS and nCTS by S/W) Example

13.3.5.1 RX operation with FIFO

1. Select receive mode (Interrupt or DMA mode).
2. The value of RX FIFO count in FSTATUSn register is checked, and the user must set the value of RTS_ACTIVE (MCONn [0]) to '1' (active nRTS) when the value is less than 15, and the user must set the value to '0' (inactive nRTS) when the value is equal to or bigger than 15.
3. Repeat item 2.

13.3.5.2 TX operation with FIFO

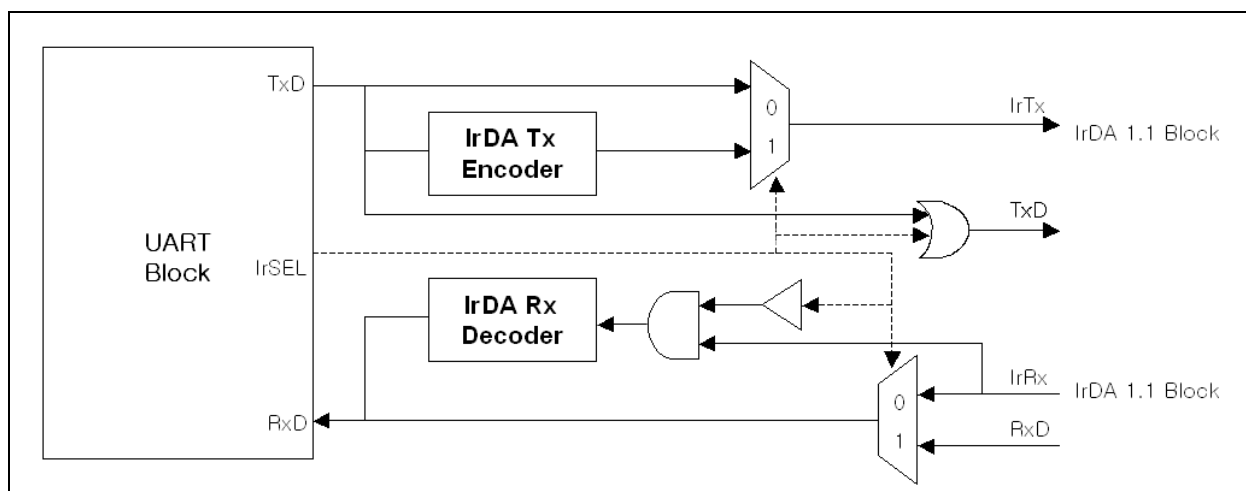
1. Select transmit mode (Interrupt or DMA mode).
2. The value of CTS (MSTATUSn [0]) is checked. The user writes the data to the TX FIFO register when the value is '1' (nCTS is activated).

13.3.6 SIR (Serial Infra-Red) Mode

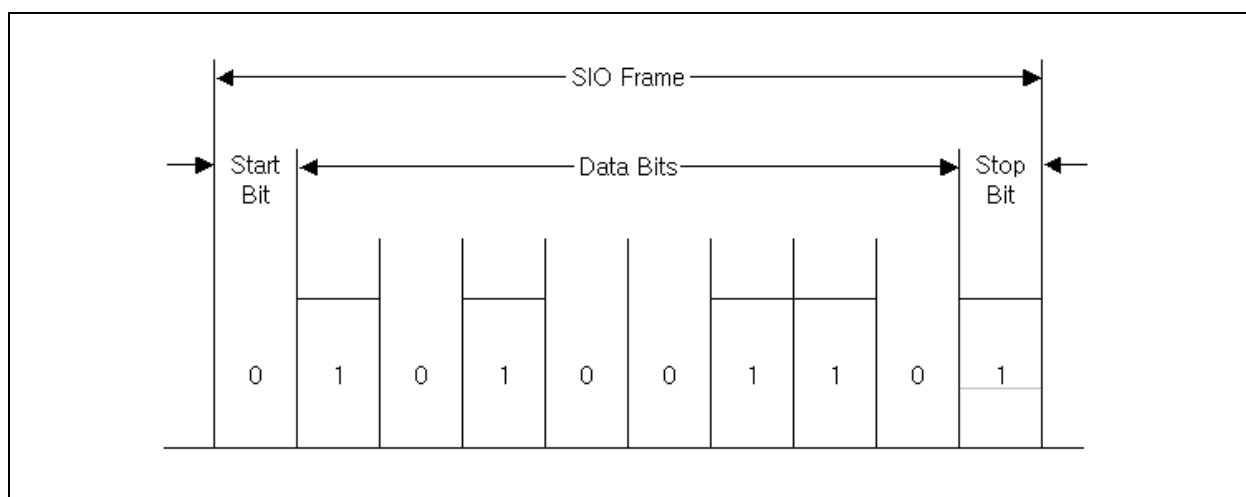
SIR (Serial Infra-red) transmission and reception are supported by the VR3D UART, and can be selected by setting the SIR (Serial Infra-red mode bit) in UART Line Control Register (LCON1 or LCON2) and SIR signal control mode in the Serial Infra-red mode set register (UCONn) depending upon the IrDA transceiver type.

When the IrLAP (Infra-red Link Access Protocol) and above application layer implements MIR/FIR, and above application layer, the IrDA SIR mode bit in the UART Line Control Register (LCON2) should be selected first, and the IrDA operation speed should be negotiated with another IrDA system in the IrLAP layer. The IrDA SIR mode should be deselected after the negotiation. Then the UART block operates in the normal TX/RX mode and the IrDA 1.1 block operates in the MIR/FIR mode. Figure **IrDA Function Block Diagram** shows the implementation of the Sir mode, and in chapter 10-5 IrDA is the MIR/FIR..

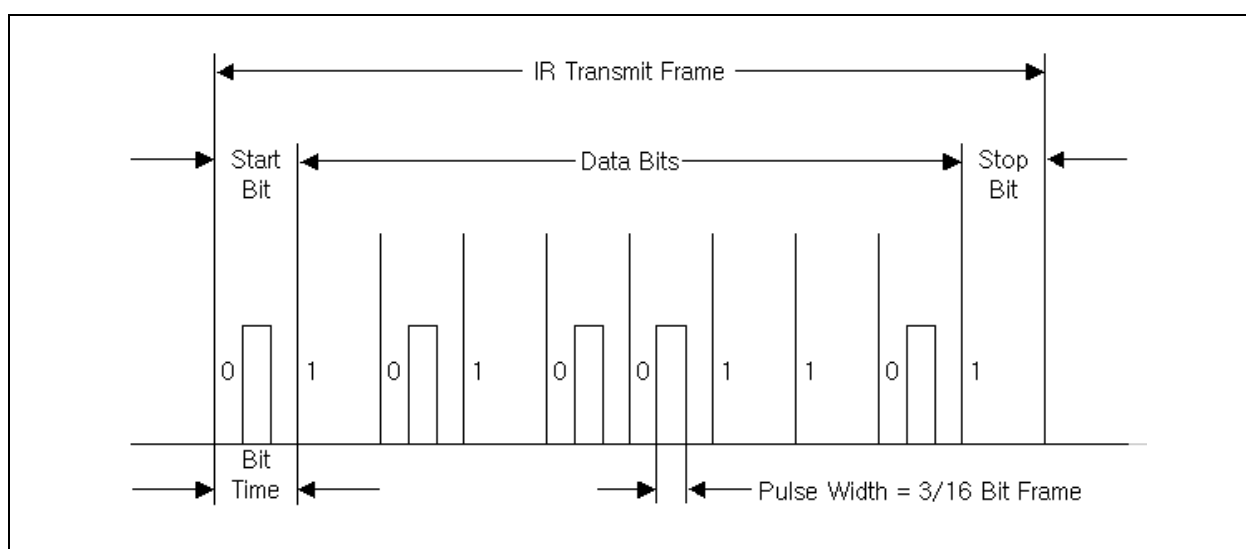
The transmit period is pulsed at the normal serial transmit rate (when the transmit data bit is zero) of 3/16, in IR transmit mode.. The receive must detect the 3/6 pulsed period to recognize a zero value, in the IR receive mode (refer to the frame timing diagrams shown in Figures 10-5 and 10-6).



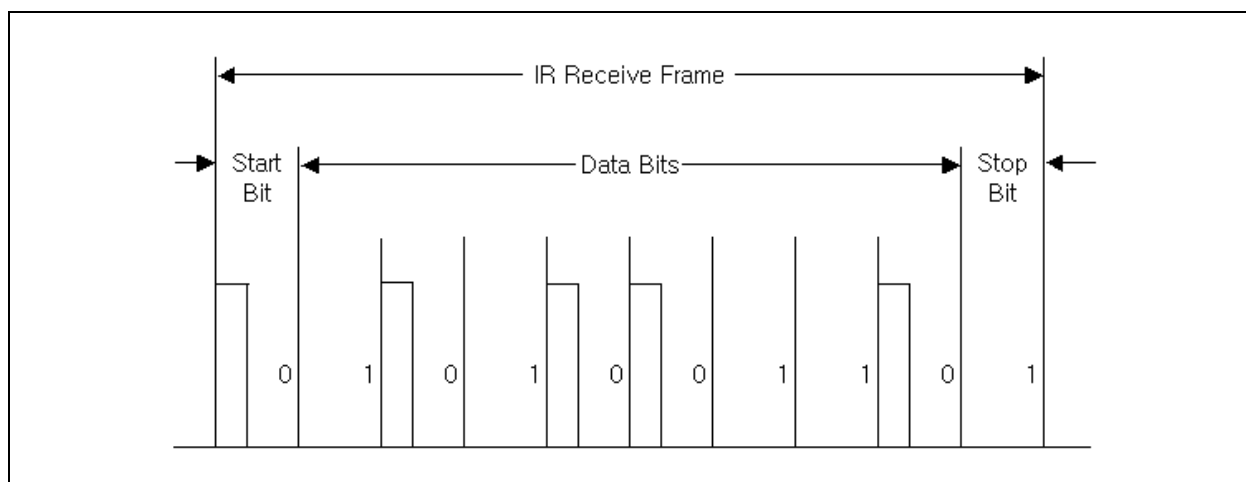
IrDA Function Block Diagram



Serial I/O Frame Timing Diagram (Normal UART)



Infra-Red Transmit Mode Frame Timing Diagram



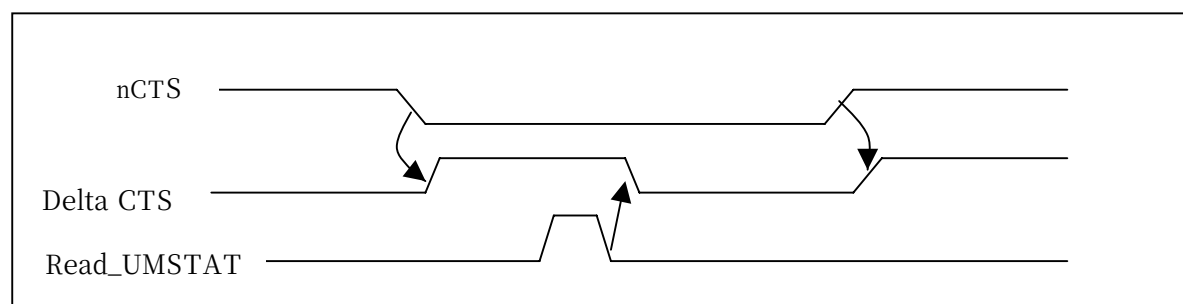
Infra-Red Receive Mode Frame Timing Diagram

13.3.7 Loop-back Mode

To aid in isolating faults in the communication link, the VR3D UART provides a test mode referred to as the loopback mode. The transmitted data is immediately received in this mode. The processor is able to verify the internal transmit and receives the data path of each SIO channel due to this feature. Setting the loopback-bit in the UART control register (UCONn) will select this mode.

13.3.8 Modem Status Register Read

When the CTS is Active and the Delta CTS is High, the Modem Status Register becomes readable.



nCTS and Delta CTS Timing Diagram

13.3.9 RS-232C interface

The user needs nRTS, nCTS, nDSR, nDTR, DCD and nRI signals to connect to modem interface (not equal null modem).

13.3.10 Interrupt/DMA Request Generation

Each VR3D UART consists of seven status (TX/RX/Error) signals, including Overrun error, Parity error, Frame error, Break, Receive buffer data ready, Transmit buffer empty, and Transmit shifter empty, all of which are indicated by the corresponding UART status register (TRSSTATUSn/ESTATUSn).

The break condition, frame error, parity error, and overrun error are referred to as the receive error status, each of which can cause the receive error status interrupt request, if the RX ERRSTATUS (receive-error-status-interrupt-enable bit) is set to one in the control register UCONn. Reading ESTATUSn can identify the signal causing the request when a receive-error-status –interrupt-request is detected.

When the data of the receiver shifter is transferred by the receiver to the receive FIFO and RX FIFO Trigger Level is reached by the number of received data, RX interrupt is generated if Receive mode in control register is selected as interrupt request or polling mode. RX interrupt will be caused by transferring the data of the receive shifter to the receive FIFO under the Interrupt request and polling mode in the Non-FIFO mode.

When the data is transferred by the transmitter from its transmit FIFO to its transmit shifter and RX FIFO Trigger Level is reached by the number of data left in transmit FIFO, RX interrupt is generated if Transmit mode in UCONn (UART control register) is selected as Interrupt request or polling mode.

TX interrupt will be caused by transferring data from the transmit FIFO to the transmit shifter under the interrupt request and polling mode.

When the Receive and Transmit modes in UCONn (UART control register) are selected as the DMA request mode, then instead of RX or TX interrupt in the situation mentioned above, DMA request is occurred.

Type	FIFO Mode	Non-FIFO Mode
RX interrupt	Each time receive data reaches the trigger level of receive FIFO, the RX interrupt will be generated. When the number of data in FIFO does not reaches RX FIFO trigger Level and does not receive data during 3 word time(This interval follows the setting of Word Length bit), the RX interrupt will be generated(receive time out).	Each time receive data becomes full, the receive holding register generates an interrupt.
TX interrupt	Each time the transmit data reaches the trigger level of transmit FIFO(TX FIFO trigger Level), the TX interrupt will be generated.	Each time the transmit data becomes empty, the transmit holding register generates an interrupt.
Error interrupt	Frame error, parity error, and break signal are detected, and will generate an error interrupt. When it gets to the top of the receive FIFO without reading out data in it, the error interrupt will be generated(overflow error).	All errors generate an error interrupt immediately. However if another error occurs at the same time, only one interrupt is generated.

Interrupts in Connection with FIFO

13.3.11 UART Error Status FIFO

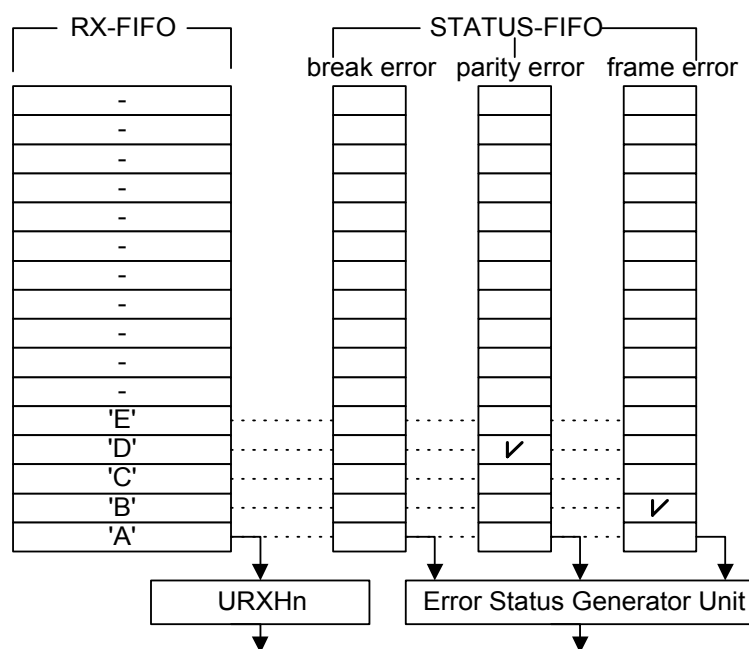
UART has the status FIFO in addition to the RX FIFO register. The status FIFO indicates which data, among FIFO registers, is received with an error. Only when the error having data is ready to read out, the error interrupt will be issued. The RHBn (Buffer or Received data) with an error and ESTATUSn must be read out to clear the status FIFO. Even if the RHBn is read out instead of the ESTATUSn, the Error Status IO is cleared.

For example,

The fact that the UART FIFO receives A, B, C, D, E characters sequentially and the frame error occurs while receiving 'B', and the parity error occurs while receiving 'D' is assumed.

The error interrupt will not be generated even though the actual UART error occurred because the character, which was received with an error, has not been read yet. The error interrupt will be occurred when the character is read out. The address of the Error Status FIFO has been synchronized with the address of the Rx-FIFO.

Time	Sequence flow	Error interrupt	Note
#0	When no character is read out	—	
#1	After A is read out	The frame error(in B) interrupt occurs	The 'B' has to be read out
#2	After B is read out	—	
#3	After C is read out	The parity error(in D) interrupt occurs	The 'D' has to be read out
#4	After D is read out	—	
#5	After E is read out	—	



A Case showing UART Receiving 5 Characters with 2 Errors

13.3.12 Baud-Rate Generation

The serial clock for transmitter and receiver are provided by each UART's baud-rate generator. The baud-rate generating source clock can be selected with the MP250F's internal system clock. Dividing the source clock by 16 and a 16-bit divisor specified in the UART baud-rate divisor register (BRDn) generate the baud-rate clock. The following formula can determine the BRDn:

$$\text{BRDn} = (\text{int}) (\text{PCLK}/(\text{bps} \times 16))$$

where the divisor is from 1 to $(2^{16}-1)$. For example, when the baud-rate is 115,200 bps and PCLK is 40 MHz, BRDn is as follows:

$$\begin{aligned} \text{BRDn} &= (\text{int})(40,000,000/(115,200 \times 16)) - 1 \\ &= (\text{int})(21.7) - 1 \\ &= 21 - 1 = 20 \end{aligned}$$

13.4 UART Registers

13.4.1 UART0 Line Control Register (LCONREG0)

Address : C000 1200h

Bit	R/W	Description	Default Value
7	RW	Reserved	0b
6	RW	Infra-red operation mode. '0' : Normal mode, '1' : IR mode	0b
5 : 3	RW	Parity mode. '0xx' : No parity, '100' : Odd parity, '101' : Even parity, '110' : Parity forced/checked as 1, '111' : Parity forced/checked as 0	000b
2	RW	Number of stop bits per frame. '0' : 1 bit, '1' : 2 bits	0b
1 : 0	RW	Number of data bits per frame. '00' : 5 bits, '01' : 6 bits, '10' : 7 bits, '11' : 8 bits	00b

13.4.2 UART0 Operation Control Register (UONREG0)

Address : C000 1202h

Bit	R/W	Description	Default Value
9	RW	Tx interrupt request type. '0' : Pulse type, '1' : Level when empty	0b
8	RW	Rx interrupt request type. '0' : Pulse type, '1' : Level when empty	0b
7	RW	Rx time out interrupt receive enable. '0' : Disable, '1' : Enable	0b
6	RW	Rx error status interrupt enable. '0' : Do not generate, '1' : Generate	0b
5	RW	Loop-back mode for test purpose only. '0' : Normal operation, '1' : Test purpose only	0b
4	RW	Send break signal. '0' : Normal transmit, '1' : Send break signal	0b
3 : 2	RW	Transmit mode. '00' : Disable, '01' : Interrupt/Polling, '10' : DMA mode	
1 : 0	RW	Receive mode. '00' : Disable, '01' : Interrupt/Polling, '10' : DMA mode	

13.4.3 UART0 FIFO Control Register (FCONREG0)

Address : C000 1204h

Bit	R/W	Description	Default Value
7 : 6	RW	Transmit FIFO trigger level. '00' : Empty, '01' : 4 byte '10' : 8 byte, '11' : 12 byte	00b
5 : 4	RW	Receive FIFO trigger level. '00' : Empty, '01' : 4 byte '10' : 8 byte, '11' : 12 byte	00b
3	RW	Reserved	0b
2	RW	Tx FIFO reset. '0' : Normal, '1' : Tx FIFO reset	0b
1	RW	Rx FIFO reset. '0' : Normal, '1' : Rx FIFO reset	0b
0	RW	FIFO enable '0' : FIFO disable, '1' : FIFO mode	0b

13.4.4 UART0 Modem Control Register (MCONREG0)

Address : C000 1206h

Bit	R/W	Description	Default Value
7 : 5	RW	Reserved	000b
4	RW	AFC (Auto Flow Control) '0' : Disable, '1' : Enable	0b
3	RW	Modem interrupt enable. '0' : Disable, '1' : Enable	0b
2	RW	Reserved	0b
1	RW	DTR(Data Terminal Ready) output. '1' : Active nDTR	0b
0	RW	RTS(Request To Send) output. '1' : Active nRTS	0b

13.4.5 UART0 Tx/Rx Status Register (TRSTATUS0)

Address : C000 1208h

Bit	R/W	Description	Default Value
2	R	Transmit shifter empty. '1' : Empty	0b
1	R	Transmit buffer empty. '1' : Empty	0b
0	R	Receive buffer data ready. '0' : Empty, '1' : Data Ready	0b

13.4.6 UART0 Error Status Register (ESTATUS0)

Address : C000 120Ah

Bit	R/W	Description	Default Value
3	R	Break detect.	0b

Bit	R/W	Description	Default Value
		'0' : No break receive, '1' : Break receive	
2	R	Frame error. '0' : No error receive, '1' : Frame error receive	0b
1	R	Parity error. '0' : No error receive, '1' : Parity error receive	0b
0	R	Overrun error. '0' : No error receive, '1' : Overrun error receive	0b

13.4.7 UART0 FIFO Status Register (FSTATUS0)

Address : C000 120Ch

Bit	R/W	Description	Default Value
10	R	Error in Rx FIFO detect. '0' : No error receive, '1' : Error receive	0b
9	R	Tx FIFO full. '0' : No Full, '1' : Full	0b
8	R	Rx FIFO full. '0' : No Full, '1' : Full	0b
7 : 4	R	Tx FIFO count. Number of data in Tx FIFO.	0000b
3 : 0	R	Rx FIFO count. Number of data in Rx FIFO.	0000b

13.4.8 UART0 Modem Status Register (MSTATUS0)

Address : C000 120Eh

Bit	R/W	Description	Default Value
7	R	Delta DCD (Data Carrier Detect) input. '0' : Has not changed	0b
6	R	Delta RI (Ring Indicator) input.	0b
5	R	Delta DSR (Data Set Ready) input.	0b
4	R	Delta CTS (Clear To Send) input.	0b
3	R	DCD (Data Carrier Detect) input.	0b
2	R	RI (Ring Indicator) input.	0b
1	R	DSR (Data Set Ready) input.	0b
0	R	CTS (Clear To Send) input.	0b

13.4.9 UART0 Transmit Holding(Buffer) Register & FIFO Register (THBREG0)

Address : C000 1210h

Bit	R/W	Description	Default Value
7 : 0	R	Transmit Holding register & FIFO register	00h

13.4.10 UART0 Receive Holding(Buffer) Register & FIFO Register (RHBREG0)

Address : C000 1212h

Bit	R/W	Description	Default Value
7 : 0	R	Receive Holding register & FIFO register	00h

13.4.11 UART0 Baud Rate Division Register (BRDREG0)

Address : C000 1214h

Bit	R/W	Description	Default Value
15 : 0	R	Baud rate division	0000h

13.4.12 UART1 Related Registers

Address : C000 1220h ~ C000 1234h

The same register set exist at address C000 1200h ~ C000 1214h for UART0.

13.4.13 Interrupt Status Register (INTSTATREG)

Address : C000 1240h

Bit	R/W	Description	Default Value
15:8	R	Reserved	0b
7	R/W	UART Channel 1 Modem Interrupt	0b
6	R/W	UART Channel 1 Error Interrupt	0b
5	R/W	UART Channel 1 Receive Interrupt	0b
4	R/W	UART Channel 1 Transmit Interrupt	0b
3	R/W	UART Channel 0 Modem Interrupt	0b
2	R/W	UART Channel 0 Error Interrupt	0b
1	R/W	UART Channel 0 Receive Interrupt	0b
0	R/W	UART Channel 0 Transmit Interrupt	0b

14. PWM

14.1 Overview

The VR3D has four Pulse Width Modulators (PWM). Each PWM operates respectively and is controlled by its own register set. Each channel outputs the pulse width modulated signal via an external pin. Four PWMs have the same circuit architectures and, hereinafter, will be described as PWMn (n = 0, 1, 2, 3)

Each PWM Channel has a 7-bit Prescaler. The Prescaler receives the Clock provided by the Power Management and divides the Clock with the value specified by the PWMnPERIOD, and then produces the Clock for carrying out the PWM operation.

In addition, each PWM Channel has a Counter composed of 10 bits. The CPU can produce the desired signal by specifying the period and the duty of a PWM Waveform in the PWMnPERIOD and PWMnDUTY registers.

When the specified value in the PWMnPERIOD corresponds to the DUTY Counter value, the PWM resets the Counter and sets the PWM Signal to '0'.

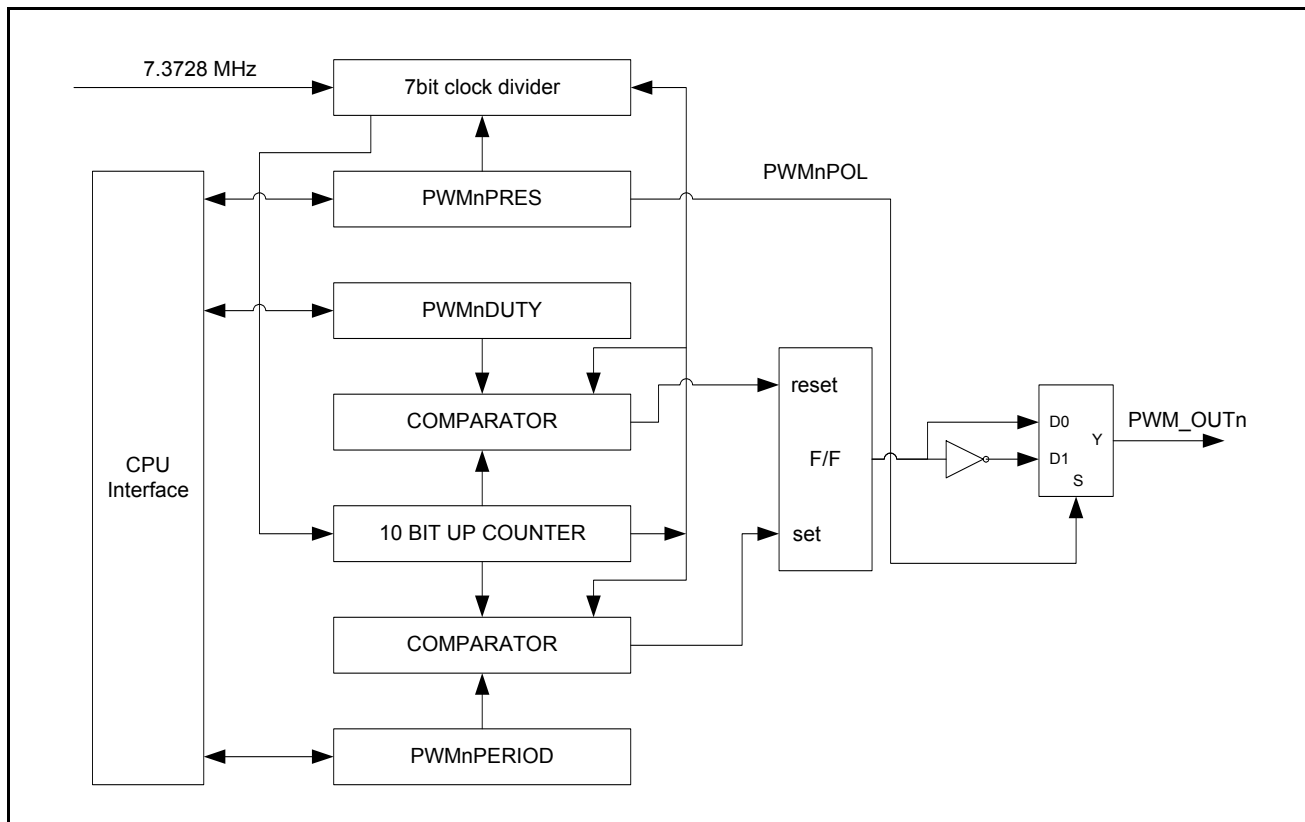
When the specified value in the PWMnDUTY corresponds to the DUTY Counter value, the PWM sets the PWM Signal to '1'.

The PWMnPOL determines whether to Invert or Bypass the value before the PWM Signal produced in the pertinent Channel goes out through the external PAD.

14.1.1 Features

- 4 Pulse Width Modulator channels
- 7-Bit Clock Divider & 10-Bit Period Counter
- 10-Bit Duty Counter

14.1.2 Block Diagram



PWM Block Diagram

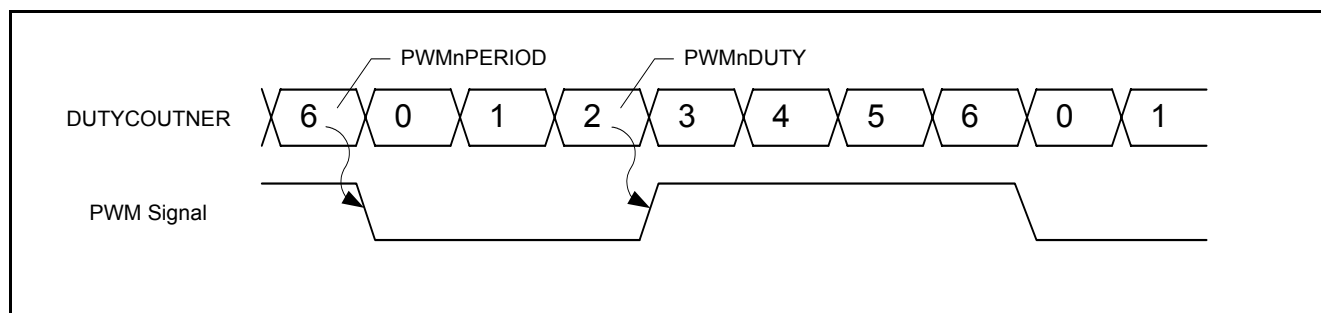
14.2 PWM Signal

As shown below, the 10-bit DUTY Counter is reset by the PWMnPERIOD, and the PWM Signal is changed to '0'.

When the PWMnDUTY register value corresponds to the DUTY Counter value, the PWM Signal is set to '1'.

If the value specified by the PWMnDUTY is bigger than the PWMnPERIOD value, the PWM Signal is always "0".

If the PWMnDUTY is set to "0", the PWM Signal is always "1".



PWM Signal Timing Diagram

14.3 PWM Controller Registers

14.3.1 PWM Prescale0/1 Register (PWMPRES01)

Address : C000 0B00h

Bit	R/W	Description	Default Value
15	RW	PRI1 : Priority of channel #1	0
14:8	RW	PRESCALE1 : Prescale for channel #1	0
7	RW	PRI0 : Priority of channel #0	0
6:0	RW	PRESCALE0 : Prescale for channel #0	0

14.3.2 PWM0 Duty Cycle (PWM0DUTY)

Address : C000 0B04h

Bit	R/W	Description	Default Value
15:10	R	Reserved	0
9:0	RW	PWMDUTY0 : Duty Cycle	0

14.3.3 PWM1 Duty Cycle (PWM1DUTY)

Address : C000 0B06h

Bit	R/W	Description	Default Value
15:10	R	Reserved	0
9:0	RW	PWMDUTY1 : Duty Cycle	0

14.3.4 PWM0 Period Cycle (PWM0PERID)

Address : C000 0B0Ch

Bit	R/W	Description	Default Value
15:10	R	Reserved	0
9:0	RW	PWM0PERID : Period Cycle	0

14.3.5 PWM1 Period Cycle (PWM1PERID)

Address : C000 0B0Eh

Bit	R/W	Description	Default Value
15:10	R	Reserved	0
9:0	RW	PWM1PERID : Period Cycle	0

15. I2C

15.1 Overview

The MP2520F Microprocessor can support a multi-master I2C-bus serial interface. Information between bus masters and peripheral devices that are connected to the 12C-bus is carried by a dedicated serial data line (SDA) and a serial clock line (SCL). The SDA and SCL lines are bi-directional.

In multi-master I2C-bus mode Serial data to or from slave devices can be received or transmitted by multiple MP2520F Microprocessors. The master MP2520F, by which a data transfer can be initiated over the 12C-bus, is responsible for terminating the transfer. This I2C-bus in MP2520F uses the standard bus arbitration procedure.

Values must be written to the following registers to control multi-master 12C-bus operations:

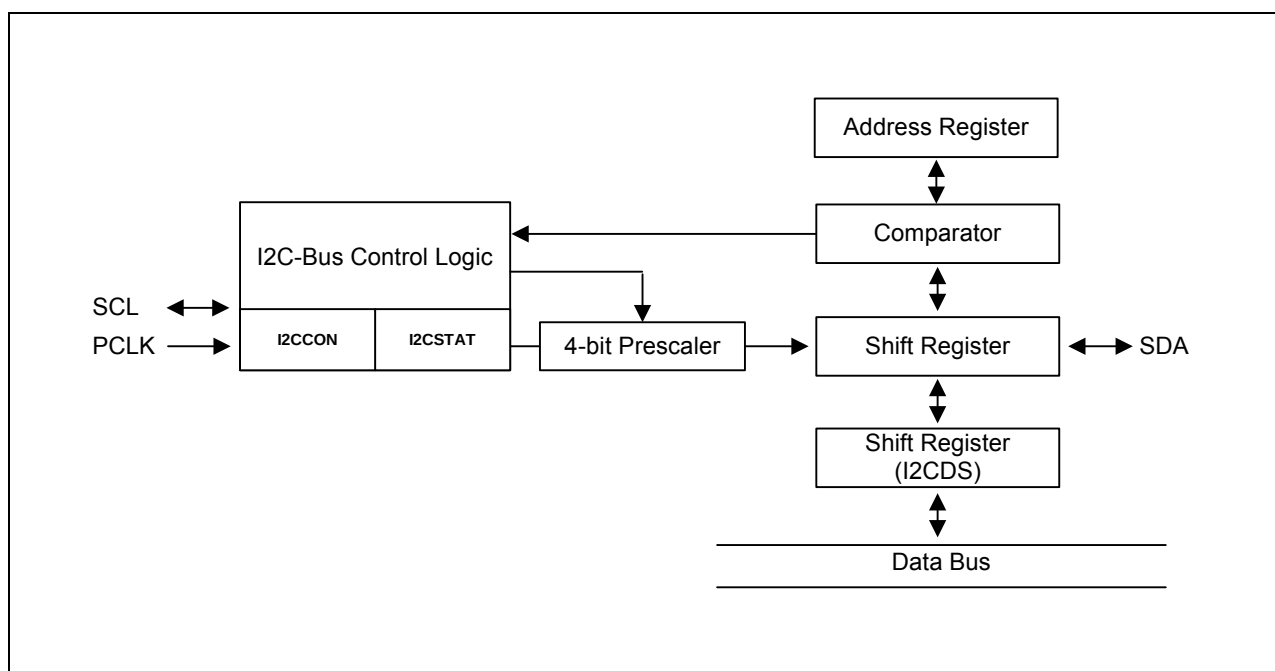
- I2CCON, Multi-master I2C-bus Control Register
- I2CSTAT, Multi-master I2C-bus Control/Status Register
- I2CDS, Multi-master I2C-bus TX/RX Data Shift Register
- I2CADD, Multi-master I2C-bus Address Register

The SDA and SCL lines should both be at High level when the 12C-bus is free. A Start condition can be initiated by a High-to-Low transition of SDA. A Stop condition can be initiated by a Low-to-High transition of SDA while SCL remains steady at High Level.

Start and Stop conditions can always be generated by Master devices. The slave device that the bus master device has selected can be determined by a 7-bit address value in the first data byte, which is put onto the bus after the Start condition has been initiated. The direction of the transfer (read or write) is determined by the 8th bit.

Each SDA line data byte should total eight bits. The number of bytes that can be received or sent during the bus transfer operation is not limited. An acknowledge (ACK) bit should immediately follow every byte, and data is always sent from the most significant bit (MSB) first.

15.1.1 Block Diagram



I2C Block Diagram

15.2 Operation

15.2.1 THE I2C-BUS INTERFACE

There are four operation modes in the MP2520F I2C-bus interface:

- Master transmitter mode
- Master receive mode
- Slave transmitter mode
- Slave receive mode

The following describes functional relationships among these operating modes:

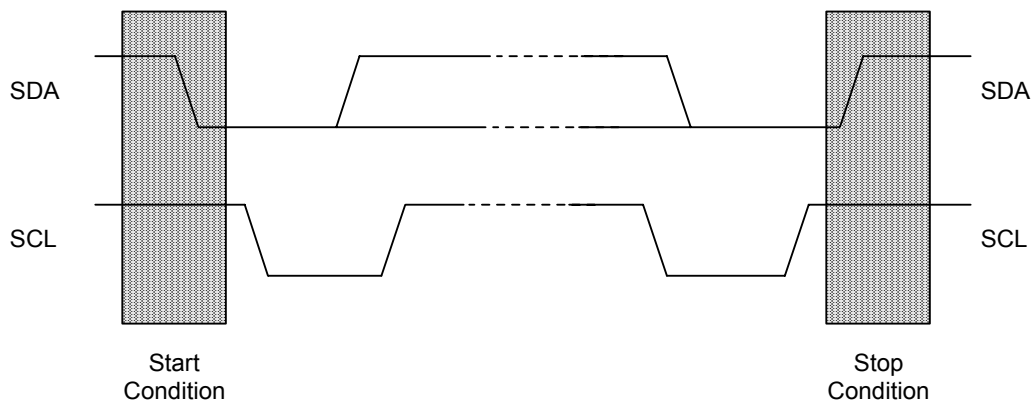
17.2.1.1. START AND STOP CONDITIONS

The I2C bus is usually in slave mode when it is inactive. That is, the interface should be in slave mode before detecting a Start condition on the SDA line. (A High-to-Low transition of the SDA line can be initiated by a Start condition while the clock signal of SCL is High). A data transfer on the SDA line can be initiated and SCL signal generated when the interface state is changed to master mode.

A single byte serial data over the SDA line can be transferred after a Start condition, and the data transfer can be terminated by a Stop condition. A Stop condition is a Low-to-High transition of the SDA line while SCL is High. Stop and Start conditions are always generated by the master. When a Start condition is generated, the I2C-bus is busy. The I2C-bus will be free again within a few clocks after a Stop condition.

A slave address is sent to notify the slave device when a master initiates a Start condition. A 7-bit address and a 1-bit transfer direction indicator (write or read) are included in the single byte of the address field. A write operation (transmit operation) is indicated when bit 8 is 0; request for data read (receive operation) is indicated when bit 8 is 1.

The transfer operation will be completed by the master through transmitting a Stop condition. Another Start condition and a slave address should be generated when the master wants to continue data transmission to bus. So the read-write operation can be performed in various formats.

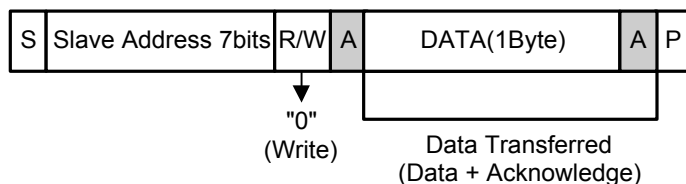


Start and Stop Condition

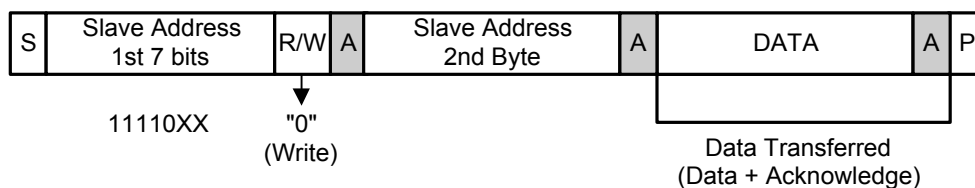
17.2.1.2. DATA TRANSFER FORMAT

Every Data will be transmitted on the SDA line byte should be 8 bits long. The number of bytes that can be transmitted per transfer is unlimited. The address field should be contained in the first byte following a Start condition. When the I2C-bus is operating in master mode, the master can transmit the address field. An acknowledge (ACK) bit should follow each byte. The MSB bit of the serial data and addresses are always sent first.

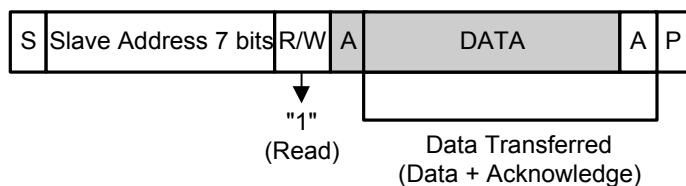
Write Mode Format with 7-bit Addresses



Write Mode Format with 10-bit Addresses



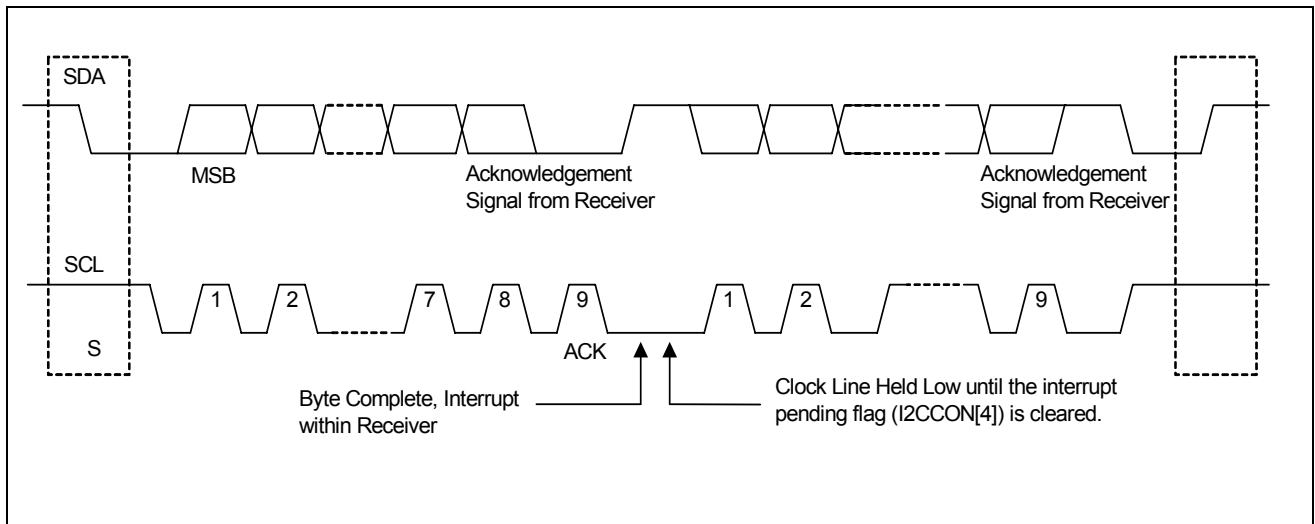
Read Mode Format with 7-bit Addresses



NOTES:

1. S: Start, rS: Repeat Start, P: Stop, A: Acknowledge
2. □ : From Master to Slave, ■ : from Slave to Master

I2C-Bus Interface Data Format



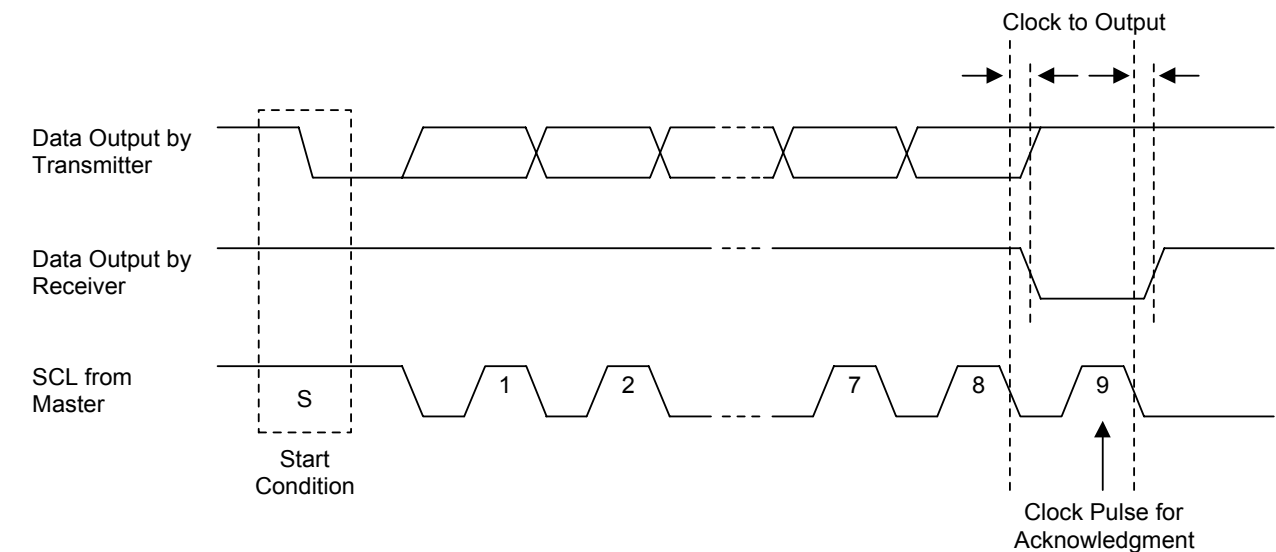
Data Transfer on the I2C-Bus

17.2.1.3. ACK SIGNAL TRANSMISSION

An ACK should be sent by the receiver to the transmitter Master to finalize a single byte transfer operation. The ACK pulse should occur at the ninth clock of the SCL line. For the single byte data transfer eight clocks are required. The clock pulse required to transmit the ACK bit should be generated by the master.

The transmitter should release the SDA line by making the SDA line High when the ACK clock pulse is received, should release the SDA line. The SDA line should also be driven Low by the receiver during the ACK clock pulse so that SDA is Low during the High period of ninth SCL pulse.

Software (I2CSTAT) can enable or disable the ACK bit transmit function. But completion of a single byte data transfer operation requires the ACK pulse on the ninth clock of SCL.



Acknowledge on the I2C-Bus

17.2.2 READ-WRITE OPERATION

Until new data is written to the I2CDS (I2C-bus Data Shift Register) the I2C-bus interface will wait after the transmission of the data in transmitter mode. The SCL line will be held Low until the new data is written. The SCL line will be released after the new data is written to the I2CDS Register. The interrupt should be held by the MP2520F to identify completion of the current data transfer. New data should be written to I2CDS again after the interrupt is received by the CPU.

Until the I2CDS Register is read, the I2C-bus interface will wait after data is received in receive mode. The SCL line will be held Low until the new data is read. The SCL line will be released after the new data is read from the I2CDS Register. The interrupt should be held by the MP2520F to identify the completion of the new data reception. The data should be read from I2CDS after the interrupt request is received by the CPU.

17.2.2.1 BUS ARBITRATION PROCEDURES

To prevent conflict between two masters on the bus, arbitration takes place on the SDA line. A data transfer by a master with an SDA High level will not be initiated if another master with an SDA active Low level is not detected because the current level on the bus does not correspond to its own level. Until the SDA line turns High, the arbitration procedure will be extended.

But each master should evaluate whether or not the mastership is allocated to itself when each master simultaneously lowers the SDA line. The address bits are detected by each master for the purpose of evaluation. The address bit is also detected on the SDA line while the save address is generated by each master because the lowering of the SDA line is stronger than maintaining High on the line. For instance, a Low is generated by one master as the first address bit, while the other master is maintaining High – in this case, both masters will detect Low on the bus because Low is stronger than High even if the first master is trying to maintain High on the line. When this happens, the Low (as the first bit of the address) generating master will get the mastership and the High (as the first bit of the address) generating master should withdraw from mastership. When Low is generated by both masters as the first bit of the address, there should be arbitration for the second address bit again. This arbitration will continue to the end of the last address bit.

17.2.2.2 ABORT CONDITIONS

The SDA line level should be maintained at High when a slave receiver cannot acknowledge the slave address confirmation. In this case, to abort the transfer, the master should generate a Stop condition.

The end of the slave transmit operation should be signaled when a master receiver is involved in the aborted transfer by canceling the generation of an ACK after the last data byte is received from the slave. To allow a master to generate a Stop condition, the slave transmitter should then release the SDA.

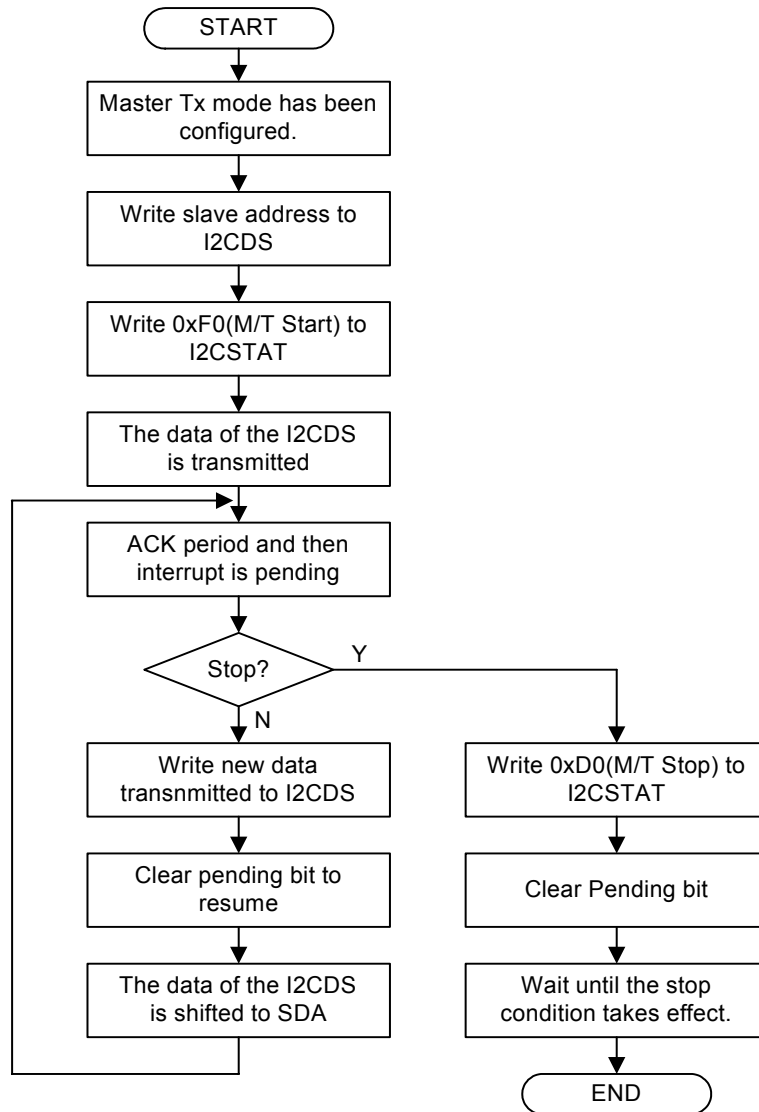
17.2.2.3 CONFIGURING THE I2C-BUS

In the I2CCON Register, the 4-bit prescaler value can be programmed to control the frequency of the serial clock (SCL). The I2C-bus interface address is stored in the I2C-bus address register, I2CADD. (By default, the value of I2C-bus interface address is unknown.)

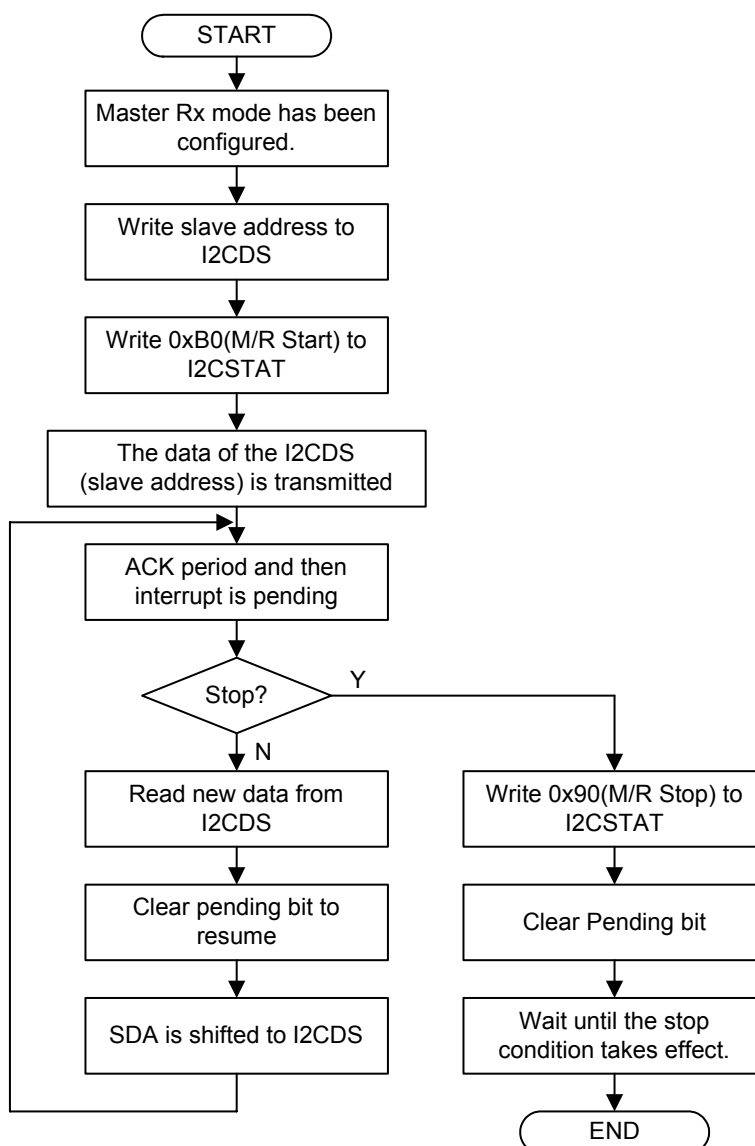
17.2.3 FLOWCHARTS OF THE OPERATIONS IN EACH MODE

Before any of the I2C TX/RX operations, the following steps should take place:

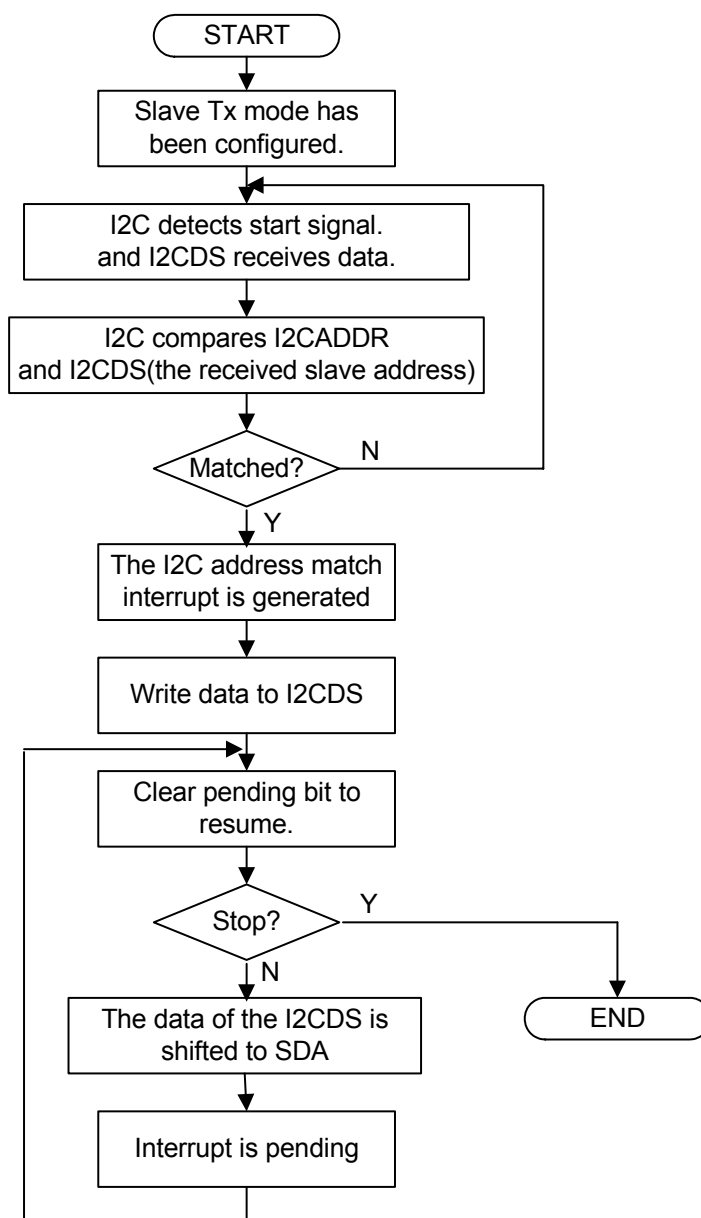
- 1) Write own slave address on the I2CADD register if needed.
- 2) Set I2CCON Register.
 - a) Enable interrupt
 - b) Define SCL period
- 3) Set I2CSTAT to enable Serial Output.



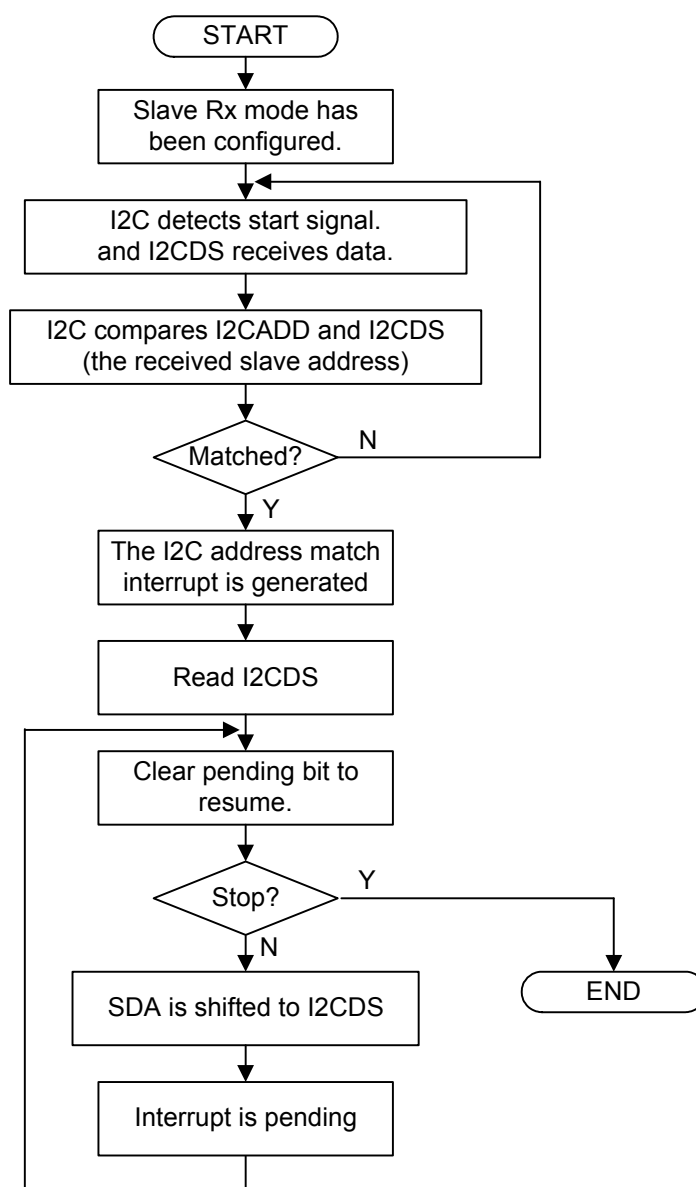
Operations for Master/Transmitter Mode



Operations for Master/Receiver Mode



Operations for Slave/Transmitter Mode



Operations for Slave / Receiver Mode

15.3 I2C Controller Register

15.3.1 I2C Bus Control Register (I2CCONREG)

Address: C000 0D00h

Bit	R/W	Description	Default Value
[7]	R/W	Acknowledge enable 0 = Disable ACK generation 1 = Enable ACK generation	0
[6]	R/W	TX clock source selection 0 = PCLK/16 1 = PCLK/512	0
[5]	R/W	Tx/Rx Interrupt enable 0 = Disable interrupt 1 = Enable interrupt	0
[4]	R/W	Interrupt pending flag	0
[3:0]	R/W	Tranmit clock value Tx clock = PCLK/(I2CCONREG[3:0]+1)	

15.3.2 I2C Bus Control/Status Register (I2CSTATREG)

Address: C000 0D02h

Bit	R/W	Description	Default Value
[7:6]	R/W	Mode selection 00 : Slave receive mode 01 : Slave transmit mode 10 : Master receive mode 11 : Master transmit mode	0
[5]	R/W	Busy signal status/START STOP condition 1 : Busy 0 : Idle	0
[4]	R/W	Serial output enable	0
[3]	R/W	Arbitration status flag	0
[2]	R/W	Address-as-slave status flag	0
[1]	R/W	Address zero status flag	0
[0]	R/W	Last-received bit status flag	0

15.3.3 I2C Bus Address Register (I2CADDRREG)

Address: C000 0D04h

Bit	R/W	Description	Default Value
[7:0]	R/W	Slave address	0

15.3.4 I2C Bus Transmit/Receive Data Shift Register (I2CDSREG)

Address: C000 0D08h

Bit	R/W	Description	Default Value
[7:0]	R/W	Data shift	0

16. ADC & TPC

16.1 Overview

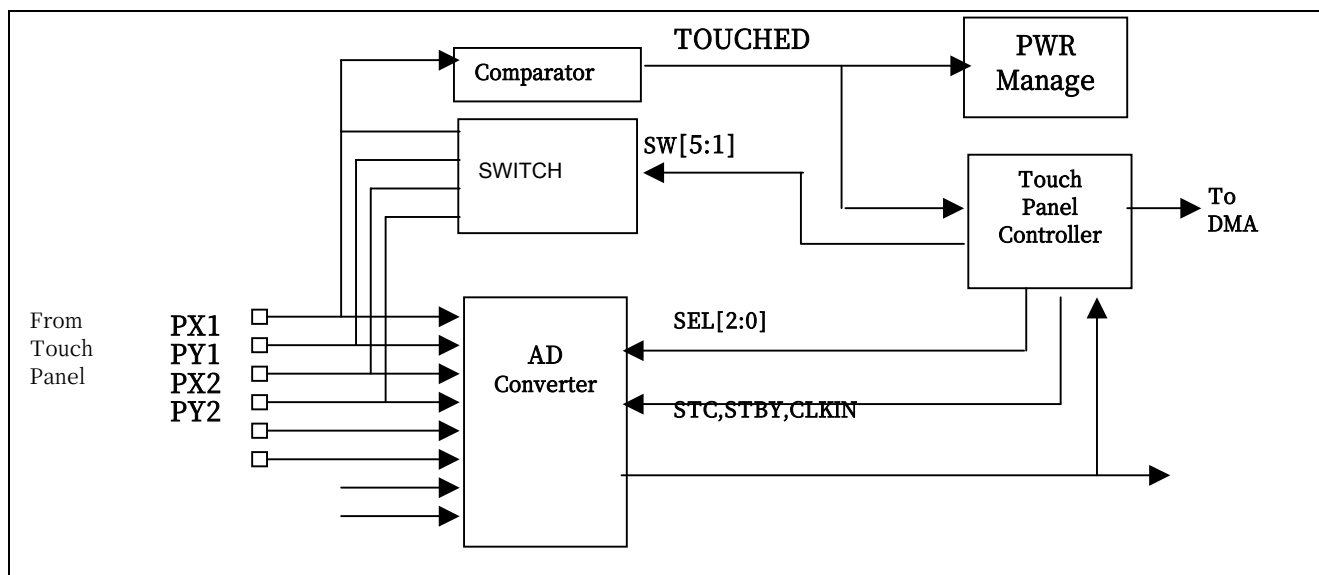
The Touch Panel Controller (TCP) controls the switch, comparator, and ADC to read the X, Y, position of the Touch Panel touched. It also reads the data from any ADC Channel. All data is transmitted into external memory using the DMA. Refer to ADC System Block Diagram.

16.1.1 Features:

The TPC module offers the following features:

- 10-bit ADC for touch panel and voltage detection
- On-chip voltage reference for ADC
- 16x16-bit FIFO for X, Y, Z, U1 and U2
- Embedded touch panel circuitry (Switch and Comparator)
- Supports auto, manual, and Touched Detection Enable Mode (TDEM)
- Programmable pen down and pen-up interrupt to interrupt handler
- Provides data-ready and FIFO-full interrupts

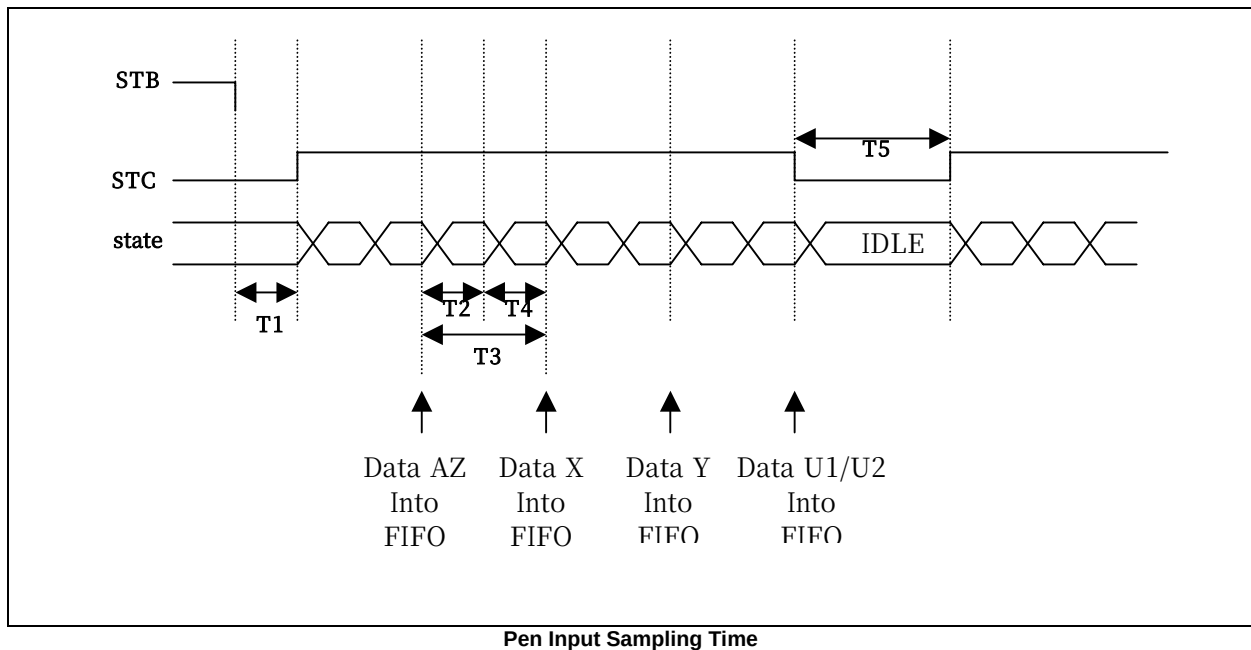
16.1.2 Block Diagram



This block consists of three analog blocks and one control block.

The Comparator detects pen down. The switch creates the bias required for TOUCH PAD when it reads each coordinate. The AD Converter (ADC) converts analog signals entered through a PAD into digital signals. Finally, the TPC (Touch Panel Controller) controls the analog blocks and transfers the data attained from the ADC to the outside through the DMA.

16.2 Operation



16.2.1 CLOCK CONTROL

The clock uses the PCLK. In addition, the ADC uses the clock divided through the Prescaler. The frequency can be divided from a minimum of two times to a maximum of 256 times. The APSV in Table Clock Control Register determines the division of the clock. When the PADE in Table Control Register is set to '1' and the APSY is set to 'non-0', the clock occurs.

Address : 0xC000_1754

Bit	R/W	Symbol	Description	Reset Value
[15:8]	R/W	TPSNS	Touched Input Sensitivity	0
[7:0]	R/W	APSV	Prescaler Value	0

Clock Control Register(TPC_CLK_CNTL)

Address : 0xC000_1740

Bit	R/W	Symbol	Description	Reset Value
[15]	R/W	ADCEN	AD Converter Enable	
[14]	R/W	PADE	Touch Panel Enable	0b
[13]	R/W	AUTO	AUTO Mode Enable	0b
[12]	R/W	U1E	U1 Channel Enable	0b
[11]	R/W	U2E	U2 Channel Enable	0b
[10]	R/W	MOD	Touch Panel Enable	0b
[9]	R/W	AZE	Auto-ZERO Enable	0b
[8:7]	R		Reserved	000b
[6]	R/W	COMPD	Comparator Power Down Mode	1'b1
[5]	R/W	TDEM	Touch Detection Enable Mode	0b
[4]	R/W	FIFOCLR	FIFO Clear	0bj
[3]	R	TOUCHED	Touched Status	0b
[2]	R	FIFOFULL	FIFO Full Status	0b
[1]	R	FIFOEMPTY	FIFO Empty Status	0b

[0]	R	STBY	ADC Status	0b
-----	---	------	------------	----

Control Register(TPC_CNTL)

16.2.2 SAMPLING RATE CONTROL

A Data Sampling Rate is determined by the parameter to control the ADC. As shown in figure Pen Input Sampling Time, four parameters (T1, T2, T4 and T5) are used to control the ADC Sampling Rate. These parameters are listed in Table Sampling Rate Control Parameters

Register	Parameter	Description	Min/Max
BGCNT	T1	Bandgap warm up time	0/255
DSCNT	T2	Transistor switching and input selecting set up time	0/255
DMCNT	T4	The Decimation ratio of the second-state FIR	4/255
IDLECNT	T5	The number of clocks between the last capture and the first capture of two pen input points	0/255

Table 27-3. Sampling Rate Control Parameters

T2 and T4 out of the above parameters determine the time required for the sampling of each data. The T2 (DSCNT) indicates the required time to switch the ADC analog channel. In addition, the T4 indicates the required time to convert analog signals after sampling into digital signals. The ADC of the MMSP-2 needs 5 cycles to convert analog signals into digital signals: the minimum value of the DMCNT must be 4.

The Registers to specify these parameters are listed in Table Timing Parameter Register 1 and 2.

Address : 0xC000_175C

Bit	R/W	Symbol	Description	Reset Value
[15:8]	R/W	DMCNT	The decimation ratio of the second-stage FIR (DSCNT>=1)	0
[7:0]	R/W	DSCNT	The input signal data set up time after transistor switching circuit and input select are settled	0

Timing Parameter Register 1(TPC_TIME_PARM1)

Address : 0xC000_1760

Bit	R/W	Symbol	Description	Reset Value
[15:8]	R/W	BGCNT	Bandgap warm up time	0
[7:0]	R/W	IDLECNT	The number of clocks between the last capture and the first capture of two pen input points	0

Timing Parameter Register 2(TPC_TIME_PARM2)

The Sampling rate to read X and Y coordinates from the Touch PAD is determined by the number of the selected channels, the DILECNT, the DSCNT, and the DMCNT. Table Timing Parameter Register 1 shows the way to determine the sampling frequency according to the data types filled in FIFO. The FIFO is always filled with data in order of AZ, X, Y, U1 and U2.

The # of Data Type	Equation	FIFO Data Format
5	$FCLK/[5x(DSCNT+DMCNT+1)+IDLECNT]$	AZ, X, Y, U1, U2,...
4	$FCLK/[4x(DSCNT+DMCNT+1)+IDLECNT]$	AZ, X, Y, U1,... or AZ, X, Y, U2,...
3	$FCLK/[3x(DSCNT+DMCNT+1)+IDLECNT]$	AZ, X, Y, AZ, X, Y,... or X, Y, U1,... or X, Y, U2,...
2	$FCLK/[2x(DSCNT+DMCNT+1)+IDLECNT]$	X, Y, X, Y,... or U1, U2, U1, U2,...
1	$FCLK/[(DSCNT+DMCNT+1)+IDLECNT]$	U1, U1,... or U2, U2,...

Sampling Rate Calculation

16.2.3 MANUAL MODE

PADE	AUTO	TDEM	U1E	U2E	MOD	AZE	Operation
0	X	X	X	X	X	X	No Valid Data
1	0	X	0	0	0	X	No Valid Data
			0	0	1	0	Read X, Y
			0	0	1	1	Read X, Y, AZ
			1	0	1	0	Read X, Y, U1
			1	0	1	1	Read X, Y, AZ, U1
			0	1	1	0	Read X, Y, U2
			0	1	1	1	Read X, Y, AZ, U2
			1	1	1	0	Read X, Y, U1, U2
			1	0	0	x	Read U1
			0	1	0	x	Read U2
			1	1	0	x	Read U1, U2

Manual Mode Operation

When the AUTO in Table Control Register is set to '0', Manual Mode is executed. In addition, when the PADE and the ADCEN is set to '1', Sampling is initiated. The sampling proceeds in order of AZ, X, Y, U1 and U2 channels, and the channel to read is selected by the values of U1E, U2E, MOD and AZE.

Transformed data are stored in the TP_X_VALUE, TP_Y_VALUE, TP_AZ_VALUE, TP_U1_VALUE and TP_U2_VALUE registers. In order to complete the sampling, the STBY value in Table Control Register must be '1'.

Table Manual Mode Operation shows the operation according to register values in Manual Mode.

After reading the transformed data, PADE, U1 and U2 must be set to '0' to read data again.

The registers to store the transformed values are listed in Table X Value Register to Table U2 Value Register.

Address : 0xC000_1768

Bit	R/W	Symbol	Description	Reset Value
[15:11]	R/W	READ_X	Switch Control for Read X	5'h5
[10]	R		Reserved	0
[9:0]	R	X_VALUE	X value	0

X Value Register(TPC_X_VALUE)

Address : 0xC000_176C

Bit	R/W	Symbol	Description	Reset Value
[15:11]	R/W	READ_Y	Switch Control for Read Y	5'ha
[10]	R		Reserved	0
[9:0]	R	Y_VALUE	Y value	0

Y Value Register(TPC_Y_VALUE)

Address : 0xC000_1770

Bit	R/W	Symbol	Description	Reset Value
[15:11]	R/W	AUTO_ZERO	Switch Control for Read Auto-zero	5'h5
[10]	R		Reserved	0
[9:0]	R	AZ_VALUE	Auto-zero value	0

Auto Zero Value Register(TPC_Y_VALUE)

Address : 0xC000_1774

Bit	R/W	Symbol	Description	Reset Value
[15:11]	R/W	SW_DFALT	Switch Control for Read U1,	5'h04
[10]	R		Reserved	0
[9:0]	R	U1_VALUE	U1 value	0

U1 Value Register(TPC_U1_VALUE)

Address : 0xC000_1778

[15:10]	R		Reserved	0
[9:0]	R	U2_VALUE	U2 value	0

U2 Value Register(TPC_U2_VALUE)

16.2.4 AUTO MODE

PADE	AUTO	TDEM	U1E	U2E	MOD	AZE	Data Format in FIFO
0	X	X	X	X	X	X	No Valid Data
1	1	0	0	0	0	X	No Valid Data
			0	0	1	0	X, Y, X, Y,...
			0	0	1	1	AZ ¹ , X ² , Y ³ , AZX, Y,...
			1	0	1	0	X, Y, U1, X, Y, U1,...
			1	0	1	1	AZ, X, Y, U1, AZ, X, Y, U1,...
			0	1	1	0	X, Y, U2, X, Y, U2,...
			0	1	1	1	AZ, X, Y, U2, AZ, X, Y, U2,...
			1	1	1	0	AZ, X, Y, U1U2,...
			1	0	0	x	U1, U1, U1,...
			0	1	0	x	U2, U2, U2,...
			1	1	0	x	U1, U2, U1, U2,...

AUTO Mode Operation

When the AUTO bit is set to 1, the TPC operates in AUTO mode. In this mode, the sampling operation is carried out continuously, and all sampled data is first stored into FIFO. Whenever FIFO is full, data is transmitted into external memory using the DMA. Data stored in FIFO have information about the sampled channel, and data type (X, Y, AZ, U1 and U2), and sampled data. For more information about data format, refer to Table Auto Mode Operation.

16.2.5 SWITCHING CONTROL

The switch block has five switches. Switches are controlled differently for each mode. READ_X, READ_Y, AUTO_ZERO, and SW_DFALT registers contain control information for each mode. For more information, Refer to TABLE X Value Register to U2 Value Register.

Operation	SW[5:1]	Read Channel
READ_X	5'b00101	1/3
READ_Y	5'b01010	0/2
AUTO_Z	5'b00101	0

¹ AZ : Auto Zero Value

² X : X Value

³ Y : Y Value

Default	5'b01000	0
READ_U1	5'b01000	0 ~ 7
READ_U2	5'b01000	0 ~ 7

Switch Default Control

READ_X Value

In order to read the data on the X coordinate, set SW1 and SW3 to 'ON' and then read data from channel 1/3.

READ_Y Value

In order to read the data on the Y coordinate, set SW2 and SW4 to 'ON' and then read data from channel 0/2.

READ_AZ Value

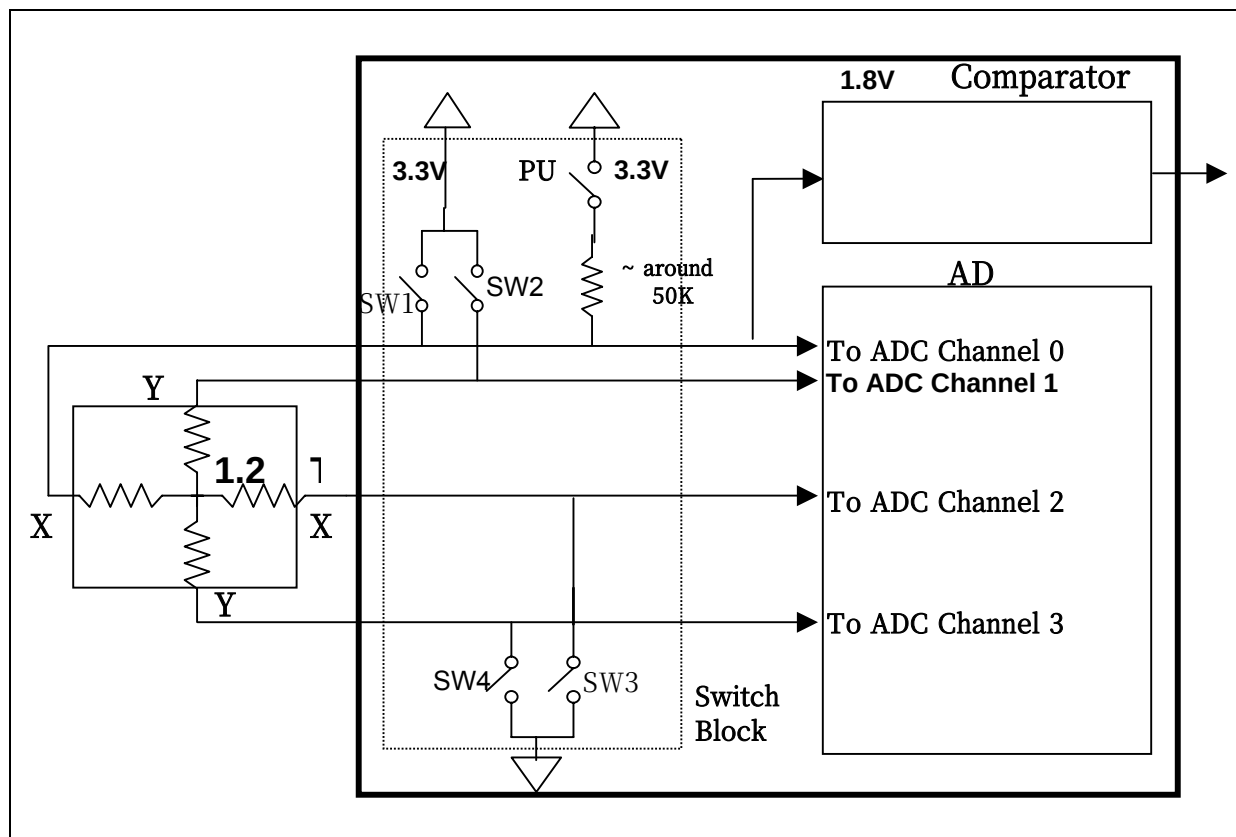
This mode is used to measure the voltage between switches. For example, you may read data from channel 2 after connecting SW1 to SW3 to measure the voltage between both ends of SW2. Likewise, you may read data from channel 3 after connecting SW2 to SW4 to measure the voltage between the both ends of SW3.

READ U1 Value

Read data from a channel out of ADC channels.

READ U2 Value

Read data from a channel out of ADC channels.



Switching Control

16.2.6 INTERRUPT

The TPC has five interrupt sources. Three are for the TOUCH PANEL. The other two sources are for auxiliary input. Each Interrupt is enabled and controlled by the TPC_INTR Register.

- 1) Touch Interrupt

The TOUCH Interrupt occurs when the comparator output (TOUCH) signal is toggled from 1 to 0, and the zero state is continued for some period specified by the TPSNS Register. For example, when the TPSNS is zero, an interrupt is generated as soon as it detected by the system clock.

2) Touch panel Compare Interrupt

The Touch Panel Compare Interrupt is generated when data sampled is compared to some threshold value that is specified by the TPTHRES Register. The Compare condition is controlled by the TPCC bit.

If TPCC is 0, the interrupt occurs when sampled data is less than TPTHRES, and if TPCC is 1, the interrupt occurs when sampled data is greater than TPTHRES.

3) Pen Data Interrupt

The PEN Data Interrupt occurs when at least one valid data sample is stored in the FIFO.

4) U1 Compare Interrupt

The U1 compare interrupt is generated when data sampled is compared to some threshold value that is specified by the U1THRES Register. The compare condition is controlled by the U1CC bit.

5) U2 Compare Interrupt

The U2 Compare Interrupt is generated when data sampled is compared to some threshold value that is specified by the U2THRES Register. The compare condition is controlled by the U2CC bit.

16.2.7 FIFO and Data FORMAT

In the case of non-Manual Mode, all data are stored in the FIFO. In addition, the data are stored in an external memory via the DMA. The data to store in the external memory consist of the source channel of the data and the type of the data as well as the values transformed from analog values. The size of data output from the ADC is 10 bits. This 10-bit data, the ADC channel information, and the data type are stored together in the FIFO. The data formats by the pertinent DATA type are listed in Table **Data Format**

ASEL[2]/TOUCHED	ASEL[1:0]	DTYPE[2:0]	ADO[9:0]
-----------------	-----------	------------	----------

Data Format

ASEL[2:0] : Analog Input Channel.

TOUCHES : The TOUCHED bit is used only when Comparator Output DTYPE is related to a Touch PAD. This bit can be used as the criterion to evaluate the validity of the data entered by the software.

DTYPE	AZ	X	Y	U1	U2
Value	1	2	3	4	5

Data Type

16.3 ADC Control Register

16.3.1 A/D CONVERTER CONTROL (ADCCON)

Address : C000 1700h (16bit Access Only)

Bit	R/W	Description	Default Value
15	R	FLAG : A/D Converter state flag. It takes max. 16PCLKs for the FLAG bit to be effective after the A/D Conversion is started. So, program must not check the FLAG not just after the A/D conversion is started. 0 : A/D Conversion in progress 1 : End of A/D Conversion	1
14	RW	APEN : Prescale Enable	1
13:6	RW	APSV : A/D Conversion clock prescale	00
5:3	RW	ASEL : Analog Input select	0
2	RW	STBY : System power down 0 : normal operation 1 : Power down mode	1
1	R/W	ASRM : A/D Conversion Start-by-Read mode	0
0	RW	ADEN : A/D Conversion Enable 0 : No Operation 1 : A/D conversion starts and this bit is cleared after the start-up	0

16.3.2 A/D CONVERTER DATA (ADCDAT)

After A/D Conversion is completed, the converted data is read by the ADCDAT register. ADCDAT has to be read after the conversion is completed.

Address : C000 1704h (16bit Access Only)

Bit	R/W	Description	Default Value
15:10	R	Reserved	x
9:0	R	ADCDAT : A/D Conversion output	x

16.3.3 TP CONTROL REGISTER (TPC_CNTL)

Address : C000 1740h

Bit	R/W	Description	Default Value
15	R/W	AD_COV_EN :AD Converter Enable – Control the ADC. This bit must be enabled to sample data from ADC	0
14	R/W	PADE :Touch Panel Enable : Controls the touch panel Controller. This bit must be enabled during the sampling. In The manual mode(AUTO=0). This bit must be cleared to read data sampled	0
13	R/W	AUTO_MD :Auto Mode Enable : When Enabled. Data sampled is stored in FIFO automatically. When disabled. Data sampled is stored into TPC_X_VALUE, etc.	0
12	R/W	U1_EN :U1 Channel Enable- Controls the sampling for the U1 Channel	0
11	R/W	U2_EN :U1 Channel Enable- Controls the sampling for the U2 Channel	0
10	R/W	MOD : Touch Panel Enable – Controls the sampling for the X,Y Channel	0
9	R/W	AUTO_ZERO :Auto-Zero Enable – Controls the sampling for the AUTO-Zero Channel	0
8:7	R	Reserved	X
6	R/W	CO_PW_MD : Comparator Power-Down Mode- When enabled the comparator is in power-down mode. In order to detect the TOUCH. This bit must be disabled.	1
5	R/W	TDET_MD : Touch detection enable mode : This bit is valid when AUTO=1. In AUTO mode, If this bit is enabled, just the sampling for X,Y and AZ is started when comparator output is active.	0
4	R/W	FF_CLR : FIFO Clear- When enabled. FIFO is cleared.	0
3	R	TOUCHED : Touched Status – Reads the comparator output	0
2	R	FF_FULL : FIFO Full Status	0
1	R	FF_EMPT : FIFO Empty Status	0
0	R	STBY : ADC Status- Reads the ADC Status. When this bit is set. ADC is busy. When this bit is cleared. ADC is in idle state.	0

16.3.4 TP INTERRUPT CONTROL REGISTER (TPC_INTR)

Address : C000 1744h

Bit	R/W	Description	Default Value
31:13	R	Reserved	x
12	R/W	U1CCE :Compare interrupt Enable for U1 channel – this bit controls the compare interrupt for the U1 channel	0
11	R/W	U2CCE :Compare interrupt Enable for U2 channel – this bit controls the compare interrupt for the U2 channel	0
10:8	R/W	TP_INT_EN :Touch panel interrupt enable [10] : Touch interrupt – enable/disable touch interrupt signal [9] : Touch panel compare interrupt – enable/disable compare interrupt for touch panel	000b

Bit	R/W	Description	Default Value
		[8] : pen data interrupt : enable/disable Pen data interrupt	
7:5	R	Reserved	x
4	R	U1_IF : U1 Compare interrupt Flag- indicated the compare interrupt is pending for U1 channel	0
3	R	U2_IF : U2 Compare interrupt Flag- indicated the compare interrupt is pending for U2 channel	0
2:0	R	Touch pad Interrupt Flag [2] : Touch Interrupt Flag [1] : Touch panel compare interrupt flag [0] : Pen data interrupt flag	000b

16.3.5 TP COMPARE REGISTER FOR TOUCH PANNEL (TPC_COMP_TP)

Address : C000 1748h

Bit	R/W	Description	Default Value
31:11	R	Reserved	x
10	R/W	TP_CC :Touch Panel Compare Control 0 : Trigger when the compare value is greater then the sample 1 : Trigger when the sample is grater then the compare value	0
9:0	R/W	TP_THRES : Touch Panel Threshold level	0

16.3.6 TP COMPARE REGISTER FOR U1 (TPC_COMP_U1)

Address : C000 174Ch

Bit	R/W	Description	Default Value
31:11	R	Reserved	x
10	R/W	U1_CC :Compare Control for U1 channelU1 0 : Trigger when the compare value is greater then the sample 1 : Trigger when the sample is grater then the compare value	0
9:0	R/W	U1_THRES : U1 channel Threshold level – contains the value to be compared with the sampled data in the U1 channel	128

16.3.7 TP COMPARE REGISTER FOR U2 (TPC_COMP_U2)

Address : C000 1750h

Bit	R/W	Description	Default Value
31:11	R	Reserved	x
10	R/W	U2_CC :Compare Control for U2 channelU2 0 : Trigger when the compare value is greater then the sample 1 : Trigger when the sample is grater then the compare value	0
9:0	R/W	U2_THRES : U2 channel Threshold level – contains the value to be compared with the sampled data in the U2 channel	128

16.3.8 TP CLOCK CONTROL REGISTER (TPC_CLK_CNTL)

Address : C000 1754h

Bit	R/W	Description	Default Value
31:16	R	Reserved	x
15:8	R/W	TP_SNS : Touched input Sensitivity- This register sets the sensitivity for the	0

Bit	R/W	Description	Default Value
		comparator output. It reduce the glitch effect.	
7:0	R/W	A_PS_VAL : Prescaler Value-sets the divide ratio for the prescaler When zero. Noclock is generated for the ADC. When one. Divider ratio is two. So we can set from fclk/2 to fclk/512	0

16.3.9 TP CHANNEL SELECTION REGISTER (TPC_CH_SEL)

Address : C000 1758h

Bit	R/W	Description	Default Value
31:16	R	Reserved	x
14:12	R/W	U2_SEL : Channel Selection for U2	0
11:9	R/W	U1_SEL : Channel Selection for U1	0
8:6	R/W	Y_SEL : Channel Selection for Y-Read	0
5:3	R/W	X_SEL : Channel Selection for X-Read	0
2:0	R/W	AZ_SEL : Channel Selection for AUTO Zero	0

16.3.10 TP TIMING PARAMTER REGISTER1 (TPC_TIME_PARAM1)

Address : C000 175Ch

Bit	R/W	Description	Default Value
31:16	R	Reserved	x
15:8	R/W	BG_CNT : Band-gap warm up time (Refer to xx-x)	0
7:0	R/W	IDLE_CNT : The number of clocks between the last capture and the first capture of two pen input point	0

16.3.11 TP TIMING PARAMTER REGISTER2 (TPC_TIME_PARAM2)

Address : C000 1760h

Bit	R/W	Description	Default Value
31:16	R	Reserved	X
15:8	R/W	DS_CNT : The input signal data set up time after transistor switching circuit and input select are settled.	0
7:0	R/W	DM_CNT : The decimation ratio of the second-stage FIR (DSCNT>=1)	0

16.3.12 TP TIMING PARAMTER REGISTER3 (TPC_TIME_PARAM3)

Address : C000 1764h

Bit	R/W	Description	Default Value
31:4	R	Reserved	X
3:0	R/W	ADC_TIME : ADC Conversion Time- Sets the cycle needed to convert the data sample This value must be greater than four. This is generated from the conversion time of ADC embedded	4

16.3.13 X VALUE REGISTER (TPC_X_VALUE)

Address : C000 1768h

Bit	R/W	Description	Default Value
-----	-----	-------------	---------------

Bit	R/W	Description	Default Value
31:16	R	Reserved	x
15:11	R/W	READ_X : Switch control for Read X	5'h5
10	R	Reserved	X
9:0	R	X_VALUE : X Value	X

16.3.14 Y VALUE REGISTER (TPC_Y_VALUE)

Address : C000 176Ch

Bit	R/W	Description	Default Value
31:10	R	Reserved	X
15:11	R/W	READ_Y : Switch control for Read Y	5'ha
10	R	Reserved	X
9:0	R	Y_VALUE : Y Value	x

16.3.15 AZ VALUE REGISTER (TPC_AZ_VALUE)

Address : C000 1770h

Bit	R/W	Description	Default Value
31:16	R	Reserved	x
15:11	R/W	AUTO_ZERO : Switch control for Read Auto-Zero	5'h5
10	R	Reserved	x
9:0	R	AZ_VALUE : AZ Value	x

16.3.16 U1 VALUE REGISTER (TPC_U1_VALUE)

Address : C000 1774h

Bit	R/W	Description	Default Value
31:16	R	Reserved	x
15:11	R/W	SW_DFLAT : Switch control for Read U1	5'h04
10	R	Reserved	X
9:0	R	U1_VALUE : U1 Value	0

16.3.17 U2 VALUE REGISTER (TPC_U2_VALUE)

Address : C000 1778h

Bit	R/W	Description	Default Value
31:10	R	Reserved	x
9:0	R	U2_VALUE : U2 Value	x

17. SD/MMC Controller

17.1 Overview

Multi-Media Card (MMC), Secure Digital (SD) memory card and SDIO device are supported by the SD/MMC controller. MMC specification version 2.11, SD memory card specification version 1.0 and SDIO card specification version 1.0 are appropriate for the SD/MMC controller.

17.1.1 Feature

- Compatible with the Multi-Media Card System Specification, version 2.11
- Compatible with the SD Memory Card Specification, version 1.0 and SDIO Card Specification, version 1.0
- Supports clock speeds up to 10 MHz in MMC mode
- Supports clock speeds up to 25 MHz in SD mode
- Contains an Internal Clock Prescaler
- Contains 64 bytes of FIFO for data receive/transmit
- Supports PIO and DMA mode data transfer
- Supports 4-bit data bus

17.2 PIN Description

The external SD/MMC card is connected to three (single data line) or six (4 data line) signals.

Name	Direction	Description
SDICLK	O	SD/MMC Interface Clock - Varies due to clock prescaler
SDICMD	B	SD/MMC Command/Response
SDIDAT[3:0]	B	SD/MMC Data

17.3 SD/MMC Controller Operation

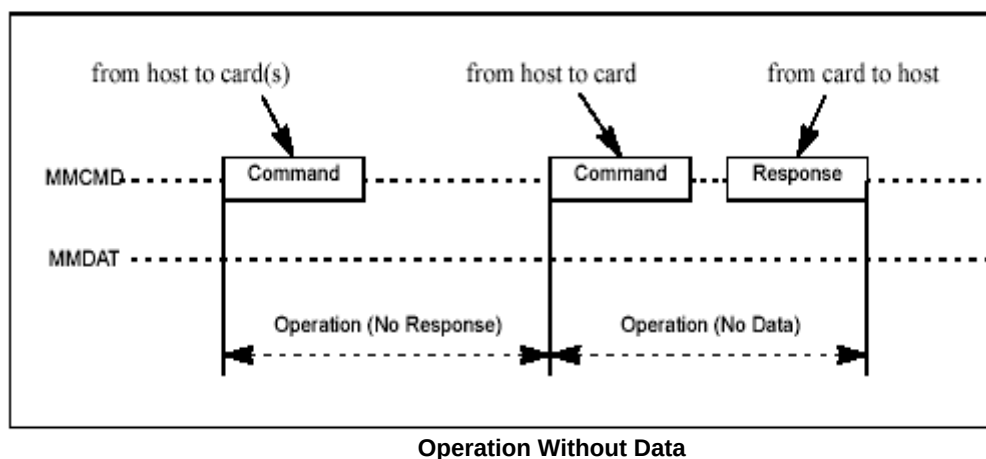
The clock frequency for the transmission of the controller and the device can be adjusted by the clock prescale register (SPIDER) of the SD controller. The samplings of commands and data are synchronized by this clock.

17.3.1 Bus protocol

Communication over the SD bus is based on command and data bit streams which are initiated by a start bit and terminated by a stop bit.

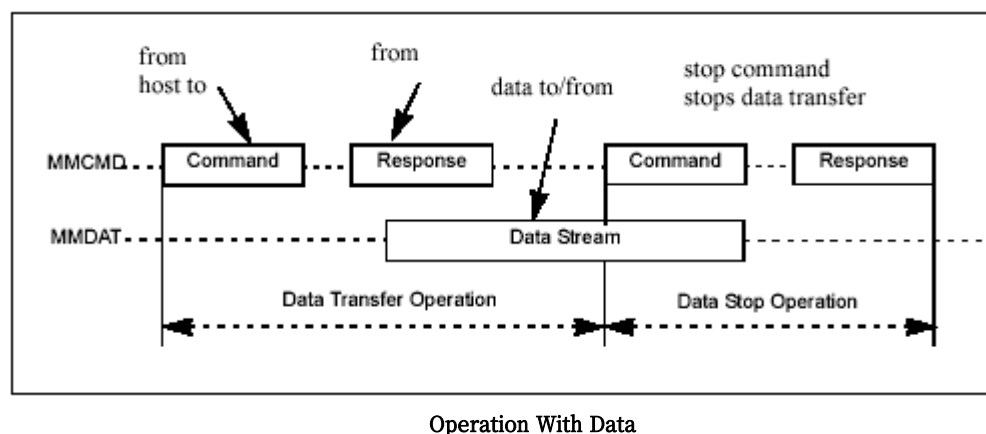
- **Command:** A command is an operation starting token. It's sent from the controller to a card, and is transferred serially on the CMD line.
- **Response:** A response is a token that is sent from a card as an answer to a previously received command to the controller, and is transferred serially on the CND line.
- **Data:** Data can be transferred from the controller to the card or vice versa, and it is transferred via the DAT line.

Operation without data token is as follows:



Data transfers from/to the card are done in block. CRC bits always follow Data blocks. Defines the single and multiple block operations. For faster write operations, the multiple block operation mode is better. The controller terminates a multiple block transmission when a stop command follows on the CMD line. The host can configure data transfer to use single or multiple data lines.

The following is the operation with data token:



17.3.2 Initialization

6 Pins of GPIO[L5:0] is used by the SD/MMC controller. For the SD/MMC interface function, configure these six pins as GPIO Alt Function 1.

1. Set **GPIO L AltFnSelL** register (C000 1036h) to xxxx_1010_1010_1010b.

To program the SDI modules, follow these basic steps :

1. Set **SDICON** and **SDIntMsk0/1** to configure properly with clock and interrupt enable.
2. Set **SDIPRE** to configure with a proper value.
3. Wait 74 SDCLK clock cycle in order to initialize the card.

For the sequences to execute their actual operations, refer the operation sequence examples in [28.3.5](#).

17.3.3 Error Detection

The controller checks the following errors and stores the results into the Command Status Register (**SDICmdSta**) and the Data Status Register (**SDIDatSta**).

- Response CRC error: Command response shows the calculation of CRC error.
- Response time out: No response begins before the specific number of clocks
- Write data CRC error: A CRC status error on the data is returned by card.
- Read data CRC error: The data shows the calculation of CRC error.
- Read time out: No read data operation begins before the specified number of clocks.

17.3.4 Clocks

The Processor enables/disables the SD bus clock (SDICLK) by means of the bit 0 (ENCLK) of the Control Register (**SDICON**). The SD bus clock frequency is adjusted by the value of the Prescaler Register (**SDIPRE**). The clock frequency is determined by the following expression:

$$\text{Bus Clock Frequency} = \text{PCLK} / (2 * (\text{Prescaler value} + 1))$$

17.3.5 SDIO Operation

There are SDIO Interrupt Receiving and Read Wait Request Generation in the types of the SDIO operation. These two types of operations are enabled by the RCV_IO_INT bit (i.e. bit [3]) and the RWAIT_EN bit (i.e. bit [2]) of the **SDICON** register respectively. Each step is per the following:

17.3.5.1 SDIO Interrupt

In 1-bit mode, an interrupt is received into the SDDAT1 pin. In 4-bit mode, it is shared by received data and interrupt and the range of interrupt detection is as follows:

- 1) Single Block: Time between A and B
 A : 2 clocks after completion of a data packet
 B : Completion of sending end bit of the next with-data command
- 2) Multi Block and SDIDatConH[5] = 0: Time between A and B, restart from C
 A : 2 clocks after completion of a data packet
 B : 2 clocks after A
 C : 2 clocks after end bit of abort command response
- 3) Multi Block and SDIDatConH[5] = 1: Time between C and B, restart from A
 A : 2 clocks after completion of a data packet
 B : 2 clocks after A

17.3.5.2 Read Wait Request

Regardless of bit mode, the Read Wait Signal is transmitted into the SDDAT2 pin in case of the following:

- 1) 2 clocks after end of a data block in read multiple operations
- 2) Transmission ends when the processor writes '1' to SDIDatSta[10]

17.3.6 Operation Sequence Example

17.3.6.1 Multiple Block Write with polling and single data line

The write block operation sequence with a single data line, multiple blocks and a polling case is as follows:

1. CMD7 selection of specific sdcard.
 - 1) Set **SDICmdArgH** & **SDICmdArgL** at 0000h.
 - 2) Set **SDICmdCon** at 0747h (wait response, immediate command start, command id = 7).
 - 3) Wait until sdcard finishes responding (check **SDICmdSta**[9]), and clear the status bit.
2. Clear command and data status register.
 - 1) Write **SDICmdSta** 0200h (clear response finish bit).
 - 2) Write **SDIDatSta** 07ffh (clear all status bits).
3. Set **SDIDatConL** & **SDIDatConH** to the proper value.
 - 1) Set **SDIDatConL** as (3000h | BlkNum) - polling, normal transfer, Tx start.
 - 2) Set **SDIDatConH** as 0012h - Tx starts after response, block mode, single data line.
4. Clear **SDICmdArgL** and **SDICmdArgH** - no argument for write block.
5. Set **SDICmdCon** as 0359h - short response, wait response, immediate start, command id=25.
6. Wait until response finish. (**SDICmdSta**[9] == 1), then write 1e00h to **SDICmdSta** to clear command status.
7. Write data for Tx to **SDIDAT** until all (BlkNum x BlkSize) bytes are transferred.
 When Tx FIFO is available (**SDIFSTA**[13]==1), data can be written.
8. Wait until DatFin active (**SDIDatSta**[4]), then force stop. --- Set the STOP bit of **SDIDatCon**[14].
9. Initiate the stop command as follows:
 - 1) Write **SDICmdArgL** as 0000h.

- 2) Write **SDICmdArgL** as 0000h.
- 3) Write **SDICmdCon** as 134ch.
10. When Read wait request occurs (**SDIDatSta**[10]==1), write '1' to this bit to clear.
11. Clear **SDIDatConL** and **SDIDatConH** by writing '1' to all bits.

17.3.6.2 Multiple Block Write with DMA and wide bus

Write block operation sequence with wide bus, multi block and DMA transfer case is as follows.

1. Select specific sdcard by CMD7.
 - 1) Set **SDICmdArgH** & **SDICmdArgL** as 0000h.
 - 2) Set **SDICmdCon** as 0747h (wait response, immediate command start, command id = 7).
 - 3) Wait until response from sdcard finishes (check **SDICmdSta**[9]), then clear status bit.
2. Clear command and data status register.
 - 1) Write **SDICmdSta** 0200h (clear response finish bit).
 - 2) Write **SDIDatSta** 07ffh (clear all status bits).
3. Set **SDIDatConL** & **SDIDatConH** to proper value.
 - 1) Set **SDIDatConL** as (b000h | BlkNum) - DMA, normal transfer, Tx start.
 - 2) Set **SDIDatConH** as 0013h - Tx start after response, block mode, wide bus.
4. Clear **SDICmdArgL** and **SDICmdArgH** - no argument for write block.
5. Set **SDICmdCon** as 0359h - short response, wait response, immediate start, command id=25.
6. Wait until response finishes. (**SDICmdSta**[9] == 1), then write 1e00h to **SDICmdSta** to clear command status.
7. Enable Tx DMA channel.
8. Wait until DatFin active (**SDIDatSta**[4]), then force stop. --- Set STOP bit of **SDIDatCon**[14].
9. Initiate stop command as follows.
 - 1) Write **SDICmdArgL** as 0000h.
 - 2) Write **SDICmdArgL** as 0000h.
 - 3) Write **SDICmdCon** as 134ch.
10. If Read wait request occurs (**SDIDatSta**[10] == 1), then Write '1' to this bit to clear.
11. Clear **SDIDatConL** and **SDIDatConH** by writing '1' to all bits.

17.3.6.3 Multiple Block Read with polling and single data line

The block operation sequence with single data line, multiple blocks and polling case is as follows:

1. Select specific sdcard by CMD7.
 - 1) Set **SDICmdArgH** & **SDICmdArgL** as 0000h.
 - 2) Set **SDICmdCon** as 0747h (wait response, immediate command start, command id = 7).
 - 3) Wait until response from sdcard finish (check **SDICmdSta**[9]), then clear status bit.
2. Clear command and data status register.
 - 1) Write **SDICmdSta** 0200h (clear response finish bit).
 - 2) Write **SDIDatSta** 07ffh (clear all status bits).
3. Set **SDIDatConL** & **SDIDatConH** to the proper value.
 - 1) Set **SDIDatConL** as (2000h | BlkNum) - polling, normal transfer, Rx start.
 - 2) Set **SDIDatConH** as 000ah - Rx start after response, block mode, single data line.
4. Clear **SDICmdArgL** and **SDICmdArgH** - no argument for write block.
5. Set **SDICmdCon** as 0352h - short response, wait response, immediate start, command id=18.
6. Wait until response finishes (**SDICmdSta**[9] == 1), then write 1e00h to **SDICmdSta** to clear the command status.
7. Read data for Rx from **SDIDAT** until (BlkNum x BlkSize) bytes are all transferred.
Data can be read when Rx FIFO is available (**SDIFSTA**[12]==1).
8. Wait until DatFin active (**SDIDatSta**[4]), then force stop. --- Set the STOP bit of **SDIDatCon**[14].
9. Initiate stop command as follows:
 - 1) Write **SDICmdArgL** as 0000h.

- 2) Write **SDICmdArgL** as 0000h.
- 3) Write **SDICmdCon** as 134ch.
10. If Read wait request occurs (**SDIDatSta**[10] == 1), then Write '1' to this bit to clear.
11. Clear **SDIDatConL** and **SDIDatConH** by writing '1' to all bits.

17.3.6.4 Multiple Block Read with DMA and wide data line

The read block operation sequence with data line, multiple block and DMA case is as follows:

1. Select specific sdcard by CMD7.
 - 1) Set **SDICmdArgH** & **SDICmdArgL** as 0000h.
 - 2) Set **SDICmdCon** as 0747h (wait response, immediate command start, command id = 7).
 - 3) Wait until response from sdcard finish (check **SDICmdSta**[9]), then clear status bit.
2. Clear command and data status register.
 - 1) Write **SDICmdSta** 0200h (clear response finish bit).
 - 2) Write **SDIDatSta** 07ffh (clear all status bits).
3. Set **SDIDatConL** & **SDIDatConH** to the proper value.
 - 1) Set **SDIDatConL** as (a000h | BlkNum) - DMA, normal transfer, Rx start.
 - 2) Set **SDIDatConH** as 000bh - Rx start after response, block mode, single data line.
4. Clear **SDICmdArgL** and **SDICmdArgH** - no argument for write block.
5. Set **SDICmdCon** as 0352h - short response, wait response, immediate start, command id=18.
6. Wait until response finishes (**SDICmdSta**[9] == 1), then write 1e00h to **SDICmdSta** to clear the command status.
7. Enable Rx DMA channel.
8. Wait until DatFin active (**SDIDatSta**[4]), then force stop. --- Set STOP bit of **SDIDatCon**[14].
9. Initiate stop command as follows:
 - 1) Write **SDICmdArgL** as 0000h.
 - 2) Write **SDICmdArgL** as 0000h.
 - 3) Write **SDICmdCon** as 134ch.
10. To clear this bit,when Read wait request shows (**SDIDatSta**[10]==1), write '1' to it.
11. Clear **SDIDatConL** and **SDIDatConH** by writing '1' to all bits.

17.4 SD/MMC Controller Registers

17.4.1 SDI Control Register (SDICON)

Address : C000 1500h

Bit	R/W	Description	Default Value
15:12	-	Reserved	x
11	R/W	RspCrcInt Determines SDI generate an interrupt if response CRC check fails 0 = disable, 1 = interrupt enable	0b
10	R/W	CmdSentInt Determines SDI generate an interrupt if command sent(no response required) 0 = disable, 1 = interrupt enable	0b
9	R/W	CmdToutInt Determines SDI generate an interrupt if command response timeout occurs 0 = disable, 1 = interrupt enable	0b
8	R/W	RspEndInt Determines SDI generate an interrupt if command response received 0 = disable, 1 = interrupt enable	0b
7	R	CrcStaInt Determines SDI generate an interrupt if CRC status error occurs 0 = disable, 1 = interrupt enable	0b

Bit	R/W	Description	Default Value
6	R/W	DatCrcInt Determines SDI generate an interrupt if data receive CRC failed 0 = disable, 1 = interrupt enable	0b
5	R/W	DatToutInt Determines SDI generate an interrupt if data receive timeout occurs 0 = disable, 1 = interrupt enable	0b
4	R/W	DatFinInt Determines SDI generate an interrupt if data counter is zero 0 = disable, 1 = interrupt enable	0b
3	R/W	BusyFinInt Determines SDI generate an interrupt if only busy check completes 0 = disable, 1 = interrupt enable	0b
2	R/W	SBitErrInt Determines SDI generate an interrupt if start bit error detect 0 = disable, 1 = interrupt enable	0b
1	R/W	FRST Determines resetting FIFO. 0 = normal mode, 1 = FIFO reset	0b
0	R/W	ENCLK Determines whether FSCLK Out enable or not 0 = disable(prescaler off), 1 = clock enable	0b

17.4.2 SDI Baud Rate Prescaler Register (SDIPRE)

Address : C000 1502h

Bit	R/W	Description	Default Value
15:0	-	Reserved.	xxh
7:0	R/W	Determines SDI clock(SDCLK) rate as above equation. Baud rate = PCLK / 2 / (Prescaler value + 1)	0x00

17.4.3 SDI Command Argument Register (SDICmdArg)

Address : C000 1504h

Bit	R/W	Description	Default Value
31 : 0	R/W	Command Argument	00000000h

17.4.4 SDI Command Control Register(SDICmdCon)

Address : C000 1508h

Bit	R/W	Description	Default Value
15:11	-	Reserved	x
10	R/W	LongRsp Determines whether host receives a 136-bit long response or not 0 = short response, 1 = long response	0
9	R/W	WaitRsp Determines whether host waits for a response or not 0 = no response, 1 = wait response	0
8	R/W	CMST Determines whether command operation starts or not 0 = command ready, 1 = command start	0
7:0	R/W	CmdIndex Command index with start 2bit(8bit)	0

17.4.5 SDI Command Status Register (SDICmdSta)

Address : C000 150Ah

Bit	R/W	Description	Default Value
15:13	-	Reserved	x
12	R	RspCrc CRC check failed when command response received. This flag is cleared by setting to one this bit. 0 = not detect, 1 = crc fail	0

Bit	R/W	Description	Default Value
11	R	CmdSent Command sent(not concerned with response). This flag is cleared by setting to one this bit. 0 = not detect, 1 = command end	0
10	R	CmdTout Command response timeout(64clk). This flag is cleared by setting to one this bit. 0 = not detect, 1 = timeout	0
9	R	RspFin Command response received. This flag is cleared by setting to one this bit. 0 = not detect, 1 = response end	0
8	R	CmdOn Command transfer in progress 0 = not detect, 1 = in progress	0
7:0	R	RspIndex Response index 6bit with start 2bit(8bit)	0

17.4.6 SDI Response Register 0 (SDIRSP0)

Address : C000 150Ch

Bit	R/W	Description	Default Value
31 : 0	R	Card status[31:0](short), card status[127:96](long)	00000000h

17.4.7 SDI Response Register 1 (SDIRSP1)

Address : C000 1510h

Bit	R/W	Description	Default Value
31:24	R	CRC7(with end bit, short), card status[95:88](long)	x
23:0	R	unused(short), card status[87:64](long)	0

17.4.8 SDI Response Register 2 (SDIRSP2)

Address : C000 1514h

Bit	R/W	Description	Default Value
31 : 0	R/W	unused(short), card status[63:32](long)	00000000h

17.4.9 SDI Response Register 3 (SDIRSP3)

Address : C000 1518h

Bit	R/W	Description	Default Value
31 : 0	R/W	unused(short), card status[31:0](long)	00000000h

17.4.10 SDI Data Timer Register (SDIDTimer)

Address : C000 151Ch

Bit	R/W	Description	Default Value
15 : 0	R/W	Data timeout period(0~65535 cycle)	0000h

17.4.11 SDI Block Size Register (SDIBSize)

Address : C000 151Eh

Bit	R/W	Description	Default Value
15:12		Reserved	x
11:0	R/W	Block Size value(0~4095 word) , don't care when stream mode	0

17.4.12 SDI Data Control Register (SDIDatCon)

Address : C000 1520h

Bit	R/W	Description	Default Value
31:21	R	Reserved	x
20	R/W	TARSP Determines when data transmit start after response receive or not 0 = directly after DatMode set, 1 = after response receive(DatMode is already set)	0
19	R/W	RACMD Determines when data receive start after command sent or not 0 = directly after DatMode set, 1 = after command sent (DatMode is already set)	
18	R/W	BACMD Determines when busy receive start after command sent or not 0 = directly after DatMode set, 1 = after command sent (DatMode is already set)	
17	R/W	BlkMode Data transfer mode 0 = stream data transfer, 1 = block data transfer	
16	R/W	WideBus Determines enable wide bus mode 0 = standard bus mode(only SDIDAT[0] used), 1 = wide bus mode(SDIDAT[3:0] used)	
15	R	Reserved	0
14	R/W	STOP Determines whether data transfer stop by force or not 0 = normal, 1 = stop by force	0
13:12	R/W	DatMode Determines which direction of data transfer 00 = ready, 01 = only busy check start 10 = data receive start, 11 = data transmit start	0
11:0	R/W	BlkNum Block Number(0~4095), don't care when stream mode	0

17.4.13 SDI Data Remain Counter Register (SDIDatCnt)

Address : C000 1524h

Bit	R/W	Description	Default Value
31:24	R	Reserved	x
23:12	R	BlkNumCnt Remaining Block number	0
11:0	R	BlkCnt Remaining data of 1 block	0

17.4.14 SDI Data Status Register (SDIDatSta)

Address : C000 1528h

Bit	R/W	Description	Default Value
15:8	R	Reserved	x
7	R/W	CrcSta CRC Status error when data block sent(CRC check failed). This flag is cleared by setting to one this bit. 0 = not detect, 1 = crc status fail	0
6	R/W	DatCrc Data block received error(CRC check failed). This flag is cleared by setting to one this bit. 0 = not detect, 1 = receive crc fail	
5	R/W	DatTou Data receive timeout. This flag is cleared by setting to one this bit. 0 = not detect, 1 = timeout	
4	R/W	DatFin Data transfer completes(data counter is zero). This flag is cleared by setting to one this bit. 0 = not detect, 1 = data finish detect	
3	R/W	BusyFin Only busy check finish. This flag is cleared by setting to one this bit	

Bit	R/W	Description	Default Value
		0 = not detect, 1 = busy finish detect	
2	R/W	SbitErr Start bit is not detected on all data signals in wide bus mode. This flag is cleared by setting to one this bit. 0 = not detect, 1 = command end	0
1	R/W	TxDatOn Data transmit in progress 0 = not active, 1 = data Tx in progress	0
0	R/W	RxDatOn Data receive in progress 0 = not active, 1 = data Rx in progress	0

17.4.15 SDI FIFO Status Register (SDIFSTA)

Address : C000 152Ah

Bit	R/W	Description	Default Value
15:7	R	Reserved	x
6	R	TFDET This bit indicates that FIFO data is available for transmit when DatMode is data transmit mode. 0 = not detect(FIFO full), 1 = detect($0 \leq \text{FIFO} \leq 15$)	0
5	R	RFDET This bit indicates that FIFO data is available for receive when DatMode is data receive mode. 0 = not detect(FIFO empty), 1 = detect($1 \leq \text{FIFO} \leq 16$)	0
4	R	FULL This bit automatically set to 1 whenever FIFO is full. 0 = $0 \leq \text{FIFO} \leq 15$, 1 = Full(16word)	0
3:0	R	FFCNT Number of data(word) in FIFO	0

17.4.16 SDI Data Register (SDIDAT)

Address : C000 152Ch

Bit	R/W	Description	Default Value
31 : 0	R/W	This field contains the data to be transmitted or received over the SDI channel	00000000h

18. NGPIO Controller

18.1 Overview

NGPIO Controller is used for getting the data which external cpu send. NGPIO Data bus (8/16bit) is connected to SRAM bus of external CPU I/F, and NGPIOLAT(or EIRQ3) signal is connected to the chip select signal which is made with address, write enable signal.

When External CPU write data to SRAM bus, The data on SRAM bus is written to FIFO in NGPIO Controller, then The ARM CPU of VR3D can check of validating the data by checking NGPIOSTATE

This Block stores 16bit NGPIO data, sampled at NGPIOLAT signal Rising/Falling Edge, in internal FIFO. It has CPU interface for CPU's reading.

18.2 NGPIO Controller Register

18.2.1 NGPIO Data Register (NGPIODATA)

Address : C000 1100h

Bit	R/W	Description	Default Value
15 : 0	R	NGPIODATA : Read NGPIO FIFO value	0h

18.2.2 NGPIO Control Register (NGPIOCTRL)

Address : C000 1102h

Bit	R/W	Description	Default Value
15 : 5	R	Reserved	-
4	R/W	FIFOENB : FIFO Enable. 0 : Disable 1 : Enable	0
R	R/W	SELWRENB : Set Write Enable Signal 0 : NGPIOLAT 1 : EIRQ3	0
2	R/W	FALLENB : Set Write Enable Falling Edge Detection 0 : Disable 1 : Enable	0
1	R/W	RISEENB : Set Write Enable Raising Edge Detection 0 : Disable 1 : Enable	0
0	R/W	ClrFIFO : Set 1 to clear FIFO.	0

18.2.3 NGPIO Status Register (NGPIOSTATE)

Address : C000 1104h

[illegible]

19. SSP/SPI

19.1 Overview

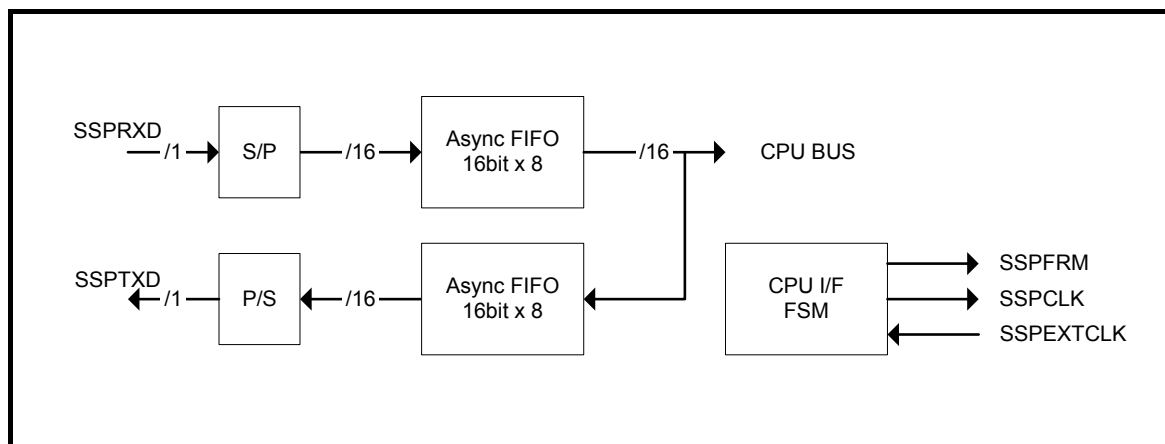
The SSP/SPI is a full-duplex synchronous serial interface. It supports Synchronous Serial Protocol(SSP) and Serial Peripheral Interface(SPI) protocol. It can connect to a variety of external converters, serial memory, and many other devices which use serial protocols for transferring data.

There are 4 I/O pin signals associated with SSP/SPI transfers: the SSPCLK, the SSPRXD data receive line, the SSPTXD data transfer line, and SSPFRM (Frame Indicator in SSP mode, Chip Select in SPI mode).

19.1.1 Features

- SPI Protocol, SSP Protocol compatible
- 16-bit Shift Register for transmit
- 16-bit Shift Register for receive
- 5-bit Pre-scale counter
- Polling, Interrupt, and DMA transfer mode

19.1.2 Block Diagram



SSP/SPI Block Diagram

19.2 Pin Description

The Pins for the communication between the SSP/SPI Block and external peripherals are listed in the following table. These pins are used as the GPIO or other Alternative functions

Name	Direction	Description
SSPCLK	O	Serial Bit Clock (SSPCLK) provides a clock signal to external devices for synchronizing the transferred Serial Data.
SSPFRM	O	Notify the start and end of Serial Data.
SSPRXD	I	Use to receive Data from external devices.
SSPTXD	O	Use to send Data to external devices.

19.3 Operation

The SSI/SPI Block transfers Serial Data from/to external devices via FIFO in the SSP/SPI Block. The transfer operation is initiated

by CPU, using the programmed I/O system or the DMA, to/from the system memory. The SSI/SPI Data transfer is performed in full duplex.

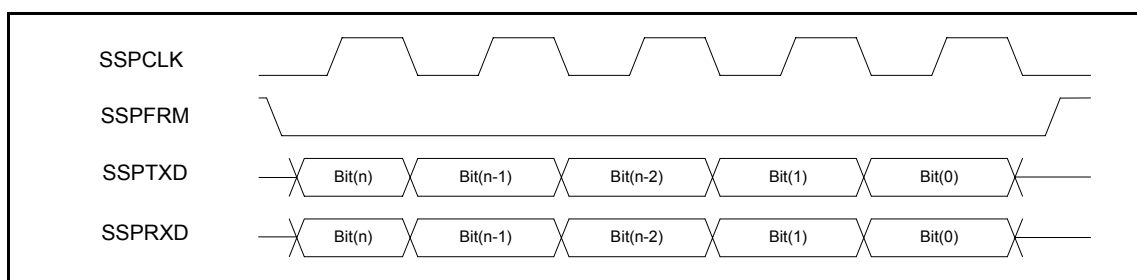
When the SSI/SPI sends Data to the PIO Mode, the transfer operation is completed by Reading 'Read FIFO' or Writing to 'Write FIFO' by means of a program.

In the case of communication with DMA Mode,- In DMA transfer mode, the DMAENB bit in the DMA and the SSPSPICONT0 bit are set to 1 to enable the DMA transfer

For Data reception using the SSI/SPI Block, the Read operation is automatically performed until FIFO is be FULL by writing one Data to TX FIFO. If the RXONLY bit is cleared, the Read operation is performed as many times as Data was written to TX FIFO.

Signal waveforms by SSP Mode and SPI Mode are as shown below.

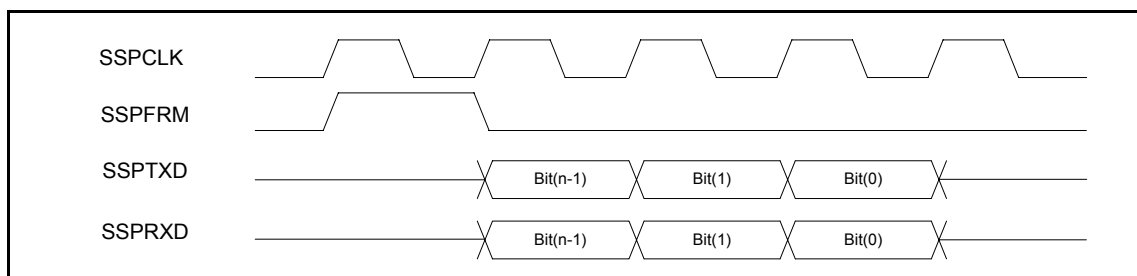
19.3.1 SPI Mode



To enable the data transfer mode, SSPFRM signal is required to be low when SSPCLK is on the rising edge. The MSB of stored data will be transferred first. MSB Data bits out are output first. The input of the Data read is performed in a similar way.

The Polarity and Phase of SSPCLK can be converted by the use of a programming.

19.3.2 SSP Mode



When Data output is ready in the SSP Block, the SSPFRM is out for 1 Clock. In the next clock, the Data is out via the SSD TXD. The MSB out of the Data bit is output first. The input of the Data read is performed in a similar way. The Word length can be determined by the SSPSPICONT 0 Register.

The Polarity and Phase are converted by the use of a programming.

19.4 SSP/SPI Controller Register

19.4.1 SSP/SPI Control Register0 (SSPSPICONT0)

Address : C000 3F00h

Bit	R/W	Description	Reset Value
[4:0]	R/W	Set internal clock divide value. Value should be bigger than 0 ex) DIVCNT=1 : 2devide DIVCNT=2 : 3devide	2
[8:5]	R/W	Set Bit width of Data. (n+1)bit, MAX 16biy	7
[9]	R/W	Use External Clock 0 : Internal Clock (3.6864Mhz) 1 : I_SSPEXTCLK.	0
[10]	R/W	Initialize internal FIFO. It should toggle once initially 0 : IDLE 1 : RESET	0

[11]	R/W	SPI/SPP의 Enable 0 : Disable 1 : Enable	0
[12]	R/W	Set SIO Mode. 0 : PIO Mode 1 : DMA Mode	0
[13]	R/W	Set SIO Mode Writing data to Tx Buffer, automatically transfer until Rx FIFO Full. To stop transfer Reset FFCLR.Will stop after finishing current access. 0 : Read When Writeing 1 : Read Only만 수행	0
[14]	R/W	Make TXD Pin Hi-Z status 0 : Always 1 : Only When Transmitting	1
[15]	R	-	0

19.4.2 SSP/SPI Control Register1 (SSPSPICONT1)

Address : C000 3F02h

Bit	R/W	Description	Reset Value
[1:0]	R/W	Set SIO Mode 00 : SSP 01 : SPI 1x: Reserved	X
[2]	R/W	Set Polarity of External out SSPCLK 0 : Inverting 1 : Normal	X
[3]	R/W	Set Phase of External out SSPCLK 0 : 90degrees 1 : 0degrees	X
[15:4]	R	-	0

19.4.3 SSP/SPI Data Register (SSPSPIDATA)

Address : C000 3F04h

Bit	R/W	Description	Reset Value
[7:0]	R/W	Data to be transmitted when write. Read transmitted data when Read	X
[15:8]	R	-	0

19.4.4 SSP/SPI Status Register (SSPSPISTAT)

Address : C000 3F06h

[illegible]

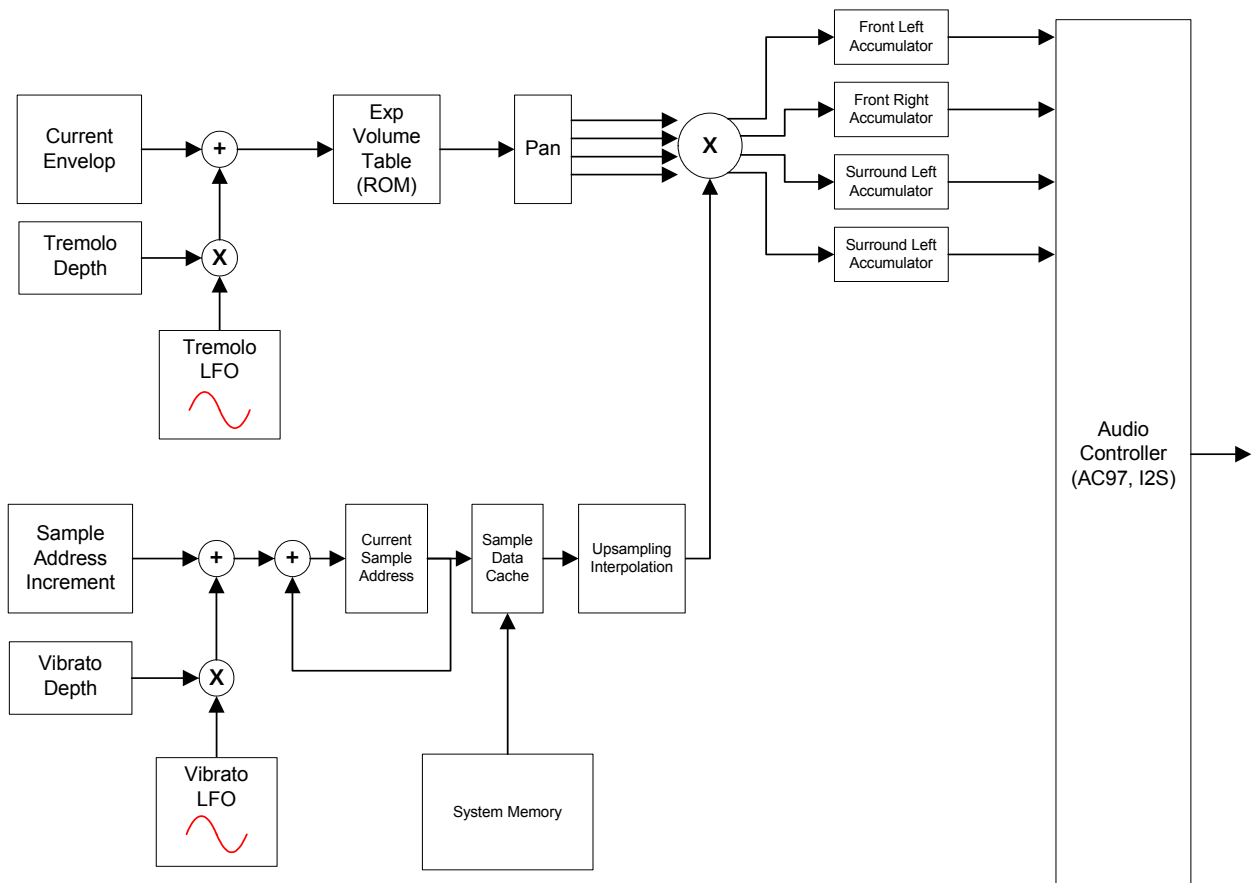
20. Wavetable Synthesizer

20.1 Overview

VR3D has two wavetable synthesizers (WTS). Two WTS are called as WTS#0 and WTS#1. Each WTS can synthesis 32 wave channels. So VR3D can play 64 wave channels. WTS in VR3D has following features

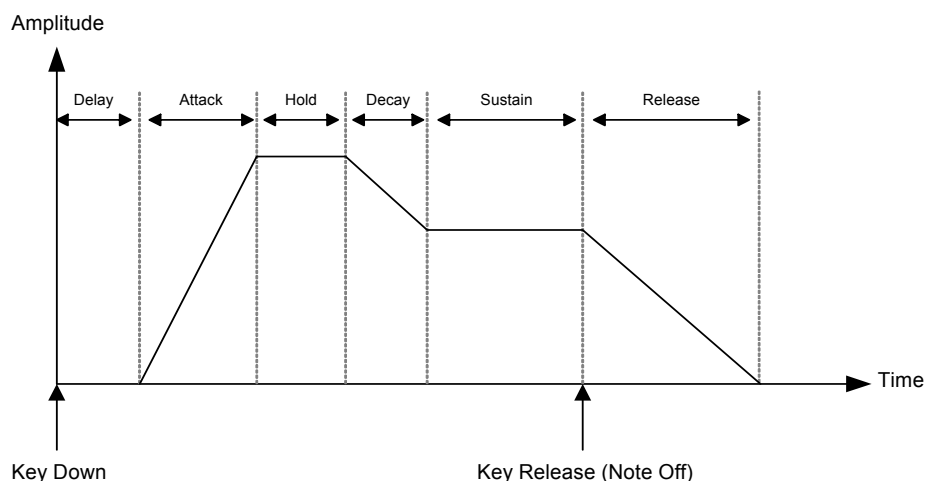
- Two WTS Supports up to 64 wave channels (One WTS supports 32 wave channels)
- Including Vibrato and Tremolo.
- 4 Channel Speaker panning.
- Support speech format (u-LAW) for wave font.

20.1.1 Block Diagram



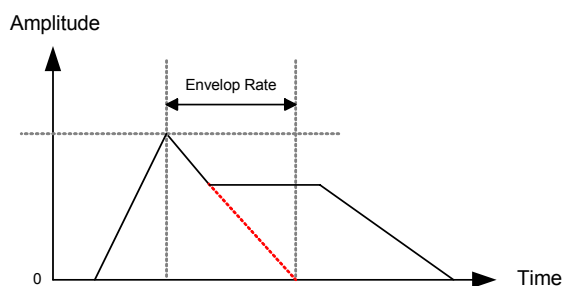
20.2 Envelope Shaping

Envelope shaping model used in wavetable synthesizer is DAHDSR (Delay-Attack-Hold-Decay-Sustain-Release). In DAHDSR model envelope consists of six stages like following figure. ENV bit in **WTPARAM** (Channel Parameter Register) should be set to 1 to enable envelope shaping. When key is downed, Envelope Volume field in **WTPARAM** must be set to zero because envelope starts from zero.



20.2.1 Encoded Envelop Rate

Gradient of envelope is represented by envelope rate. It is time taken to reach its maximum value from 0. If envelope has negative gradient, it is time taken to reach 0 from maximum. Following figure represents envelop rate of decay stage.



Envelope rate is a value between 0.001 and 17.0 seconds. Envelope rate must be encoded to 8bit data. Following C code shows encoding method.

```
#define MAX_ENVELOP          102
#define ASSUME_ZERO          0.000001f
#define MAX_ENVRATE          17.0f
#define LIM_ENVELOP_INC      3.9f
#define ENVELOP_INC_FIXED    13

typedef unsigned char    U8;
typedef unsigned int     U32;

// EnvelopRate: [0.001 ~ 17.0 seconds] or zero (delay and hold stages)
// PlayRate: [8000~48000 Hz]

U8 EncodeEnvRate(float EnvelopRate, int PlayRate)
{

```

```

float    EnvelopInc;
U32      Mantisa;
U8       EncodedEnvRate;
int      cZero;

if( EnvelopRate < ASSUME_ZERO ) return 0xff;
EnvelopInc = MAX_ENVELOP / ( EnvelopRate * PlayRate );

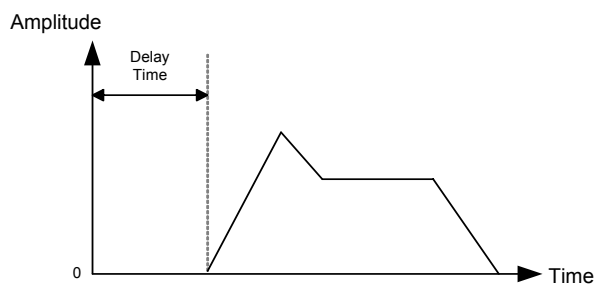
if( EnvelopInc < LIM_ENVELOP_INC )
{
    Mantisa = (U32)(EnvelopInc * (1<<ENVELOP_INC_FIXED));
    cZero=0;
    while( Mantisa >= 64 )
    {
        Mantisa = Mantisa>>3;
        cZero++;
    }
    EncodedEnvRate = (U8)(cZero<<6 | Mantisa);
}
else if( EnvelopInc < MAX_ENVELOP ) EncodedEnvRate = 0xfe;
else EncodedEnvRate = 0xff;

return EncodedEnvRate;
}

```

20.2.2 Delay and Hold Time

Delay stage is encoded like envelope rate. But delay stage has no envelope changes. So, use delay time instead of envelope rate to encoding. (See following figure). Normally delay time would be zero. If delay time is zero, it must be encoded to 0xff.



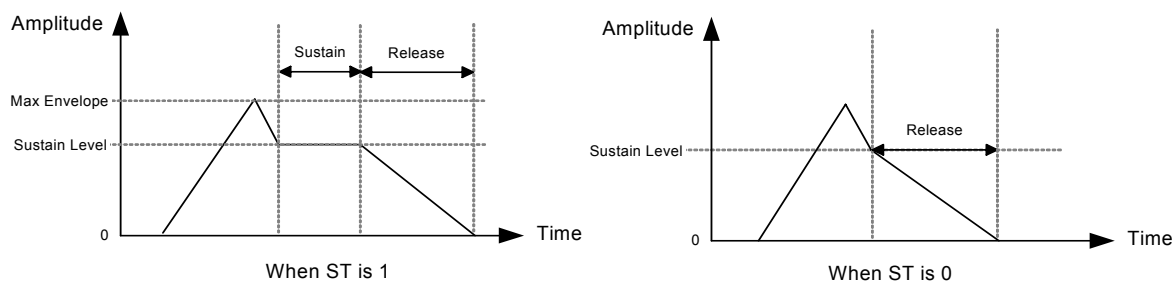
Hold time encoding is the same as delay time.

20.2.3 Sustain

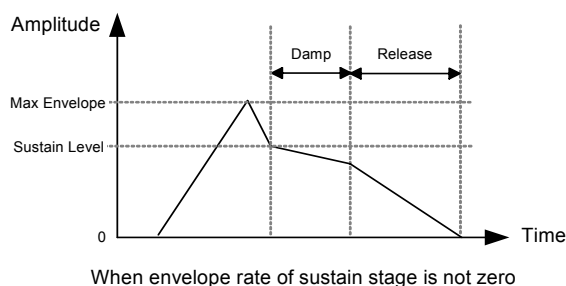
Sustain level indicates the end of decay stage. If envelope in decay stage arrives sustain level, it changes to sustain stage or release stage.

In the following figure, 'Max Envelope' is 102. So, sustain level must be a value between 0 and 102.

ST bit in **WTPARAM** (Channel Parameter Register) determines whether envelope goes to sustained stage or not. If ST is 1, envelope is sustained until key is released. If ST is 0, envelope goes to release stage from decay stage.



It is possible to have non-zero envelope rate in sustain stage. It is called damped sustain.



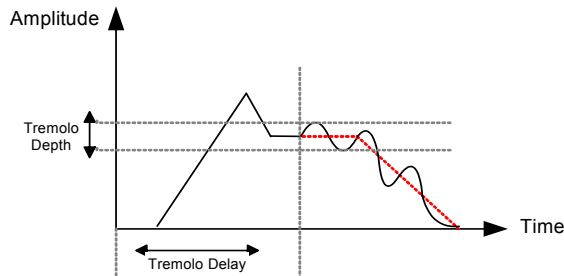
To exit sustain and go to release stage, set bits of **WTNOTE** register to zero.

20.2.4 Non-Envelope Mode

Non-envelope mode is used to play wave samples directly without envelope shaping. It is useful to play voice. To enable non-envelope mode, set ENV bit in **WTPARAM** to zero.

20.3 Tremolo and Vibrato

Tremolo modulates envelope. It is AM (Amplitude Modulation). To use tremolo, set TE (Tremolo Enable) bit in **WTPARAM** to 1. TS (Tremolo Start) bit in **WTPARAM** determines whether tremolo delay is applied or not. If it is set to 0, then tremolo delay is applied. If it is set to 1, tremolo starts without delay.



Vibrato modulates frequency of sample data. It is FM (Frequency Modulation). To use vibrato, set VE (Vibrato Enable) bit in **WTPARAM** to 1. VS (Vibrato Start) bit in **WTPARAM** determines whether vibrato delay is applied or not. If it is set to 0, then vibrato delay is applied. If it is set to 1, vibrato starts without delay.

20.4 Channel Control

20.4.1 Status On/Off

To start playing certain channel, set status in **WTSTATUS** register. When envelope in release stage reaches zero, status of the channel becomes zero automatically. Status determines whether a channel is active or not.

20.4.2 Note On/Off

Note on/off is controlled via **WTNOTE**. Turn on note of the channel before turn on status of a channel. In sustain stage, envelope will not be changed until note is off. After note is off, envelope go to release stage. If envelope of a channel doesn't have sustained stage, note on/off will not affect.

20.4.3 Interrupt

If envelope of a channel becomes zero in release stage, status of the channel will be off automatically. And interrupt occurs. Interrupt is controlled via **WTINTENB** and **WTINTPND**.

20.4.4 Cache

Each channel has cache to read wave sample data. Cache must be cleared before turning on status. Set **WTCLRCACHE** to clear cache.

20.5 Sample Data

20.5.1 Segment

Segment is 2M-byte aligned memory. Sample data of one instrument must be in one-segment. It cannot be laid across two segments.

20.5.2 Sample Data Format

Wavetable synthesizer of VR3D supports following sample data format.

- Linear signed PCM (8bit, 16bit)
- u-LAW (8bit)

u-LAW implemented in VR3D is not compatible with CCITT G.711. Following C code shows u-LAW encoding and decoding of VR3D.

Encoding.

```
typedef unsigned char    U8;
typedef unsigned short   U16;
typedef unsigned int     U32;
typedef signed char      S8;
typedef signed short     S16;
typedef signed int       S32;

U8  Normalize(S16 *value)
{
    U8      numShiftBits= 0, msb, nextmsb;

    msb = (*value & 0x8000) >> 15;
    nextmsb = (*value & 0x4000) >> 14;
    while (msb== nextmsb)
    {
        *value <<= 1;
        numShiftBits ++;
        msb= (*value & 0x8000) >> 15;
        nextmsb= (*value & 0x4000) >> 14;
    }
    return numShiftBits;
}

U8  EncodeMuLAW (S16 data)
{
    U8      newSample, X, YYY, ZZZZ, numberOfShiftBits;

    data >>= 2;
```

```

if (data <= -8159) newSample = 0x00;
else if (data >= 8159) newSample = 0x1F;
else
{
    X= (data & 0x2000) >> (6+3);
    if (data & 0x2000) data= -data;
    data <<= 2;
    data += 0x84;
    numberOfShiftBits= Normalize(&data);
    ZZZZ= (data & 0x3c00) >> 10;
    YYY= (7-numberofShiftBits) <<5;
    if (YYY== 0x00) X= (X ^ 0x10);
    else if (!X) ZZZZ= (~ZZZZ) & 0xf;
    newSample= YYY | X | ZZZZ;
    newSample= ~newSample;
}
return newSample;
}

```

Decoding.

```

S16 DecodeMuLAW(U8 data)
{
    U16 en= (U16)(data & 0x0f), de= 0;
    switch(data & 0xf0) {
        case 0x00: de= 0x8000 | (en << 10);      break;
        case 0x20: de= 0xc000 | (en << 9);       break;
        case 0x40: de= 0xe000 | (en << 8);       break;
        case 0x60: de= 0xf000 | (en << 7);       break;
        case 0x80: de= 0xf800 | (en << 6);       break;
        case 0xa0: de= 0xfc00 | (en << 5);       break;
        case 0xc0: de= 0xfe00 | (en << 4);       break;
        case 0xf0: de= 0xff80 | (en << 3);       break;

        case 0x10: de= 0x4000 | (en << 10);      break;
        case 0x30: de= 0x2000 | (en << 9);       break;
        case 0x50: de= 0x1000 | (en << 8);       break;
        case 0x70: de= 0x0800 | (en << 7);       break;
        case 0x90: de= 0x0400 | (en << 6);       break;
        case 0xb0: de= 0x0200 | (en << 5);       break;
        case 0xd0: de= 0x0100 | (en << 4);       break;
    }
}

```



```
        case 0xe0: de= 0x0000 | (en << 3);        break;
    }
    return (S16)de;
}
```

20.6 Performance

As system clock varies, available channels are varied. Faster system clock is, more channels can be played.

$$\text{Available channels} = (\text{system clock}) / [60 \times (\text{play rate})]$$

If two wavetable synthesizers are used, number of available channels is doubled. Following table shows performance of wavetable synthesizer (WTS).

System clock	Play rate	Single Wavetable	Dual Wavetable
18 Mhz	8000 Hz	32	64
	16000 Hz	16	32
	22050 Hz	12	24
	44100 Hz	6	12
50 Mhz	8000 Hz	32	64
	16000 Hz	32	64
	22050 Hz	32	64
	44100 Hz	16	32
100 Mhz	8000 Hz	32	64
	16000 Hz	32	64
	22050 Hz	32	64
	44100 Hz	32	64

20.7 Registers Map

Channel Control Registers (E000_6000h ~ E000_6010h / E000_8000~E000_8010h)

Register Name	offset	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WTSTATUS	00h	Channel Status (31 – 0)																															
		S																														Channel Number	
WTNOTE	04h	Channel Note (31 – 0)																															
		S																														Channel Number	
WTINTENB	08h	Note Off Interrupt Enable (31 – 0)																															
WTINTPND	0Ch	Note Off Interrupt Pending (31 – 0)																															
WTCLRCACHE	10h																														Channel Number		

Instrument Parameter Registers (E000_6800h ~ E000_6D00h / E000_8800~E000_8D00h)

Register Name	offset	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																	
WTINSPARAM	00h			Sample Address (21.9 fixed point)																																														
	04h			Sustain Level												Envelope Volume (7.16 fixed point)																																		
	08h	Tremolo LFO Rate								Tremolo Depth								Tremolo Phase/Delay Count																																
	0Ch	Vibrato LFO Rate								Vibrato Depth								Vibrato Phase/Delay Count																																
	10h	Delay Rate										Attack Rate										Sample Address Increment (7.9 fixed point)																												
	14h	Segment																Loop Begin Address																																
	18h	ENV	ST			LP	LD	PP	P8	UL	EnvStage						Loop End Address																																	
	1Ch	Hold Rate										Decay Rate												TE	TS	Tremolo Delay																								
	20h	Sustain Rate										Release Rate												VE	VS	Vibrato Delay																								
	24h			Surround Left Volume												Surround Right Volume														Front Left Volume												Front Right Volume								

20.7.1 Status On/Off Register (WTSTATUS)

Address: E000_6000h (WTS#0), E000_8000h (WTS#1)

Read

Bit	R/W	Description	Default Value
31	R	Status of 31 st channel. '0': channel is busy '1': channel is idle	0
30	R	Status of 30 th channel.	0
...	R	...	0
1	R	Status of 1 st channel.	0
0	R	Status of 0 th channel.	0

Write

Bit	R/W	Description	Default Value
31	W	Set channel status. '1': set channel enable '0': set channel disable	-
30:5		Reserved	-
4:0	W	Channel number	-

20.7.2 Note On/Off Register (WTNOTE)

Address: E000_6004h (WTS#0), E000_8004h (WTS#1)

Read

Bit	R/W	Description	Default Value
31	R	Note status of 31 st channel. '1': note on '0': note off	0
30	R	Note status of 30 th channel.	0
...	R	...	0
1	R	Note status of 1 st channel.	0
0	R	Note status of 0 th channel.	0

Write

Bit	R/W	Description	Default Value
31	W	Set note status. '1': note on channel '0': note off channel	-
30:5		Reserved	-
4:0	W	Channel number	-

20.7.3 Status Off Interrupt Enable Register (WTINTENB)

Address: E000_6008h (WTS#0), E000_8008h (WTS#1)

Bit	R/W	Description	Default Value
31	R/W	Interrupt enable flag of 31 st channel.	0
30	R/W	Interrupt enable flag of 30 th channel.	0
...	R/W	...	0
1	R/W	Interrupt enable flag of 1 st channel.	0
0	R/W	Interrupt enable flag of 0 th channel. If envelope of a channel becomes zero in release stage, status of the channel will be 'idle' automatically. And interrupt occurs.	0

20.7.4 Status Off Interrupt Pending Register (WTINTPND)

Address: E000_600Ch (WTS#0), E000_800Ch (WTS#1)

Bit	R/W	Description	Default Value
31	R/W	Pending bit of 31 st channel	0
30	R/W	Pending bit of 30 th channel	0
...	R/W	...	0
1	R/W	Pending bit of 1 st channel.	0
0	R/W	Pending bit of 0 th channel. Write '1' to clear each bit.	0

20.7.5 Clear Cache Register (WTCLRCACHE)

Address: E000_6010h (WTS#0), E000_8010h (WTS#1)

Bit	R/W	Description	Default Value
31:5		Reserved	-
4:0	W	Channel number Before turning on status of a channel (WTSTATUS), clear corresponding cache.	-

20.7.6 Instrument Parameter Registers (WTINSPARAM)

Address (WTS#0) : E000_6800h + [28h x (channel number) + Offset]

Address (WTS#1) : E000_8800h + [28h x (channel number) + Offset]

One wavetable synthesizer has 10 instrument parameter registers for each channel. (Total 320 registers in one wavetable synthesizer). Each register is 32 bit. Following tables shows 10 parameter registers for a channel.

Parameter 0 (Offset: 00h)

Bit	R/W	Description	Default Value
31:30		Reserved	
29:0	W	Sample Address (21.9 fixed point)	

Parameter 1 (Offset: 04h)

Bit	R/W	Description	Default Value
31		Reserved	
30:24	W	Sustain Level	
23		Reserved	
22:0	W	Envelope Volume (7.16 fixed point) Set Envelope Volume to 0 when using envelope shaping.	

Parameter 2 (Offset: 08h)

Bit	R/W	Description	Default Value
31:25	W	Tremolo LFO Rate Tremolo LFO Rate = [1024 x 64 x (LFO frequency)] / (play rate) LFO frequency ranges [1 ~50Hz] Play rate ranges [8000 ~ 48000Hz]	
24:19	W	Tremolo Depth Tremolo Depth is Amplitude of LFO Maximum Tremolo Depth is 15. Don't set larger value than 15.	
19:0	W	Tremolo Phase/Delay Count Always set this field to zero.	

Parameter 3 (Offset: 0Ch)

Bit	R/W	Description	Default Value
31:25	W	Vibrato LFO Rate Vibrato LFO Rate = [1024 x 64 x (LFO frequency)] / (play rate) LFO frequency ranges [1 ~ 50Hz] Play rate ranges [8000 ~ 48000Hz]	
24:19	W	Vibrato Depth Vibrato Depth is Amplitude of LFO.	

Bit	R/W	Description	Default Value
		Maximum Vibrato Depth is 63.	
19:0	W	Vibrato Phase/Delay Count Always set this field to zero.	

Parameter 4 (Offset: 10h)

Bit	R/W	Description	Default Value
31:24	W	Delay Rate (Encoded Envelope Rate)	
23:16	W	Attack Rate (Encoded Envelope Rate)	
15:0	W	Sample Address Increment (7.9 fixed point)	

Parameter 5 (Offset: 14h)

Bit	R/W	Description	Default Value
31:21		Segment Address Segment is 2M Byte aligned memory. Sample data of one instrument must be laid in one segment. It cannot be laid across two segments.	
20		Reserved	
19:0		Loop Begin Address Loop Begin Address = [{ (sample begin address) & 1FFFFh } / 2]	

Parameter 6 (Offset: 18h)

Bit	R/W	Description	Default Value
31	W	ENV (Envelope) '1': use Envelope Model (instrument mode) '0': don't use Envelope Model (wave mode)	
30	W	ST (Sustain) '1': sustain mode '0': damped mode	
29		Reserved.	
28	W	LP (Looping) '1': loop sample data '0': don't loop sample data.	
27	W	LD (Sample Increment Direction) In typical cases, LD is positive. Programmer must set this value to 1. '1': positive '0': negative	
26	W	PP (Ping-Pong) '1': Ping-Pong enable '0': Ping-Pong disable	
25	W	P8 (PCM 8bit) '1': use 8 bit linear signed PCM sample	
24	W	UL (uLAW) '1': use uLAW encoded sample data.	
23:21	W	Envelope Stage '0': Delay '1': Attack '2': Hold '3': Decay	

Bit	R/W	Description	Default Value
		'4': Sustain '5': Release	
20		Reserved	
19:0	W	Loop End Address Loop End Address = [{ (sample end address) & 1FFFFFFh } / 2]	

Parameter 7 (Offset: 1Ch)

Bit	R/W	Description	Default Value
31:24	W	Hold Rate (Encoded Envelope Rate)	
23:16	W	Decay Rate (Encoded Envelope Rate)	
15		Reserved	
14	W	TE (Tremolo Enable)	
13	W	TS (Tremolo Start)	
12:0	W	Tremolo Delay Tremolo Delay = [Delay x (Play Rate)] / 32 Delay ranges [0 ~ 5 seconds] Play Rate ranges [8000 ~ 48000Hz]	

Parameter 8 (Offset: 20h)

Bit	R/W	Description	Default Value
31:24	W	Sustain Rate (Encoded Envelope Rate)	
23:16	W	Release Rate (Encoded Envelope Rate)	
15		Reserved	
14	W	VE (Vibrato Enable)	
13	W	VS (Vibrato Start)	
12:0	W	Vibrato Delay Vibrato Delay = [Delay x (Play Rate)] / 32 Delay ranges [0 ~ 5 seconds] Play Rate ranges [8000 ~ 48000Hz]	

Parameter 9 (Offset: 24h)

Bit	R/W	Description	Default Value
31		Reserved	
30:24	W	Surround Left Volume	
23		Reserved	
22:16	W	Surround Right Volume	
15		Reserved	
14:8	W	Front Left Volume	
7		Reserved	
6:0	W	Front Right Volume	

21. Audio Controller

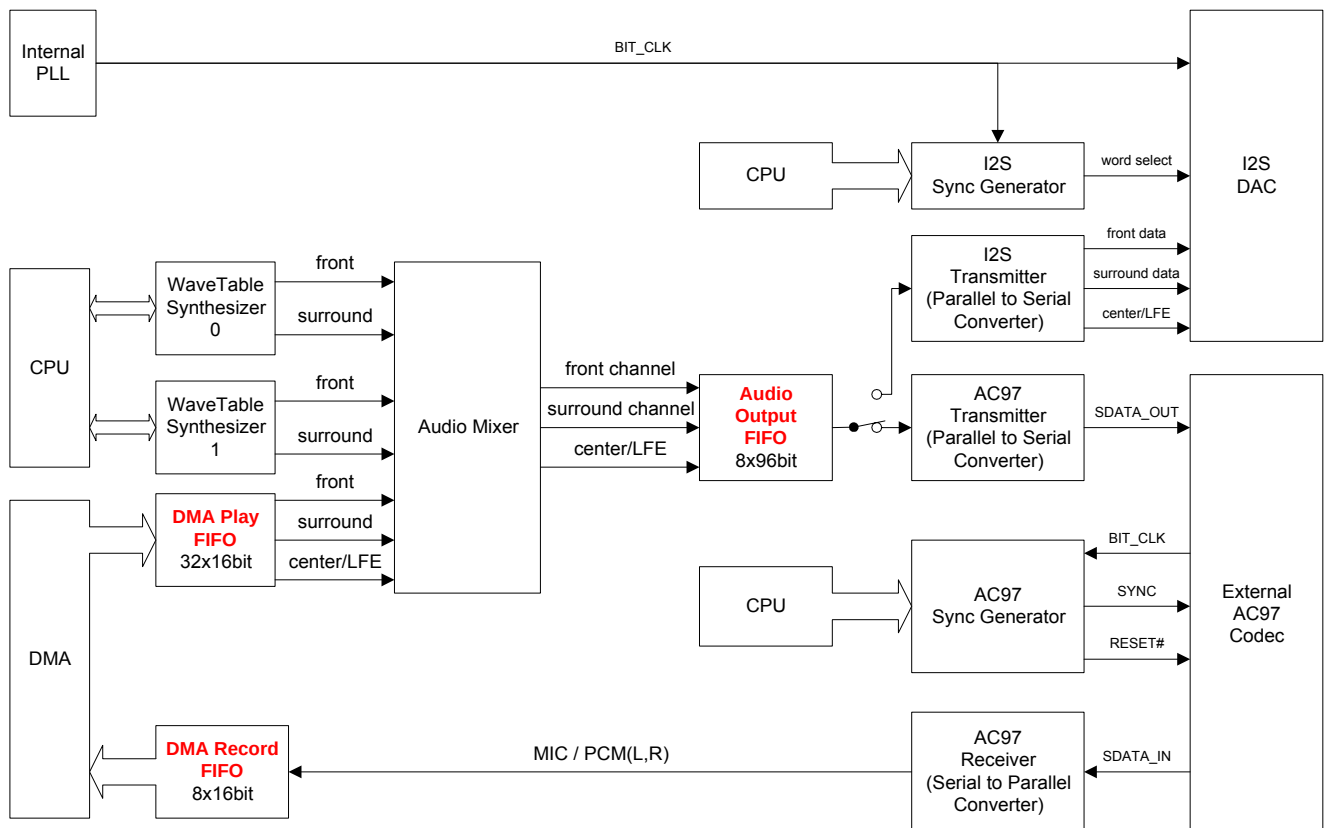
21.1 Overview

Audio Controller controls audio playing, mixing, codec interfacing and recording.

Audio Controller performs followings

- Convert sound data to I2S/AC97
- AC97 codec access
- Audio playing using DMA
- Audio recording using DMA
- Audio playing using wavetable synthesizer (See wavetable synthesizer document)
- Mixing sound data from DMA and wavetable synthesizer

21.1.1 Block Diagram



21.2 Initialize

21.2.1 Initialize for I2S

First, set GPIO configuration to use I2S. (See GPIO document for more details). Second, set internal PLL to generate I2S BIT_CLK. (See PLL document for more details). Third, set **ACCONTROL** as follows

```
#define BIT(n) (1<<n)

ACCONTROL = BIT(1) | BIT(0);
```

BIT(0) in **ACCONTROL** means 'codec select'. It determines whether **audio output FIFO** is connected to I2S or AC97.

BIT(1) in **ACCONTROL** means 'I2S bus enable'. If this bit is off, then I2S bus (BIT_CLK, WS, DATA) will be off. This bit may be off in power down mode.

21.2.2 Initialize for AC97

First, set GPIO configuration to use AC97. (See GPIO document for more details). Second, set **ACCONTROL** and **AC97RESET** as follows

```
#define BIT(n) (1<<n)

ACCONTROL = 0;

AC97RESET = BIT(0);
```

BIT(0) in **AC97RESET** means 'cold reset'. This bit is active low. At boot time, it is set to low. You must set this bit to high to activate AC97 Codec.

BIT(1) in **AC97RESET** means 'warm reset' It is active high. (See [AC97 Power Down](#) for more details)

After you set **AC97RESET**, you have to wait for AC97 codec ready. Codec ready can be checked via **AC97STATUS**. It can also be checked by interrupt. (**ACINTENB** and **ACINTPND**).

```
#define BIT(n) (1<<n)

while( (AC97STATUS & BIT(1)) == 0 )
{
    If you wait for a long time then exit loop;
}
```

BIT(1) in **AC97STATUS** means 'codec ready'. If codec is not ready for a long time (about 100usec), there is problem in AC97 Codec or ACLINK is not active.

21.3 AC97 Access

21.3.1 Write Command to AC97

You can write command to AC97 register via **AC97WRITE**. Following example is to set master volume mute

```
#define BIT(n) (1<<n)

#define AC97WRITE(n) ((volatile unsigned short*)(0xc0002900 + (n)))

AC97WRITE(0x2) = 0x8000; // write to AC97 master volume register

while( (AC97STATUS & BIT(0)) == 0 )
{
    If you wait for a long time then exit loop;
}
```

BIT(0) in **AC97STATUS** means 'command write done'. After you write to **AC97WRITE**, you should wait till writing is done. If command is not written for a long time(about 100usec). there is problem in AC97 Codec or ACLINK is not active.

21.3.2 Read Status from AC97

You can read AC97 status register via **AC97RADDR** and **AC97RDATA**. Following example is to read master volume

```
AC97RADDR = 0x02; // address of AC97 master volume register

while( (AC97STATUS & BIT(2)) == 0 )
{
    If you wait for a long time then exit loop;
}

MasterVolume = AC97RDATA;
```

BIT(2) in **AC97STATUS** means 'status read done'. After you write to **AC97RADDR**, you should wait till reading is done. If status is not read for a long time(about 100usec). there is problem in AC97 Codec or ACLINK is not active.

21.3.3 AC97 Power Down and Wake Up

If you set PR4 of AC97 power down control register, then AC97 go to power down mode. (Please see AC97 codec spec.). To wake up AC97, you must warm reset AC97.

See example.

```
AC97RESET = BIT(1) | BIT(0); // activate warm reset

wait for about 1 usec.

AC97RESET = BIT(0); // deactivate warm reset

while( (AC97STATUS & BIT(1)) == 0 ) // wait for codec ready.
{
    If you wait for a long time then exit loop;
}
```

BIT(1) in **AC97RESET** means 'warm reset'. Don't remove BIT(0) otherwise AC97 will go to cold reset.

21.4 Audio Playing and Recording

21.4.1 Audio Mixer

Audio channels from DMA or wavetable synthesizer can be selectively mixed. Set **ACMIXENB** to select desired audio channels. If you want to play stereo sound using DMA, you should set only SL and SR bit in **ACMIXENB**. (Other bits are off)

EN bit in **ACCONTROL** let Audio Mixer run. If EN bit is off, audio playing will be stopped. You can **ACCONTROL** to pause audio. When you use I2S DAC, don't forget to set BIT(1) and BIT(0) in **ACCONTROL**. See following example.

when using I2S DAC

```
ACCONTROL = BIT(2) | BIT(1) | BIT(0);    // Enable Audio Mixer when using I2S DAC
ACCONTROL = BIT(1) | BIT(0);            // Disable Audio Mixer when using I2S DAC
```

when using AC97 Codec

```
ACCONTROL = BIT(2);                      // Enable Audio Mixer when using AC97 Codec
ACCONTROL = 0;                          // Disable Audio Mixer when using AC97 Codec
```

BIT(2) in **ACCONTROL** means 'enable Audio Mixer'.

21.4.2 Playing DMA Channel

Example to play stereo by two DMA channels

```
(1) ACFFCONTROL = BIT(0);                // Set 'DMA Play FIFO' to clear
(2) ACFFCONTROL = 0;                    // Set 'DMA Play FIFO' not to clear
(3) Program DMA controller to play audio. // See DMA controller for more details.
(4) ACMIXENB = BIT(1) | BIT(0);          // Set Left/Right channel to be mixed
(5) ACCONTROL = BIT(2);                  // Enable Audio Mixer (Caution! This assumes not I2S but AC97)
```

Because **ACCONTROL** has I2S control bits, please carefully set **ACCONTROL**.

Example to stop audio playing

```
(1) ACCONTROL = 0;                      // Disable Audio Mixer (Caution! This assumes not I2S but AC97)
(2) ACMIXENB = 0;                      // All channels will be not mixed
```

21.4.3 Playing Wavetable Channel

Example to play only one Wavetable channel

```
(1) ACMIXENB = BIT(6);                  // Enable Wavetable Synthesizer #0
(2) ACCONTROL = BIT(2);                  // Enable Audio Mixer (Caution! This assumes not I2S but AC97)
```

Example to stop audio playing

```
(1) ACMIXENB = 0;                      // All channels will be not mixed
(2) ACCONTROL = 0;                      // Disable Audio Mixer (Caution! This assumes not I2S but AC97)
```

In fact, you don't need to reset **ACCONTROL** to stop audio playing. When you want to stop only some channels while other channels are played, don't touch **ACCONTROL**.

It is possible to play both DMA channels and Wavetable channels.

21.4.4 Recording

Recording is available when you use AC97 Codec. (We don't support I2S ADC). Mono(MIC) and stereo(PCM) can be recorded.

Example to record by two DMA channels

- (1) Clear DMA recording FIFO (See **ACFFCONTROL**)
- (2) Program DMA controller to start recording.
- (3) Write AC97 Codec register to start recording. (See **AC97WRITE**)
- (4) wait for recording end
- (5) Write AC97 Codec register to stop recording (See **AC97WRITE**)
- (6) Program DMA controller to stop recording.

There is only two sound recording formats that 'Audio Controller' supports. See following tables.

Mono Audio Recording Format

Memory Address (Byte Address)	Data (16bit)
N + 0	Sample[0]
N + 2	Sample[1]
...	...

Stereo Audio Recording Format

Memory Address (Byte Address)	Data (32bit)	
	MSB 16 bit	LSB 16 bit
N + 0	Sample[0].Right	Sample[0].Left
N + 4	Sample[1].Right	Sample[1].Left
		...

21.5 Registers Map (C000_2800h ~ C000_2980h)

Register Name	offset		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ACINTENB	00h														OC	SD	CR	CD
ACINTPND	02h														OC	SD	CR	CD
ACMIXENB	04h										W1	W0	LO	CE	SR	SL	FR	FL
ACFFCONTROL	06h														R	OF	PF	RF
ACCONTROL	08h															EN	ZE	CO
ACPERIOD	0Ah																	
AC97RESET	0Ch																WR	CR
AC97STATUS	80h															SD	CR	CD
AC97RADDR	82h																	0
AC97RDATA	84h																	
AC97WRITE	100h ~ 180h																	

21.5.1 Interrupt Enable Register (ACINTENB)

Address : C000_2800h

Bit	R/W	Description	Default Value
15: 4		Reserved	
3	R/W	Enables interrupt for audio output counter Programmers don't need to know this register. It's for debugging hardware. Please always set this bit to '0'.	0
2	R/W	Enables interrupt for AC97 status read done.	0
1	R/W	Enables interrupt for AC97 codec ready.	0
0	R/W	Enables interrupt for AC97 command write done.	0

21.5.2 Interrupt Pending Register (ACINTPND)

Address : C000_2802h

Bit	R/W	Description	Default Value
15: 4		Reserved	
3	R/W	Pending for sound output count. .	0
2	R/W	AC97 status read done.	0
1	R/W	AC97 codec ready.	0
0	R/W	AC97 command write done.	0

21.5.3 Channel Mixing Enable Register (ACMIXENB)

Address : C000_2804h

Bit	R/W	Description	Default Value
-----	-----	-------------	---------------

Bit	R/W	Description	Default Value
15:8		Reserved.	
7	W	WaveTable1 output to be mixed. (Front 2 channel and surround 2 channel)	0
6	W	WaveTable0 output to be mixed. (Front 2 channel and surround 2 channel)	0
5	W	LFE DMA channel to be mixed	0
4	W	Center DMA channel to be mixed	0
3	W	Surround right DMA channel to be mixed	0
2	W	Surround left DMA channel to be mixed	0
1	W	Right DMA channel to be mixed	0
0	W	Left DMA channel to be mixed	0

21.5.4 FIFO Control Register (ACFFCONTROL)

Address : C000_2806h

Bit	R/W	Description	Default Value
15:4		Reserved	
3	R	Audio mixer is running. '1' : running '0' : idle This flag is used for hardware debugging. If EN flag in ACCONTROL is set, then audio mixer would be running. Programmers don't need to know about this flag.	0
2	R/W	Clear Audio Output FIFO '1' : Clear '0' : Not Clear This flag is maintains its value. After you clear FIFO, you must back this bit to '0'	0
1	R/W	Clear DMA Record FIFO This flag is maintains its value. After you clear FIFO, you must back this bit to '0'	0
0	R/W	Clear DMA Play FIFO This flag is maintains its value. After you clear FIFO, you must back this bit to '0'	0

21.5.5 Audio Control Register (ACCONTROL)

Address : C000_2808h

Bit	R/W	Description	Default Value
15:3		Reserved	
2	R/W	Enable Audio Mixing	0
1	R/W	I2S Enable '0' : disable I2S BUS (default) '1' : enable I2S BUS This bit controls I2S BUS shut down. This bit must be set to '0' when you use AC97 Codec	0
0	R/W	Codec select '0' : AC97 (default) '1' : I2S DAC This bit controls path from OUTFIFO to AC97/I2S.	0

21.5.6 Audio Output Counter Period Register (ACPERIOD)

Address : C000_280Ah

Bit	R/W	Description	Default Value
15:0	W	Period. Audio Controller internally counts how many times audio sample is generated. If counter becomes larger than period, then interrupt will be occur and counter will be reset. Programmers don't need to know this register. It's for debugging hardware.	1

21.5.7 AC97 Reset Control Register (AC97RESET)

Address : C000_280Ch

Bit	R/W	Description	Default Value
15:2		Reserved	
1	W	Warm Reset '1' : reset After you power down AC97 codec, this bit used for waking up AC97.	0
0	W	Cold Reset '0' : reset AC97. This bit must be set to '1' when you enable AC97 Codec	0

21.5.8 AC97 Status Register (AC97STATUS)

Address : C000_2880h

Bit	R/W	Description	Default Value
15:3		Reserved	
2	R	AC97 status read done.	1
1	R	AC97 codec ready.	
0	R	AC97 command write done.	1

21.5.9 AC97 Read Address Register (AC97RADDR)

Address : C000_2882h

Bit	R/W	Description	Default Value
15:7		Reserved	
6:0	W	Read address to AC97 codec	

21.5.10 AC97 Read Data Register (AC97RDATA)

Address : C000_2884h

Bit	R/W	Description	Default Value
15:0	R	Read data from AC97 codec	

21.5.11 AC97 Write Registers (AC97WRITE)

Address : C000_2900h ~ C000_2980h

Bit	R/W	Description	Default Value
15:0	W	Write data to AC97 codec	

22. VLD/VLC

The VLD/VLC unit in MMSP-2 receives input video stream(decode) or run/level data(encode) from external memory, and stores decoded run/level data (decode) or video stream (encode) to external memory

22.1 Overview

There are three functional blocks in VLD/VLC unit, SP(Stream Processor), Huffman, and RVLD. SP manipulates in/out bitstream data and are controlled directly by host CPU for header data or by Huffman/RVLD for coefficient data. Huffman block excutes all coefficient VLD/VLC function except for MPEG4 RVLD. RVLD block executes decoding of MPEG4 RVLC code, except for feature reading RVLCs backward. SP is able to be used independently by host CPU for bitstream manipulation for any purpose.

22.1.1 Feature

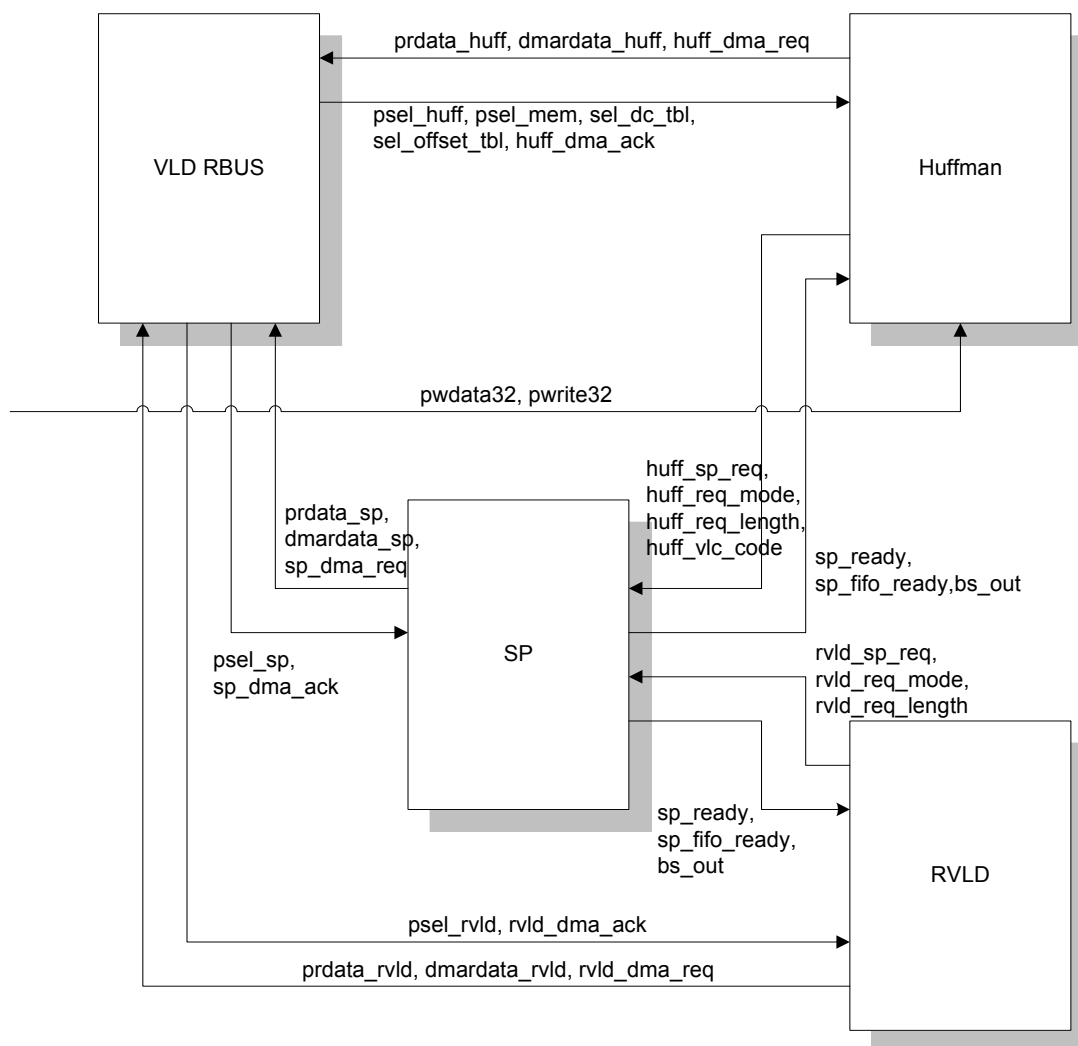
SP(Stream Processor)

- Getbit, Flushbit, Putbit operation with 0 ~ 16 bit length
- Showbit operation with 0 ~ 16 and 32 bit length

VLD/VLC

- MPEG1, MPEG2 and MPEG4 (includes RVLC) decode
- H.263 baseline and Annex I,T decode
- JPEG decode
- MPEG2 and MPEG4 encode
- H.263 baseline encode
- JPEG encode

22.1.2 Block Diagram and Description



- There are two 16word x 32bit FIFOs in SP, VLD/C, RVLD block each used as in/out data buffer.
- SP has 16-bit barrel shifter.
- VLD/VLC has one 32 word x 12 bit, one 32 word x 16 bit, one 64 word x 17 bit register file and 368 word x 21 SRAM for VLC table

22.2 SP (Stream Processor) Operation

The operations deal with bit stream data is executed by SP. When decode, SP reads input video stream data from external memory using DMA, and transfers aligned stream data as required length to CPU or VLD part. When encode, SP receives bit stream data from CPU or VLC part, aligns the data and store to external memory through DMA.

SP offers 3 functional commands in decode mode, and 1 functional command in encode mode.

Decode commands

1) Showbit

Returns bit stream data from current bit pointer with requested length, does not change any pointer value.

2) Flushbit

Returns no data, only changes pointer by requested value.

3) Getbit

Returns bit stream data from current pointer with requested length, also changes pointer value.

Encode commands

1) Putbit

Pushes and aligns bit stream data at current pointer, also changes pointer value.

22.2.1 SP Initialization

SP uses a DMA channel for stream data read/write. In both case, SP issues DMA request. The allowed data format of external memory is 32-bit word little-endian mode. DMA controller has to be properly configured before SP is turned on and initialized.

SP is not activated unless bit 0 of SP_CONTROL register (SP on bit) is activated. All control/config registers and FIFO in SP are remained their initial value during SP is off. CPU is able to write/read SP registers after setting the SP on bit to 1. After setting properly SP_CONFIG and SP_STATUS_ENA registers, CPU must set bit 1 of SP_CONTROL register to initialize FIFO and SP state machine.

Before VLD/VLC is enabled, SP has to be properly configured and enabled. When JPEG decode mode, JPEG decode valid bit of SP_CONFIG_REG is to be set to 1. This bit is used for images which have small size image in user data area.

In the case of CODEC mode, in which encode/decode switching is occurred, SP has to be initialized when encode/decode mode switching is occurred.

22.2.2 Using SP Command

For using bit stream manipulation feature of SP, the following rules are to be satisfied.

A. Decode mode

- 1) Do not request putbit commands. Putbit command is for encode operation, so requesting putbit when decode mode corrupts input stream data.
- 2) When CPU wants showbit or getbit operation, CPU should write SP_SHIFT_CMD register with appropriate value and then read SP_SHIFT_RESP register for getting required bit stream data.

B. Encode mode

- 1) Do not request showbit, getbit and flushbit commands. These commands are for decode operation, so requesting these commands when encode mode corrupts output stream data.
- 2) When CPU wants putbit operation, CPU should write SP_PUT_DAT register with bit stream data and then write SP_SHIFT_CMD register with appropriate value.

22.2.3 Encode Stream Arrangement

The encode stream data consists of stream from CPU and VLC part. This stream data stored in SP FIFO is transferred to external memory via DMA by 16 words (64 bytes) with each DMA request routine. So, there is remaining bit stream data which is still in shift register or FIFO and does not transferred to external memory. This bit stream data also should be transferred to external memory and

the process is accomplished by following steps.

- 1) Byte align of stream data using SP_BITPTR[2:0]
- 2) Flush out word-aligned data by setting SP_CONTROL[2] to 1
- 3) Read and store byte aligned data stored in SP_BREG0 ~ SP_BREG2 and SP_HWREG according to the value of SP_BYTE_PTR and SP_BITPTR[3].
- 4) Read external memory address to which remaining data should be written from DMA controller.
- 5) Write remaining data to external memory.

If SP is in codec mode, remaining data of encode mode should not be written directly to external memory. Instead, the remaining data has to be written to SP shift register by the CPU via putbit command. Therefore, sequence 5) of the above is changed as following.

- 5) Write remaining data to SP using putbit command.

22.3 Huffman(VLD/VLC) Operation

The VLD/VLC operations except MPEG4 RVLD are executed in this Huffman block. When decode, Huffman block receives aligned coded video stream data from SP, decodes them, and writes decoded symbol to external memory through DMA. When encode, Huffman block reads encoded symbol data from external memory through DMA, encodes them and puts the coded stream data to SP.

22.3.1 Huffman Initialization

Huffman block uses a DMA channel for reading/writing coded symbol data. In both cases, Huffman block issues DMA request. The allowed data format of external memory is 32-bit word little endian mode. DMA controller has to be properly configured before Huffman block is turned on and initialized.

Huffman block is not activated unless bit 0 of HUFF_CONTROL register (Huff on) is activated. All control/config registers and FIFO in Huffman block are remained their initial value during Huffman block is off. CPU is able to write/read Huffman registers after setting the Huff on bit to 1. After setting properly HUFF_CONFIG and HUFF_STATUS_ENA registers, the CPU downloads VLD/VLC table data. Next, CPU must set bit 1 of HUFF_CONTROL register to initialize FIFO and state machine.

Before Huffman block is enabled, SP has to be properly configured and enabled. In the case of CODEC mode, in which encode/decode switching is occurred, Huffman block has to be initialized when encode/decode mode switching is occurred.

22.3.2 Using VLD/VLC

Huffman block is controlled by the CPU as macro block base. Following is macro block loop sequence.

- 1) Write macro block parameter to HUFF_MB_PARAM register.
- 2) Set HUFF_MB_START register to start a macro block decode/encode operation.
- 3) Check current macro block end flag (bit 1 of HUFF_STATUS register).

If the video stream to be decoded is MPEG4 and using data partition feature, the control method of Huffman block is different from other cases. In inter macro block there is no change needed, but in intra macro block the control of Huffman block is changed to block base and the CPU has to write decoded DC symbol directly to Huffman block decoded symbol register. The sequence needed to support this mode is

- 1) Write HUFF_MB_PARAM register as data partition bit to 1, DC VLC selection bit to 0, coded block pattern to 001h (only one block is coded)
- 2) Write decoded DC symbol directly to Huffman decoded symbol register refer to HUFF_STATUS register bit 6.
- 3) Set HUFF_MB_START register to 1 to start a block decode.
- 4) Check current block end flag (bit 1 of HUFF_STATUS register).

22.3.3 Flush Out Decoded Symbol

The decoded symbol data are stored in Huffman block FIFO and transferred to external memory via DMA by 16 words (64 bytes) with each DMA request routine. So, there are remaining decoded symbol data still in decoded symbol register or FIFO. These symbol data also should be transferred to external memory and this is accomplished by setting bit 2 of HUFF_CONTROL register to 1.

22.3.4 Symbol Syntax

Mode	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Normal	E	L	DC	Run [5:0]						S	Level [5:0]					
DC	E	L	DC	S	Level [11:0]											
Escape	E	L	DC	Run [5:0]						Not used						
	S	Level														

E : Escape flag

L : Last symbol of block flag

DC : DC flag

S : Sign

22.4 RVLD Operation

The RVLD block executes decoding of MPEG4 RVLD video streams only. It receives aligned coded video stream data from SP, decodes them, and writes decoded symbol to external memory through DMA. Huffman block and RVLD block are to be exclusively enabled, that is only one block can be enabled at one time. The RVLD block supports MPEG4 RVLC coded bit stream, but not reverse decoding optional feature.

22.4.1 RVLD Initialization

RVLD block uses a DMA channel for writing decode symbol data and issues DMA request. The data format allowed for external memory is 32-bit word little-endian mode. DMA controller has to be properly configured before RVLD is turned on and initialized.

RVLD is not activated unless bit 0 of RVLD _CONTROL register(RVLD on) is activated. All control/config registers and FIFO in RVLD are remained their initial value during RVLD block is off. CPU is able to write/read RVLD registers after setting the RVLD on bit to 1. After setting properly RVLD_STATUS_ENA registers, the CPU must set bit 1 of RVLD _CONTROL register to initialize FIFO and state machine. Before RVLD is enabled, SP has to be properly configured and enabled.

22.4.2 Using RVLD

The RVLD block is controlled by the CPU as macro block base or block base. In inter macro block, RVLD is controlled as macro block base. In case of intra macro block, both macro block and block base control are possible.

A. Macro block base control

Prior to start decoding macro block, the CPU must write decoded DC symbol data to RVLD DC register0 to 5. RVLD inserts these DC symbol data at appropriate position automatically. The sequence are

- 1) Write decoded DC symbol data to RVLD_DE_REG0 ~5. (0 corresponds to Y block 0, 4 and 5 correspond to Cr, Cb block each)
- 2) Write macro block parameter. Bit 7 of parameter register is must be 0 (meaning macro block mode).
- 3) Set bit 1 of RVLD_OP_START register. This means macro block start.
- 4) Check current macro block end flag (bit 1 of RVLD_STATUS register).

B. BLOCK BASE CONTROL

In this mode, CPU has to write decoded DC symbol directly to RVLD decoded symbol register block by block.

- 1) Write macro block parameter. Bit 7 of parameter register is must be 1 (meaning block mode).
- 2) Write decoded DC symbol data to RVLD_DEC register.
- 3) Set bit 0 of RVLD_OP_START register. This means block start.
- 4) Check current block end flag (bit 2 of RVLD_STATUS register).
- 5) repeat 2) ~ 4) until all block of current macro block are processed.

22.4.3 Flush Out Decoded Symbol

The decoded symbol data are stored in RVLD block FIFO and transfered to external memory via DMA by 16 words(64 bytes) with each DMA request routine. So, there are remaining decoded symbol data still in decoded symbol register or FIFO. These symbol data also should be transfered to external memory and this is accomplished by setting bit 2 of RVLD_CONTROL register to 1.

22.5 SP (Stream Processor) Registers Map

22.5.1 SP Control Reg (SP_CONTROL)

Address: E000_2000h

Bit	R/W	Description	Default Value
2	RW	FIFO flush out. This bit is automatically cleared. Flush out all data in internal buffer. Used for finishing a frame (encode only) 1 : Flush out FIFO data, 0 : normal operation	0b
1	RW	SP init. This bit is automatically cleared. 1 : Initialize FIFO and go to stand-by state	0b
0	RW	SP on 1 : SP operation enable, 0 : SP off	0b

22.5.2 SP Config Reg (SP_CONFIG)

Address : E000_2004h

Bit	R/W	Description	Default Value
4	RW	JPEG decode enable 1 : JPEG decode enable, 0 : disable	0b
3	RW	MPEG4 RVLD on 1 : MPEG4 RVLD, 0 : else	0b
2	RW	Enc/nDec 1 : Enc, 0 : Dec	0b
1	RW	JPEG flag 1 : JPEG, 0 : else	0b
0	RW	Header/nDecode 1 : header (contrlloed by CPU), 0 : decode coefficient (controlled by Huffman)	0b

22.5.3 SP Status Enable Reg (SP_STATUS_ENA)

Address : E000_2008h

Bit	R/W	Description	Default Value
2	RW	FIFO under-run status enable 1 : status enable, 0 : status disable	0b
1	RW	FIFO over-run status enable 1 : status enable, 0 : status disable	0b
0	RW	SP ready status enable 1 : status enable, 0 : stclsatus disable	0b

22.5.4 SP Status Reg (SP_STATUS)

Address : E000_200Ch

Bit	R/W	Description	Default Value
3	RW	1 : Encode DMA in progress, 0 : not in progress If DMA is not in progress, Encode Flush out can be executed.	0b
2	RW	1 : FIFO under-run, 0 : not	0b
1	RW	1 : FIFO over-run, 0 : not	0b
0	RW	1 : SP ready for operation, 0 : not ready	0b

22.5.5 SP Ready Reg (SP_READY)

Address : E000_2010h

Bit	R/W	Description	Default Value
0	R	1 : SP ready for operation, 0 : not ready	0b

22.5.6 SP Shift Command Reg (SP_SHIFT_CMD)

Address : E000_2014h

Bit	R/W	Description	Default Value
6 : 5	RW	CPU command mode 0 = showbit, 1 = flushbit, 2 = getbit, 3 = putbit	0h
4 : 0	RW	data length. 0 ~ 16 : valid for all commands, 17 ~ 31 : only for 32-bit showbit	0h

22.5.7 SP Shift Response Reg (SP_SHIFT_RESP)

Address : E000_2018h

Bit	R/W	Description	Default Value
31 : 0	R	CPU read command (showbit, getbit) response data	0h

22.5.8 SP Encode Word Counter Reg High (SP_TOTAL_CNTH)

Address : E000_201Ch

Bit	R/W	Description	Default Value
7 : 0	RW	Total word counter [23:16] when encode mode	0h

22.5.9 SP Encode Word Counter Reg Low (SP_TOTAL_CNTL)

Address : E000_2020h

Bit	R/W	Description	Default Value
15 : 0	RW	Total word counter [15:0] when encode mode	0h

22.5.10 SP Byte Pointer Reg (SP_BYTE_PTR)

Address : E000_2024h

Bit	R/W	Description	Default Value
1 : 0	R	Current byte pointer of a word in processing. 0 is first byte of word	0h

22.5.11 SP Put Data Reg (SP_PUT_DAT)

Address : E000_2028h

Bit	R/W	Description	Default Value
15 : 0	RW	Put bitstream data for encode	0h

22.5.12 SP Bit Pointer Reg (SP_BITBTR)

Address : E000_202Ch

Bit	R/W	Description	Default Value
4 : 0	R	Pointer of next bit operation start position in HWORDREG0. 0 = MS bit, 15 = LS bit	0h

22.5.13 SP Half Word Reg (SP_HWREG)*Address : E000_2030h*

Bit	R/W	Description	Default Value
15 : 0	R	Current processing half word	0h

22.5.14 SP Byte Reg 0 (SP_BREG0)*Address : E000_2034h*

Bit	R/W	Description	Default Value
7 : 0	R	Previously processed byte register 0 when encode mode	0h

22.5.15 SP Byte Reg 1 (SP_BREG1)*Address : E000_2038h*

Bit	R/W	Description	Default Value
7 : 0	R	Previously processed byte register 1 when encode mode	0h

22.5.16 SP Byte Reg 2 (SP_BREG2)*Address : E000_203Ch*

Bit	R/W	Description	Default Value
7 : 0	R	Previously processed byte register 2 when encode mode	0h

22.6 Huffman Register Map

22.6.1 Huffman Control Reg (HUFF_CONTROL)

Address : E000_2200h

Bit	R/W	Description	Default Value
2	RW	FIFO flush out 1 : Flush out FIFO data, 0 : normal operation	0b
1	RW	Initialize FIFO and state machine. config registers are not changed 1 : Initialize FIFO and go to stand-by state	0b
0	RW	Huffman on 1 : Huffman operation enable, 0 : Huffman off	0b

22.6.2 Huffman Config Reg (HUFF_CONFIG)

Address : E000_2204h

Bit	R/W	Description	Default Value
13	RW	Code error detection enable 1 : enable, 0 : disable	0b
12:9	RW	Number of C block in a macro block. Used for various color format (4:2:0, 4:4:4, etc)	0b
8:6	RW	Number of Y block in a macro block. Used for various color format (4:2:0, 4:4:4, etc)	0b
5	RW	Enc/nDec 1 = Enc, 0 = Dec	0b
4	RW	JPEG flag	0b
3	RW	MPEG1 flag	0b
2	RW	MPEG2 flag	0b
1	RW	H263 flag	0b
0	RW	MPEG4 flag	0b

Bit 4 to Bit 0 is to be enabled exclusively, that is only one bit of them is able to be value 1.

22.6.3 Huffman Status Enable Reg (HUFF_STATUS_ENA)

Address : E000_2208h

Bit	R/W	Description	Default Value
3	RW	Decode buffer over run status enable 1 : status enable, 0 : status disable	0b
2	RW	Encode buffer under run status enable 1 : status enable, 0 : status disable	0b
1	RW	Current MB end status enable 1 : status enable, 0 : status disable	0b
0	RW	Code error status enable 1 : status enable, 0 : stclsatus disable	0b

22.6.4 Huffman Status Reg (HUFF_STATUS)

Address : E000_220Ch

Bit	R/W	Description	Default Value
4	R	Huffman DMA in progrss flag	0b

Bit	R/W	Description	Default Value
3	RW	Decode buffer over run	0b
2	RW	Encode buffer under run	0b
1	RW	Current MB end	0b
0	RW	Code error occurred (Dec)	0b

22.6.5 Huffman MB Start Reg (HUFF_MB_START)

Address : E000_2210h

Bit	R/W	Description	Default Value
0	RW	Enable new macro block processing start. This bit is automatically cleared. 1 = Start new macro block	0b

22.6.6 Huffman MB Parameter Reg (HUFF_MB_PARAM)

Address : E000_2214h

Bit	R/W	Description	Default Value
15	RW	MPEG4 decode : data partitioned macro block flag JPEG encode : insert dummy macro block for macro block alignment Else, ignored	0h
14	RW	Intra MB/nInter MB 1 = Intra MB, 0 = Inter MB	0b
13	RW	MPEG2 table 0/1 selection 1 = MPEG2 table 1, 0 = MPEG2 table 0	0b
12	RW	MPEG4 decode : DC VLC method selection 1 : use DC method, 0 = use AC method H263 decode : Advanced Intra MB 1 = advanced intra MB, 0 = normal MB	0b
11 : 8	RW	CBP of Cr block (1 = coded, 0 = not coded) [11] : Cr block 3 coded or not, [10] : Cr block 2 coded or not [9] : Cr block 1 coded or not, [8] : Cr block 0 coded or not	0h
7:4	RW	CBP of Cb block [7] : Cb block 3 coded or not, [6] : Cb block 2 coded or not [5] : Cb block 1 coded or not, [4] : Cb block 0 coded or not	0h
3:0	RW	CBP of Y block [3] : Y block 3 coded or not, [2] : Y block 2 coded or not [1] : Y block 1 coded or not, [0] : Y block 0 coded or not	0h

22.6.7 Huffman Decoded Symbol Reg High (HUFF_DECH)

Address : E000_2220h

Bit	R/W	Description	Default Value
15:0	RW	Decoded run, level symbol data MS half word	0h

22.6.8 Huffman Decoded Symbol Reg Low (HUFF_DECL)

Address : E000_2224h

Bit	R/W	Description	Default Value
15:0	RW	Decoded run, level symbol data LS half word	0h

22.6.9 Huffman Decode Word Counter Reg Low (HUFF_TOTAL_CNTL)

Address : E000_222Ch

Bit	R/W	Description	Default Value
15 : 0	RW	Total decoded word counter [15:0]	0h

22.6.10 Huffman Decode Word Counter Reg High (HUFF_TOTAL_CNTH)

Address : E000_2230h

Bit	R/W	Description	Default Value
7 : 0	RW	Total decoded word counter [23:16]	0h

22.7 RVLD Register Map

22.7.1 RVLD Control Reg (RVLD_CONTROL)

Address : E000_2400h

Bit	R/W	Description	Default Value
2	RW	FIFO flush out 1 : Flush out FIFO data, 0 : normal operation	0b
1	RW	Initialize FIFO and state machine. config registers are not changed 1 : Initialize FIFO and go to stand-by state	0b
0	RW	RVLD on 1 : RVLD operation enable, 0 : RVLD off	0b

Huffman and RVLD have to be enabled exclusively. That is one of them can be activated at one time.

22.7.2 RVLD Status Enable Reg (RVLD_STATUS_ENA)

Address : E000_2404h

Bit	R/W	Description	Default Value
3	RW	Decode buffer over run status enable 1 : status enable, 0 : status disable	0b
2	RW	Current block end status enable (if block mode)	0b
1	RW	Current MB end status enable (if MB mode)	0b
0	RW	Reserved. Must be set to 0.	0b

22.7.3 RVLD Status Reg (RVLD_STATUS)

Address : E000_2408h

Bit	R/W	Description	Default Value
4	RW	RVLD DMA in progress flag	0b
3	RW	Decode buffer over run	0b
2	RW	Block end flag when block mode	0b
1	RW	Current MB end flag when MB mode	0b
0	RW	Reserved	0b

22.7.4 RVLD OP Start Reg (RVLD_OP_START)

Address : E000_240Ch

Bit	R/W	Description	Default Value
1	RW	Enable new macro block processing start. This bit is automatically cleared. 1 = Start new macro block	0b
0	RW	Enable new block processing start(block mode). This bit is automatically cleared. 1 = Start new block	0b

22.7.5 RVLD MB Parameter Reg (RVLD_MB_PARAM)

Address : E000_2410h

Bit	R/W	Description	Default Value
7	RW	Block mode enable. When block mode, CPU must write DC symbol to buffer directly. 1 = block mode enable	0h
6	RW	Intra MB/nInter MB	0b

Bit	R/W	Description	Default Value
		1 = Intra MB, 0 = Inter MB	
5	RW	CBP of Cr block 1 = coded, 0 = not coded	0b
4	RW	CBP of Cb block 1 = coded, 0 = not coded	0b
3:0	RW	CBP of Y block [3] : Y block 3 coded or not, [2] : Y block 2 coded or not [1] : Y block 1 coded or not, [0] : Y block 0 coded or not	0h

22.7.6 RVLD Decoded Symbol Reg (RVLD_DEC)

Address : E000_2414h

Bit	R/W	Description	Default Value
15:0	RW	Decoded run, level symbol data MS half word	0h

22.7.7 RVLD DC Reg 0 (RVLD_DC_REG0)

Address : E000_2418h

Bit	R/W	Description	Default Value
15:0	W	Block 0 DC data decoded by CPU. This data is used with MB mode and inserted appropriate position of run/level data buffer.	0h

22.7.8 RVLD DC Reg 1 (RVLD_DC_REG1)

Address : E000_241Ch

Bit	R/W	Description	Default Value
15:0	W	Block 1 DC data decoded by CPU.	0h

22.7.9 RVLD DC Reg 2 (RVLD_DC_REG2)

Address : E000_2420h

Bit	R/W	Description	Default Value
15:0	W	Block 2 DC data decoded by CPU.	0h

22.7.10 RVLD DC Reg 3 (RVLD_DC_REG3)

Address : E000_2424h

Bit	R/W	Description	Default Value
15:0	W	Block 3 DC data decoded by CPU.	0h

22.7.11 RVLD DC Reg 4 (RVLD_DC_REG4)

Address : E000_2428h

Bit	R/W	Description	Default Value
15:0	W	Block 4 DC data decoded by CPU.	0h

22.7.12 RVLD DC Reg 5 (RVLD_DC_REG5)

Address : E000_242Ch

Bit	R/W	Description	Default Value
-----	-----	-------------	---------------

Bit	R/W	Description	Default Value
15:0	W	Block 5 DC data decoded by CPU.	0h

22.7.13 RVLD Decode Word Counter Reg Low (RVLD_TOTAL_CNTL)

Address : E000_2430h

Bit	R/W	Description	Default Value
15 : 0	RW	Total decoded word counter [15:0]	0h

22.7.14 RVLD Decode Word Counter Reg High (RVLD_TOTAL_CNTH)

Address : E000_2434h

Bit	R/W	Description	Default Value
3 : 0	RW	Total decoded word counter [19:16]	0h

22.8 Table Registers

22.8.1 DC Table Registers

Address : E000_2300h ~ E000_237Ch

22.8.2 Offset Table Registers (decode only)

Address : E000_2380h ~ E000_23FCh

22.8.3 AC Table Registers (E00h ~ EBCh used only JPEG decode)

Address : E000_2800h ~ E000_2EBCh

23. MPEG4 Picture Decoder

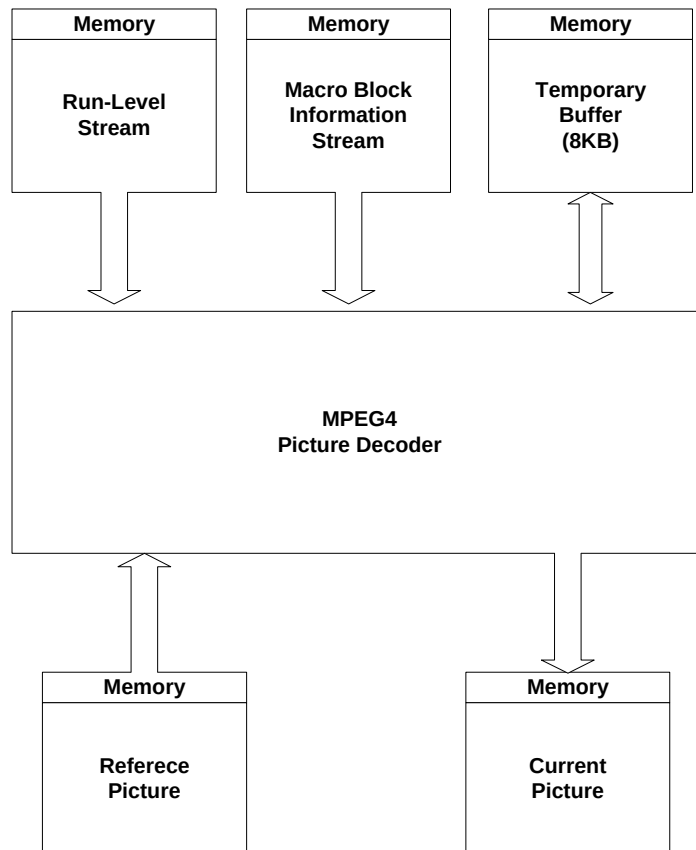
23.1 Overview

MPEG4 decoding process consists of VLD Processor and MPEG4 Picture Decoder. VLD Processor decodes huffman coded bit stream, and MPEG4 Picture Decoder performs following

- Inverse Quantization
- Inverse DCT
- Motion Prediction/Compensation

MPEG4 Picture Decoder requires Run-Level data which is generated from VLD Processor. It is not necessary that you know about VLD Processor because Run-Level data can be generated by programmer (but slow). Please refer to VLD Processor part for more information. This document only explains MPEG4 Picture Decoder. Now we support only MPEG4 Simple Profile. In this document we call MPEG4 Picture Decoder as MPEG4DEC.

23.1.1 Block Diagram



23.2 Decoding

23.2.1 Temporary Buffer

MPEG4DEC need temporary memory (8KB) for AC-Prediction. Programmer must set 8-byte aligned address to **MP4TEMPBUF**. It is not necessary to set this register every time. You may set **MP4TEMPBUF** just once at boot time

```
MP4TEMPBUF = malloc(8192); // assume 8-byte address
```

23.2.2 Decode a Frame (one Picture)

Steps of decoding are

- (1) Set Run-Level Stream Address (**MP4RLSTART**)
- (2) Set Macro Block Information Stream Address (**MP4MBSTART**)
- (3) Set Macro Block Number (**MP4MBNUM**)
- (4) Set Resolution & Round Control (**MP4RESMISC**)
- (5) Set Reference Picture Address (**MP4REFY**, **MP4REFCB**, **MP4REFCR**)
- (6) Set Current Picture Address (**MP4CURY**, **MP4CURCB**, **MP4CURCR**)
- (6) Set Interrupt Enable and Clear Pending (**MP4INTENB**, **MP4INTPND**)
- (7) Start Decode (**MP4START**)
- (8) Wait till decode finishes. (**MP4STATUS** or **MP4INTPND**)

23.2.3 Run-Level Stream

Run-Level Stream is 16 bit data array in memory. It contains data for one picture. After Picture Decoder start, it read **Run-Level Stream** from memory and converts it to picture.

Run-Level Stream can be generated by VLD Processor (fast) or can be generated by programming CPU (slow). If you use VLD, you may skip this section

If a block is skipped by cbp(coded block pattern), the block must be coded 0x4000 in stream. If Macro Block is skipped by SK flag(See Macro Block Information Stream), stream is not coded. Stream start address is set to **MP4RLSTART** register. **MP4RLSTART** must be set to 64 bit aligned address.

Stream Format

Mode	Type	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NORMAL	A	0	L	0	RUN[5:0]						S	LEVEL[5:0]					
DC	B	0	L	1	S	LEVEL[11:0]											
ESCAPE	C	1	L	0	RUN[5:0]						R						
	D	S	LEVEL[14:0]														

Name	Description
R	Reserved field must be set to 0
ESCAPE	Level is too big. So Run (Type C) and level (Type D) are coded separately. (D must immediately follow C)
L (Last)	Indicate that this data is last data of the block
S (Sign)	sign of level
LEVEL	unsigned value for level

Stream Example

0x0801	// Run-Level = (16, 1)
0x4000	// end of Y[0] Block
0x8F80	// Escape
0x0002	// (31, 2)
0x0041	// (0, -1)
0x0501	// (10, 1)
0x0042	// (0, -2)
0x4000	// end of Y[1] Block
0x4000	// skip Y[2] block
0x3004	// DC = -4
0x0101	// (2, 1)
0x0241	// (4, -1)
0x4000	// end of Y[3] block
0x4000	// skip Cb block
0x4000	// skip Cr block
...	
...	
...	

23.2.4 Macro Block Information Stream

Macro Block Information Stream is 16 bit data stream in memory. It is generated by programmer. All Macro Block Information is stored in memory before MPEG4DEC Start.

Stream start and end address are set to **MP4MBSTART** register. **MP4MBSTART** must be set to 64 bit aligned address.

Stream Format

Mode	Type	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MB_BEGIN	A	E	0	0	0	R	SK	R	AC	R			QSCALE				
MB_MOTION	B	E	1	0	0	R				FP	4MV	R	MV Cnt				
	C	0	R		MVX												
	D	E	R		MVY												

Name	Description
R	Reserved field must be set to 0
E	End of this Macro Block Information.
SK	Skip flag of Macro Block
AC	AC Prediction Enable
QSCALE	Quantization Scale
FP	Forward Prediction
4MV	This Macro Block have 5 Motion Vector (4 vectors for Y and 1 vectors for CbCr)

Name	Description
MV Cnt	Number of (MVX, MVY) pairs - MV Cnt is 1 when 4MV Mode is disabled - MV Cnt is 5 when 4MV Mode is enabled
MVX, MVY	Motion Vector (12bit 2's complement)

Stream Example

0x0005	// AC-prediction disable, QScale=5
0x4081	// Forward prediction, 4MV disable
0x0002	
0x8003	// motion vector = (2,3), 1 st Macro Block is ended
0x8405	// 2 nd Macro Block is skipped, QScale is ignored
0x0005	// AC-prediction disable, QScale=5
0x40D5	// Forward prediction, 4MV enable, MBCnt=5, QScale=5
0x1FFD	
0x0001	// motion vector for Y[0] is (-3,1)
0x3FFF	
0x0000	// motion vector for Y[1] is (-1,0)
0x5FFD	
0x0002	// motion vector for Y[2] is (-3,2)
0x7FFF	//
0x0001	// motion vector for Y[3] is (-1,1)
0x9FFF	
0x8001	// motion vector for Cb,Cr is (-1,1), 2 nd Macro Block is ended
...	
...	
...	

23.2.5 Macro Block Number

Total Macro Block number of a picture is set to **MP4MBNUM** Register.

Macro block number calculation formula is

$$\mathbf{MP4MBNUM} = (\text{Horizontal Resolution} \times \text{Vertical Resolution} / 256) - 1$$

If picture size is 640x480, then **MP4MBNUM** must be set to 1199 (640x480/256 – 1).

23.2.6 Resolution

Picture resolution is set to **MP4RESMISC** Register. MPEG4DEC support up to 2048x1024 resolution. But large images are decoded slowly. See [Frame Rate](#) for more details

23.2.7 Rounding Control

MPEG standard stream specifies rounding_control at every frame. Set this value to **MP4RESMISC** register.

23.2.8 Short Header (H.263 compatible)

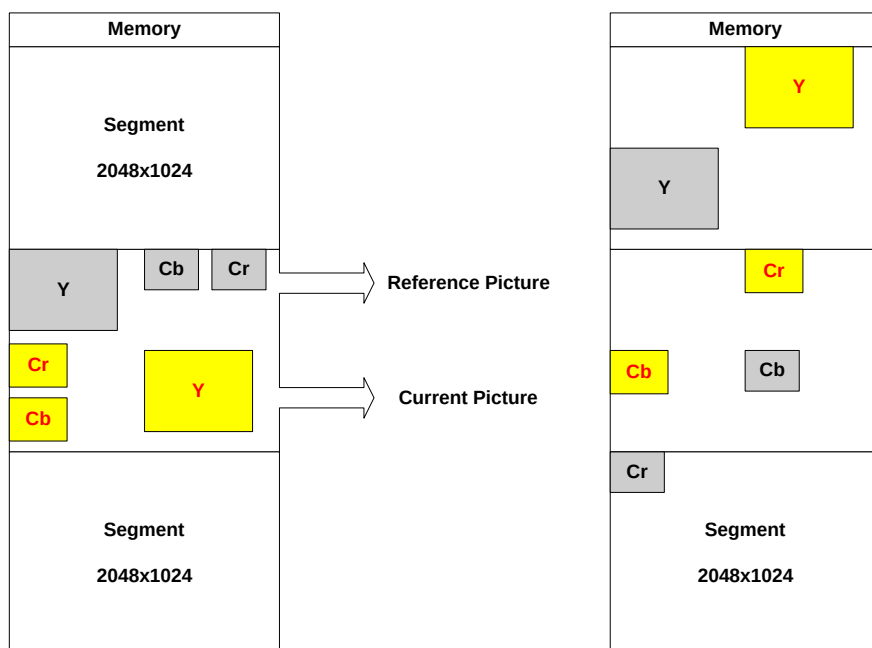
MPEG4 Simple Profile supports short header. But it would be off in most cases. It is set to **MP4RESMISC** Register

23.2.9 Picture Coordinate

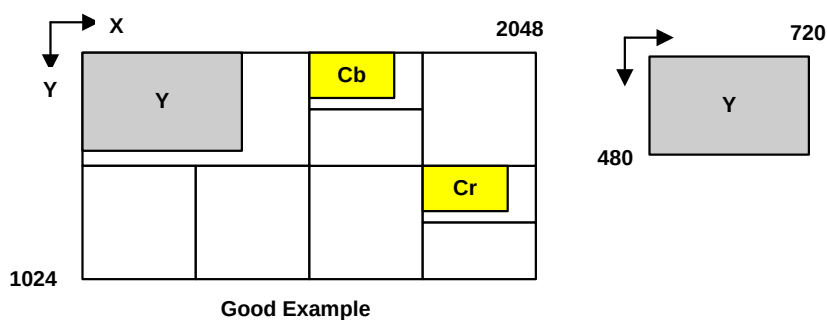
MPEG4 Simple Profile uses only forward prediction. So two pictures are used for MPEG4 decoding. (one is reference picture, one is current picture). Reference picture coordinate is set to **MP4REFY**, **MP4REFCB**, **MP4REFCR**, and current picture coordinate is set to **MP4CURY**, **MP4CURCB**, **MP4CURCR**. When decoding I-Picture, programmer don't need to set reference picture coordinate.

Segment is 2M byte aligned physical memory space. Y, Cb and Cr pictures can be placed in one segment or in difference segments.

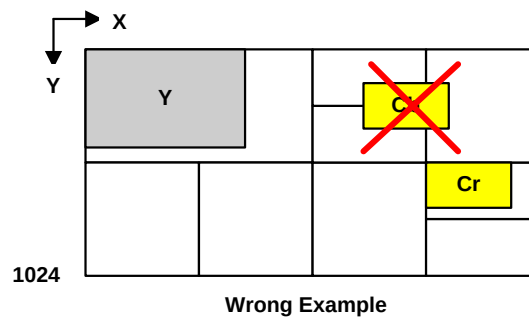
Two possible memory configurations are



All picture is must be placed at the 2's power in segment.



Don't place a picture across 2's power aligned line.



23.2.10 Interrupt and Run

MPEG4DEC have three interrupt. One of them indicates end of decoding. End of decoding can be known by interrupt or polling.

Interrupt example

```
MP4INTENB = 1;           // interrupt enable
MP4INTPND = 1;           // clear interrupt
MP4START = 1;            // start decoding
....                     // When decoding finishes, interrupt will occur
```

Polling example

```
MP4START = 1;           // start decoding
while(MP4STATUS & 1);   // wait for end
```

23.2.11 Frame Rate (Performance)

The formula for frame rate is

$$\text{frame rate} = \text{system clock} / [(\text{cycles per macro block}) \times (\text{number of macro blocks in a picture})]$$

Decoding a macro block consumes about 1200 cycles.

$$\text{number of macro blocks in a picture} = (\text{horizontal resolution} \times \text{vertical resolution}) / 256$$

So we can predict performance as following

$$\text{frame rate} = \text{system clock} / (4.6 \times \text{horizontal resolution} \times \text{vertical resolution})$$

example) decoding 640 x 480 picture at 110 Mhz

$$110\text{M} / (4.6 \times 640 \times 480) = 77.8 \text{ frames / sec}$$

23.2.12 Direct DCT

To more flexible for various codec, MPEG4DEC allows you to access internal DCT directly. So we call this mode direct DCT.

Direct DCT supports both forward DCT(FDCT) and inverse DCT(IDCT). To select forward or inverse, set DT flag in **MP4DCTMODE**.

For effective usage, you must understand internal 3 buffer configuration. MPEG4DEC have 3 buffers for IDCT. These are for writing, reading and IDCT. These buffers can be read or written during the other buffer doing IDCT. To change the path, set BTYPE flag in **MP4DCTMODE**.

See next figures for understanding.

Buffer BTYPE	Buffer A	Buffer B	Buffer C
0	Write	Read	IDCT
1	IDCT	Write	Read
2	Read	IDCT	Write

Table. Buffer configuration as BTYPE changes

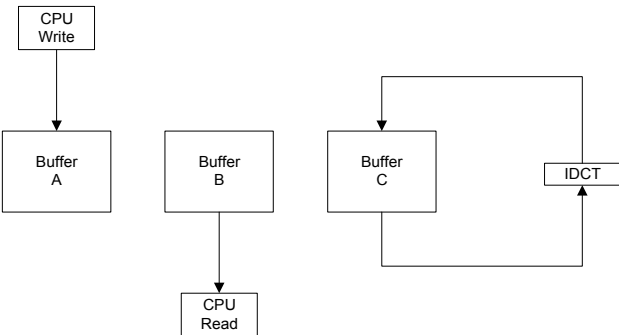


Figure. When BTYPE equals 0

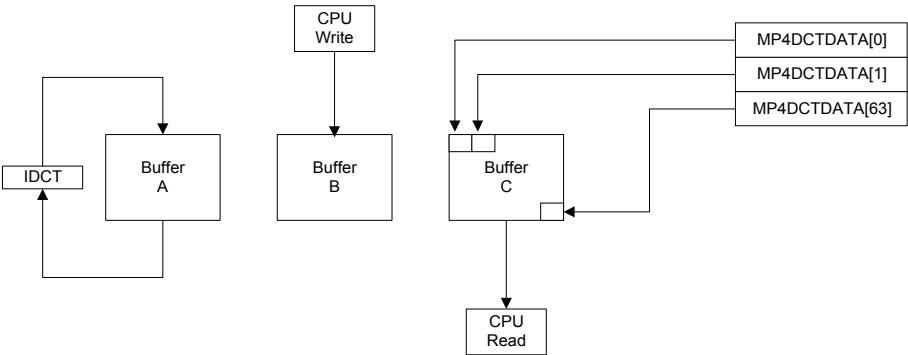


Figure. When BTYPE equals 1

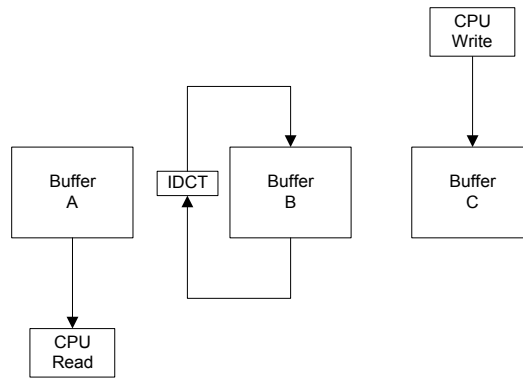


Figure. When BTYPE equals 2

3 buffer structure allows to do IDCT in pipeline.

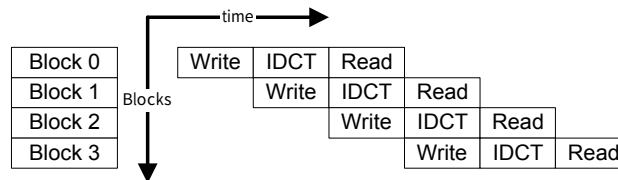


Figure. Possible pipeline

Buffer can be read or written via **MP4DCTDATA**. See example.

```
#define MP4DCTDATA ((volatile signed short*)(0xc0003080))
```

n^{th} component of buffer is **MP4DCTDATA[n]**

One buffer contains 64 components.

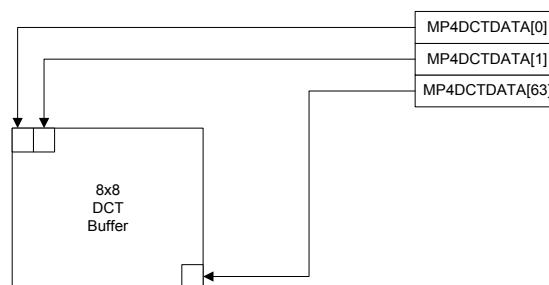


Figure. DCT buffer index

23.2.13 Direct MP

To more flexible for various codec, MPEG4DEC allows you to access internal Motion Predictor (MP) directly. So we call this mode direct MP

For effective usage, you must understand internal 2 buffer configuration. MPEG4DEC have 2 buffers for MP. These are for reading

and MP. It is possible to read one buffer during the other buffer doing MP. To change the path, set BT flag in **MP4MPMODE**.

If a macro block is not coded, the macro block is skipped. Then set SK flag in **MP4MPMODE**.

Generally one macro block have one motion vector. But MPEG4 Simple Profile supports motion vectors per block. (5 motion vectors for a macro block). This mode is called as 4MV. If MPEG4 stream is coded by 4MV, set M4 flag in **MP4MPMODE**

See next figures for understanding

Buffer BT	Buffer A	Buffer B
0	MP	Read
1	Read	MP

Table. Buffer configuration as BT changes

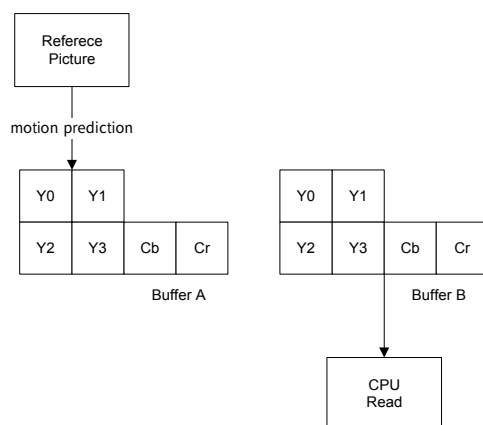


Figure. When BT equals 0

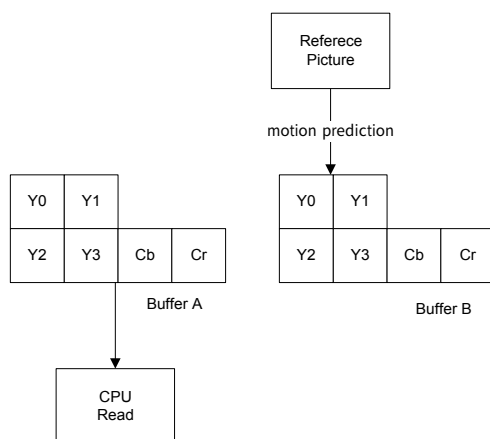


Figure. When BT equals 1

Motion predicted color can be read via **MP4MPDATA**. See example.

```
#define MP4MPDATA ((volatile unsigned short*)(0xc0003100))

nth component of buffer is MP4MPDATA[n]
```

Ignore upper 8 bits of **MP4MPDATA[n]**. It is always filled with 0's.

One buffer contains 384 components

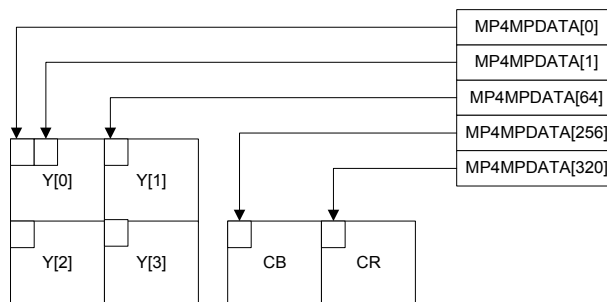


Figure. MP buffer index

Currently decoded macro block position is set to **MP4BLKPOS**.

In 4MV mode, five motion vectors are set to **MP4MVY0**, **MP4MVY1**, **MP4MVY2**, **MP4MVY3**, **MP4MVC**. In non-4MV mode, one motion vector is set to **MP4MVY0**.

23.3 Registers Map (C000_3000h ~ C000_33FFh)

Register Name	offset	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0														
MP4MBSTART	00h	Macro Block Information Stream Start Address																														0	0	0													
	04h	Reserved																																													
MP4RLSTART	08h	Run-Level Stream Start Address																														0	0	0													
	0Ch	Reserved																																													
MP4TEMPBUF	10h	Temporary Buffer Address																														0	0	0													
MP4CURY	14h	Segment										Y Coordinate in Segment										X Coordinate in Segment										0															
MP4CURCB	18h	Segment										Y Coordinate in Segment										X Coordinate in Segment										0															
MP4CURCR	1Ch	Segment										Y Coordinate in Segment										X Coordinate in Segment										0															
MP4REFY	20h	Segment										Y Coordinate in Segment										X Coordinate in Segment										0															
MP4REFCB	24h	Segment										Y Coordinate in Segment										X Coordinate in Segment										0															
MP4REFCR	28h	Segment										Y Coordinate in Segment										X Coordinate in Segment										0															
MP4MBNUM	2Ch																			Number of Macro Blocks – 1																											
MP4RESMISC	30h											S	R	Vertical Resolution – 1										1	Horizontal Resolution – 1								1														
MP4START	34h																													M	P	D	C	P													
MP4STATUS	38h																													M	P	D	C	P													
MP4INTENB	3Ch																													M	P	D	C	P													
MP4INTPND	40h																													M	P	D	C	P													
MP4DIRENB	44h																													M	P	D	C	P													
MP4DCTMODE	48h																															BTYPE		D	T												
MP4MPMODE	4Ch																															B	S	4	M												
MP4BLKPOS	50h											Macro Block Y Position										0	0	0	0									Macro Block X Position										0	0	0	0
MP4MVY0	54h											y component of motion vector																				x component of motion vector															
MP4MVY1	58h											y component of motion vector																				x component of motion vector															
MP4MVY2	5Ch											y component of motion vector																				x component of motion vector															
MP4MVY3	60h											y component of motion vector																				x component of motion vector															
MP4MVC	64h											y component of motion vector																				x component of motion vector															
MP4DCTDATA	80h ~ FFh																			DCT coefficient (64 component)																											
MP4MPDATA	100h ~ 3FFh																									motion predicted color value 6 blocks x 64 component																					

23.3.1 Macro Block Information Stream Start Address Register (MP4MBSTART)

Address : C000 3000h

Bit	R/W	Description	Default Value
31: 0	W	8 byte aligned address [2:0] is ignored (filled with 0's)	

23.3.2 Run-Level Stream Start Address (MP4RLSTART)

Address : C000 3008h

Bit	R/W	Description	Default Value
31: 0	W	8 byte aligned address [2:0] is ignored (filled with 0's)	

23.3.3 Temporary Buffer Address (MP4TEMPBUF)

Address : C000 3010h

Bit	R/W	Description	Default Value
31: 0	W	8 byte aligned address [2:0] is ignored (filled with 0's)	

23.3.4 Current Y-Picture Address/Coordinate Register (MP4CURY)

Address : C000 3014h

Bit	R/W	Description	Default Value
31:21	W	Segment - Segment is 2M byte space.in memory - One segment consists of 2048x1024 pixels (8bpp) - One picture cannot be stored across two segments - maxmum picture size is 2048x1024	
20:11	W	y coordinate y coordinate is ranged 0~1023	
10: 0	W	x coordinate x coordinate is ranged 0~2046 (even number) LSB is ignored (filled with 0)	

23.3.5 Current Cb-Picture Address/Coordinate Register (MP4CURCB)

Address : C000 3018h

Bit	R/W	Description	Default Value
31:0	W	Same as Current Y-Picture Address/Coordinate Register(MP4CURY)	-

23.3.6 Current Cr-Picture Address/Coordinate Register (MP4CURCR)

Address : C000 301Ch

Bit	R/W	Description	Default Value
31:0	W	Same as Current Y-Picture Address/Coordinate Register(MP4CURY)	-

23.3.7 Reference Y-Picture Address/Coordinate Register (MP4REFY)

Address : C000 3020h

Bit	R/W	Description	Default Value
31:0	W	Same as Current Y-Picture Address/Coordinate Register(MP4CURY)	-

23.3.8 Reference Cb-Picture Address/Coordinate Register (MP4REFCB)

Address : C000 3024h

Bit	R/W	Description	Default Value
31:0	W	Same as Current Y-Picture Address/Coordinate Register(MP4CURY)	-

23.3.9 Reference Cr-Picture Address/Coordinate Register (MP4REFCR)

Address : C000 3028h

Bit	R/W	Description	Default Value
31:0	W	Same as Current Y-Picture Address/Coordinate Register(MP4CURY)	-

23.3.10 Macro Blocks Number Register (MP4MBNUM)

Address : : C000 302Ch

Bit	R/W	Description	Default Value
31:13		Reserved	-
12:0	W	Number of Marco Blocks – 1 This value must be (Horizontal Resolution x Vertical Resolution / 256) – 1	-

23.3.11 Resolution and Miscellaneous Register (MP4RESMISC)

Address : : C000 3030h

Bit	R/W	Description	Default Value
31:22		Reserved	-
22	W	Short Header (H.263)	-
21	W	Rounding Control	-
20:11	W	Vertical Resolution – 1 Bit [11] is ignored	-
10:0	W	Horizontal Resolution – 1 Bit [0] is ignored	-

23.3.12 Start Register (MP4START)

Address : C000 3034h

Bit	R/W	Description	Default Value
31:3		Reserved	
2	W	Start MP(motion prediction)	
1	W	Start DCT/IDCT	
0	W	Start MPEG decoding	

23.3.13 Status Register (MP4STATUS)

Address : C000 3038h

Bit	R/W	Description	Default Value
31:2		Reserved	
2	R	MP Busy '1' : Busy '0' : Idle	0
1	R	DCT/IDCT Busy	0
0	R	MPEG decoding Busy	0

23.3.14 Interrupt Enable Register (MP4INTENB)

Address : C000 303Ch

Bit	R/W	Description	Default Value
31:1		Reserved	
2	R/W	MP interrupt enable '1' : Enable '0' : Disable	0
1	R/W	DCT/IDCT interrupt enable	0
0	R/W	MPEG decoding interrupt enable	0

23.3.15 Interrupt Pending Register (MP4INTPND)

Address : C000 3040h

Bit	R/W	Description	Default Value
31:1		Reserved	
2	R/W	MP interrupt pending [R] 0: no interrupt is generated 1: interrupt caused [W] 1: interrupt pending bit clear	0
1	R/W	DCT/IDCT interrupt pending	0
0	R/W	MPEG decoding interrupt pending	0

23.3.16 Direct Enable Register (MP4DIRENB)

Address : C000 3044h

Bit	R/W	Description	Default Value
31:1		Reserved	
2	W	Enable Direct MP	0
1	W	Enable Direct DCT/IDCT	0
0		Reserved	

23.3.17 Direct DCT Mode Register (MP4DCTMODE)

Address : C000 3048h

Bit	R/W	Description	Default Value
31:3		Reserved	
2:1	W	Buffer Type (See Direct DCT)	
0	W	DCT Type '0' : Forward DCT '1' : Inverse DCT	

23.3.18 Direct MP Mode Register (MP4MPMODE)

Address : C000 304Ch

Bit	R/W	Description	Default Value
31:3		Reserved	
2	W	Buffer Type (See Direct MP)	
1	W	Skip Macro Block If this flag is set, motion prediction result is the same as 0 motion vector	
0	W	4MV enable	

23.3.19 Direct MP Macro Block Position Register (MP4BLKPOS)

Address : C000 3050h

Bit	R/W	Description	Default Value
31:26		Reserved	
25:16	W	Macro Block Y Position Bit[19:16] is ignored	
15:11		Reserved.	
10:0	W	Macro Block X Position Bit[3:0] is ignored	

23.3.20 Direct MP Y0 Motion Vector Register (MP4MVY0)

Address : C000 3054h

Bit	R/W	Description	Default Value
31:29		Reserved	
28:16	W	Y component of Motion Vector	
15:13		Reserved	
12:0	W	X component of Motion Vector	

23.3.21 Direct MP Y1 Motion Vector Register (MP4MVY1)

Address : C000 3058h

Bit	R/W	Description	Default Value
-----	-----	-------------	---------------

Bit	R/W	Description	Default Value
31:0	W	same as MP4MVY0	

23.3.22 Direct MP Y2 Motion Vector Register (MP4MVY2)

Address : C000 305Ch

Bit	R/W	Description	Default Value
31:0	W	same as MP4MVY0	

23.3.23 Direct MP Y3 Motion Vector Register (MP4MVY3)

Address : C000 3060h

Bit	R/W	Description	Default Value
31:0	W	same as MP4MVY0	

23.3.24 Direct MP CB,CR Motion Vector Register (MP4MVC)

Address : C000 3064h

Bit	R/W	Description	Default Value
31:0	W	same as MP4MVY0	

23.3.25 Direct DCT Data Access Register (MP4DCTDATA)

Address : C000 3080h ~ C000 30FEh

Bit	R/W	Description	Default Value
15:0	R/W	DCT coefficient data. Caution : This registers are not 32bit but 16bit. C000 3080h : 1 st component of DCT data C000 3082h : 2 nd component of DCT data ... C000 30FEh : 64 nd component of DCT data	

23.3.26 Direct MP Data Access Register (MP4MPDATA)

Address : C000 3100h ~ C000 33FEh

Bit	R/W	Description	Default Value
15:8	R	filled with '0'	
7:0	R	motion predicted color value Caution : This registers are not 32bit but 16bit. C000 3100h : 1 st component of Y[0] Block C000 3180h : 1 st component of Y[1] Block C000 3200h : 1 st component of Y[2] Block C000 3280h : 1 st component of Y[3] Block C000 3300h : 1 st component of CB Block C000 3380h : 1 st component of CR Block C000 33FEh : 64 nd component of MP data (CR Block)	

24. 2D Graphic Controller

24.1 Overview

VR3D has 2D Graphic Controller to accelerate GUI and optimized for graphic base O/S such as WindowsCE or Linux

- 32-deep source image FIFO
- Dual-bank Command Register Set provides efficient interface between CPU and graphics controller.
- Hardware support for font color expansion, font caching and image caching
- Graphic acceleration support for 16 bits per pixel mode.
- Graphic acceleration function include :
 - Rectangular source –copy block transfer (BitBLTs)
 - Transparent source-copy BitBLTs
 - Masked source-copy BitBLTs
 - Monochrome-to-color expansion on source and pattern data
 - Pattern and Rectangle Fills
 - 256 Raster Ops (ROPs)

24.2 Operations

24.2.1 Raster-Ops

A Raster operation (Raster-Op or ROP) is a graphic operation that is performed on a rectangular array of pixels. The 2D Graphic Controller supports 3-operand (source, pattern, destination). And Its operand is capable of supporting all most raster operations as defined for operating system based on GUI.

Below are basically provided ROPs.

- Bit Block-Level Transfer (BitBLT) : move Rectangular region to other area.
- Area Fill : Fill Rectangular area with certain color.
- Pattern Fill : Fill certain Rectangular area with stored image in internal 8x8 Pattern Register.
- Memory-to-Screen Transfer : Copy data from Source image storing memory to display memory.

3 input source needed for operate ROP.

- Source Data : Source Data is located in memory which stores CPU generated image or in frame buffer which is allocated for physical memory. Source data in memory can be transferred through DMA or CPU. If it is physical memory mapped data, 2D Graphics controller can access memory directly. Source Data could be either monochrome or color data. Monochrome source data expands to foreground or background.
- Pattern Data : Pattern Data is stored pattern register in 2D Graphics controller. The pattern Size of pattern register is 8x8. Pattern data could be either monochrome or color data. Monochrome source data expands to foreground or background.
- Destination Data : Destination Data must be stored in Display memory.

24.2.2 Transparency

Data through ROPs is designated as transparent. If the color is same as transparency color of 2D Graphics controller, it is not stored in memory. Transparency color function can be Enable/Disable by CPU.

24.2.3 Monochrome to Color Expansion

2D Graphic Controller support 8bit or 16bit per pixel. If Source Image or Pattern Image is Monochrome(1BPP), Data is changed according to destination color format and ROP is performs. 2D Graphic Controller operates color expansion according to foreground and background color Register value in the Controller. If monochrome source image is font data, it can be used as font expansion in some O/S.

24.2.4 Fraction Control

2D Graphic controller performs 32bit pixel processing internally regardless color format of Source / Destination / Patter Image. Maximum performance can be maintained regardless color format through this function.

If image is not aligned as word(32bit), Fraction parameter should be set. Fraction parameter defines offset value of image by 32bit units.

Fraction parameter consists of 5bits and unit is bit.

For example,

When color format of Destination Image is 16bpp. Fraction parameter according to X coordinates of destination:

$$\text{Fraction Parameter} = (x\%2)*16$$

When color format of Source Image is 1bpp. Fraction parameter according to X coordinates of Destination:

$$\text{Fraction Parameter} = (x\%32)*1$$

24.2.5 FIFO Control

If source image data is stored in virtual address and/or cache of CPU, CPU should transfer source image data to source FIFO register of 2D graphics controller to operate ROPs.

Transferred data should be units of word(32bit). The number of transferred data is defined by source image color format, coordinates of X axis and length of X axis.

Below is source FIFO transfer routine.

```
GRP2D_UTIL_PushFIFO(
    GRP2D_COLORFMT    ColorFormat,          // [in] Color Format.
    S32                Stride,                // [in] number of byte per line of Source region
    U32                Width, U32 Height,     // [in] Width and Height
    U32*               pVirAddr32,           // [in] Virtual Address of Image
    U32                Frac,                 // [in] Define Fraction value. (unit:: bit, 0~31)
    GRP2D_DIR          XDir                  // [in] Direction of x coordinates
)
{
    U32    x, y, NumOfWord, BitNum;
    U32*   pEdgeAddr;
    U32    *pReadAddr;

    if (XDir==GRP2D_DIR_POSITIVE )          BitNum= (int)Frac & 0x1f;
    else {
        switch(ColorFormat) {
            case GRP2D_COLORFMT_1BPP:      BitNum= ((S32)Frac-(Width-1)) & 0x1f; break;
            case GRP2D_COLORFMT_2BPP:      BitNum= ((S32)Frac-(Width-1)*8) & 0x1f;      break;
            case GRP2D_COLORFMT_8BPP:      BitNum= ((S32)Frac-(Width-1)*16) & 0x1f;      break;
            default:                        MES_NEVER_GET_HERE();                      break;
        }
    }

    switch(ColorFormat) {
        case GRP2D_COLORFMT_1BPP:          NumOfWord= (Width+31+BitNum)/32; break;
        case GRP2D_COLORFMT_2BPP:          NumOfWord= (Width*2+31+BitNum)/32;      break;
        case GRP2D_COLORFMT_8BPP:          NumOfWord= (Width*8+31+BitNum)/32;      break;
        case GRP2D_COLORFMT_16BPP:         NumOfWord= (Width*16+31+BitNum)/32;     break;
        default:                            MES_NEVER_GET_HERE();                  break;
    }

    pEdgeAddr = pVirAddr32;

    // Dummy Writing
    NumOfWord++;

    for (y=0; y<Height; y++) {
        pReadAddr = (U32*)((U32)pEdgeAddr & 0xfffffc);
        if (XDir==GRP2D_DIR_POSITIVE) {
            for (x=0; x<NumOfWord; x++) {
                GRP2D.SRCFIFO= *pReadAddr;
                pReadAddr++;
            }
        } else {
            for (x=0; x<NumOfWord; x++) {
                GRP2D.SRCFIFO= *pReadAddr;
                pReadAddr--;
            }
        }
        pEdgeAddr += Stride;
    }
}
```

24.2.6 Dual Register Set

Controller register of 2D graphics controller consists of 2 banks. 1 bank contains current working command and the other is for receiving new command.

From this register, CPU can set new command to register when graphic controller is rendering. It will upgrade performance by avoiding wait stage of CPU even during graphics controller's working. New command in register will not affect graphics controller before input of run command.

24.2.7 FIFO Flush

When 2D Graphic completes the bit block transfer, The several last pixel are not drawn to destination buffer because of not flushing out data of output fifo. These pixel is drawn when next draw command begin to be execute. If you want that last pixel is written to buffer immediately, you should execute the dummy draw command not affecting operation.

24.3 2D Graphic Controller Register

24.3.1 Destination Control Register (MESGDSTCTRL)

Address : E000 4000h

Bit	R/W	Description	Default Value
31 : 7	R	Reserved	-
6	R/W	DSTENB : Perform ROP operation reading Destination area 0 : Disable 1 : Read Operation	0
5	R/W	DSTBPP : Set Color Format of Destination area 0 : 8bpp 1 : 16bpp	0
4 : 0	R/W	DSTFRAC : Fraction Parameter : bit unit	0

Set destination data.

DSTENB defines whether referring destination value or not when ROPs operation. It is useful to minimize destination memory access If destination color value is not valid according to ROP code.

24.3.2 Destination Address Register (MESGDSTADDR)

Address : E000 4004h

Bit	R/W	Description	Default Value
31 : 0	R/W	Set Base Address of Destination Image	0h

24.3.3 Destination Stride Register (MESGDSTSTRIDE)

Address : E000 4008h

Bit	R/W	Description	Default Value
31 : 0	R/W	Set Address increment ratio following Y coordinates increment in Destination area	0h

24.3.4 Source Control Register (MESGSRCTRL)

Address : E000 400Ch

Bit	R/W	Description	Default Value
31 : 8	R	Reserved	-
8	R/W	INVIDEO : Set Source Image location 0 : Controlled by CPU 1 : Frame Buffer	
7	R/W	SRCENB : Set ROP operation reading Source Image area 0 : Disable 1 : Read Operation	
6 : 5	R/W	SRCBPP : Set Color Format of Source Image area	0

Bit	R/W	Description	Default Value
		00 : 8bpp 10 : 1bpp 01 : 16bpp	
4 : 0	R/W	SRCFRAC : Unit bit	0

INVIDEO bit defines source image storing memory. If it exists in frame buffer, 2D graphics controller read source image automatically and performs ROPs. If it exists in system memory which controlled by CPU, CPU should write image into source FIFO when performs ROPs

24.3.5 Source Address Register (MESGSRCADDR)

Address : E000 4010h

Bit	R/W	Description	Default Value
31 : 0	R/W	Set Base Address of Source area	0h

24.3.6 Source Stride Register (MESGSRCSTRIDE)

Address : E000 4014h

Bit	R/W	Description	Default Value
31 : 0	R/W	Set Address increment ratio following Y coordinates increment in Source area	0h

24.3.7 Source Foreground Color Register (MESGSRCFORCOLOR)

Address : E000 4018h

Bit	R/W	Description	Default Value
31 : 16	R	Reserved	-
15 : 0	R/W	MESGSRCFORCOLOR : RGB5:6:5 Format or Indexed Color Valid when Source Image Color Format is 1BPP	0

Performs monochrome color expansion function. If Destination color is 8bpp, Palette index value should be set.

24.3.8 Source Background Color Register (MESGSRCBACKCOLOR)

Address : E000 401Ch

Bit	R/W	Description	Default Value
31 : 16	R	Reserved	-
15 : 0	R/W	MESGSRCBACKCOLOR : RGB5:6:5 Format Valid when Source Image Color Format is 1BPP	0

Performs monochrome color expansion function. If Destination color is 8bpp, Palette index value should be set..

24.3.9 Pattern Control Register (MESGPATCTRL)

Address : E000 4020h

Bit	R/W	Description	Default Value
31 : 7	R	Reserved	-
6	R/W	PATMONO : Set Patten type 0 : Color 1 : Mono	0
5	R/W	PATENB : Set Pattern Image Usage 0 : Disable 1 : Enable	0
4 : 3	R/W	PATBPP : Set Color Format of Pattern Image area	0

Bit	R/W	Description	Default Value
		00 : 8bpp 10 : 1bpp 01 : 16bpp	
2 : 0	R/W	YOFFSET : Set Y Offset of Pattern Unit : Pixel	0

PATMONO is set when fill rectangular region with color set by MESGFORCOLOR Register.

YOFFSET sets offset value of Y coordinates when fill pattern image in destination area.

24.3.10 Pattern Foreground Color Register (MESGFORCOLOR)

Address : E000 4024h

Bit	R/W	Description	Default Value
31 : 16	R	Reserved	-
15 : 0	R/W	MESGFORCOLOR : RGB5:6:5 Format Valid when Pattern Image Color Format is 1BPP	0

24.3.11 Pattern Background Color Register (MESGBACKCOLOR)

Address : E000 4028h

Bit	R/W	Description	Default Value
31 : 16	R	Reserved	-
15 : 0	R/W	MESGBACKCOLOR : RGB5:6:5 Format Valid when Pattern Image Color Format is 1BPP	0

24.3.12 Size Register (MESGSIZE)

Address : E000 402Ch

Bit	R/W	Description	Default Value
31 : 27	R	Reserved	-
26 : 16	R/W	HEIGHT : Set height. Unit Pixel.	0
10 : 0	R/W	WIDTH : Set width. Unit Pixel.	0

Set length of X and Y axis of destination area. Each length should be set to meet inside display boundary..

24.3.13 Control Register (MESGCTRL)

Address : E000 4030h

Bit	R/W	Description	Default Value
31 : 16	R/W	TRANSPARENCYCOLOR : Set transparency Color	-
15 : 12	R	Reserved	0
11	R/W	TRANSPARENCYENB : Determine to use transparency Color check 0 : Disable 1 : Enable	0
10	R/W	FFCLR : Clear Source Input FIFO	0
9	R/W	YDIR : Set Y coordinates Fill direction 0 : Negative 1 : Positive	0
8	R/W	XDIR : Set X coordinates Fill direction 0 : Negative 1 : Positive	0
7 : 0	R/W	OPCODE : ROP3 This defines the 8-bit raster operation (ROP). All 256 possible ROPs are supported.	0

TRANSPARENCYENB decide whether updates destination area value when the result of ROPs has same value as TRANSPARENCYCOLOR value.

FFCLR initializes FIFO to transfer source image.

XDIR and YDIR define direction for updating rectangular area. If YDIR is positive, rectangular area will be updated from top to bottom and if it is negative, rectangular area will be updated from bottom to top. With this function, display scroll function can be supported in various direction and it also can be use as image update of BMP

24.3.14 Status Register (MESGSTATUS)

Address : E000 4034h

Bit	R/W	Description	Default Value
31 : 1	R	Reserved	-
0	R/W	RUN/BUSY : Read operation status of 2D Graphic Engine 0 : Idle 1 : Busy	0

If RUN/BUSY set to 1, 2D graphics controller performs according to parameter register setting. CPU could check 2D graphics controller operating status by checking this bit.

24.3.15 FIFO Status Register (MESGFIFOSTATUS)

Address : E000 4038h

Bit	R/W	Description	Default Value
31	R	FULL : Check Source FIFO status 0 : Not Full 1 : Full	0
30 : 5	R	Reserved	0
4 : 0	R/W	REMAIN : Remain word of Source FIFO.	0

This register has status of FIFO when transferring image using source FIFO. FULL bit shows whether FIFO is full or not while REMAIN bit shows remaining area of source FIFO. The depth of FIFO is 32.

If CPU updates FIFO when full, 2D graphics controller automatically hold CPU and CPU performs next operation after securing FIFO area. To avoid CPU hold, check FIFO status before using FIFO when using source FIFO. Generally speed of 2D graphics controller taking data from FIFO is faster than CPU writing data to FIFO but care must be taken because it can be different according to memory controller busy status

24.3.16 Source FIFO Register (MESGFIFO)

Address : E000 403Ch

Bit	R/W	Description	Default Value
31 : 0	W	Source FIFO Data.	-

24.3.17 Pattern Register (MESGPAT)

Address : E000 4080h ~ E000 40FCh

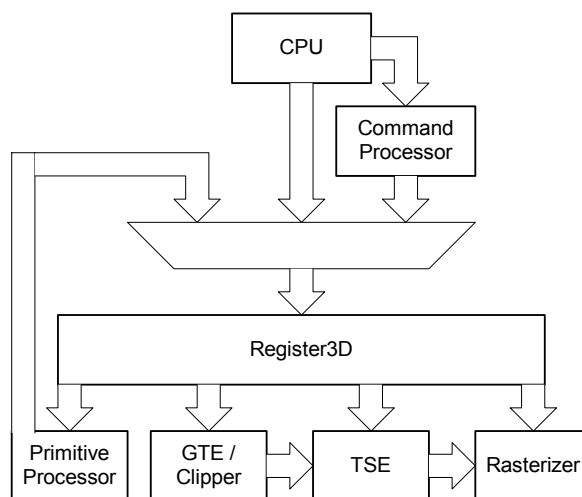
Bit	R/W	Description	Default Value
31 : 0	W	Pattern Value	-

It stores pattern image. It stores 8x8 pattern image regardless color format. Pattern Image should be stored from offset 0

25. 3D Graphic Engine

25.1 Overview

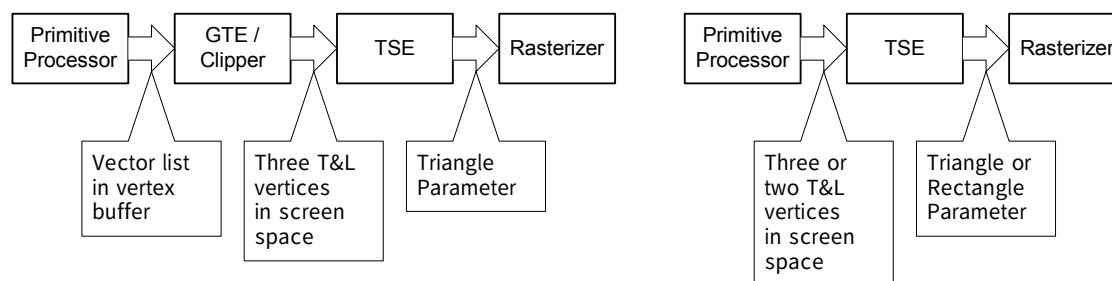
3D core of VRender3D consists of several functional blocks (sub-module).



< 3D Core control flow >

CPU controls all operation of Sub-module or CPU can request Command processor to control them.

Below picture shows data flow of primitive rendering using vertex buffer. Left side is for hardware vertex transform/view volume clipping and right side is for using Transformed & Lit vertex (in screen space).



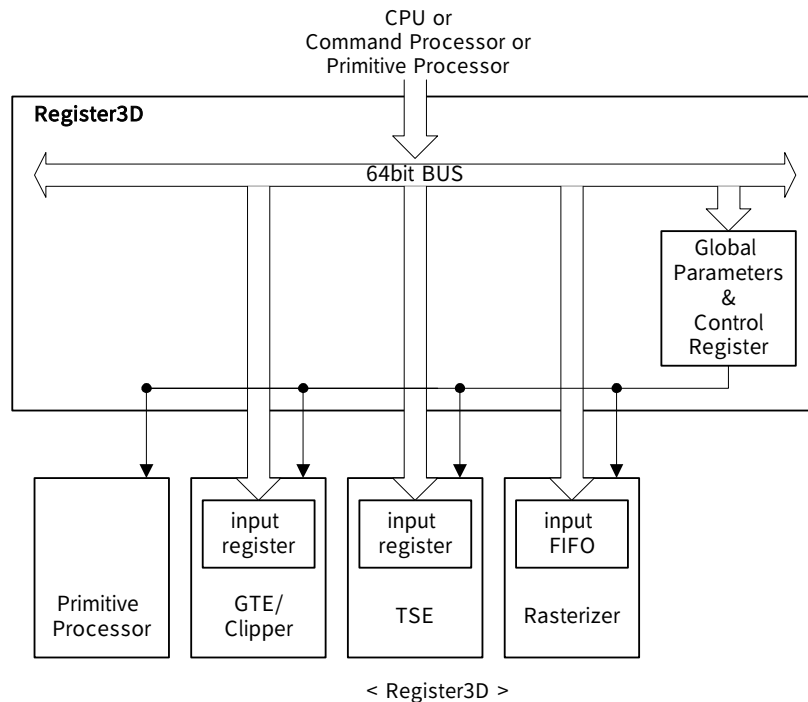
< 3D Core data flow >

Sub-modules, related primitive rendering, operate with parameters and control signals from Register3D. As special purpose DAM [Command Processor](#), which could write 3D core register, also operate 3D core.

25.1.1 Register3D

Register3D stores parameter for Sub-module and sends operation signal to Sub-module when external device writes in control register (**GRP3D_CONTROL**). [Command Processor](#) and [Primitive Processor](#) can control sub-modules through Register3D.

Access priority is Primitive Processor > Command Processor > CPU

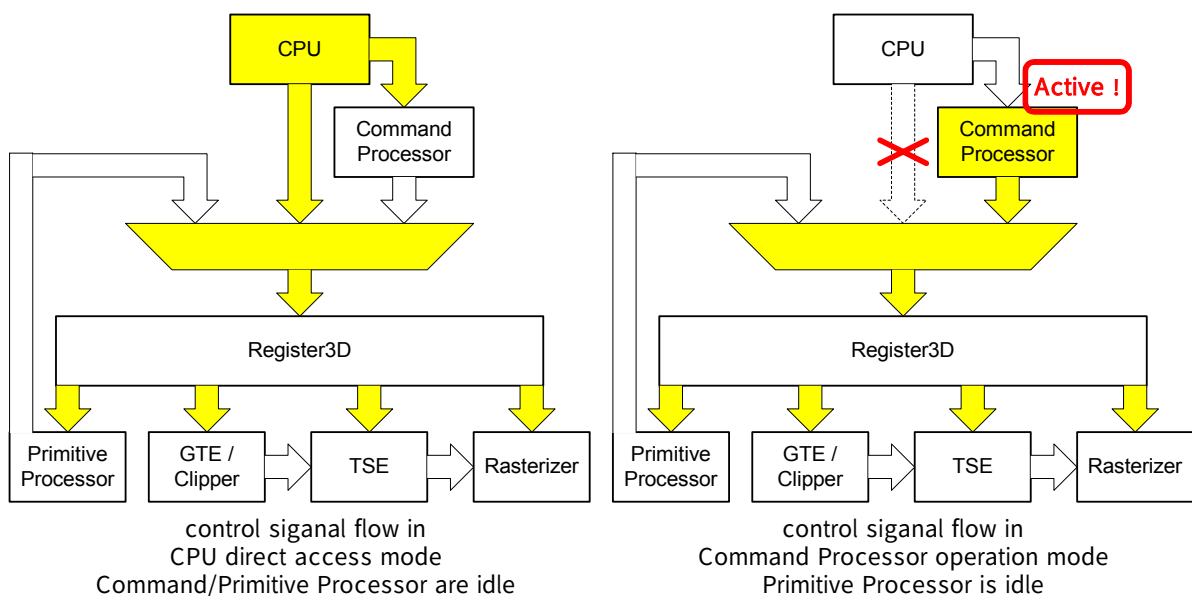


25.1.2 Command Processor

Command Processor module read Command buffer and operate sub-module. It is special purpose DMA, which reads data from memory and copy to Register3D. It can operate Sub-module or generate interrupt after copy operation or it can be wait stage(flow control) after checking sub-module or vertical retrace status before copy operation. Command Processor can operate all 3D operation independently-without interference of CPU.

Command buffer (or command queue) is memory area, which stores command. It contains command, which will be copied to Register3D. It also stores other additional information. CPU assigns command buffer area (refer to [Command buffer](#) section)

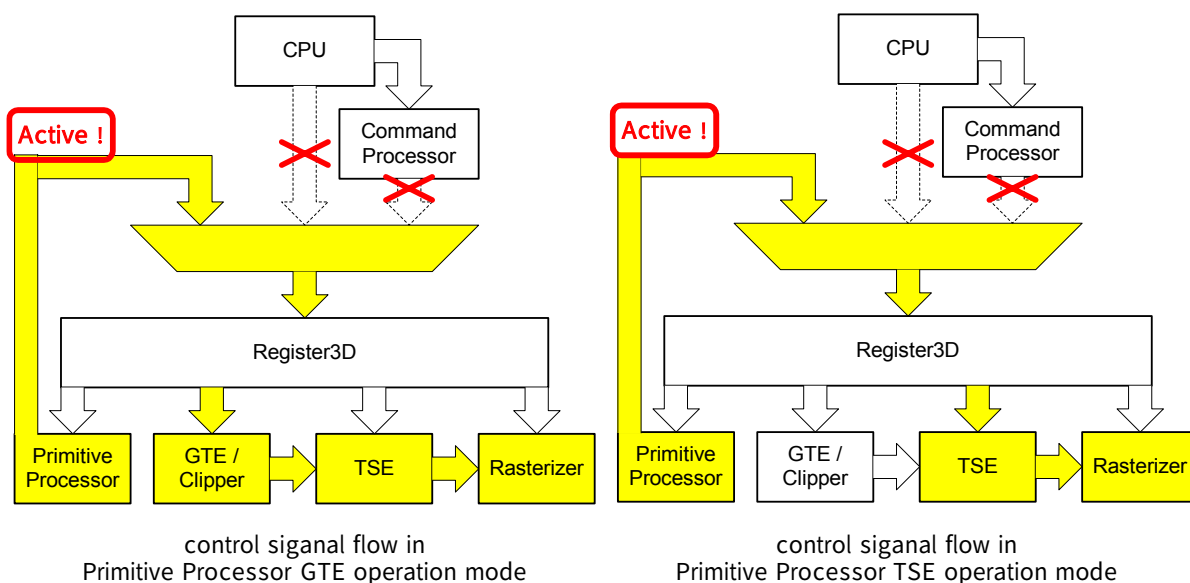
CPU controls Command Processor directly without Register3D. For Register3D, Command Processor has priority to CPU. CPU cannot access Register3D when Command Processor access Register3D.



25.1.3 Primitive Processor

Primitive Processor drive GTE/TSE to render sequence of geometric sequence. Basic operation of Primitive Processor is loading vertex data from memory to input register of [GTE](#)(or [TSE](#)) and drive GTE(or TSE). [Vertex buffer](#) is memory storing vertex data and [Index buffer](#) is memory storing Index data.

Primitive Processor has priority for Register3D to CPU or Command Processor. CPU or Command Processor cannot access Register3D when Command Processor access Register3D.

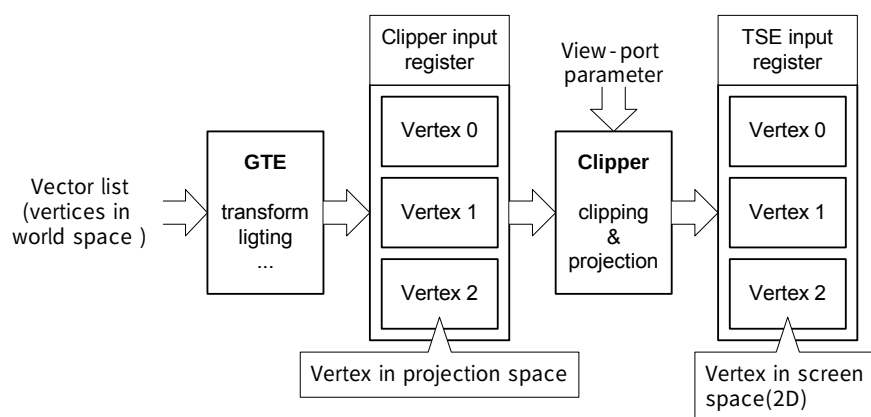


25.1.4 GTE/Clipper (Geometry Transformation Engine/ triangle Clipper)

GTE/Clipper makes triangle from Vertex data and operates clipping/projection. After Register3D input vector list in GTE Input register, GTE program calculate vector list and deliver it to input register of Clipper.

GTE is a vector processing unit, which can handle matrix calculation and vector calculation. GTE also can handle coordinates, color and texture coordinates calculation. GTE can be used as Floating point Coprocessor because CPU can read the result of calculation.

Clipper operates clipping and projection of triangle following request from GTE and request rendering to TSE.



< GTE/Clipper data flow >

25.1.5 TSE (Triangle Setup Engine)

TSE convert vertex data in Input Register to fit into [Rasterizer](#). It calculates several calculation for Rasterizer drawing triangle/rectangle using internal three vertex input register. TSE is controlled from Register3D or GTE/Clipper.

25.1.6 Rasterizer

Rasterizer has below two operations.

- Render triangle or rectangle to TSE output
Shading/Texture mapping/ fog blending/ alpha blending/ z-buffering/...
- Memory fill
Fill certain memory area(rectangle) with certain value.
Fast z-buffer/screen/texture clear

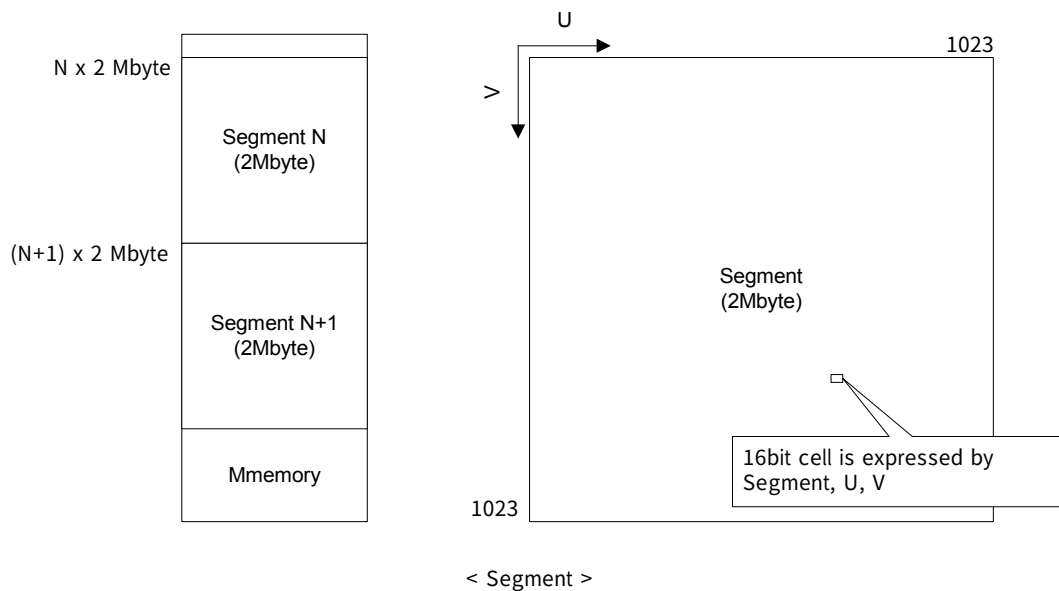
Rasterizer is controlled from Register3D and TSE. Rasterizer has 1-depth FIFO to input next parameter during operation.

25.2 Memory structure for VRender3D

VRender3D applies Non-linear addressing for a few resources for maximum performance. Below are terminology description for VRender3D memory addressing

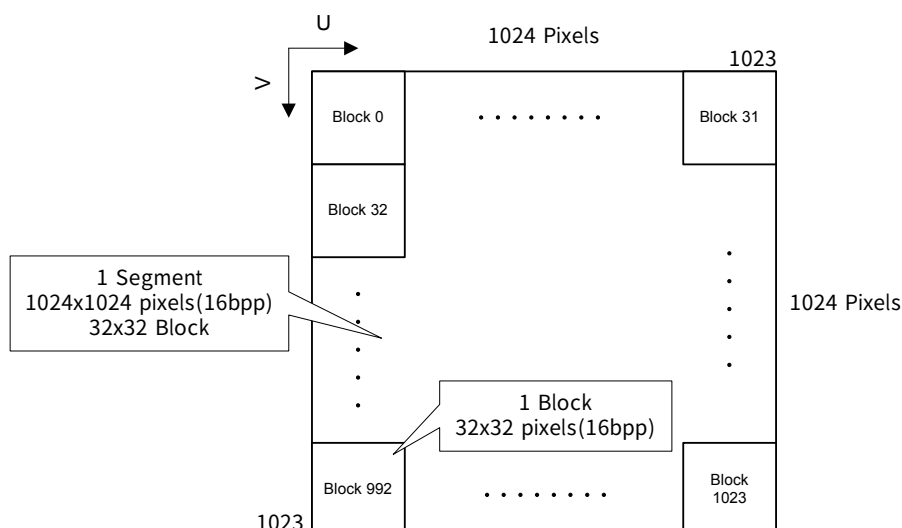
25.2.1 Segment

Segment means 2Mbyte memory area whose starting address is multiple of 2Mbyte. Segment has 2 dimension architecture. (1024x1024x16bit). Resource addresses using Non-linear addressing are assigned as form of segment and offset(U,V).



25.2.2 Block

To reduce memory latency, VRender3D use Block Memory address instead of linear address. One Segment consists of 32x32 Blocks and one Block consists of 32x32x16bit(2Kbyte).



31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Segment [10:0]											Block index in segment [9:0]										Block address [10:0]										

< 32bit address described in segment and block >

25.2.3 Block addressing Mode

25.2.3.1 Display mode

Display mode is addressing mode, which directly display to LCD/CRT. Linear addressing is used in Block.

16bit cell address is calculated as below in Display addressing mode.

U,V is 16bpp pixel coordinate in segment ($0 \leq U \leq 1023$, $0 \leq V \leq 1023$)

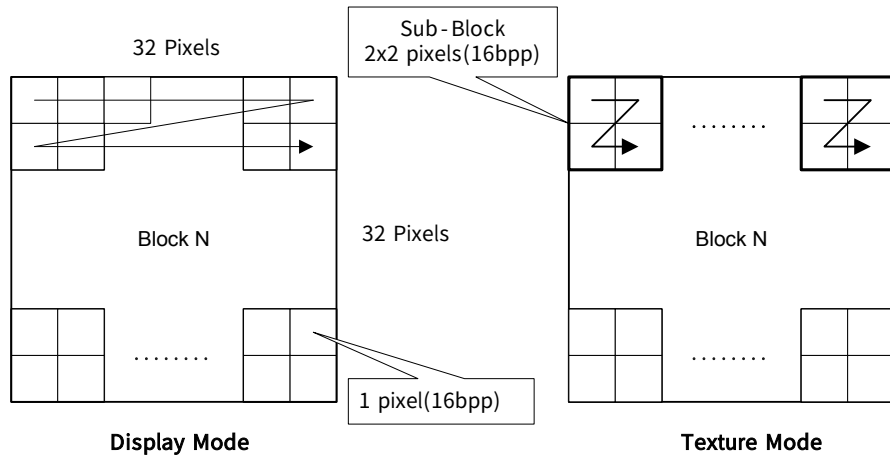
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Segment [10:0]											V[9:5]					U[9:5]					V[4:0]					U[4:0]					0
											Block index in segment [9:0]										Block address [10:0]										

25.2.3.2 Texture mode

Texture mode is addressing mode which could be used as Texture/Palette. It divides Block into 16x16 sub-block of 2x2x16bit. 16bit cell address is calculated as below in Texture addressing mode.

U,V is 16bpp pixel coordinate in segment ($0 \leq U \leq 1023$, $0 \leq V \leq 1023$)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Segment [10:0]											V[9:5]					U[9:5]					V[4:1]				U[4:1]				V[0]	U[0]	0
											Block index in segment [9:0]										Block address [10:0]										



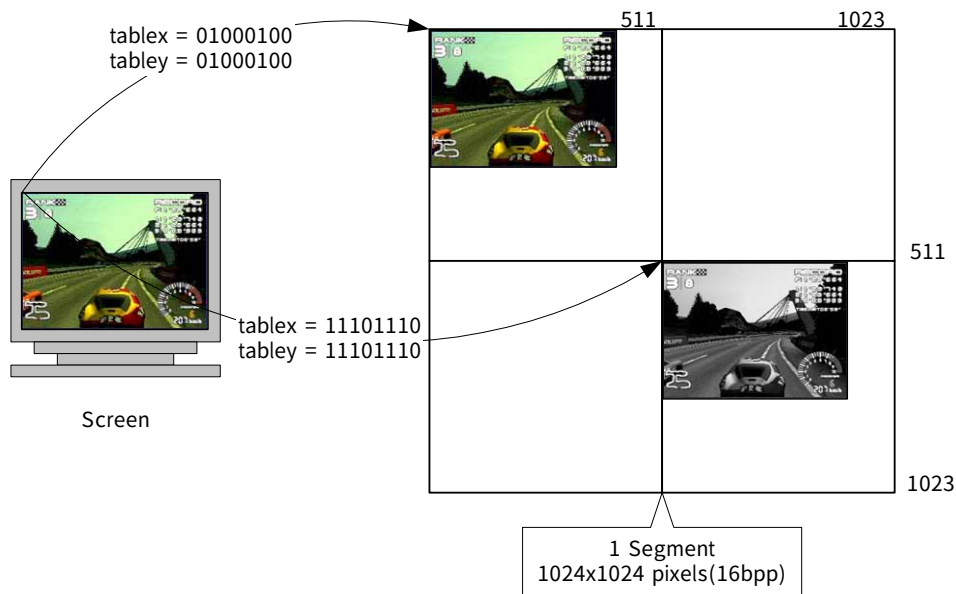
25.2.4 Address converting table

Z-buffer / Display buffer/ Render target are assigned as segment. It needs converting work to mapping screen to segment. Address converting table is 8bit table to convert screen address(x,y,16bit pixel coordinate) to segment address(U,V).

```

If xy[9:8] == 0, UV[9:0] = { table[1:0] , xy[7:0] }
If xy[9:8] == 1, UV[9:0] = { table[3:2] , xy[7:0] }
If xy[9:8] == 2, UV[9:0] = { table[5:4] , xy[7:0] }
If xy[9:8] == 3, UV[9:0] = { table[7:6] , xy[7:0] }

```



< Address converting (Screen to Segment) >

25.2.5 Resources using Non-linear addressing

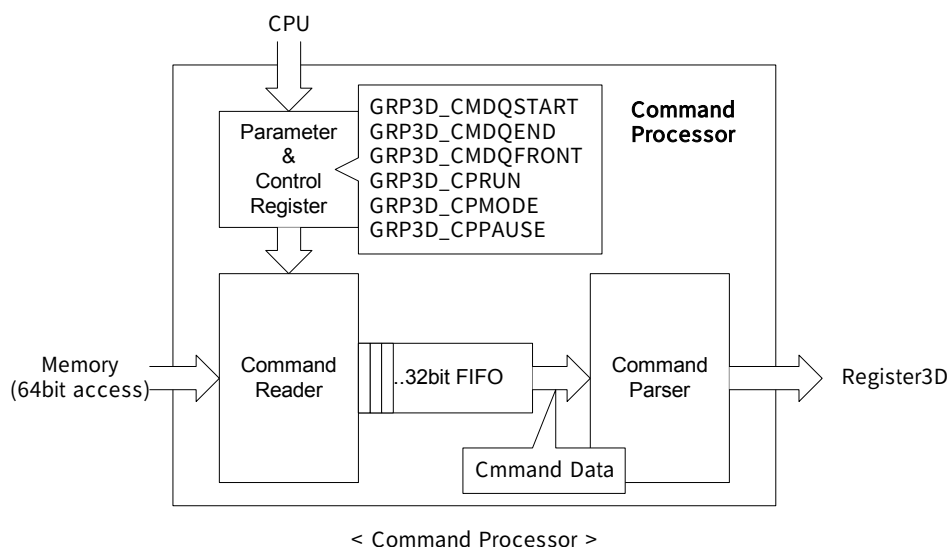
Block addressing mode	Directly displayed to LCD/CRT	Used as Texture	Resources
Display addressing mode	O	X	Render target Depth buffer

Block addressing mode	Directly displayed to LCD/CRT	Used as Texture	Resources
Texture addressing mode	X	O	Render target Texture Palette

25.3 Command Processor

Command Processor operates 3D Core in place of CPU. CPU makes command list and deliver it to Command Processor, Command Processor operate 3D operation independent of CPU.

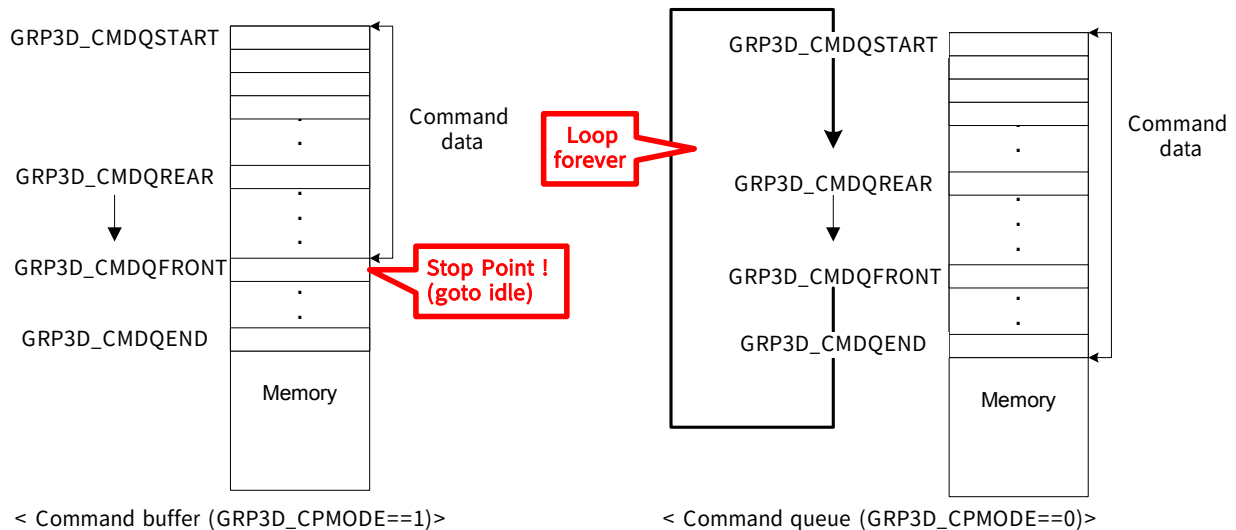
Command Processor has two parts. One is Command Reader, which reads data from memory while the other is Command Parser, which manages data.



25.3.1 Command buffer/queue

Command data is 32bit data delivered to Command Parser. The memory contains Command data is Command buffer or command queue. Command buffer and Command queue are named by Command Processor operation mode (**GRP3D_CPMODE**). Command buffer/queue can be located freely in memory.

Below picture shows the concept and the difference of Command buffer and Command queue.



< Command buffer (GRP3D_CPMODE==1)>

< Command queue (GRP3D_CPMODE==0)>

Command buffer/queue is memory area, which assigned by three register – **GRP3D_CMDQSTART/GRP3D_CMDQEND/GRP3D_CMDQFRONT**. Each register indicates memory starting point, end point and end of valid data. They must satisfy below condition.

$$GRP3D_CMDQSTART \leq GRP3D_CMDQFRONT \leq GRP3D_CMDQEND$$

GRP3D_CMDQREAR is a Read only register indicating read address and automatically increase or initialize by Command Reader. Basic operation of Command Reader is reading memory until **GRP3D_CMDQREAR = GRP3D_CMDQFRONT**.

Command Processor Mode (GRP3D_CPMODE)	Description
Command buffer mode (GRP3D_CPMODE==1)	Memory content operates one time and Command Processor stops. Command Reader goes idle stage after reading memory area, which is assigned by GRP3D_CMDQSTART and GRP3D_CMDQFRONT . (GRP3D_CMDQSTART ≤ address < GRP3D_CMDQFRONT)
Command queue mode (GRP3D_CPMODE==0)	Command Reader reads memory area, which assigned by GRP3D_CMDQSTART and GRP3D_CMDQEND . (GRP3D_CMDQSTART ≤ address ≤ GRP3D_CMDQEND) If GRP3D_CMDQREAR is same as GRP3D_CMDQFRONT , wait for GRP3D_CMDQFRONT increasing. Use Command buffer as infinity memory area through writing new data at the already read.

25.3.1.1 Command

Command is basic unit of Command buffer/queue. One Command consists of several Command data. Command header as first Command data has command ID, data and address for Register3D and other information.

The number of Command is 12 and each Command is distinguished by first 4bit.

CMDID	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-------	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	---	---	---	---	---	---	---	---	---	---

Name	CMDID	Target register	CMDSIZE (32bit)	Command type
CMDID_CONTROL	4	GRP3D_CONTROL	1	Optimized command Write Command into Register3D. It is very efficient way because it does not have extra header compare to Register fill command(CMDID_ REGFILL). Optimized command waits for sub-module idle before writing but does not wait for Vertical retrace.(fixed flow control) After writing, it automatically operate sub-module following Command.
CMDID_RENDERSTATE	5	GRP3D_RENDERSTATE	1	
CMDID_ALPHABLEND	6	GRP3D_ALPHABLEND	1	
CMDID_TEXSEGMENT	7	GRP3D_TEXSEGMENT	1	
CMDID_MAPPARAM	8	GRP3D_TPCOLOR GRP3D_MIPMAPBIAS	2	
CMDID_LUTFILL	9	GRP3D_LUTPARAM	1	

Name	CMDID	Target register	CMDSIZE (32bit)	Command type
CMDID_TEXINFO0	10	<i>GRP3D_TEXINFO0_0</i> <i>GRP3D_TEXINFO0_1</i>	2	CMDID_TEXSEGMENT : Rasterizer(texture cache clear) CMDID_LUTFILL : Rasterizer (LUT fill) CMDID_PRIMITIVE : Primitive Processor
CMDID_TEXINFO1	11	<i>GRP3D_TEXINFO1_0</i> <i>GRP3D_TEXINFO1_1</i>	2	
CMDID_TEXBLEND0	12	<i>GRP3D_TEXBLEND0</i>	1	
CMDID_TEXBLEND1	13	<i>GRP3D_TEXBLEND1</i>	1	
CMDID_PRIMITIVE	14	<i>GRP3D_PRIMCON</i> <i>GRP3D_PRIMIDX</i> <i>GRP3D_PRIMVTX0</i> <i>GRP3D_PRIMVTX1</i> <i>GRP3D_PRIMVTX2</i> <i>GRP3D_PRIMVTX3</i>	6	
CMDID_REGFILL	15	-	Number of 32bit data + 1	Register fill command It operates flow control following Command header, writing following data into Register3D and invoke interrupt. Command header includes data to write in Register3D, register address and flow-control flag for Command processor's flow control. (synchronous page flipping uses this command) All operation of 3D Core can be handled with this command but it could be not efficient for small data because every command has 32bit header.

25.3.1.2 Register fill command (CMDID_ REGFILL)

It operates flow control to follow Command header and copy following command data to Register3D.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1				DATACOUNT32[8:0]									①	②	③	④	⑤	⑥		REGADDR32[8:0]								
DATA[0]																															
DATA [1]																															
...																															
DATA [DATACOUNT32]																															

DATACOUNT32

Data count

Number of 32bit data = DATACOUNT32 + 1

① NOWAIT_RAST

No-wait check flag for Rasterizer. if this flag is 0, Command processor wait for Rasterizer idle.

② NOWAIT_TSE

No-wait check flag for TSE. if this flag is 0, Command processor wait for TSE idle.

③ NOWAIT_GTE

No-wait check flag for GTE. if this flag is 0, Command processor wait for GTE idle.

④ WAIT_SYNC

Wait check flag for Vertical retrace. if this flag is 1, Command processor wait for vertical retrace.

⑤ IGNOREFIELD

Ignore field flag for Interlace mode. if display controller is not Interlace mode, this bit is ignored.

0 : vertical retrace happens only in odd field.

1 : vertical retrace happens regardless of field.

⑥ INTERRUPT

When *GRP3D_REGFILLINTR*=1, Request interrupt after copy.

REGADDR32

Register address = E000_0000h + (REGADDR32 * 4)
 REGADDR32 must be greater than 7

DATA[0] ~ DATA[DATACOUNT32]

32bit Data list.

Command processor operation

1. wait sub-module idle (NOWAIT_RAST, NOWAIT_TSE, NOWAIT_GTE)
2. wait vertical retrace (WAIT_SYNC, IGNOREFIELD)
3. Register copy
 for(i=0; i < DATACOUNT32+1; i++) (unsigned long*)(E000_0000h)[REGADDR32 + i] = DATA[i];
4. If(**GRP3D_REGFILLINTR**) Interrupt request

25.3.1.3 Optimized command

Wait for all Sub-module go idle stage, copy Command to Register3D.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																												
CMDID				DATA[0]																																																							
DATA[1]																																																											
...																																																											

CMDID

Command ID

4<= CMDID <=14

DATA[0] ~

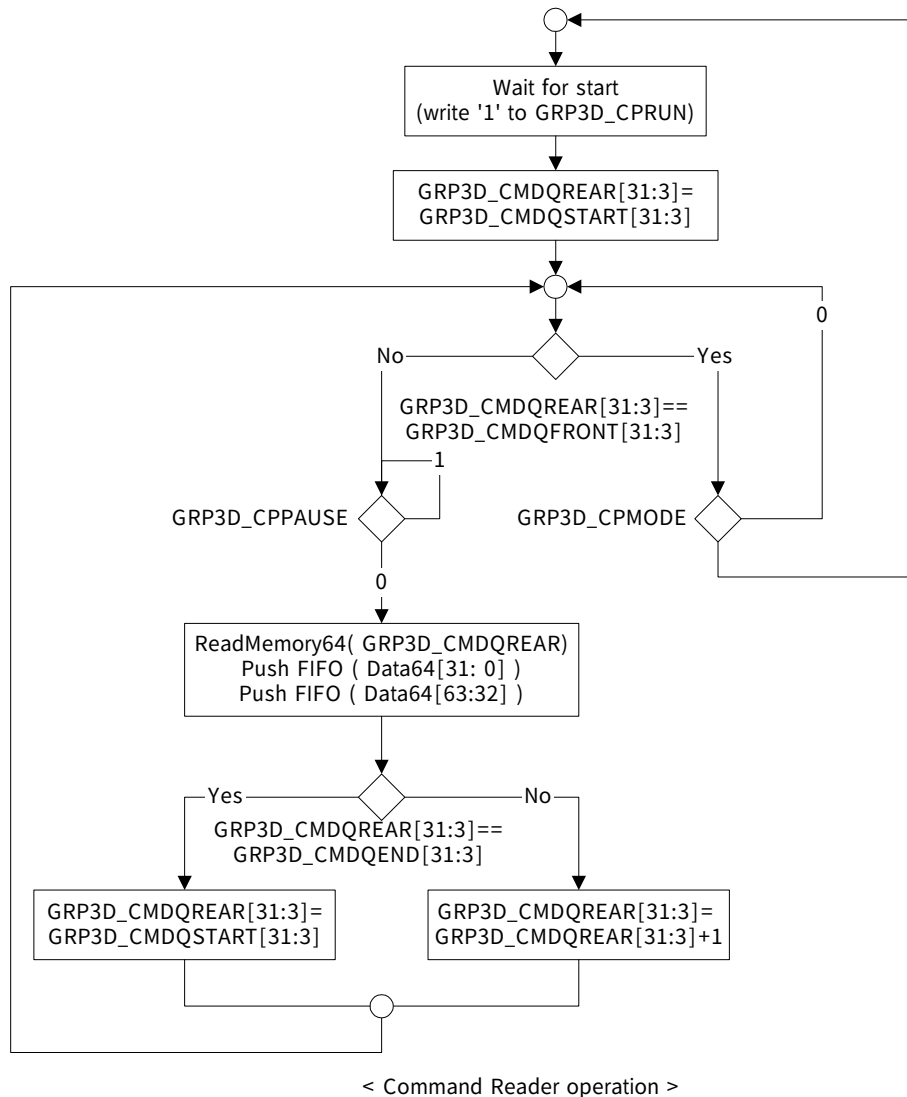
data list which will be copied to Register3D

Command processor operation

1. wait sub-module idle
2. Register copy
 for(i=0; i < CMDSIZE; i++) (unsigned long*)(E000_0000h)[CMDID*2+ i] = DATA[i];
3. Run sub-module
 if(CMDID == CMDID_TEXSEGMENT) Rasterizer(texture cache clear)
 if(CMDID == CMDID_LUTFILL) Rasterizer (LUT fill)
 if(CMDID == CMDID_PRIMITIVE) Primitive Processor

25.3.2 Command Reader

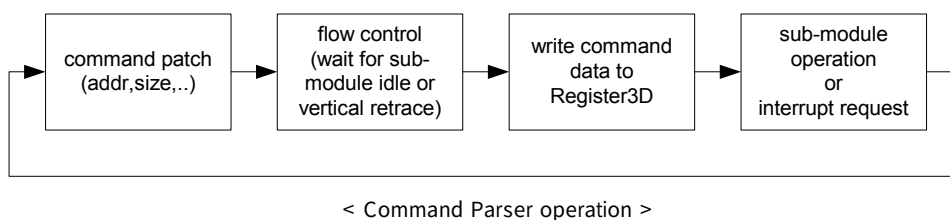
Command Reader read Command data from Command buffer/queue and delivers them to Command Parser. Input is Command buffer/queue assigned by CPU. Output is 32bit input FIFO of Command Parser.



Command Reader reads memory until **GRP3D_CMDQREAR = GRP3D_CMDQFRONT** and (1)wait for CPU's changing **GRP3D_CMDQFRONT** or (2)stop operating. (1) stage is called Command queue mode and (2) stage is called Command buffer mode. (Command Processor Mode: **GRP3D_CPMODE**)

25.3.3 Command Parser

After Command Reader read 32bit data from memory, Command Parser writes 32bit data to Register3D. Command Parser operate flow control or sub-module depending on condition.



Below is pseudo-code of Command Parser operation.

```

while(1)
{
    // command patch
  
```

```

CMDID = CommandData[31:28];

if( CMDID < 4 )
{
    interrupt ( Invalid Command );
    Core down;
}

if( CMDID == 15 )
{
    start_address32 = CommandData[8:0];
    num_of_data     = CommandData[24:16] + 1;
    wait_Rasterizer = ~ CommandData[15];
    wait_TSE        = ~ CommandData[14];
    wait_GTE        = ~ CommandData[13];
    wait_VSync      = CommandData[12];
    IgnoreField     = CommandData[11];
    RegfillIntEnb   = CommandData[10];
    Pop FIFO; // get next 32bit data
}
else
{
    start_address32 = CMDID*2;
    num_of_data     = CMD_SIZE[ CMDID ];
    wait_Rasterizer = true;
    wait_TSE        = true;
    wait_GTE        = true;
    wait_VSync      = false;
}

// flow control
wait for idle state of Primitive Processor
if( wait_Rasterizer ) { wait for idle state of Rasterizer }
if( wait_TSE )        { wait for idle state of TSE }
if( wait_GTE )        { wait for idle state of GTE }
if( wait_VSync )      { wait for Vertical retrace ( with IgnoreField ) }

// write command data to Register3D
for( i=0; i < num_of_data ; i++ )
{
    Register3D_Write32bit( start_address32 + i, CommandData );
    Pop FIFO; // get next 32bit data
}

// sub-module operation
if( CMDID == 7 ) { Texture cache reset } // CMDID_TEXSEGMENT
if( CMDID == 9 ) { Fill LUT              } // CMDID_LUTFILL
if( CMDID == 14 ) { Run Primitive        } // CMDID_PRIMITIVE

// Register fill interrupt
if( CMDID == 15 && RegfillIntEnb ) { interrupt ( Register fill ); }
}

```

25.4 TSE

TSE module performs calculation on vertex for Rasterizer. Rasterizer needs lot of information such as screen coordinate, vertex color, start value of texture coordinate, etc, to draw triangle or rectangle but Rasterizer could not use vertex data directly. TSE module convert vertex data into the form which Rasterizer can use.

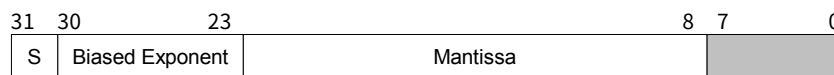
25.4.1 24bit floating point number

TSE uses 24bit floating point number internally. It is upper 24bit of standard 32bit floating point number.

All floating point number used in Vrender3D is 24bit and all floating point number register stores 24bit floating point number. But application software can write 32bit floating point number because 32bit register area is assigned for floating point number register.



CPU : 32bit floating point number



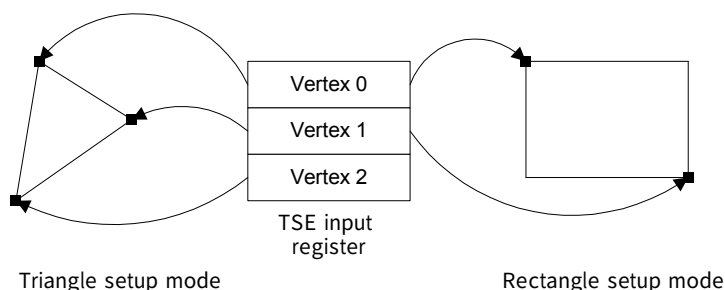
3D Core : 24bit floating point number register

25.4.2 TSE operation mode

TSE has two operation modes. One is triangle setup mode using three vertex the other is rectangle setup mode using two vertex.

TSE operation mode is defined by TSE control module.

Controlled by	TSE operation mode
CPU or Command Processor	GRP3DTSEMODE register 1: Triangle setup mode 0: Rectangle setup mode
Primitive Processor	GRP3D_PRIMTYPE register 0,1,2: Triangle setup mode 3: Rectangle setup mode (RECTLIST)
GTE/Clipper	Triangle setup mode



< TSE operation mode >

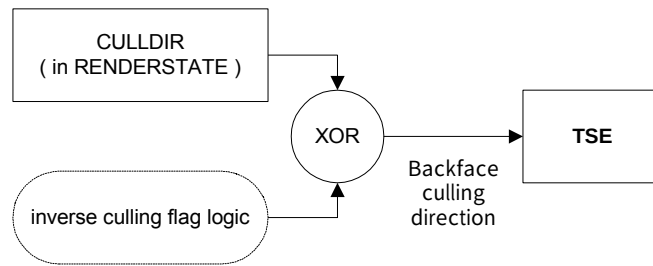
25.4.3 Culling direction and inverse culling flag

Backface culling is affected by backface culling direction

- 1: CW (Cull back faces with clockwise vertices)
- 0: CCW

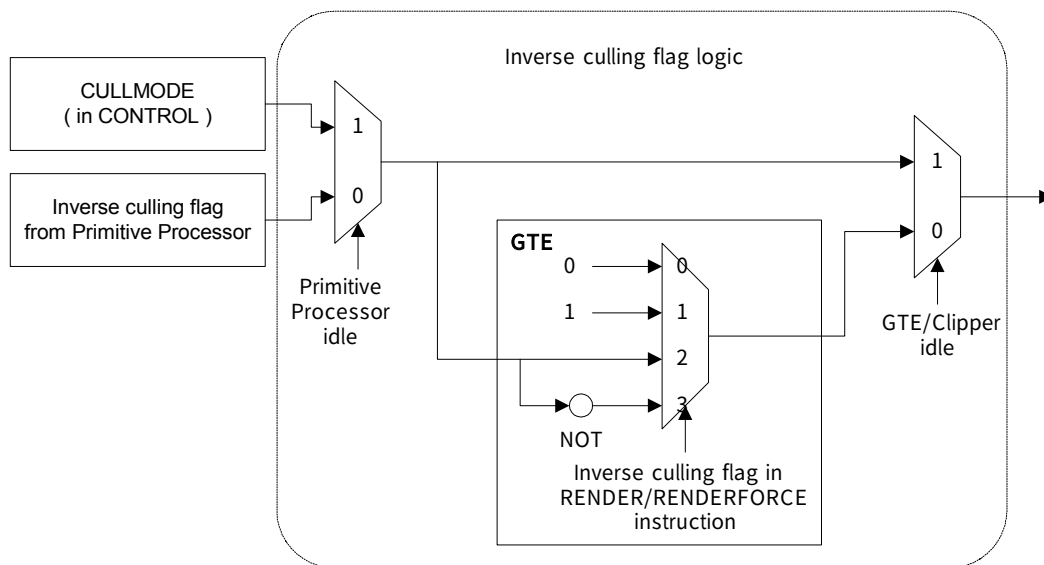
Backface culling direction in the **GRP3D_RENDERSTATE** is global variable bit. Use Inverse culling flag for temporary changing culling direction if necessary. For example, when render triangle strip, culling direction should be changed once per each triangle.

Inverse culling flag inverses **GRP3D_CULLDIR** effect.



< Backface culling direction >

Inverse culling flag made by **GRP3D_CONTROL** register or Primitive processor, can be modified by GTE instruction when GTE is used.



< Inverse culling flag logic >

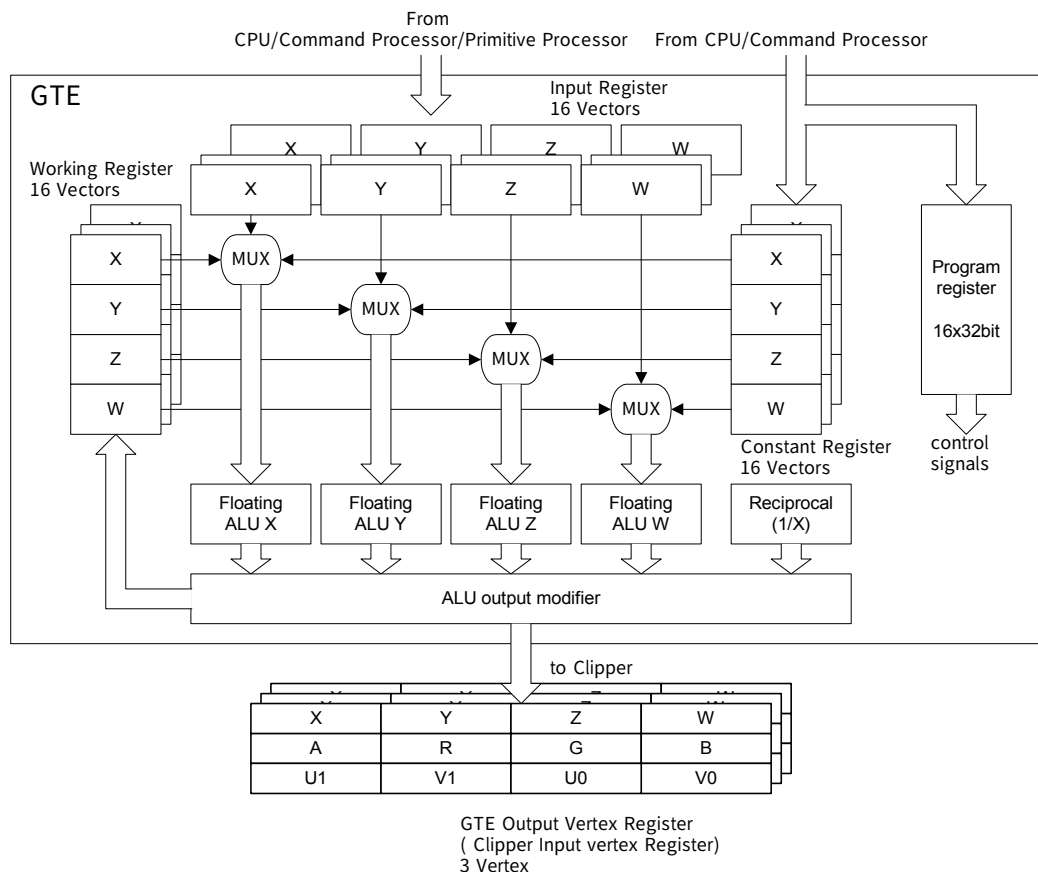
25.4.4 TSE input vertex format

TSE has input register(**GRP3D_TSEINPUT**) for three vertex, each vertex consists of twelve floating point number.

Name	TSE operation mode	Range
X	The x-coordinate of vertex in screen space	-1023 ~ +1023
Y	The y-coordinate of vertex in screen space	-1023 ~ +1023
Z	The z-coordinate of vertex in screen space,	0 ~ 1
W	Reciprocal homogeneous w	$2^{-16} \sim 2^{12}$
A	The alpha component of vertex color	0 ~ 255
R	The red component of vertex color	0 ~ 255
G	The green component of vertex color	0 ~ 255
B	The blue component of vertex color	0 ~ 255
U0	The u-coordinate of vertex in texture (texture stage 0)	If texture addressing mode is clamp (- Texture size) ~ (Texture size x 2) If texture addressing mode is repeat $-2^{13} \sim 2^{13}$
V0	The v-coordinate of vertex in texture (texture stage 0)	
U1	The u-coordinate of vertex in texture (texture stage 1)	
V1	The v-coordinate of vertex in texture (texture stage 1)	

25.5 GTE

GTE module creates or changes vertex or triangle and transfers triangle, which converted into projection space, to Clipper. GTE is Vector processing units, which can handle four floating point number at the same time. General purpose of GTE is multiplying vertex and transform matrix(world/view/projection) and converting into projection space and transfer to Clipper. GTE is programmable and can operate other operation.

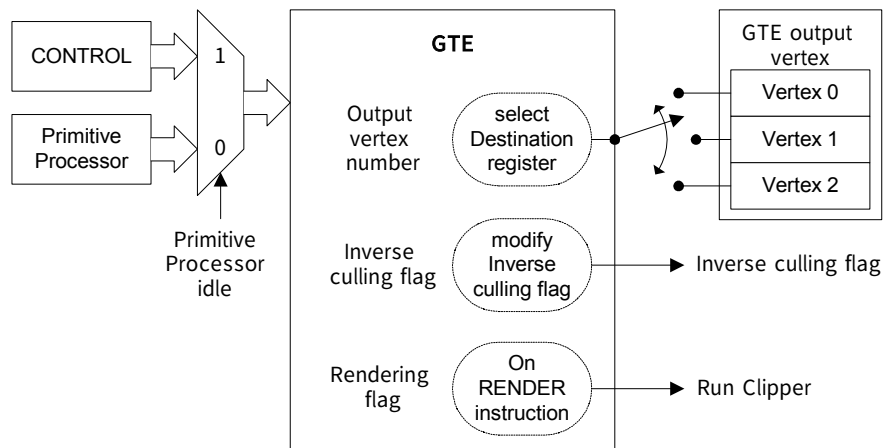


GTE consists four ALU for add/multiply and one divider and registers.

25.5.1 GTE input parameters

GTE receives three input parameters from **GRP3D_CONTROL** register or [Primitive processor](#).

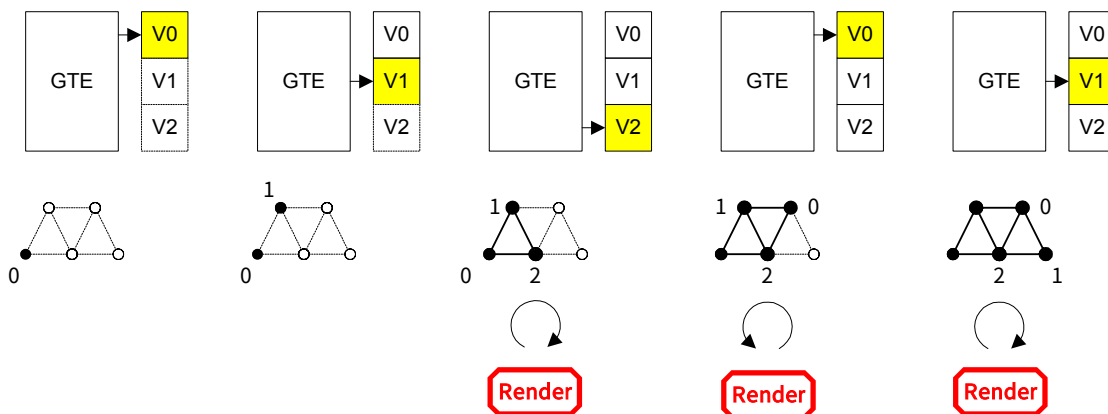
GTE input parameter	Description	Range
Output vertex number	Assign vertex of Output register. It can be use in GTE instruction or it can be disregard.	0,1 or 2
Inverse culling flag	Inverse culling flag is transferred to TSE after modified by GTE instruction. (RENDER,RENDERFORCE)	0 or 1
Rendering flag	Decide Rendering go/no go of RENDER instruction. RENDER instruction operate Clipper when '1'. 0: RENDER is treated as NOP 1: RENDER is treated as RENDERFORCE	0 or 1



< GTE input parameters >

// Render triangle-strip

```
// Load GTE() - Load vertex to GTE input vector register
// Run GTE (output vertex number, inverse culling flag, rendering flag)
Load GTE()
Run GTE (0, x, false) // process vertex 0
Load GTE()
Run GTE (1, x, false) // process vertex 1
Load GTE()
Run GTE (2, 0, true) // process vertex 2 & rendering
Load GTE()
Run GTE (0, 1, true) // process vertex 0 & rendering
Load GTE()
Run GTE (1, 0, true) // process vertex 1 & rendering
...
```



< Render triangle strip >

25.5.2 Registers

Register name	R/W (GTE program)	Size	Description
Input register	R	16 vectors (vector : 4 float)	Stores geometric data to perform. CPU/Command Processor/Primitive Processor write new data when GTE operation end.

Register name	R/W (GTE program)	Size	Description
Constant register	R	16 vectors	Stores non-frequent update data. CPU/Command Processor writes data. It normally stores Transform matrix, light information.
Working register	R/W	16 vectors	Stores interim result of calculation. It cannot be accessed from outside. It preserves data unless GTE program changes the value.
Output register (Clipper input register)	W	3 vertices (vertex: 12 float)	Stores final result of calculation. It can be accessed from CPU only. It preserves data unless GTE program changes the value.
Program register	-	64 instructions	Stores GTE program. CPU/Command Processor write data.

25.5.3 GTE instruction

Categories		Instructions
Arithmetic instructions	Add and multiply	ADD SUB MAD MUL DP3 DP4 DP3CLAMP DP4CLAMP
	Math	RSQRT SETRCP GETRCP
	Move	MOV0 MOV1
	Set On	SET CLR SGE SLT
	Minimum/maximum	MIN MAX
Control instructions	Loop	NOP DELAY EXIT SETLOOP LOOP
	Rendering	RENDER RENDERFORCE

25.5.3.1 Instruction format

GTE instruction is 32bit and each instruction can be distinguished by upper 5bit(instruction ID).

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Instruction ID					Instruction body																										

25.5.3.2 Instruction cycle

The GTE instruction cycle has three sub-operations.

Sub-operation	Description
I	Instruction fetch (Read instruction)
A	ALU operation
E	Execute instruction (register writing, jump, ...)

Below table is instruction set by ALU operation.

ALU Group(An)	Instructions
A0	RSQRT
A1	DP4 DP4CLAMP
A2	DP3 DP3CLAMP
A3	ADD SUB MAD SGE SLT MIN MAX MUL
Not use ALU	NOP DELAY EXIT SETLOOP LOOP RENDER RENDERFORCE SET CLR MOV0 MOV1 SETRCP GETRCP

Below table is Instruction by pipeline timing.

Pipeline Group (Pn)	Clock cycles											Instructions
	0	1	2	3	4	5	6	7	8	9	10	
P11	I	A	A	A	A	A	A	A	A	A	E	RSQRT

Pipeline Group (Pn)	Clock cycles											Instructions
	0	1	2	3	4	5	6	7	8	9	10	
P7	I	A	A		A		E					DP4 DP4CLAMP DP3 DP3CLAMP
P4	I	A		E								ADD SUB MAD SGE SLT MIN MAX
P3	I	A	E									MUL
P2	I	E										MOV0 MOV1 SETRCP
P1	I,E											NOP DELAY EXIT SETLOOP LOOP RENDER RENDERFORCE SET CLR GETRCP

Please observe below 4 rules when program GTE because GTE Instruction has variety of pipeline timing and use ALU in different ways.

- One destination write(E) for one cycle. (write crush)
- ALU operation(An) in same clock should be same group. (ALU crush)
- source0 read(I) and destination write(E) in same clock can not access Working register at the same time. (read/write crush)
- RSQRT has at least 9 delay.

Rule 1.

Clock cycles	0	1	2	3
ADD	I	A3		E
MUL		I	A3	E

ADD D, S0, S1
MUL D, S0, S1

write address crush !

Clock cycles	0	1	2	3	4
ADD	I	A3		E	
MUL			I	A3	E

ADD D, S0, S1, delay 1
MUL D, S0, S1

D: Destination
S0: Source 0
S1: Source 1
W : Working vector register

Rule 2.

Clock cycles	0	1	2	3	4	5	6	7
DP3	I	A2	A2		A2		E	
DP4		I	A1	A1		A1		E

DP3 D, S0, S1
DP4 D, S0, S1

use ALU crush !

Clock cycles	0	1	2	3	4	5	6	7	8	9	10
DP3	I	A2	A2		A2		E				
DP4					I	A1	A1		A1		E

DP3 D, S0, S1, delay 3
DP4 D, S0, S1

Rule 3.

Clock cycles	0	1	2
MOV0	I	E	
MOV0		I	E

MOV0 W, S0, -
MOV0 D, W, -

Working register read/write crush !

Clock cycles	0	1	2	3
MOV0	I	E		
MOV0			I	E

MOV0 W, S0, -, delay 1
MOV0 D, W, -

'D = W = S0'

Clock cycles	0	1	2
MOV0	I	E	
MOV1		I	E

MOV0 W, S0, -
MOV1 D, -, W

no problems (S1) but, read/write operations are in same time (W = S0, D = W). W should be not equal to D

Rule 4

Clock cycles	0	1	2	3	4	5	6	7	8	9	10
RSQRT	I	A0	A0	A0	A0	A0	A0	A0	A0	A0	E
DELAY									I,E		
											I

RSQRT D, S0, delay 7
DELAY 1

< Example of GTE programming rules >

25.5.3.3 Arithmetic instruction

Arithmetic instruction handles and stores data of Vector register.

Instruction	Instruction ID	Description	An	Pn
RSQRT	6	$\text{Dst} = 1.0f / \sqrt{ \text{Src0.w} }$	A0	P11
DP4	7	$\text{Dst} = \text{Src0.x} * \text{Src1.x} + \text{Src0.y} * \text{Src1.y} + \text{Src0.z} * \text{Src1.z} + \text{Src0.w} * \text{Src1.w}$	A1	P7
DP3	8	$\text{Dst} = \text{Src0.x} * \text{Src1.x} + \text{Src0.y} * \text{Src1.y} + \text{Src0.z} * \text{Src1.z}$	A2	P7
DP4CLAMP	9	$\text{value} = \text{Src0.x} * \text{Src1.x} + \text{Src0.y} * \text{Src1.y} + \text{Src0.z} * \text{Src1.z} + \text{Src0.w} * \text{Src1.w}$ $\text{Dst} = (\text{value} < 0) ? 0 : \text{value}$	A1	P7
DP3CLAMP	10	$\text{value} = \text{Src0.x} * \text{Src1.x} + \text{Src0.y} * \text{Src1.y} + \text{Src0.z} * \text{Src1.z}$ $\text{Dst} = (\text{value} < 0) ? 0 : \text{value}$	A2	P7
MAD	11	$\text{Dst} = \text{multiplier output} + \text{Src1}$	A3	P4
ADD	12	$\text{Dst} = \text{Src0} + \text{Src1}$	A3	P4
SUB	13	$\text{Dst} = \text{Src0} - \text{Src1}$	A3	P4
SGE	14	$\text{Dst} = (\text{Src0} \geq \text{Src1}) ? 1.0f : 0.0f$	A3	P4
SLT	15	$\text{Dst} = (\text{Src0} < \text{Src1}) ? 1.0f : 0.0f$	A3	P4
MAX	16	$\text{Dst} = (\text{Src0} \geq \text{Src1}) ? \text{Src0} : \text{Src1}$	A3	P4
MIN	17	$\text{Dst} = (\text{Src0} < \text{Src1}) ? \text{Src0} : \text{Src1}$	A3	P4
MUL	18	$\text{Dst} = \text{Src0} * \text{Src1}$	A3	P3
SETRCP	19	Start calculation $1.0f / \text{Src0.modifier.x}$ (put Src0.modifier.x to reciprocal module)	-	P2
MOV0	20	$\text{Dst} = \text{Src0}$	-	P2
MOV1	21	$\text{Dst} = \text{Src1}$	-	P2
GETRCP	22	$\text{Dst} = \text{reciprocal module output}$	-	P1
SET	23	$\text{Dst} = 1.0f$	-	P1
CLR	24	$\text{Dst} = 0.0f$	-	P1

Dst : Destination register

Src0 : the first argument register

Src1 : the second argument register

modifier : refer to Arithmetic instruction format

25.5.3.4 Arithmetic instruction format

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Instruction ID					Destination register					Source register 0					Source register 1					Modifier					Write enable				Delay		
					DR	Vector address				SR0	Vector address				SR1	Vector address									W	Z	Y	X			
						OVnum		OVcomp																							

Instruction ID

Instruction identifier

Destination register

Assign storing register of calculation result.

If DR is 0, store calculation result at Working register.

If DR is 1, store calculation result at Output register.

If calculation result stored in Output register, Vector address is split in to two part and managed.

Output Vertex number (OVnum)	Description
0	GTE output vertex 0
1	GTE output vertex 1
2	GTE output vertex 2
3	GTE output vertex number from CPU/Command processor/Primitive processor

Output Vertex component (OVcomp)	Description
0	Output position vector (x,y,z,w)
1	Output color vector (a,r,g,b)
2	Output texture coordinates (u0,v0,u1,v1)

Source register 0

Assign register for Floating ALU source0

If SR0 is 0, use Working register as source0.

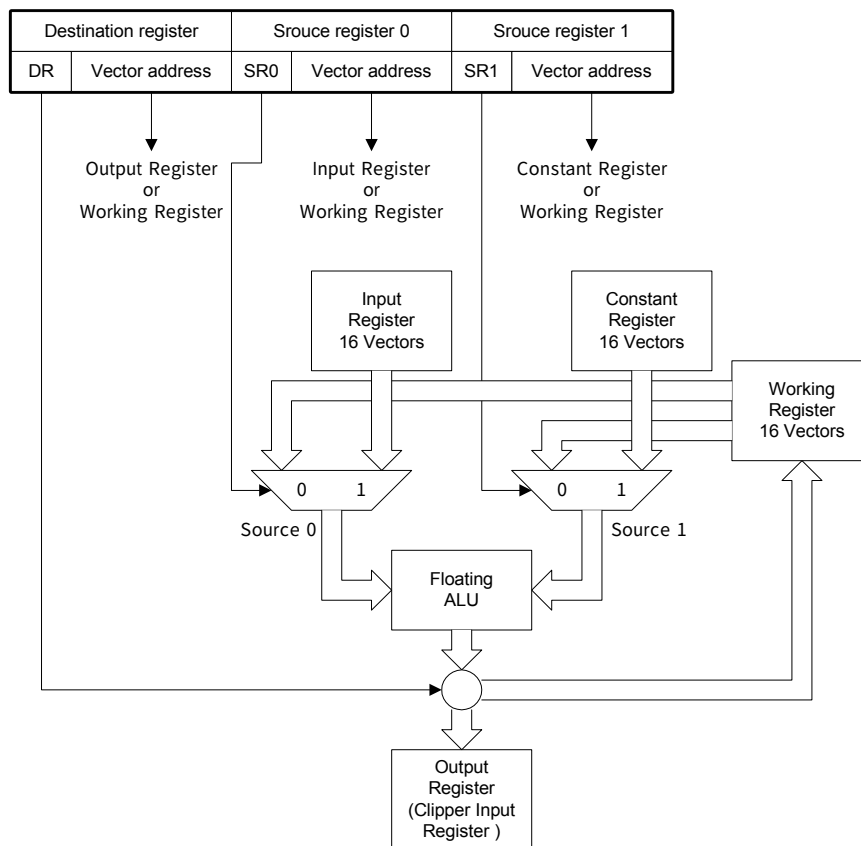
If SR0 is 1, use Input register as source0.

Source register 1

Assign register for Floating ALU source1

If SR1 is 0, use Working register as source1.

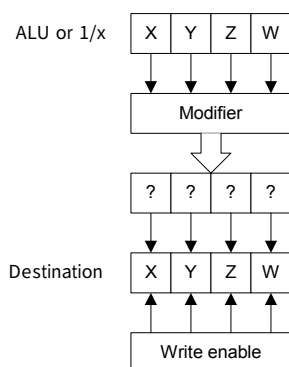
If SR1 is 1, use Constant register as source1.



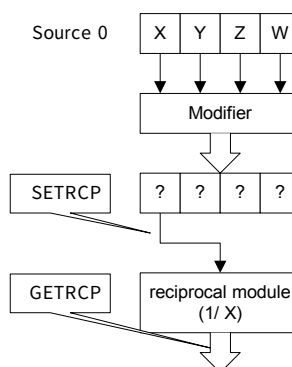
< Source & Destination in GTE instruction >

Modifier

Floating ALU output component modifier / Source0 component modifier for SETRCP



< ALU output component modifier & Destination write enable >



< Source0 component modifier for SETRCP >

Modifier	
0	xyzw
1	yzwx
2	zwx y
3	wxyz
4	xywz
5	ywzx
6	zxyw
7	wzxy
8	xzyw
9	ywxz
10	zywx
11	wxzy
12	xwyz
13	yzxw
14	zxwy
15	wyzx
16	xzwy
17	yxzw
18	zwyx
19	wyxz
20	xwzy
21	yxwz
22	zyxw
23	wzyx

Write enable

Destination component write enable

If this value = '0', Destination component value is not changed.

Delay

Clock delay between instruction patch and next instruction patch. If '0', GET patch instruction at every clock.

Clock cycles	0	1	2	3	4	5	6	7	8	9	10
DP3	I	A2	A2		A2		E				
DP4					I	A1	A1		A1		E

DP3 D, S0, S1, delay 3
DP4 D, S0, S1

< Example of Arithmetic instruction with DELAY >

25.5.3.5 Control instructions

Instruction	Instruction ID	Description	An	Pn
NOP	0	No-operation	-	P1
DELAY	25	No-operation with delay	-	P1
EXIT	1	Exit GTE program	-	P1
SETLOOP	2	Set loop point & loop count	-	P1
LOOP	3	Jump to loop point as many as SETLOOP assigned.	-	P1
RENDER	4	Operate Clipper to render triangle if Rendering flag = '1'.	-	P1
RENDERFORCE	5	Operate Clipper regardless Rendering flag status.	-	P1

25.5.3.6 Control instruction format & cycles

NOP

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0																															

ADD D, S0, S1
NOP
MUL D, S0, S1

Clock cycles	0	1	2	3	4
ADD	I	A3		E	
NOP		I,E			
MUL			I	A3	E

< Example of NOP >

DELAY

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
25																												Delay			

Clock cycles	0	1	2	3	4	5	6	7	8	9	10
DP3	I	A2	A2		A2		E				
DELAY		I,E									
DP4					I	A1	A1		A1		E

DP3 D, S0, S1
DELAY 2
DP4 D, S0, S1

< Example of DELAY >

EXIT

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1																															

SETLOOP

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
2																									LPID			Loop count				

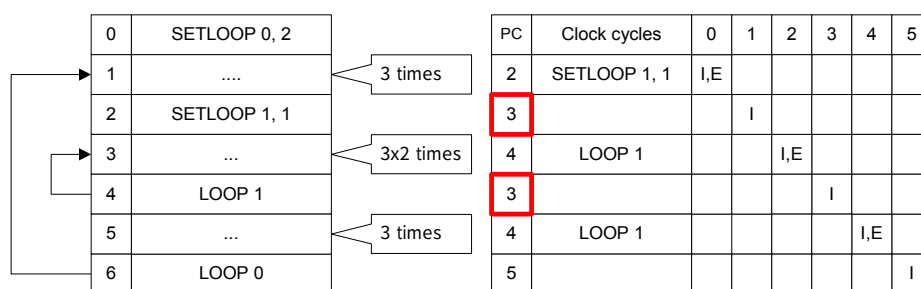
LPID : Loop point ID (0~2)

Loop count : Number of jump at LOOP instruction. If '0', it does not jump.

LOOP

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
3																							LPID								

LPID : Loop point ID (0~2)



< Example of SETLOOP/LOOP >

RENDER

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
4																												INVCULL			

INVCULL : inverse culling flag (refer to '[Culling direction and inverse culling flag](#)')

if rendering flag is '0'

RENDER
EXIT

Clock cycles	0	1	2	3
RENDER	I,E			
EXIT		I,E		
State	busy		idle	

if rendering flag is '1'

Clock cycles	0	Clipper startup (21 clocks)			22	23	24
RENDER	I,E	...					
EXIT					I,E		
State	busy					idle	

< Example of RENDER & EXIT >

RENDERFORCE

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
5																												INVCULL			

INVCULL : inverse culling flag (refer to '[Culling direction and inverse culling flag](#)')

RENDERFORCE EXIT		Rendering flag is ignored							
		Clock cycles	0	Clipper startup (21 clocks)			22	23	24
		RENDERFORCE	I,E	...					
		EXIT					I,E		
		State	busy					idle	

< Example of RENDERFORCE & EXIT >

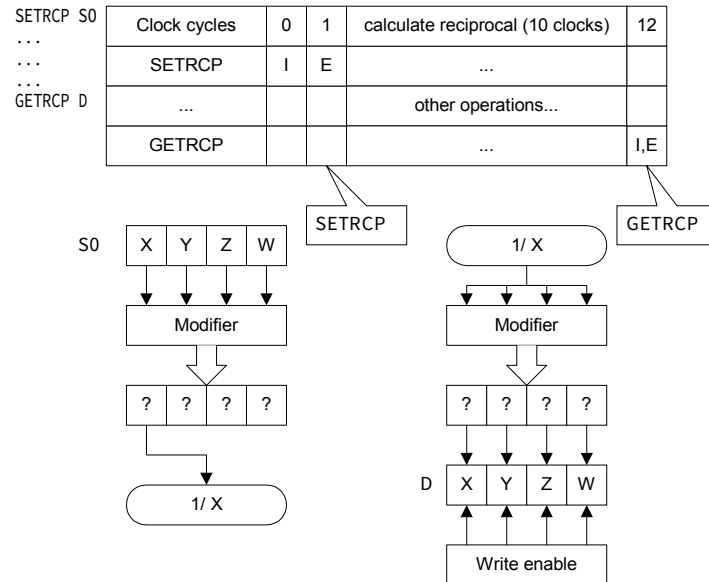
25.5.3.7 SETRCP/GETRCP

1/x calculation cannot support vector processing because GTE has one reciprocal module(1/x). Only one floating point number can be put in reciprocal module (SETRCP), and it needs plenty of time(10clock) to get result.(GETRCP)

SETRCP : Input one of component(x, y, x, w) of Source 0 register into reciprocal module using modifier.

GETRCP: Write reciprocal module output to Destination register. (using modifier, write enable)

Reciprocal module output reciprocal of input floating point number 10 clocks after SETRCP and hold output before next SETRCP operation. Other calculation can proceed after SETRCP because reciprocal module operates independent of ALU. If new SETRCP starts 10 clocks before previous SETRCP, previous SETRCP goes invalid.



< Source0 component modifier for SETRCP >

25.5.4 Example

Simple transform

Output Position = IREG[0] * CREG[0~3]

Output Color = IREG[1] // a,r,g,b = x,y,z,w

Output Tex = IREG[2] // u0,v0,u1,v1 = x,y,z,w

Where

IREG[n] : Input vector register n

WREG[n] : Working vector register n

CREG[n] : Constant vector register n

OREG[n] : Output vertex (vertex number n , if n is 3, it is treated as output vertex number from GTE input parameter)

Initial state

IREG[0] : vertex position

IREG[1] : vertex color

IREG[2] : texture coordinates

CREG[0~3] : total matrix (world x view x projection x clipping)

Description				Instruction code
MOV0	OREG[3].xyzw,	IREG[1]		0x A7620078
DP4	OREG[3].x__,	IREG[0],	CREG[0]	0x 3F210008
DP4	OREG[3]._y__,	IREG[0],	CREG[1]	0x 3F211010
DP4	OREG[3].__z__,	IREG[0],	CREG[2]	0x 3F212020
DP4	OREG[3].__w,	IREG[0],	CREG[3]	0x 3F213040
MOV0	OREG[3].xyzw,	IREG[2],	delay(4)	0x A7A4007C
RENDER				0x 20000002

Description	Instruction code
EXIT	0x 08000000

[illegible]

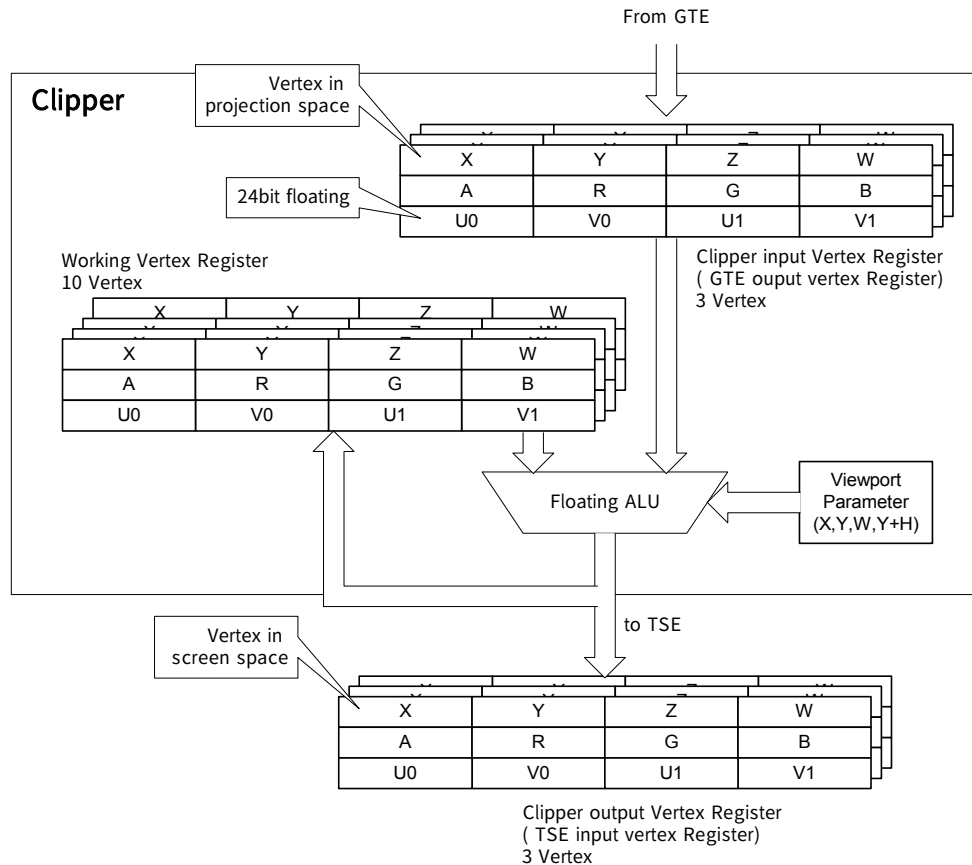
< Pipeline timing of simple transform example >

25.6 Clipper

Clipper convert triangle in projection space(GTE output) into screen space(TSE input).

Clipper gets one triangle from GTE. Clipper drops triangle, which totally clipped. Clipper transfers triangle, which totally located inside of clipping volume. Clipper clips triangle and transfer to TSE, which is across clipping volume.

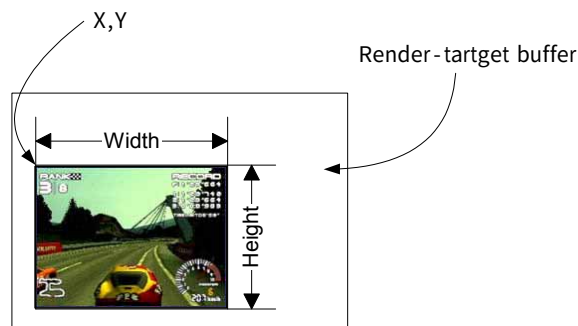
Step	Description
1. Triangle Clipping	Create triangle fan by clipping triangle of Clipper Input vertex register the clipping volume for all points (Xc, Yc, Zc, Wc) in projection space is defined as: Xc : 0 ~ Wc Yc : 0 ~ Wc Zc : 0 ~ Wc
2. Viewport Move & Scaling	The viewport scale matrix(Mvs) scales coordinate to be within the viewport window and flips the y-axis from up to down $Mvs = \begin{pmatrix} W & 0 & 0 & 0 \\ 0 & -H & 0 & 0 \\ 0 & 0 & 1 & 0 \\ X & Y+H & 0 & 1 \end{pmatrix}$ Where: X, Y : Pixel coordinates of the upper-left corner of the viewport on the render target W, H : Dimensions of the viewport on the render target surface, in pixels
3. Projection Divide	Finally, screen coordinate are computed and passed to the TSE Xs = Xvp / Wvp Ys = Yvp / Wvp Zs = Zvp / Wvp Ws = 1 / Wvp Where: (Xvp,Yvp,Zvp,Wvp) = (Xc,Yc,Zc,Wc) x Mvs (Xs,Ys,Zs,Ws) : a vertex coordinates in screen space (for TSE)



25.6.1 Viewport

viewport define the area of the screen into which a scene will be rendered. These values correspond to the destination rectangle, or viewport rectangle, as shown in the following illustration.

Refer to : `GRP3D_VPORTBOT`, `GRP3D_VPORTH`, `GRP3D_VPORTLEFT`, `GRP3D_VPORTW`



25.7 Rasterizer

25.7.1 Memory fill

Fill rectangle area of memory segment with specific value.

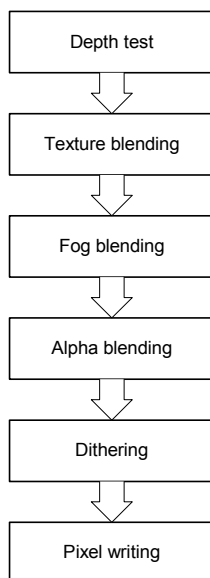
Memory fill executes high speed because it does not use Pixel pipeline. The [clipping rectangle](#) and [address converting table](#) does not have influence on Memory fill.

Memory fill is useful when clear Z-Buffer / Render target / Texture at high speed.

Refer to : `GRP3D_MFILLPARAM0`, `GRP3D_MFILLPARAM1`, `GRP3D_MFILLPARAM2`

25.7.2 Pixel Pipeline

- Depth test**
 Remove invisible pixel using Z-buffer.
 unsigned 16bit integer
 Support 8 test function
 Refer to : [Depth-buffer](#), `GRP3D_ZBINFO`
- Texture blending**
 Create Source color mixing Texture and Diffuse color
 Refer to : `GRP3D_TEXBLENDn`
- Fog blending**
 Blending Source color and Fog color using fog value.
 $\text{Source color} * (1 - f) + \text{Fog Color} * f$
 fog table stores fog value of depth corresponding 0~`GRP3D_FOGMAXZ` depth domain of camera area. If depth value of pixel is larger than `GRP3D_FOGMAXZ`, fog value sets last value of fog table.
 Refer to : `GRP3D_FOGCOLOR`, `GRP3D_FOGMAXZ`, `GRP3D_FOGTBL`
- Alpha blending**
 $\text{Source color} * \text{GRP3D_SRCBLEND} + \text{Destination Color} * \text{GRP3D_DESTBLEND}$
 Refer to : `GRP3D_ALPHABLEND(GRP3D_SRCBLEND,GRP3D_DESTBLEND)`
- Dithering**
 Convert R8G8B8 format Source color to R5G6B5 format.
 It can be enable/disable using `GRP3D_DITHERENB`.
 Refer to : `GRP3D_RENDERSTATE(GRP3D_DITHERENB)`
- Pixel writing**
 Renewal Render target and Z -buffer.



25.7.3 Render Target

- Render Target**
 Segment itself is Render target. Screen is directly mapped to memory segment through [address converting table](#).
 Maximum resolution 1024x1024 (if antialiasing: 1024x512)
 Refer to : [Address convert table](#) , `GRP3D_RENDTRG0(GRP3D_RTSEGMENT,GRP3D_RTTBLX,GRP3D_RTTBLY)`
- FSAA (Full screen antialiasing)**
 Use 2X method Full screen antialiasing. It needs twice vertical. (Maximum Render target vertical resolution goes in half)
 Refer to : [Address convert table](#) , `GRP3D_RENDTRG0(GRP3D_RTAA)`, `GRP3D_DISPINFO(GRP3D_DISPAA)`
- Renderable texture**
 Texture rendering is possible because [block addressing mode](#) of Render target can be set up.
 Refer to : [Block addressing mode](#), `GRP3D_RENDTRG0(GRP3D_RTADDRMODE)`

- **Clipping rectangle**

It is clipping rectangle exist in screen as pixel form. Pixels outside of region are not input to pixel pipeline.
Refer to : **GRP3D_RENDTRG1**, **GRP3D_RENDTRG2**

25.7.4 Depth-Buffer

- **Z-buffer**

16bit unsigned integer

Segment itself is Z-buffer. Screen is directly mapped to memory segment through [address converting table](#).

Maximum resolution 1024x1024 (if antialiasing: 1024x512)

Refer to : [Address convert table](#), **GRP3D_ZBINFO**

- **Z scale**

After Projection transform, multiply Z scale and normalized Z value and store in Z-buffer (16bit unsigned integer)
Use less than 65535.0f for precision of 24bit float point processing.

Refer to : **GRP3D_ZSCALE**

- **Z-valid buffer**

z-valid buffer is bit buffer to minimize z-buffer read.

It is 1bit correspondent to Segment 8pixel(16bit) and it occupies 64x128 pixel(16bit) rectangle area in segment.

If bit = 0, use **GRP3D_ZVVALUE** as z value of z-buffer without reading z buffer.

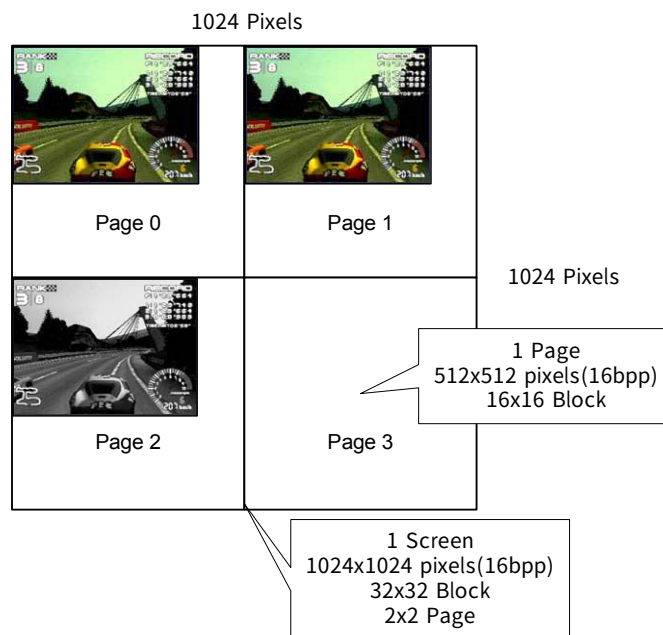
It is more efficient when certain scene has less multiple writing pixel.

Refer to : **GRP3D_ZVBINFO**

25.8 Texture

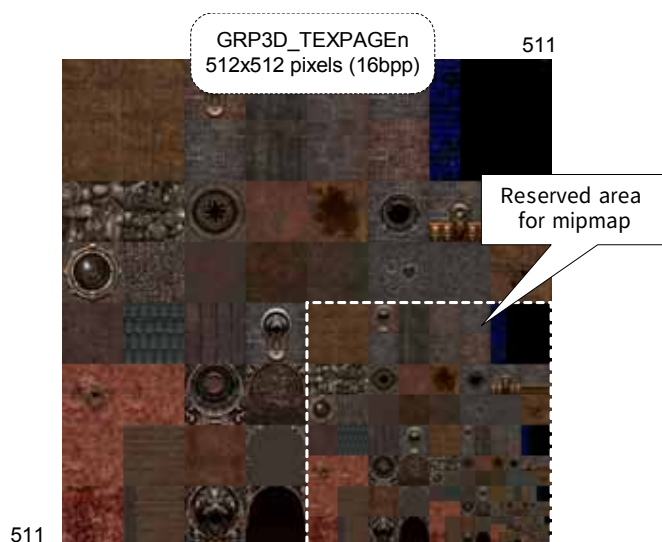
25.8.1 Page

Page is one of quadrisection of Segment.



25.8.2 Mipmap

Lower right quadrant is reserved for mipmap. If Mipmmap is used, VRender3D treats the region as other quadrant's downsampled mipmap texture.



25.8.3 Texture size

One texture(including palette) must be fully included in one page.

If mipmap is used, lower left 256x256(16bit) region of page is reserved for mipmap.

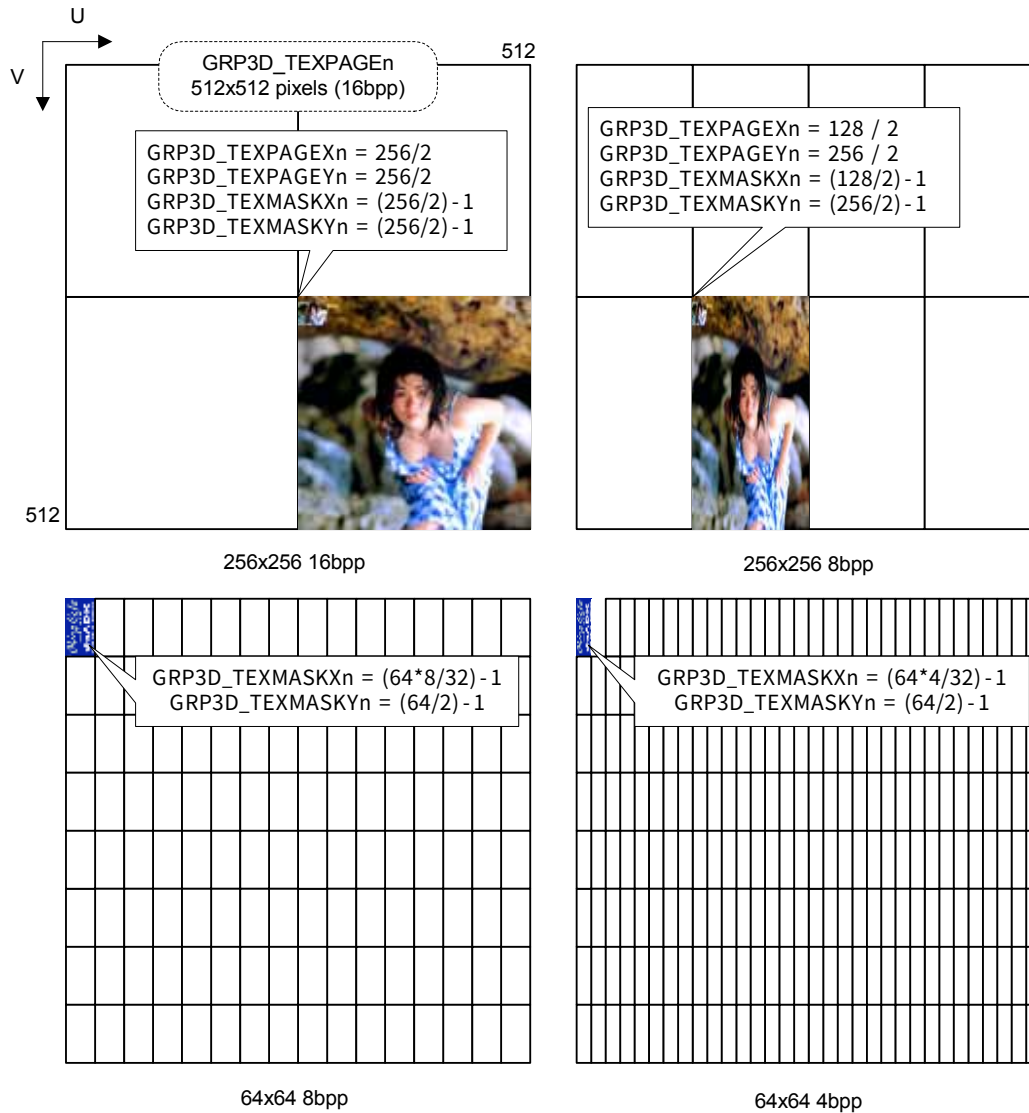
The unit size of texture is 2x2(16bit).

Color format	Maximum size (with mipmap)	Maximum size (without mipmap)	Minimum size
4bit indexed	2048x256 or 1024x512	2048x512	8x2
8bit indexed	1024x256 or 512x512	1024x512	4x2
16bit R5G6B5	512x256 or 256x512	512x512	2x2

25.8.4 Texture setting

Texture is rectangle area used in texture mapping. Texture must have widths and heights specified as square of 2, and position . specified as multiple of size.

Refer to : *GRP3D_TEXSEGMENT*, *GRP3D_TEXINFO n_0*, *GRP3D_TEXINFO n_1*



16bpp (U,V) cell address

- U,V is 16bpp pixel coordinate in page (0<=U<=511, 0<=V<=511)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GRP3D_TEXSEGMENT[10:0]											①	V[8:5]				②	U[8:5]				V[4:1]				U[4:1]				③	④	0
											Block index in segment										Block address										

① : bit 1 of GRP3D_TEXPAGE_n

② : bit 0 of GRP3D_TEXPAGE_n

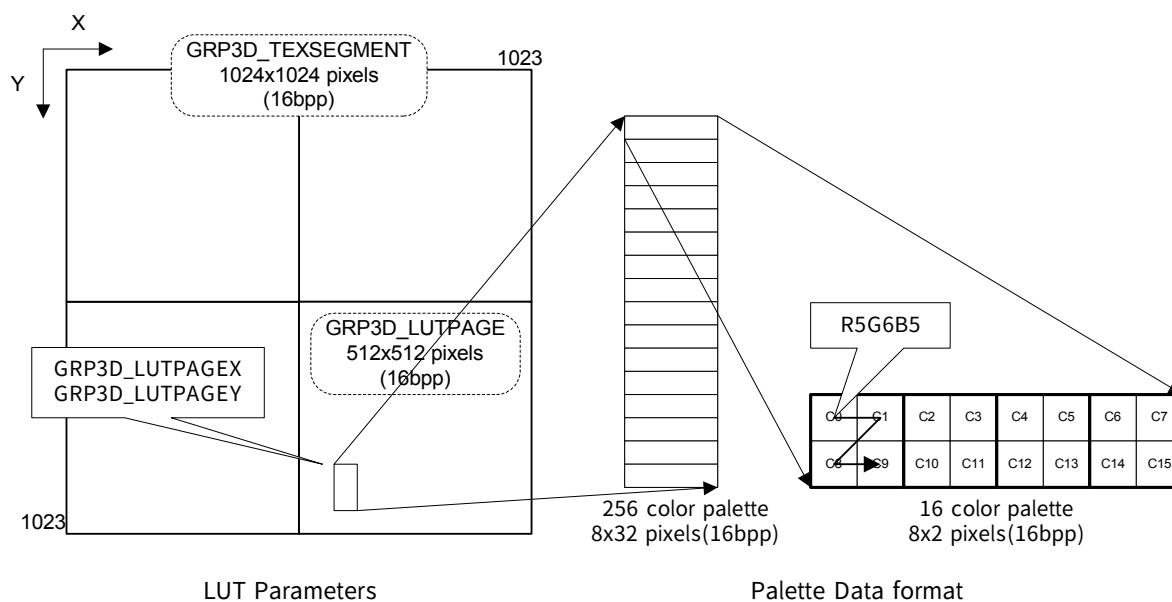
③ : bit 0 of V

④ : bit 0 of U

25.8.5 Palette

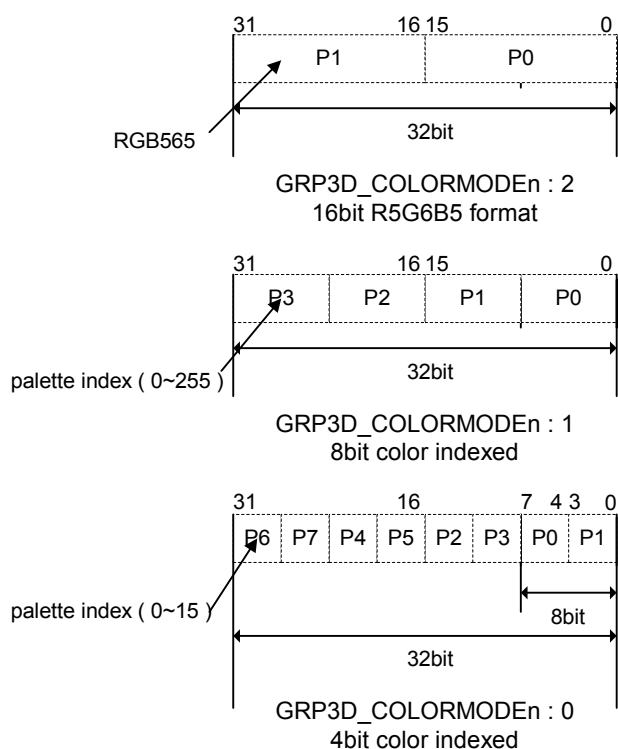
VRender3D has color lookup table(LUT) with 16bit R5G6B5 format. This LUT is loaded from memory with request from application. Palette is color data list in the memory and Palette is expressed 8x2 16bpp or 8x32 16bpp rectangle in page.

Refer to : [GRP3D_TEXSEGMENT](#), [GRP3D_LUTPARAM](#), [CMDID_LUTFILL](#)



25.8.6 Image texture

Image texture stores color data(or color index).
 The data in Image texture is used(or after through LUT) texture color.
 Refer to : *GRP3D_TEXINFO_n_0(GRP3D_COLORMODE_n)*



< Texel formats >

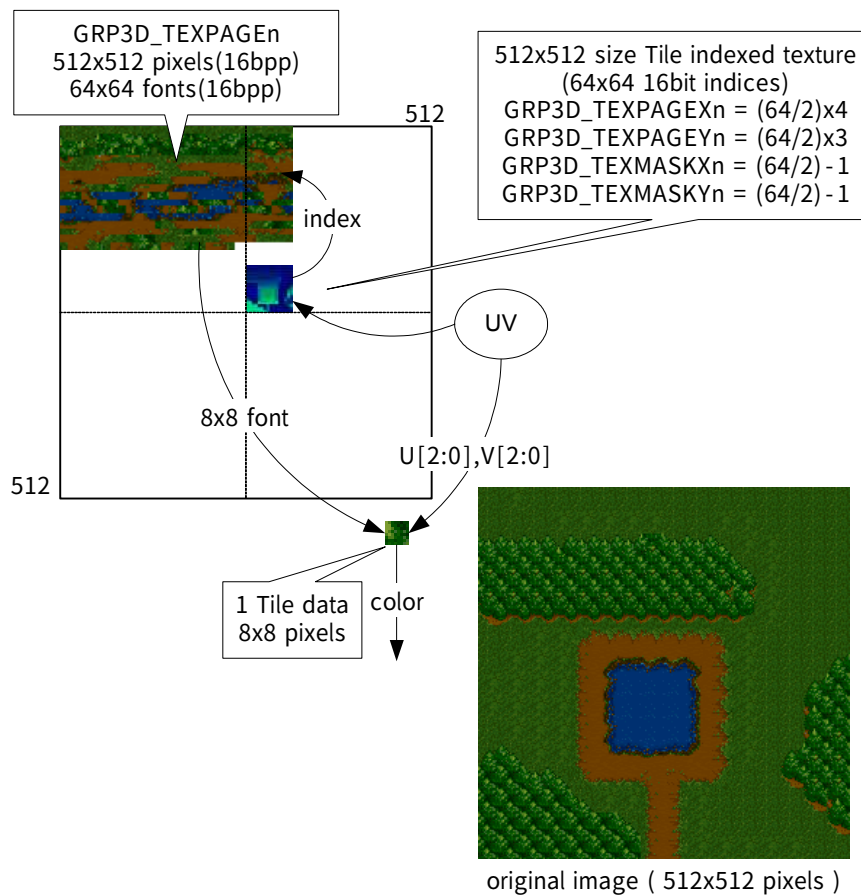
25.8.7 Tile indexed texture

Tile indexed texture stores 16bit tile index.

The data in Tile indexed texture is used in indexing tile of page.

Mipmap is not usable.

Refer to : **GRP3D_TEXINFO_n0(GRP3D_TILEINDEX_n)**



if GRP3D_TILEINDEX_n is 1, texture is tile-index map. each pixel is 16bit tile index.
Tile-index map that constructed by 64x64pixel 16bit indices is treated as 512x512 pixel texture

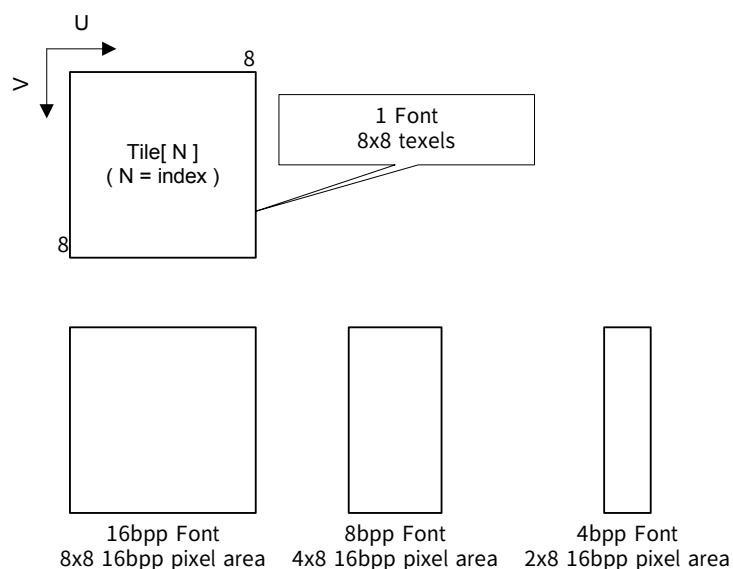
25.8.8 Tile

Tile is 8x8 image indexed by Tile indexed texture. VRender3D read color data (or color index) from Tile referring index and $tu[2:0],tv[2:0]$ when using Tile indexed texture.

Tile index number is numbering of tile by order after split one page with tile.

Color format	Tile size	Maximum index number
4bit indexed	2x8x16bit	16383 (256x64-1)
8bit indexed	4x8x16bit	8191 (128x64-1)
16bit R5G6B5	8x8x16bit	4095 (64x64-1)

Below is way of color data(or color index) address calculation with index(idx) and texture U. V read from Tile indexed texture.



16bpp (Idx, U, V) address

- U,V is pixel coordinate in font ($0 \leq U \leq 7, 0 \leq V \leq 7$)

- Idx : Font Index, ($0 \leq \text{Idx} \leq 4095$)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
			GRP3D_TEXSEGMENT								①	Idx[11:8]				②	Idx[5:2]				Idx[7:6]		V[2:1]		Idx[1:0]		U[2:1]		③	④	0	
											Block index in segment										Block address											

① : bit 1 of GRP3D_TEXPAGE

② : bit 0 of GRP3D_TEXPAGE

③ : bit 0 of V

④ : bit 0 of U

8bpp (Idx, U, V) address

- U,V is pixel coordinate in font ($0 \leq U \leq 7, 0 \leq V \leq 7$)

- Idx : Font Index, ($0 \leq \text{Idx} \leq 8191$)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
			GRP3D_TEXSEGMENT								①	Idx[12:9]				②	Idx[6:3]				Idx[8:7]		V[2:1]		Idx[2:0]		③	④	U[1:0]			
											Block index in segment										Block address											

① : bit 1 of GRP3D_TEXPAGE

② : bit 0 of GRP3D_TEXPAGE

③ : bit 2 of U

④ : bit 0 of V

4bpp (Idx, U, V) address

- U,V is pixel coordinate in font ($0 \leq U \leq 7, 0 \leq V \leq 7$)

- Idx : Font Index, ($0 \leq \text{Idx} \leq 16383$)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GRP3D_TEXSEGMENT												①	Idx[13:10]				②	Idx[7:4]				Idx[9:8]		V[2:1]		Idx[3:0]		③	U[2:1]		
												Block index in segment										Block address									

① : bit 1 of GRP3D_TEXPAGE

② : bit 0 of GRP3D_TEXPAGE

③ : bit 0 of V

palette index 4bit = U[0] ? low_4bit: high_4bit

25.8.9 Texture cache

VRender3D has 8Kbyte texture cache.

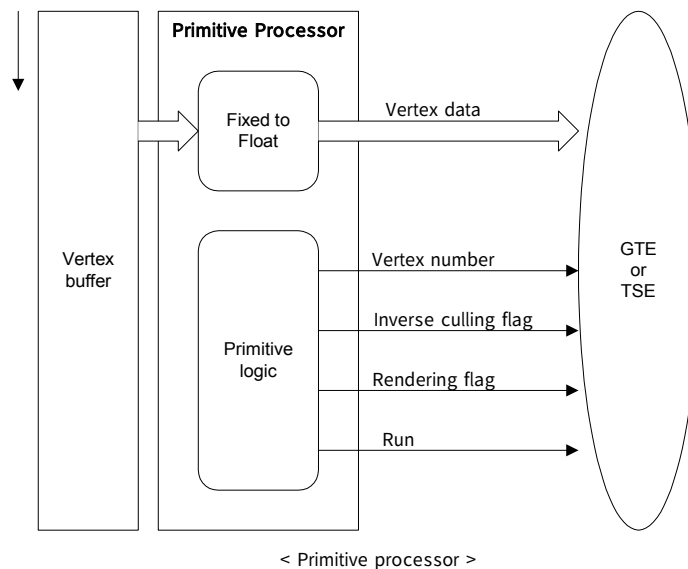
Texture cache needs reset every time when [texture segment](#) changes because it operates by segment.

Color format	Tile cache size (pixel)
4bit indexed	(32x256) x 2 way
8bit indexed	(32x128) x 2 way
16bit R5G6B5	(32x64) x 2 way

25.9 Primitive Processor

25.9.1 Overview

Primitive processor is active sub-module who renders several primitives using vertex/index buffer. Primitive processor has top priority for Register3D so CPU/Command processor cannot control other module when primitive processor is working.



The basic operation of Primitive processor is operate GTE(or TSE) after loading vertex data to [GTE](#)(or [TSE](#)) input register from vertex buffer.

25.9.2 Primitive mode

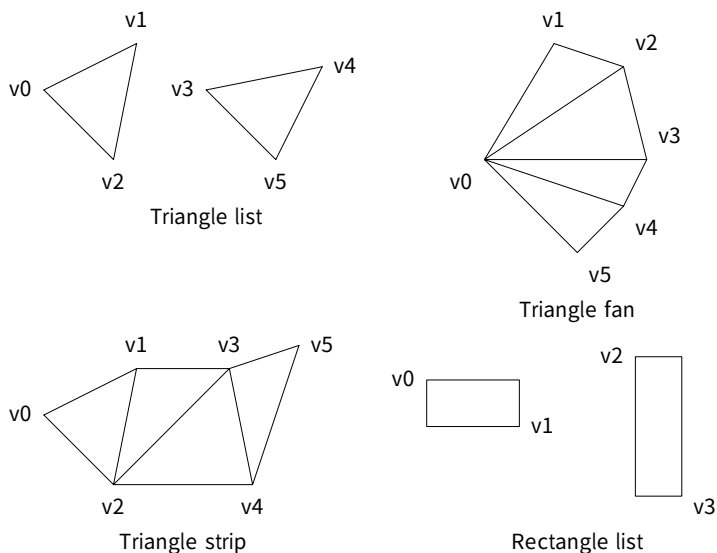
Primitive processor has two operation modes. (refer to **GRP3D_PRIMMODE**)

Primitive mode	Description
TSE mode	Use TSE to render Primitive. Vertex data format should be same as TSE input vertex format . Run TSE when Rendering is needed.
GTE mode	Use GTE to render Primitive. Vertex data should be composite to fit GTE program. Run GTE when Vertex processing is needed.

25.9.3 Primitive type

Primitive processor supports below four primitive types. (refer to **GRP3D_PRIMTYPE**)

Primitive type	Description
Triangle list	Renders the specified vertices as a sequence of isolated triangles. Each group of three vertices defines a separate triangle.
Triangle fan	Renders the vertices as a triangle fan.
Triangle strip	Renders the vertices as a triangle strip. The backface-culling flag is automatically flipped on even-numbered triangles.
Rectangle list	Renders the vertices as a rectangle. Each group of two vertices defines a separate rectangle. The first is top-left, and the second is bottom-right corner. The backface-culling is ignored. Primitive mode must be TSE mode.



< Primitive type >

// Render triangle-strip: TSE mode

```
// Load TSE(vertex number) - Load vertex to TSE input vertex register
// Run TSE( inverse culling flag ) – TSE not use rendering flag
Load TSE(0)
Load TSE(1)
Load TSE(2)
  Run TSE( 0 )
Load TSE(0)
  Run TSE( 1 )
Load TSE(1)
  Run TSE( 0 )
...
```

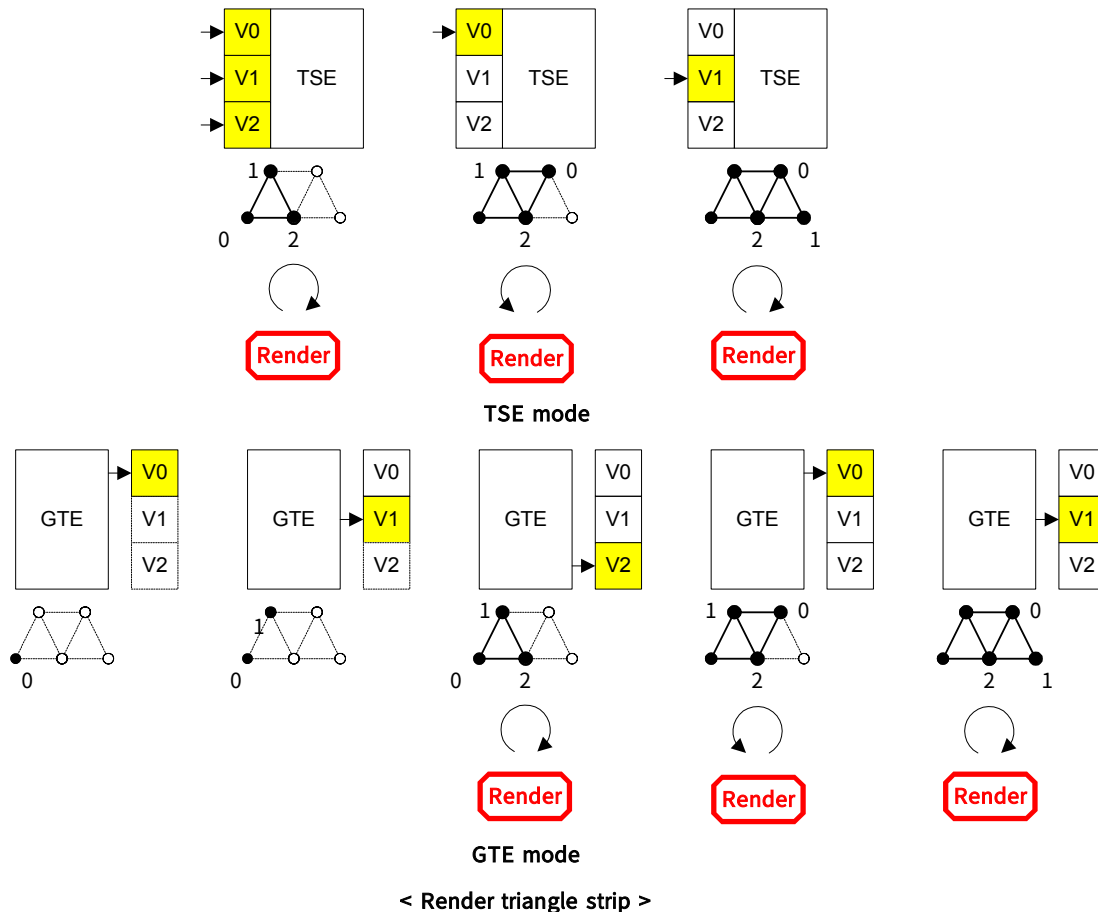
// Render triangle-strip: GTE mode

```
// Load GTE() - Load vertex to GTE input vector register
// Run GTE (output vertex number, inverse culling flag, rendering flag)
Load GTE()
  Run GTE (0, x, false) // process vertex 0
Load GTE()
  Run GTE (1, x, false) // process vertex 1
Load GTE()
  Run GTE (2, 0, true) // process vertex 2 & rendering
```

```

Load GTE()
Run GTE (0, 1, true) // process vertex 0 & rendering
Load GTE()
Run GTE (1, 0, true) // process vertex 1 & rendering
...

```



25.9.4 Vertex buffer

Vertex buffer is buffer containing data for input register of GTE/TSE. It consists of 8/16/32bit fixed point number or 32bit floating point number. Each numbers are converted in to floating point number by Primitive processor and transfer to GTE/TSE.

Vertex buffer is readable and locates anywhere in 64bit aligned memory. (refer to **GRP3D_VBFIXED**, **GRP3D_VBFORMAT**)

25.9.4.1 Vertex format

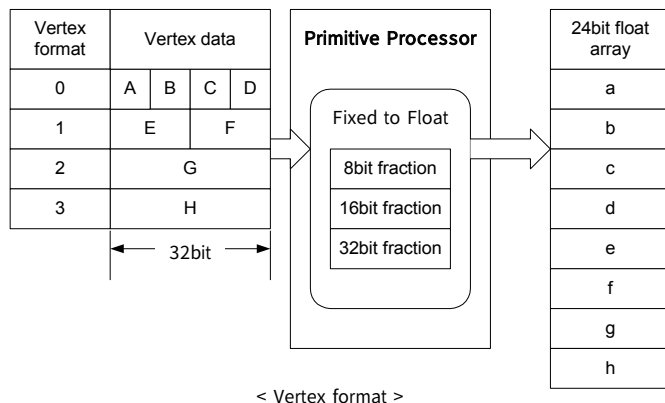
Vertex buffer consists of below four number types.

- Unsigned 8bit fixed point number
- Signed 16bit fixed point number
- Signed 32bit fixed point number
- 32bit floating point number

Vertex format is assignment of combining number type of vertex buffer in 32bit unit.

Vertex format	Description
0	4 unsigned 8bit fixed point numbers
1	2 signed 16bit fixed point numbers
2	1 signed 32bit fixed point number

Vertex format	Description
3	1 32bit floating point number

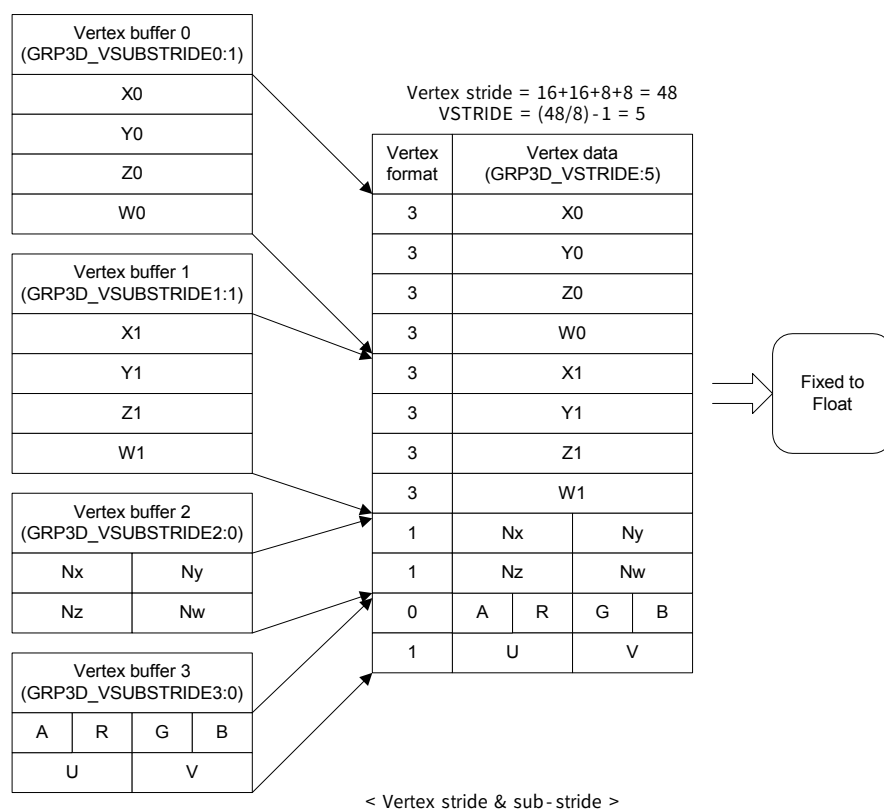


25.9.4.2 Vertex stride

Primitive processor uses maximum 4 vertex buffers. When primitive processor read from vertex buffers, it reads 4 vertex buffers in order. Vertex sub-stride is the size, which primitive processor should read from each vertex buffer. Vertex stride is summation of sub-stride of active vertex buffer.

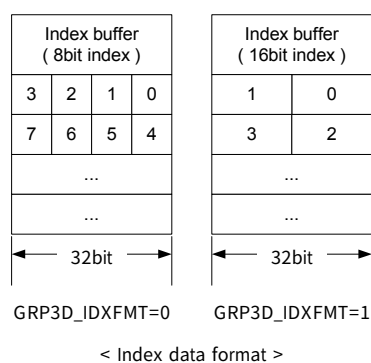
- Vertex stride** (Size of vertex: Byte/Vertex)
 $GRP3D_VSTRIDE = (\text{Vertex stride} / 8) - 1$
 Vertex stride is one of below values
 Vertex stride = sub-stride[0]
 Vertex stride = sub-stride[0] + sub-stride[1]
 Vertex stride = sub-stride[0] + sub-stride[1] + sub-stride[2]
 Vertex stride = sub-stride[0] + sub-stride[1] + sub-stride[2] + sub-stride[3]
- Vertex sub-stride** (Byte/Vertex)

Vertex sub-stride (Byte/Vertex)	$GRP3D_VSUBSTRIDEn$
8	0
16	1
32	2
64	3
128	4
256	5



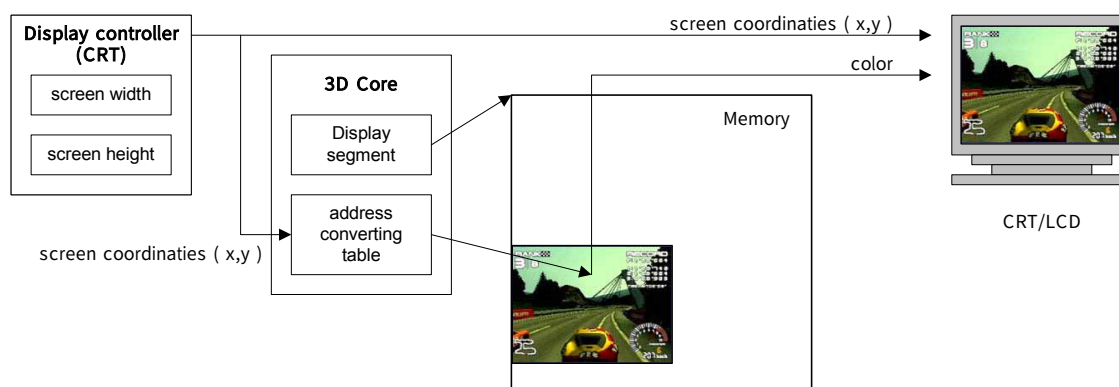
25.9.5 Index buffer

Index buffer stores vertex index. It consists of 8bit or 16bit unsigned integer array. Each index can be used for indexing vertex. It can be located anywhere in memory as long as it is readable but if it is 16bit index, it should be 16bit align. (refer to **GRP3D_IDXFMT,GRP3D_PRIMIDX**)



25.10 Display (Page flipping)

Display contents to CTR/LCD can be assigned by [segment, address converting table](#). (refer to **GRP3D_DISPINFO**)



25.10.1 Page flipping

Write suitable value to display register for Page flipping. It will display correspondent memory contents to screen. (refer to *DISPINFO*)

Display information register (<i>DISPINFO</i>)	Description
Display antialiasing flag (<i>GRP3D_DISPAA</i>)	If '1', mix 2 line and display 1 line. It should be '1' for display rendering image of target register antialiasing flag(<i>GRP3D_RTAA</i>) '1' to CRT/LCT.
Display segment (<i>GRP3D_DISPSEGMENT</i>)	Memory segment of screen contents
Address converting table (<i>GRP3D_DISPTBLX</i> , <i>GRP3D_DISPTBLY</i>)	Converting table which mapping CRT/LCD screen x, y coordinate to Memory segment. (refer to 'address converting table')

25.10.2 Synchronous page flipping

Once write value to display register, the contents of memory display to CRT/LCD. To avoid tier phenomenon, display register should be renewal to meet vertical retrace.

There are two different method to set register to meet vertical retrace depending on using Command queue/buffer.

Case	Description
Direct access	Use vertical retrace interrupt from CRT controller. After 3D core operation, receive interrupt and write value to register.
Use command queue/buffer	Use register fill command at Command queue/buffer. Using WAIT_SYNC,IGNOREFIELD option for changing register to synchronize to vertical retrace.

25.113D Core Registers

3D CORE REGISTER MAP

Address	Description
E000_0000 ~ E000_00FF	General / Control registers
E000_0100 ~ E000_017F	Fog table
E000_0180 ~ E000_01FF	Reserved
E000_0200 ~ E000_02FF	GTE program registers
E000_0300 ~ E000_03FF	GTE constant vector registers
E000_0400 ~ E000_04FF	GTE input vector registers
E000_0500 ~ E000_05FF	TSE input vertex registers

E000_0600 ~ E000_06FF	Reserved (Do not access)
E000_0700 ~ E000_078F	GTE output(Clipper input) registers
E000_0790 ~ E000_07FF	Reserved

25.11.1 General / Control register map (E000_0000h ~ E000_00FFh)

Register Name	offset	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																			
CPCONTROL	00h																																CPCONTROL																			
CMDQSTART	04h	CMDQSTART																																																		
CMDQEND	08h	CMDQEND																																																		
CMDQFRONT	0Ch	CMDQFRONT																																																		
CMDQREAR	10h	CMDQREAR																																																		
STATUS	14h												Sub-module idle flags					READVCNT																																		
INT	18h																																																			
	1Ch																																																			
CONTROL	20h	0	1	0	0																																															
	24h																																																			
RENDERSTATE	28h	0	1	0	0	1																																														
	2Ch																																																			
ALPHABLEND	30h	0	1	1	0																				DESTBLEND					SRCBLEND																						
	34h																																																			
TEXSEGMENT	38h	0	1	1	1																TEXSEGMENT																															
	3Ch																																																			
TPCOLOR	40h	1	0	0	0	0												TPCOLOR																																		
MIPMAPBIAS	44h																	MIPMAPBIAS																																		
LUTPARAM	48h	1	0	0	0	1			LUTBANK										LUTPAGEY					LUTPAGEX					0	0																						
	4Ch																																																			
TEXINFO0_0	50h	1	0	1	0				TLUTBANK																																											
TEXINFO0_1	54h	TEXMASKY					TEXMASKX					TEXPAGEY					TEXPAGEX																																			
TEXINFO1_0	58h	1	0	1	1				TLUTBANK																																											
TEXINFO1_1	5Ch	TEXMASKY					TEXMASKX					TEXPAGEY					TEXPAGEX																																			
TEXBLEND0	60h	1	1	0	0																																															
	64h																																																			
TEXBLEND1	68h	1	1	0	0	1																																														
	6Ch																																																			
PRIMCON	70h	1	1	1	0								VSTRIDE					VCOUNT																																		
PRIMIDX	74h	PRIMIDX																																																		
PRIMVTX0	78h	VTXADDR																																																		
PRIMVTX1	7Ch	VTXADDR																																																		
PRIMVTX2	80h	VTXADDR																																																		
PRIMVTX3	84h	VTXADDR																																																		
	88h																																																			
	8Ch																																																			
FOGCOLOR	90h									FOGCOLOR																																										
FOGMAXZ	94h	FOGMAXZ																																																		
DISPINFO	98h						DISPSEGMENT												DISPTBLY					DISPTBLX																												

Bit	R/W	Description	Default Value
		This address must be aligned to 8byte (64bit) (GRP3D_CMDQSTART[2:0] must be 0)	

25.11.4 Command queue end address register (GRP3D_CMDQEND)

Address : E000 0008h

Bit	R/W	Description	Default Value
31: 0	R/W	GRP3D_CMDQEND: Command queue end address (GRP3D_CMDQEND[2:0] must be 4)	-

25.11.5 Command queue front pointer register (GRP3D_CMDQFRONT)

Address : E000 000Ch

Bit	R/W	Description	Default Value
31: 0	R/W	GRP3D_CMDQFRONT: Command queue front pointer This address must be aligned to 4byte (32bit) (GRP3D_CMDQFRONT[1:0] must be 0)	-

25.11.6 Command queue rear pointer register (GRP3D_CMDQREAR)

Address : E000 0010h

Bit	R/W	Description	Default Value
31: 0	R	GRP3D_CMDQREAR: Command queue rear pointer if GRP3D_CMDQREAR == GRP3D_CMDQFRONT, command queue is empty if GRP3D_CMDQREAR-4 == GRP3D_CMDQFRONT, command queue is full	-

25.11.7 3D Core state register (GRP3D_STATUS)

Address : E000 0014h

Bit	R/W	Description	Default Value
31: 24	-	Reserved	-
23	R	GRP3D_RASTEMPTY: Empty flag of Rasterizer input FIFO	1
22	R	GRP3D_RASTFULL: Full flag of Rasterizer input FIFO	0
21	R	GRP3D_CRIDLE: Command Reader idle flag	1
20	R	GRP3D_CPIDLE: Command Processor idle flag	1
19	R	GRP3D_PRIMIDLE: Primitive Processor idle flag	1
18	R	GRP3D_RASTIDLE: Rasterizer idle flag	1
17	R	GRP3D_TSEIDLE: TSE idle flag	1
16	R	GRP3D_GTEIDLE: GTE idle flag	1
15: 0	R	GRP3D_READVCNT: Number of vertex that is loaded by Primitive processor It goes '0' status when Primitive processor starts operation.	-

25.11.8 3D Core interrupt register (GRP3D_INT)

Address : E000 0018h

Bit	R/W	Description	Default Value
31: 10	-	Reserved	-

Bit	R/W	Description	Default Value
9	R/W	GRP3D_INVCMINTR : Invalid command interrupt enable	0
8	R/W	GRP3D_REGFILLINTR : Register fill interrupt enable	0
7: 2	-	Reserved	-
1	R/W	GRP3D_INVCMDEPEND : Invalid command interrupt pending [R] 0: no interrupt is generated 1: interrupt caused because invalid command was found [W] 1: interrupt pending bit(GRP3D_INVCMDEPEND) clear	0
0	R/W	GRP3D_REGFILLPEND : Register fill interrupt pending [R] 0: no interrupt is generated 1: interrupt caused because the Register fill command(with interrupt) was processed. [W] 1: interrupt pending bit(GRP3D_REGFILLPEND) clear	0

25.11.9 3D Core sub-module control register (GRP3D_CONTROL)

Address : E000 0020h

Bit	R/W	Description	Default Value
31: 11	-	Reserved	-
10	W	GRP3D_TSEMODE : TSE operation mode 1: Triangle setup mode 0: Rectangle setup mode if GRP3D_MODULEID is not TSE(2), this bit is ignored.	-
9	W	GRP3D_CULLMODE : GTE/TSE culling mode 1: Backface culling is affected by ~(GRP3D_CULLDIR) 0: Backface culling is affected by GRP3D_CULLDIR if GRP3D_MODULEID is not GTE(1) or TSE(2), this bit is ignored.	-
8	W	GRP3D_GTERENDER : GTE rendering flag if this bit is 1, GTE render triangle when render instruction executed if GRP3D_MODULEID is not GTE(1), this flag is ignored.	-
7: 6	W	GRP3D_GTEVNUM : GTE output vertex number (0,1,2) if GRP3D_MODULEID is not GTE(1), this bits are ignored.	-
5	W	GRP3D_SPANDIR : Rasterizer span direction. if this flag is 1, span direction is right. if GRP3D_RASTMODE is not 0, this flag is ignored. if GRP3D_MODULEID is not Rasterizer(3), this bits is ignored.	-
4: 3	W	GRP3D_RASTMODE : Rasterizer operation mode 0 : Rendering 1 : memory fill (for fast z-buffer/screen clear) 2 : LUT fill 3 : texture cache clear if GRP3D_MODULEID is not Rasterizer(3), this bits is ignored.	-
2	W	GRP3D_NOP : NOP flag if this bit is 1, all bits are ignored. If this bit is 1, CMDID_CONTROL just wait sub-module idle	-
1: 0	W	GRP3D_MODULEID : Sub-module ID 0:Primitive Processor 1:GTE 2:TSE 3:Rasterizer	-

25.11.10 Render state register (GRP3D_RENDERSTATE)

Address : E000 0028h

Bit	R/W	Description	Default Value
31: 14	-	Reserved	-

Bit	R/W	Description	Default Value
13	W	GR3D_CULLDIR : Specifies how back-facing triangles are culled. 1: CW (Cull back faces with clockwise vertices) 0: CCW if GRP3D_CULLENB is 0, this bit is ignored	-
12	W	GR3D_CULLENB : Back-face culling enable. 1: Cull back faces 0: Do not cull back faces	-
11	W	GR3D_DITHERENB : Dithering enable	-
10	W	GR3D_FOGENB : Fog blending enable	-
9	W	GR3D_ZWENB : Z-buffer writing enable	-
8	W	GR3D_ZENB : Z-buffer test enable	-
7	W	GR3D_ALPHAENB : Alpha blending enable	-
6	W	GR3D_AFILTERENB : Adaptive filter-cache enable Adaptive filter-cache enhance Rasterizer performance and slightly lower Bilinear-filtering quality. if GRP3D_FILTERENB is 0, this bit is ignored	-
5	W	GR3D_FILTERENB : Bilinear filtering enable if GRP3D_TEXENB is 0, this bit must be 0	-
4	W	GR3D_SHADEENB : Shade enable Reduce Rasterizer power when vertex color is not used. if any GRP3D_TBLENDARGx is 1, this bit must be 1.	-
3	W	GR3D_TPENB : Transparency color checking enable. if GRP3D_TEXENB is 0, this bit must be 0	-
2	W	GR3D_MTEXENB : Multi-texturing enable if GRP3D_TEXENB is 0, this bit must be 0	-
1	W	GR3D_TEXENB : Texture mapping enable	-
0	W	GR3D_PCENB : Perspective correction enable if GRP3D_TEXENB is 0, this bit must be 0	-

25.11.11 Alpha blending function register (GRP3D_ALPHABLEND)

Address : E000 0030h

Bit	R/W	Description	Default Value
31: 10	-	Reserved	-
9: 5	W	GRP3D_DESTBLEND : Destination blending function 01h: Zero (00, 00, 00) 11h: One (FF, FF, FF) 02h: Source color (Rs, Gs, Bs) 12h: Inverse source color (~Rs, ~Gs, ~Bs) 04h: Source alpha (As, As, As) 14h: Inverse Source alpha (~As, ~As, ~As) 08h: Destination color (Rd, Gd, Bd) 18h: Inverse Destination color (~Rd, ~Gd, ~Bd)	-
4: 0	W	GRP3D_SRCBLEND : Source blending function	-

25.11.12 Texture segment register (GRP3D_TEXSEGMENT)

Address : E000 0038h

Bit	R/W	Description	Default Value
31: 11	-	Reserved	-
10: 0	W	GRP3D_TEXSEGMENT : Texture/Palette segment. if GRP3D_TEXSEGMENT is changed. texture cache must be cleared.	-

☞ texture cache will be automatically cleared when [CMDID_TEXSEGMENT](#) processed by Command processor

25.11.13 Transparency color register (GRP3D_TPCOLOR)

Address : E000 0040h

Bit	R/W	Description	Default Value
31: 16	-	Reserved	-
15: 0	W	GRP3D_TPCOLOR: Transparency color (R5G6B5)	-

25.11.14 Mipmap bias register (GRP3D_MIPMAPBIAS)

Address : E000 0044h

Bit	R/W	Description	Default Value
31: 16	-	Reserved	-
15: 0	W	GRP3D_MIPMAPBIAS: Mipmap bias result mipmap level = original mipmap level + bias fixed point value : sign.7.8 if this value is positive(+), mipmap level is increased if this value is negative(-), mipmap level is decreased (mipmap level '0' indicates using original texture.)	-

25.11.15 Update LUT parameter register (GRP3D_LUTPARAM)

Address : E000 0048h

Bit	R/W	Description	Default Value
31: 27	-	Reserved	-
26: 25	W	GRP3D_LUTPAGE: LUT page index in texture segment(<i>GRP3D_TEXSEGMENT</i>)	-
24: 21	W	GRP3D_LUTBANK: LUT bank for 4bit-indexed if <i>GRP3D_LUTFULL</i> is 1, this bits are ignored	-
20: 18	-	Reserved	-
17	W	GRP3D_LUTFULL: LUT full update flag if this bit is 1, update all color tables (256 color) if this bit is 0, update only 1-bank (16 color)	-
16	W	GRP3D_LUTSTAGE: LUT stage number All texture stage has one LUT (LUT is R5G6B5 256 color table) this bit indicate texture stage number	-
15: 8	W	GRP3D_LUTPAGEY: the y-coordinate of the upper-left corner of a LUT data in page <i>GRP3D_LUTPAGEY</i> = 16bpp pixel coordinate / 2	-
7: 0	W	GRP3D_LUTPAGEX: the x-coordinate of the upper-left corner of a LUT data in page <i>GRP3D_LUTPAGEX</i> = 16bpp pixel coordinate / 2 <i>GRP3D_LUTPAGEX</i> must be a multiple of 4 (0== <i>GRP3D_LUTPAGEX</i> [1:0])	-

☞ LUT will be automatically updated when [CMDID_LUTFILL](#) processed by Command processor

25.11.16 Texture stage 0 information register 0 (GRP3D_TEXINFO0_0)

Address : E000 0050h

Bit	R/W	Description	Default Value
31: 27	-	Reserved	-
26: 25	W	GRP3D_TEXPAGE0: Texture page index in texture segment(<i>GRP3D_TEXSEGMENT</i>)	-
24: 21	W	GRP3D_TLUTBANK0: LUT bank index for 4bit-indexed if <i>GRP3D_COLORMODE0</i> is not 0, this bits are ignored	-
20: 19	W	GRP3D_COLORMODE0: Color mode	-

Bit	R/W	Description	Default Value
		0: 4-bit color indexed. 1: 8-bit color indexed. 2: 16-bit RGB pixel format with 5 bits for red, 6 bits for green, and 5 bits for blue if GRP3D_TILEINDEX0 is 1, this bits indicate color mode of 8x8 tile data .	
18	W	GRP3D_MIPMAPENB0 : Mipmap enable if GRP3D_COLORMODE0 is not 2, this bit must be 0	
17	W	GRP3D_TEXADDRMODE0 : Texture addressing mode 0: Wrap Texture Address Mode. repeat the texture. 1: Clamp Texture Address Mode. it applies the texture once, then smears the color of edge pixels.	
16	W	GRP3D_TILEINDEX0 : Tile-index flag if this bit is 1, this texture is tile-index map. each pixel is 16bit tile index.	-
15: 0	-	Reserved	-

25.11.17 Texture stage 0 information register 1 (GRP3D_TEXINFO0_1)

Address : E000 0054h

Bit	R/W	Description	Default Value
31: 24	W	GRP3D_TEXMASKY0 : texture height parameter $GRP3D_TEXMASKY0 = (texture_height / 2) - 1$	-
23: 16	W	GRP3D_TEXMASKX0 : texture width parameter $GRP3D_TEXMASKX0 = (texture_width * bits_per_pixel / 32) - 1$ bits_per_pixel = 4, 8 or 16 if GRP3D_TILEINDEX0 is 1, bits_per_pixel is treated as 16.	-
15: 8	W	GRP3D_TEXPAGEY0 : the y-coordinate of the upper-left corner of texture in page $GRP3D_TEXPAGEY0 = 16bpp \text{ pixel coordinate} / 2$ GRP3D_TEXPAGEY0 must be a multiple of (texture_height / 2)	-
7: 0	W	GRP3D_TEXPAGEX0 : the x-coordinate of the upper-left corner of texture in page $GRP3D_TEXPAGEX0 = 16bpp \text{ pixel coordinate} / 2$ GRP3D_TEXPAGEX0 must be a multiple of (texture_width * bits_per_pixel / 32)	-

☞ All textures must have widths and heights specified as powers of 2

25.11.18 Texture stage 1 information register 0 (GRP3D_TEXINFO1_0)

Address : E000 0058h

Bit	R/W	Description	Default Value
31: 27	-	Reserved	-
26: 25	W	GRP3D_TEXPAGE1 : Texture page index in texture segment(GRP3D_TEXSEGMENT)	-
24: 21	W	GRP3D_TLUTBANK1 : LUT bank index for 4bit-indexed if GRP3D_COLORMODE1 is not 0, this bits are ignored	-
20: 19	W	GRP3D_COLORMODE1 : Color mode 0: 4-bit color indexed. 1: 8-bit color indexed. 2: 16-bit RGB pixel format with 5 bits for red, 6 bits for green, and 5 bits for blue if GRP3D_TILEINDEX1 is 1, this bits indicate color mode of 8x8 tile data.	-
18	W	GRP3D_MIPMAPENB1 : Mipmap enable if GRP3D_COLORMODE1 is not 2, this bit must be 0	
17	W	GRP3D_TEXADDRMODE1 : Texture addressing mode 0: Wrap Texture Address Mode. repeat the texture. 1: Clamp Texture Address Mode. it applies the texture once, then smears the color of edge pixels.	
16	W	GRP3D_TILEINDEX1 : Tile-index flag if this bit is 1, this texture is tile-index map. each pixel is 16bit tile index.	-
15: 0	-	Reserved	-

25.11.19 Texture stage 1 information register 1 (GRP3D_TEXINFO1_1)

Address : E000 005Ch

Bit	R/W	Description	Default Value
31: 24	W	GRP3D_TEXMASKY1: texture height parameter $GRP3D_TEXMASKY1 = (texture_height / 2) - 1$	-
23: 16	W	GRP3D_TEXMASKX1: texture width parameter $GRP3D_TEXMASKX1 = (texture_width * bits_per_pixel / 32) - 1$ bits_per_pixel = 4, 8 or 16	-
15: 8	W	GRP3D_TEXPAGEY1: the y-coordinate of the upper-left corner of texture in page $GRP3D_TEXPAGEY1 = 16bpp \text{ pixel coordinate} / 2$ GRP3D_TEXPAGEY1 must be a multiple of (texture_height / 2)	-
7: 0	W	GRP3D_TEXPAGEX1: the x-coordinate of the upper-left corner of texture in page $GRP3D_TEXPAGEX1 = 16bpp \text{ pixel coordinate} / 2$ GRP3D_TEXPAGEX1 must be a multiple of (texture_width * bits_per_pixel / 32)	-

☞ All textures must have widths and heights specified as powers of 2

25.11.20 Texture stage 0 blending function register (GRP3D_TEXBLEND0)

Address : E000 0060h

Bit	R/W	Description	Default Value
31: 8	-	Reserved	-
7: 6	W	GRP3D_TBLENDARG0_1: the second color argument for the stage 0: texture color for this texture stage 1: diffuse color interpolated from vertex components 2: result of the previous blending stage	-
5: 4	W	GRP3D_TBLENDARG0_0: the first color argument for the stage	-
3	-	Reserved	-
2: 0	W	GRP3D_TBLENDFUNC0: texture color blending function 0: Use second color argument, unmodified, as the output (RGB = Arg1) 1: Multiply the components of the arguments, and shift the products to the left 1 bit (effectively multiplying them by 2) for brightening $RGB = clamp((Arg0 \times Arg1) << 1)$ 2: Multiply the components of the arguments. $RGB = (Arg0 \times Arg1)$ 3: Add the components of the arguments $RGB = clamp(Arg0 + Arg1)$ 4: Multiply or Add the components of the arguments if(MSB(Arg1.R)) R = clamp(Arg0.R + LB(Arg1.R)) else R = (Arg0.R x Arg1.R) if(MSB(Arg1.G)) G = clamp(Arg0.G + LB(Arg1.G)) else G = (Arg0.G x Arg1.G) if(MSB(Arg1.B)) B = clamp(Arg0.B + LB(Arg1.B)) else B = (Arg0.B x Arg1.B) (☞ LB(x) == (RemoveMSB(x)<<1))	-

25.11.21 Texture stage 1 blending function register (GRP3D_TEXBLEND1)

Address : E000 0068h

Bit	R/W	Description	Default Value
31: 8	-	Reserved	-
7: 6	W	GRP3D_TBLENDARG1_1: the second color argument for the stage 0: texture color for this texture stage 1: diffuse color interpolated from vertex components	-

Bit	R/W	Description	Default Value
		2: result of the previous blending stage	
5: 4	W	GRP3D_TBLENDARG1_0 : the first color argument for the stage	-
3	-	Reserved	-
2: 0	W	GRP3D_TBLENDFUNC1 : texture color blending function 0: Use second color argument, unmodified, as the output (RGB = Arg1) 1: Multiply the components of the arguments, and shift the products to the left 1 bit (effectively multiplying them by 2) for brightening RGB = clamp((Arg0 x Arg1) << 1) 2: Multiply the components of the arguments. RGB = (Arg0 x Arg1) 3: Add the components of the arguments RGB = clamp(Arg0 + Arg1) 4: Multiply or Add the components of the arguments if(MSB(Arg1.R)) R = clamp(Arg0.R + LB(Arg1.R)) else R = (Arg0.R x Arg1.R) if(MSB(Arg1.G)) G = clamp(Arg0.G + LB(Arg1.G)) else G = (Arg0.G x Arg1.G) if(MSB(Arg1.B)) B = clamp(Arg0.B + LB(Arg1.B)) else B = (Arg0.B x Arg1.B) (☞ LB(x) == (RemoveMSB(x)<<1))	-

25.11.22 Primitive processor control register (GRP3D_PRIMCON)

Address : E000 0070h

Bit	R/W	Description	Default Value
31: 26	-	Reserved	-
25	W	GRP3D_PRIMMODE : Primitive mode. 0: TSE rendering mode.(use vertex in screen space) 1: GTE rendering mode.	-
24	W	GRP3D_IDXFMT : index data format. 0: Indices are 8 bits each. 1: Indices are 16 bits each. if GRP3D_IDXENB is 0, this bit is ignored	-
23	W	GRP3D_IDXENB : Indexed primitive enable. 0: Renders a sequence of nonindexed, geometric primitives of the specified type from the set of vertex. 1: Renders the indexed geometric primitive into an array of vertices.	-
22: 21	W	GRP3D_PRIMTYPE : Primitive type. 0: TRILIST Renders the specified vertices as a sequence of isolated triangles. Each group of three vertices defines a separate triangle. 1: TRIFAN Renders the vertices as a triangle fan. 2: TRISTRIP Renders the vertices as a triangle strip. The backface-culling flag is automatically flipped on even-numbered triangles. 3: RECTLIST Renders the vertices as a rectangle. Each group of two vertices defines a separate rectangle. one is top-left, and the other is bottom-right corner. The backface-culling is ignored. GRP3D_PRIMMODE must be 0.	-
20: 16	W	GRP3D_VSTRIDE : Vertex stride stride between start point for each vertex. GRP3D_VSTRIDE = (Vertex stride / 8) – 1 here Vertex stride value is one of below values Vertex stride = sub-stride[0] Vertex stride = sub-stride[0] + sub-stride[1] Vertex stride = sub-stride[0] + sub-stride[1] + sub-stride[2] Vertex stride = sub-stride[0] + sub-stride[1] + sub-stride[2] + sub-stride[3]	-
15: 0	W	GRP3D_VCOUNT : Vertex count	-

Bit	R/W	Description	Default Value
		number of vertex to primitive render. number of vertex = $GRP3D_VCOUNT + 1$ if $GRP3D_PRIMTYPE$ is 0, $GRP3D_VCOUNT$ must be (multiple of 3)-1 if $GRP3D_PRIMTYPE$ is 0,1,2, $GRP3D_VCOUNT$ must be greater than 1 if $GRP3D_PRIMTYPE$ is 3, $GRP3D_VCOUNT$ must be (multiple of 2)-1	

☞ Primitive processor will be automatically running when [CMDID_PRIMITIVE](#) processed by Command processor

25.11.23 Primitive index buffer address register (GRP3D_PRIMIDX)

Address : E000 0074h

Bit	R/W	Description	Default Value
31: 0	W	GRP3D_PRIMIDX : Index buffer address if $GRP3D_IDXFORMAT$ is 1, $GRP3D_PRIMIDX$ must be even	-

25.11.24 Primitive vertex buffer 0 register (GRP3D_PRIMVTX0)

Address : E000 0078h

Bit	R/W	Description	Default Value
31: 3	W	GRP3D_VTXADDR0 : Vertex buffer address vertex data pointer must be aligned to 8 Byte(64 bit) ☞ vertex data pointer = { $GRP3D_VTXADDR0[31:3], 000$ }	-
2: 0	W	GRP3D_VSUBSTRIDE0 : Vertex sub-stride. 0: 8 Byte/Vertex 1: 16 Byte/Vertex 2: 32 Byte/Vertex 3: 64 Byte/Vertex 4: 128 Byte/Vertex 5: 256 Byte/Vertex	-

25.11.25 Primitive vertex buffer 1 register (GRP3D_PRIMVTX1)

Address : E000 007Ch

Bit	R/W	Description	Default Value
31: 3	W	GRP3D_VTXADDR1 : Vertex buffer address vertex data pointer must be aligned to 8 Byte(64 bit) ☞ vertex data pointer = { $GRP3D_VTXADDR1[31:3], 000$ }	-
2: 0	W	GRP3D_VSUBSTRIDE1 : Vertex sub-stride. 0: 8 Byte/Vertex 1: 16 Byte/Vertex 2: 32 Byte/Vertex 3: 64 Byte/Vertex 4: 128 Byte/Vertex 5: 256 Byte/Vertex	-

25.11.26 Primitive vertex buffer 2 register (GRP3D_PRIMVTX2)

Address : E000 0080h

Bit	R/W	Description	Default Value
31: 3	W	GRP3D_VTXADDR2 : Vertex buffer address vertex data pointer must be aligned to 8 Byte(64 bit) ☞ vertex data pointer = { $GRP3D_VTXADDR2[31:3], 000$ }	-
2: 0	W	GRP3D_VSUBSTRIDE2 : Vertex sub-stride. 0: 8 Byte/Vertex	-

Bit	R/W	Description	Default Value
		1: 16 Byte/Vertex 2: 32 Byte/Vertex 3: 64 Byte/Vertex 4: 128 Byte/Vertex 5: 256 Byte/Vertex	

25.11.27 Primitive vertex buffer 3 register (GRP3D_PRIMVTX3)

Address : E000 0084h

Bit	R/W	Description	Default Value
31: 3	W	GRP3D_VTXADDR3 : Vertex buffer address vertex data pointer must be aligned to 8 Byte(64 bit) ☞ vertex data pointer = { GRP3D_VTXADDR3 [31:3],000}	-
2: 0	W	GRP3D_VSUBSTRIDE3 : Vertex sub-stride. 0: 8 Byte/Vertex 1: 16 Byte/Vertex 2: 32 Byte/Vertex 3: 64 Byte/Vertex 4: 128 Byte/Vertex 5: 256 Byte/Vertex	-

25.11.28 Fog color register (GRP3D_FOGCOLOR)

Address : E000 0090h

Bit	R/W	Description	Default Value
31: 4	-	Reserved	-
23: 0	W	GRP3D_FOGCOLOR : Fog color R8G8B8	-

25.11.29 Fog max z register (GRP3D_FOGMAXZ)

Address : E000 0094h

Bit	R/W	Description	Default Value
31: 8	W	GRP3D_FOGMAXZ : maximum z-value of applicable region of fog table . $2^{-12} < \text{GRP3D_FOGMAXZ} < 2^{16}$ if $z \geq \text{GRP3D_FOGMAXZ}$, fog blend parameter = last fog table value. 24bit floating point number. this register has 8bit offset. In this way, you can use 32bit floating point number, like this...(example in C) float Fog_max_z; ... $((\text{volatile unsigned long}^*)(0xE0000094)) = ((\text{unsigned long}^*)&\text{Fog_max_z})$	-
7: 0	-	Reserved	-

25.11.30 Display information register (GRP3D_DISPINFO)

Address : E000 0098h

Bit	R/W	Description	Default Value
31: 28	-	Reserved	-
27	W	GRP3D_DISPAA : Display antialiasing flag	-
26: 16	W	GRP3D_DISPSEGMENT : Display segment	-
15: 8	W	GRP3D_DISPTBLY : Display y-coordinate conversion table If Ydisplay [9:8] == 0, Ymemory[9:0] = { GRP3D_DISPTBLY [1:0] , Ydisplay[7:0]}	-

Bit	R/W	Description	Default Value
		If Ydisplay [9:8] == 1, Ymemory[9:0] = { GRP3D_DISPTBLY [3:2], Ydisplay[7:0]} If Ydisplay [9:8] == 2, Ymemory[9:0] = { GRP3D_DISPTBLY [5:4], Ydisplay[7:0]} If Ydisplay [9:8] == 3, Ymemory[9:0] = { GRP3D_DISPTBLY [7:6], Ydisplay[7:0]}	
7: 0	W	GRP3D_DISPTBLX : Display x-coordinate conversion table If Xdisplay [9:8] == 0, Xmemory[9:0] = { GRP3D_DISPTBLX [1:0], Xdisplay[7:0]} If Xdisplay [9:8] == 1, Xmemory[9:0] = { GRP3D_DISPTBLX [3:2], Xdisplay[7:0]} If Xdisplay [9:8] == 2, Xmemory[9:0] = { GRP3D_DISPTBLX [5:4], Xdisplay[7:0]} If Xdisplay [9:8] == 3, Xmemory[9:0] = { GRP3D_DISPTBLX [7:6], Xdisplay[7:0]}	-

25.11.31 Memory fill parameter register 0 (GRP3D_MFILLPARAM0)

Address : E000 00A0h

Bit	R/W	Description	Default Value
31: 28	-	Reserved	-
27	W	GRP3D_MFADDRMODE : Memory fill addressing mode 0: display address mode 1: texture address mode	-
26: 16	W	GRP3D_MFSEGMENT : Memory fill segment	-
15: 0	W	GRP3D_MFDATA : Memory fill data	-

25.11.32 Memory fill parameter register 1 (GRP3D_MFILLPARAM1)

Address : E000 00A4h

Bit	R/W	Description	Default Value
31: 26	-	Reserved	-
25: 16	W	GRP3D_MFTOP : the y-coordinate of the upper-left corner of rectangle in Memory fill segment	-
15: 10	-	Reserved	-
9: 0	W	GRP3D_MFLEFT : the x-coordinate of the upper-left corner of rectangle in Memory fill segment	-

25.11.33 Memory fill parameter register 2 (GRP3D_MFILLPARAM2)

Address : E000 00A8h

Bit	R/W	Description	Default Value
31: 26	-	Reserved	-
25: 16	W	GRP3D_MFBOTTOM : the y-coordinate of the lower-right corner of rectangle in Memory fill segment GRP3D_MFTOP <= Y range <= GRP3D_MFBOTTOM	-
15: 10	-	Reserved	-
9: 0	W	GRP3D_MFRIGHT : the x-coordinate of the lower-right corner of rectangle in Memory fill segment GRP3D_MFLEFT <= X range <= GRP3D_MFRIGHT	-

25.11.34 Render target register 0 (GRP3D_RENDTRG0)

Address : E000 00B0h

Bit	R/W	Description	Default Value
31: 29	-	Reserved	-

Bit	R/W	Description	Default Value
28	W	GRP3D_RTAA : Render target antialiasing flag	-
27	W	GRP3D_RTADDRMODE : Render target addressing mode 0: display address mode 1: texture address mode	-
26: 16	W	GRP3D_RTSEGMENT : Render target segment	-
15: 8	W	GRP3D_RTTBLY : Rasterizer y-coordinate conversion table If Yrasterizer [9:8] == 0, Ymemory[9:0] = { GRP3D_RTTBLY [1:0], Yrasterizer [7:0]} If Yrasterizer [9:8] == 1, Ymemory[9:0] = { GRP3D_RTTBLY [3:2], Yrasterizer [7:0]} If Yrasterizer [9:8] == 2, Ymemory[9:0] = { GRP3D_RTTBLY [5:4], Yrasterizer [7:0]} If Yrasterizer [9:8] == 3, Ymemory[9:0] = { GRP3D_RTTBLY [7:6], Yrasterizer [7:0]}	-
7: 0	W	GRP3D_RTTBLX : Rasterizer x-coordinate conversion table If Xrasterizer [9:8] == 0, Xmemory[9:0] = { GRP3D_RTTBLX [1:0], Xrasterizer [7:0]} If Xrasterizer [9:8] == 1, Xmemory[9:0] = { GRP3D_RTTBLX [3:2], Xrasterizer [7:0]} If Xrasterizer [9:8] == 2, Xmemory[9:0] = { GRP3D_RTTBLX [5:4], Xrasterizer [7:0]} If Xrasterizer [9:8] == 3, Xmemory[9:0] = { GRP3D_RTTBLX [7:6], Xrasterizer [7:0]}	-

25.11.35 Render target register 1 (GRP3D_RENDTRG1)

Address : E000 00B4h

Bit	R/W	Description	Default Value
31: 26	-	Reserved	-
25: 16	W	GRP3D_RTTOP : the y-coordinate of the upper-left corner of rectangle in Render target segment	-
15: 11	-	Reserved	-
10: 0	W	GRP3D_RTLEFT : the x-coordinate of the upper-left corner of rectangle in Render target segment	-

25.11.36 Render target register 2 (GRP3D_RENDTRG2)

Address : E000 00B8h

Bit	R/W	Description	Default Value
31: 26	-	Reserved	-
25: 16	W	GRP3D_RTBOTTOM : the y-coordinate of the lower-right corner of rectangle in Render target segment GRP3D_RTTOP <= Y range <= GRP3D_RTBOTTOM	-
15: 11	-	Reserved	-
10: 0	W	GRP3D_RTRIGHT : the x-coordinate of the lower-right corner of rectangle in Render target segment GRP3D_RTLEFT <= X range <= GRP3D_RTRIGHT	-

25.11.37 Z-buffer information register (GRP3D_ZBINFO)

Address : E000 00C0h

Bit	R/W	Description	Default Value
31	-	Reserved	-
30: 28	W	GRP3D_ZBFUNC : Z-buffer test function 0: NEVER Always fail the test 1: LESS Accept the new pixel if its value is less than the value of the current pixel 2: EQUAL Accept the new pixel if its value equals the value of the current pixel	-

Bit	R/W	Description	Default Value
		3: LESSEQUAL Accept the new pixel if its value is less than or equal to the value of the current pixel 4: GREATER Accept the new pixel if its value is greater than the value of the current pixel 5: NOTEQUAL Accept the new pixel if its value does not equal the value of the current pixel 6: GREATEREQUAL Accept the new pixel if its value is greater than or equal to the value of the current pixel 7: ALWAYS Always pass the test	
27	W	GRP3D_ZBADDRMODE : Z-buffer address mode 0: display address mode 1: texture address mode	-
26: 16	W	GRP3D_ZBSEGMENT : Z-buffer segment	-
15: 8	W	GRP3D_ZBTBLY : Rasterizer y-coordinate conversion table If Yrasterizer [9:8] == 0, Ymemory[9:0] = { GRP3D_ZBTBLY [1:0], Yrasterizer [7:0]} If Yrasterizer [9:8] == 1, Ymemory[9:0] = { GRP3D_ZBTBLY [3:2], Yrasterizer [7:0]} If Yrasterizer [9:8] == 2, Ymemory[9:0] = { GRP3D_ZBTBLY [5:4], Yrasterizer [7:0]} If Yrasterizer [9:8] == 3, Ymemory[9:0] = { GRP3D_ZBTBLY [7:6], Yrasterizer [7:0]}	-
7: 0	W	GRP3D_ZBTBLX : Rasterizer x-coordinate conversion table If Xrasterizer [9:8] == 0, Xmemory[9:0] = { GRP3D_ZBTBLX [1:0], Xrasterizer [7:0]} If Xrasterizer [9:8] == 1, Xmemory[9:0] = { GRP3D_ZBTBLX [3:2], Xrasterizer [7:0]} If Xrasterizer [9:8] == 2, Xmemory[9:0] = { GRP3D_ZBTBLX [5:4], Xrasterizer [7:0]} If Xrasterizer [9:8] == 3, Xmemory[9:0] = { GRP3D_ZBTBLX [7:6], Xrasterizer [7:0]}	-

25.11.38 Z-valid-buffer information register (GRP3D_ZVBINFO)

Address : E000 00C4h

Bit	R/W	Description	Default Value
31: 25	-	Reserved	-
24	W	GRP3D_ZVBENB : Z-valid-buffer enable Z-valid buffer is 64x128 pixel(16bit) (rectangle area) in Z-buffer segment . Z-valid buffer is for reducing Z-buffer access. It has 1valid bit per 8 pixel.	-
23: 20	W	GRP3D_ZVBTOP : the y-coordinate of the upper-left corner of Z-valid-buffer in Z-buffer segment GRP3D_ZVBTOP must be multiple of 2 (0== GRP3D_ZVBTOP [0]) 16bpp Y position in segment = { GRP3D_ZVBTOP [3:1], 7'b000_000 }	-
19: 16	W	GRP3D_ZVBLEFT : the x-coordinate of the upper-left corner of Z-valid-buffer in Z-buffer segment 16bpp X position in segment = { GRP3D_ZVBTOP [3:0], 6'b00_000 }	-
15: 0	W	GRP3D_ZVVALUE : Z-valid-buffer invalid value If Z-valid-buffer bit is 0, destination z vlue treated as GRP3D_ZVVALUE (Normally FFFFh)	-

25.11.39 Z scale register (GRP3D_ZSCALE)

Address : E000 00C8h

Bit	R/W	Description	Default Value
31: 8	W	GRP3D_ZSCALE : Z scale value TSE multiply a projected z-value by GRP3D_ZSCALE to make z-value to be 0~0fff, because A Projected z-value(TSE input z) is 0.0f ~ 1.0f 24bit floating point number. Although The format of register is 24bit floating point. User does not need to convert 32bit floating point to 24bit floating point as below. float Zscale = 65534.0f; $((\text{volatile unsigned long})(0xE00000C8)) = ((\text{unsigned long})\&(\text{Zscale}))$	-

Bit	R/W	Description	Default Value
7: 0	-	Reserved	-

25.11.40 Vertex buffer fixed point resolution register (GRP3D_VBFIXED)

Address : E000 00D0h

Bit	R/W	Description	Default Value
31: 15	-	Reserved	-
14: 10	W	GRP3D_VBFIXED32 : 32bit fixed point number resolution in Vertex buffer Real value = [signed 32bit] x 2 ^{-GRP3D_VBFIXED32}	-
9: 5	W	GRP3D_VBFIXED16 : 16bit fixed point number resolution in Vertex buffer Real value = [signed 16bit] x 2 ^{-GRP3D_VBFIXED16}	-
4: 0	W	GRP3D_VBFIXED8 : 8bit fixed point number resolution in Vertex buffer Real value = [unsigned 8bit] x 2 ^{-GRP3D_VBFIXED8}	-

25.11.41 Vertex buffer format register 0 (GRP3D_VBFORMAT0)

Address : E000 00D4h

Bit	R/W	Description	Default Value
31: 30	W	GRP3D_VBF15 : Vertex buffer format (address 3Ch~3Fh) 0: 8bit fixed point numbers (4ea) 1: 16bit fixed point numbers (2ea) 2: 32bit fixed point number (1ea) 3: 32bit floating point number (1ea)	-
29: 28	W	GRP3D_VBF14 : Vertex buffer format (address 38h~3Bh)	-
27: 26	W	GRP3D_VBF13 : Vertex buffer format (address 34h~37h)	-
25: 24	W	GRP3D_VBF12 : Vertex buffer format (address 30h~33h)	-
23: 22	W	GRP3D_VBF11 : Vertex buffer format (address 2Ch~2Fh)	-
21: 20	W	GRP3D_VBF10 : Vertex buffer format (address 28h~2Bh)	-
19: 18	W	GRP3D_VBF9 : Vertex buffer format (address 24h~27h)	-
17: 16	W	GRP3D_VBF8 : Vertex buffer format (address 20h~23h)	-
15: 14	W	GRP3D_VBF7 : Vertex buffer format (address 1Ch~1Fh)	-
13: 12	W	GRP3D_VBF6 : Vertex buffer format (address 18h~1Bh)	-
11: 10	W	GRP3D_VBF5 : Vertex buffer format (address 14h~17h)	-
9: 8	W	GRP3D_VBF4 : Vertex buffer format (address 10h~13h)	-
7: 6	W	GRP3D_VBF3 : Vertex buffer format (address 0Ch~0Fh)	-
5: 4	W	GRP3D_VBF2 : Vertex buffer format (address 08h~0Bh)	-
3: 2	W	GRP3D_VBF1 : Vertex buffer format (address 04h~07h)	-
1: 0	W	GRP3D_VBF0 : Vertex buffer format (address 00h~03h)	-

25.11.42 Vertex buffer format register 1 (GRP3D_VBFORMAT1)

Address : E000 00D8h

Bit	R/W	Description	Default Value
31: 30	W	GRP3D_VBF31 : Vertex buffer format (address 7Ch~7Fh)	-
29: 28	W	GRP3D_VBF30 : Vertex buffer format (address 78h~7Bh)	-
27: 26	W	GRP3D_VBF29 : Vertex buffer format (address 74h~77h)	-

Bit	R/W	Description	Default Value
25: 24	W	GRP3D_VBF28: Vertex buffer format (address 70h~73h)	-
23: 22	W	GRP3D_VBF27: Vertex buffer format (address 6Ch~6Fh)	-
21: 20	W	GRP3D_VBF26: Vertex buffer format (address 68h~6Bh)	-
19: 18	W	GRP3D_VBF25: Vertex buffer format (address 64h~67h)	-
17: 16	W	GRP3D_VBF24: Vertex buffer format (address 60h~63h)	-
15: 14	W	GRP3D_VBF23: Vertex buffer format (address 5Ch~5Fh)	-
13: 12	W	GRP3D_VBF22: Vertex buffer format (address 58h~5Bh)	-
11: 10	W	GRP3D_VBF21: Vertex buffer format (address 54h~57h)	-
9: 8	W	GRP3D_VBF20: Vertex buffer format (address 50h~53h)	-
7: 6	W	GRP3D_VBF19: Vertex buffer format (address 4Ch~4Fh)	-
5: 4	W	GRP3D_VBF18: Vertex buffer format (address 48h~4Bh)	-
3: 2	W	GRP3D_VBF17: Vertex buffer format (address 44h~47h)	-
1: 0	W	GRP3D_VBF16: Vertex buffer format (address 40h~43h)	-

25.11.43 Vertex buffer format register 2 (GRP3D_VBFORMAT2)

Address : E000 00DCh

Bit	R/W	Description	Default Value
31: 30	W	GRP3D_VBF47: Vertex buffer format (address BCh~BFh)	-
29: 28	W	GRP3D_VBF46: Vertex buffer format (address B8h~BBh)	-
27: 26	W	GRP3D_VBF45: Vertex buffer format (address B4h~B7h)	-
25: 24	W	GRP3D_VBF44: Vertex buffer format (address B0h~B3h)	-
23: 22	W	GRP3D_VBF43: Vertex buffer format (address ACh~AFh)	-
21: 20	W	GRP3D_VBF42: Vertex buffer format (address A8h~ABh)	-
19: 18	W	GRP3D_VBF41: Vertex buffer format (address A4h~A7h)	-
17: 16	W	GRP3D_VBF40: Vertex buffer format (address A0h~A3h)	-
15: 14	W	GRP3D_VBF39: Vertex buffer format (address 9Ch~9Fh)	-
13: 12	W	GRP3D_VBF38: Vertex buffer format (address 98h~9Bh)	-
11: 10	W	GRP3D_VBF37: Vertex buffer format (address 94h~97h)	-
9: 8	W	GRP3D_VBF36: Vertex buffer format (address 90h~93h)	-
7: 6	W	GRP3D_VBF35: Vertex buffer format (address 8Ch~8Fh)	-
5: 4	W	GRP3D_VBF34: Vertex buffer format (address 88h~8Bh)	-
3: 2	W	GRP3D_VBF33: Vertex buffer format (address 84h~87h)	-
1: 0	W	GRP3D_VBF32: Vertex buffer format (address 80h~83h)	-

25.11.44 Vertex buffer format register 3 (GRP3D_VBFORMAT3)

Address : E000 00E0h

Bit	R/W	Description	Default Value
31: 30	W	GRP3D_VBF63: Vertex buffer format (address FCh~FFh)	-
29: 28	W	GRP3D_VBF62: Vertex buffer format (address F8h~FBh)	-
27: 26	W	GRP3D_VBF61: Vertex buffer format (address F4h~F7h)	-
25: 24	W	GRP3D_VBF60: Vertex buffer format (address F0h~F3h)	-
23: 22	W	GRP3D_VBF59: Vertex buffer format (address Ech~EFh)	-

Bit	R/W	Description	Default Value
21: 20	W	GRP3D_VBF58: Vertex buffer format (address E8h~EBh)	-
19: 18	W	GRP3D_VBF57: Vertex buffer format (address E4h~E7h)	-
17: 16	W	GRP3D_VBF56: Vertex buffer format (address E0h~E3h)	-
15: 14	W	GRP3D_VBF55: Vertex buffer format (address DCh~DFh)	-
13: 12	W	GRP3D_VBF54: Vertex buffer format (address D8h~DBh)	-
11: 10	W	GRP3D_VBF53: Vertex buffer format (address D4h~D7h)	-
9: 8	W	GRP3D_VBF52: Vertex buffer format (address D0h~D3h)	-
7: 6	W	GRP3D_VBF51: Vertex buffer format (address CCh~CFh)	-
5: 4	W	GRP3D_VBF50: Vertex buffer format (address C8h~CBh)	-
3: 2	W	GRP3D_VBF49: Vertex buffer format (address C4h~C7h)	-
1: 0	W	GRP3D_VBF48: Vertex buffer format (address C0h~C3h)	-

25.11.45 Viewport bottom register (GRP3D_VPORTBOT)

Address : E000 00F0h

Bit	R/W	Description	Default Value
31: 8	W	GRP3D_VPORTBOT: the lower-bound coordinate of viewport 24bit floating point number. this register has 8bit offset. In this way, you can use 32bit floating point number, like this...(example in C) float Bottom = viewport.y + viewport.height; *((volatile unsigned long*)(0xE00000F0)) = (*(unsigned long*)&Bottom))	-
7: 0	-	Reserved	-

25.11.46 Viewport height register (GRP3D_VPORTH)

Address : E000 00F4h

Bit	R/W	Description	Default Value
31: 8	W	GRP3D_VPORTH: the height of viewport 24bit floating point number. this register has 8bit offset. In this way, you can use 32bit floating point number, like this...(example in C) float Height = viewport.height; *((volatile unsigned long*)(0xE00000F4)) = (*(unsigned long*)&Height))	-
7: 0	-	Reserved	-

25.11.47 Viewport left register (GRP3D_VPORTLEFT)

Address : E000 00F8h

Bit	R/W	Description	Default Value
31: 8	W	GRP3D_VPORTLEFT: the left-bound coordinate of viewport 24bit floating point number. this register has 8bit offset. In this way, you can use 32bit floating point number, like this...(example in C) float Left = viewport.x; *((volatile unsigned long*)(0xE00000F8)) = (*(unsigned long*)&Left))	-
7: 0	-	Reserved	-

25.11.48 Viewport width register 3 (GRP3D_VPORTW)

Address : E000 00FCh

Bit	R/W	Description	Default Value
31: 8	W	GRP3D_VPORTW : the width of viewport 24bit floating point number. this register has 8bit offset. In this way, you can use 32bit floating point number, like this...(example in C) float Width = viewport.width; *((volatile unsigned long*)(0xE00000FC)) = (*(unsigned long*)&(Width))	-
7: 0	-	Reserved	-

25.11.49 Fog table register (GRP3D_FOGTBL)

Address : E000 0100h ~ E000 017Fh

Bit	R/W	Description	Default Value
31: 30	-	Reserved	-
29: 24	W	GRP3D_FOGDIFF : difference of fog table value (1,3,5,7,...,63)	-
23: 16	W	GRP3D_FOGVAL : fog table value (1,3,5,7,...,63)	-
15: 14	-	Reserved	-
13: 8	W	GRP3D_FOGDIFF : difference of fog table value (0,2,4,8,...,62)	-
7: 0	W	GRP3D_FOGVAL : fog table value (0,2,4,8,...,62)	-

for $0 \leq n \leq 62$, $GRP3D_FOGDIFF[n] = (GRP3D_FOGVAL[n+1] - GRP3D_FOGVAL[n]) / 4$
 $GRP3D_FOGDIFF[63]$ must be 0

25.11.50 GTE Program register (GRP3D_GTECODE)

Address : E000 0200h ~ E000 02FFh

Bit	R/W	Description	Default Value
31: 0	W	GRP3D_GTECODE : GTE instruction	-

25.11.51 GTE Constant vector register (GRP3D_GTECONST)

Address : E000 0300h ~ E000 03FFh

Bit	R/W	Description	Default Value
31: 8	W	GRP3D_GTECONST : GTE constant vector 24bit floating point number.	-
7: 0	-	Reserved	-

$((float*)GRP3D_GTECONST)[n * 4 + 0]$: X of const vector[n]
 $((float*)GRP3D_GTECONST)[n * 4 + 1]$: Y of const vector[n]
 $((float*)GRP3D_GTECONST)[n * 4 + 2]$: Z of const vector[n]
 $((float*)GRP3D_GTECONST)[n * 4 + 3]$: W of const vector[n]
 ($0 \leq n \leq 15$, $GRP3D_GTECONST = E000_0300h$)

25.11.52 GTE Input vector register (GRP3D_GTEINPUT)

Address : E000 0400h ~ E000 04FFh

Bit	R/W	Description	Default Value
31: 8	W	GRP3D_GTEINPUT : GTE input vector 24bit floating point number.	-
7: 0	-	Reserved	-

$((float*)GRP3D_GTEINPUT)[n * 4 + 0]$: X of input vector[n]

$((\text{float}^*)\text{GRP3D_GTEINPUT})[n * 4 + 1]$: Y of input vector[n]
 $((\text{float}^*)\text{GRP3D_GTEINPUT})[n * 4 + 2]$: Z of input vector[n]
 $((\text{float}^*)\text{GRP3D_GTEINPUT})[n * 4 + 3]$: W of input vector[n]
 ($0 \leq n \leq 15$, $\text{GRP3D_GTEINPUT} = \text{E000_0400h}$)

25.11.53 TSE Input vertex register (GRP3D_TSEINPUT)

Address : E000 0500h ~ E000 05AFh

Bit	R/W	Description	Default Value
31: 8	W	GRP3D_TSEINPUT: TSE input (Clipper output) vertex 24bit floating point number.	-
7: 0	-	Reserved	-

$((\text{float}^*)\text{GRP3D_TSEINPUT})[n * 16 + 0]$: X of input vertex[n]
 $((\text{float}^*)\text{GRP3D_TSEINPUT})[n * 16 + 1]$: Y of input vertex[n]
 $((\text{float}^*)\text{GRP3D_TSEINPUT})[n * 16 + 2]$: Z of input vertex[n]
 $((\text{float}^*)\text{GRP3D_TSEINPUT})[n * 16 + 3]$: W of input vertex[n]
 $((\text{float}^*)\text{GRP3D_TSEINPUT})[n * 16 + 4]$: A of input vertex[n]
 $((\text{float}^*)\text{GRP3D_TSEINPUT})[n * 16 + 5]$: R of input vertex[n]
 $((\text{float}^*)\text{GRP3D_TSEINPUT})[n * 16 + 6]$: G of input vertex[n]
 $((\text{float}^*)\text{GRP3D_TSEINPUT})[n * 16 + 7]$: B of input vertex[n]
 $((\text{float}^*)\text{GRP3D_TSEINPUT})[n * 16 + 8]$: U0 of input vertex[n]
 $((\text{float}^*)\text{GRP3D_TSEINPUT})[n * 16 + 9]$: V0 of input vertex[n]
 $((\text{float}^*)\text{GRP3D_TSEINPUT})[n * 16 + 10]$: U1 of input vertex[n]
 $((\text{float}^*)\text{GRP3D_TSEINPUT})[n * 16 + 11]$: V1 of input vertex[n]
 ($0 \leq n \leq 2$, $\text{GRP3D_TSEINPUT} = \text{E000_0500h}$)

25.11.54 GTE output vertex register (GRP3D_GTEOUTPUT)

Address : E000 0700h ~ E000 078Fh

Bit	R/W	Description	Default Value
31: 8	R	GRP3D_GTEOUTPUT: GTE output (Clipper input) vertex 24bit floating point number.	-
7: 0	-	Reserved	-

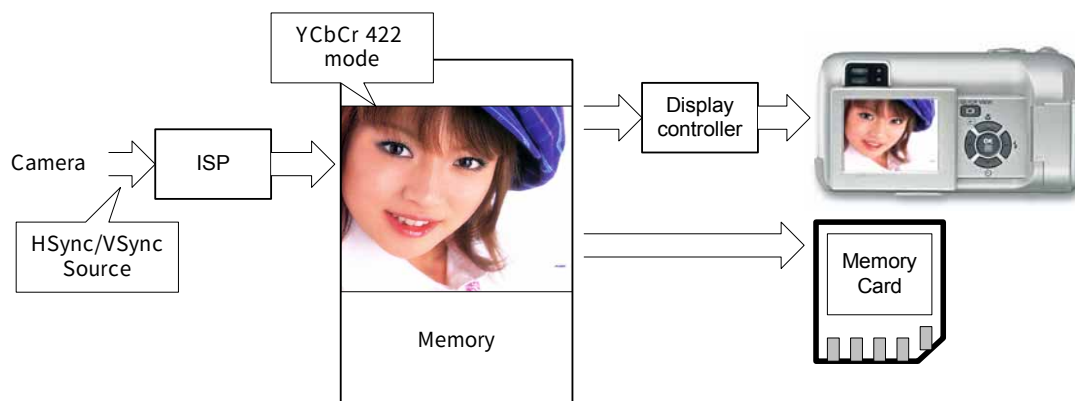
$((\text{float}^*)\text{GRP3D_GTEOUTPUT})[n * 12 + 0]$: X of input vertex[n]
 $((\text{float}^*)\text{GRP3D_GTEOUTPUT})[n * 12 + 1]$: Y of input vertex[n]
 $((\text{float}^*)\text{GRP3D_GTEOUTPUT})[n * 12 + 2]$: Z of input vertex[n]
 $((\text{float}^*)\text{GRP3D_GTEOUTPUT})[n * 12 + 3]$: W of input vertex[n]
 $((\text{float}^*)\text{GRP3D_GTEOUTPUT})[n * 12 + 4]$: A of input vertex[n]
 $((\text{float}^*)\text{GRP3D_GTEOUTPUT})[n * 12 + 5]$: R of input vertex[n]
 $((\text{float}^*)\text{GRP3D_GTEOUTPUT})[n * 12 + 6]$: G of input vertex[n]
 $((\text{float}^*)\text{GRP3D_GTEOUTPUT})[n * 12 + 7]$: B of input vertex[n]
 $((\text{float}^*)\text{GRP3D_GTEOUTPUT})[n * 12 + 8]$: U0 of input vertex[n]
 $((\text{float}^*)\text{GRP3D_GTEOUTPUT})[n * 12 + 9]$: V0 of input vertex[n]
 $((\text{float}^*)\text{GRP3D_GTEOUTPUT})[n * 12 + 10]$: U1 of input vertex[n]
 $((\text{float}^*)\text{GRP3D_GTEOUTPUT})[n * 12 + 11]$: V1 of input vertex[n]
 ($0 \leq n \leq 2$, $\text{GRP3D_GTEOUTPUT} = \text{E000_0700h}$)

26. Image Signal Processor (ISP)

26.1 Overview

ISP stores YCbCr Image from Camera & External Video to External Memory

ISP support Linear address and Block address. ISP support coarse image scale down for preview in real time.



26.1.1 memory structure

26.1.1.1 Block address mode

Output address is assigned as Segment and Base coordinate type when Target is Block address mode. Block address mode supports maximum 2million pixel image capturing at display mode. (refer to Memory structure of 3D Core for Segment/block addressing)

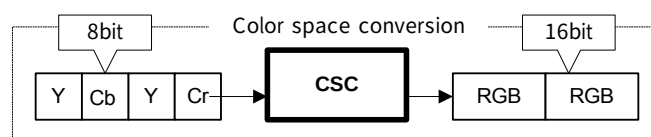
26.1.1.2 Linear address mode

Certain address and stride can be used when Target is Linear address mode.

Linear address mode support maximum 4million pixel image capture (2048x2048).

26.1.2 Pixel format

Pixel format is 422mode where Y,Cb,Cr coexist. (16 bpp)



< ISP pixel format (YCbCr 422) >

26.1.3 Scaling

Scaling supports nearest scale down. Scaling does not support scale up from camera source.

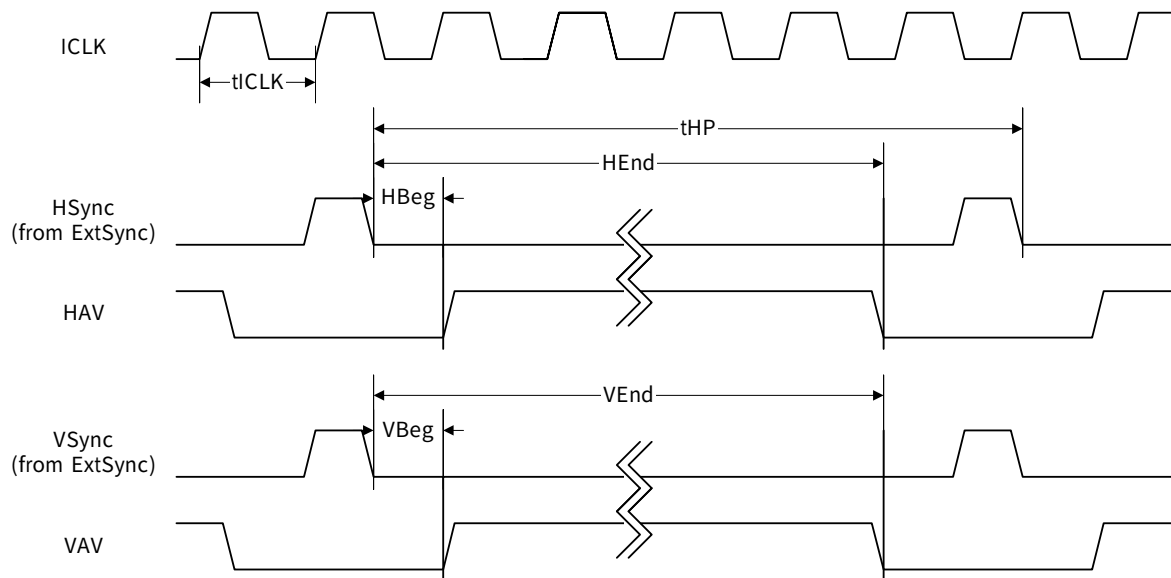
26.1.4 Input signal type

26.1.4.1 External sync (from Video decoder)

Horizontal/vertical sync signal is input from outside and active video signal is generated internally.

Parameter	Description	Unit
HBeg	Begin of horizontal active video	tICLK
HEnd	End of horizontal active video	tICLK

Parameter	Description	Unit
VBeg	Begin of vertical active video	tHP
VEnd	End of vertical active video	tHP

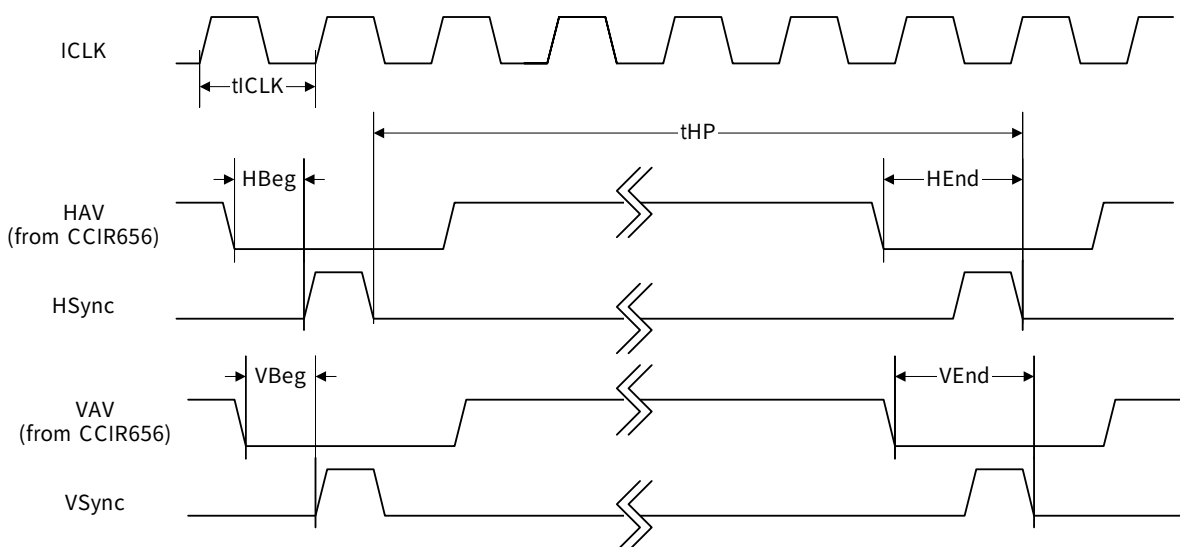


< Signal information for external sync >

26.1.4.2 CCIR656

active video signal is input from outside and Horizontal/vertical sync signal is generated internally.

parameter	Description	Unit
HBeg	Begin of horizontal sync	tICLK
HEnd	End of horizontal sync	tICLK
VBeg	Begin of vertical sync	tHP
VEnd	End of vertical sync	tHP



< Signal information for CCIR656 >

26.1.5 Capture sequence

26.1.5.1 *Image Capture for Display*

Continuously overwrite camera image to memory for screen display

- (1) ISP capture disable
- (2) ISP module disable
- (3) Camera module setup
- (4) Signal information setup (RasterizerMode, InputType, ...)
- (5) Destination setup (address mode, address, size, scale)
- (6) ISP interrupt disable
- (7) ISP module enable
- (8) ISP capture enable

26.1.5.2 *Image Capture for Still Image*

Store camera image to memory once for getting still image

- (1)~(5) Same as 'Image capture for Display'
- (6) ISP interrupt setup (interrupt enable, handler setting, pending clear ...)
- (7) ISP module enable
- (8) When ISP interrupts occur
- (9) ISP capture enable
- (10) When ISP interrupts occur
- (11) ISP capture disable
- (12) ISP module disable.

26.1.5.3 *Image Capture for Continuous Still Image*

Assign another memory after each capture for getting continuous still image

- (1)~(9) Same as 'Image Capture for Still Image'
- (10) When ISP interrupt occur, setup new destination
- (11) If finish capturing
- (12) ISP capture disable
- (13) ISP module disable.

26.2 ISP Registers

26.2.1 Register map (C000_2000h ~ C000_201Fh)

Register Name	offset	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
ISPCONTROL	00h																					ISPCONTROL															
ISPHSYNC	04h						ISPHOREND																ISPHORBEG														
ISPVSYNC	08h						ISPVEND																ISPVBEG														
ISPSIZE	0Ch						ISPHEIGHT																ISPWIDTH														
ISPDSize	10h						ISPDHEIGHT																ISPDWIDTH														
ISPSIZE_1	14h						ISPHEIGHT_1																ISPWIDTH_1														
ISPBASADDR	18h	ISPBASADDR																																			
ISPSTRIDE	1Ch	ISPSTRIDE																																			

26.2.2 ISP control register (ISPCONTROL)

Address : C000 2000h

Bit	R/W	Description	Default Value
31: 12	-	Reserved	-
11	R	ISPCAPFIELD : ISP capture field	-
10	R	ISPCAPIDLE : ISP capture idle flag	1
9	RW	ISPPEND : ISP done interrupt pending [R] 0: no interrupt is generated 1: interrupt caused because ISP done [W] 1: interrupt pending bit(ISPPEND) clear	-
8	RW	ISPINTENB : ISP done interrupt enable	0
7	RW	ISPINTERLACE : Interlace mode 0: progressive mode 1: interlace mode	-
6	RW	ISPADDRMODE : target addressing mode 0: linear addressing 1: block addressing (block addressing mode is display mode)	-
5	RW	ISPFIELDPOL : Field signal polarity 0: Invert	-
4	RW	ISPVSYNCPOL : Vertical sync polarity 0: Invert	-
3	RW	ISPHSYNCPOL : Horizontal sync polarity 0: Invert	-
2	RW	ISPISPEXTSYNC : External sync enable 0 : CCIR656 mode 1 : External sync mode (from IHSYNC, IVSYNC PAD)	0
1	RW	ISPCAPENB : ISP capture enable	0
0	RW	ISPSYNCENB : ISP sync enable	0

26.2.3 ISP horizontal sync register (ISPHSYNC)

Address : C000 2004h

Bit	R/W	Description	Default Value
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Bit	R/W	Description	Default Value
31: 27	-	Reserved	-
26:16	W	ISPHOREND : Determine end of horizontal sync or active video In CCIR656 mode, Set the end of horizontal sync In External sync mode, Set the end of horizontal active video	-
15:11	-	Reserved	-
10: 0	W	ISPHORBEG : Determine beginning of horizontal sync or active video In CCIR656 mode, Set the beginning of horizontal sync In External sync mode, Set the beginning of horizontal active video	-

26.2.4 ISP vertical sync register (ISPVSYNC)

Address : C000 2008h

Bit	R/W	Description	Default Value
31: 27	-	Reserved	-
26:16	W	ISPVEREND : Determine end of vertical sync or blank In CCIR656 mode, Set the end of vertical sync In External sync mode, Set the end of vertical blank	-
15:11	-	Reserved	-
10: 0	W	ISPVERBEG : Determine beginning of vertical sync or blank In CCIR656 mode, Set the beginning of vertical sync In External sync mode, Set the beginning of vertical blank	-

26.2.5 ISP target size register (ISPSIZE)

Address : C000 200Ch

Bit	R/W	Description	Default Value
31: 27	-	Reserved	-
26:16	W	ISPWIDTH : Target width ISPWIDTH must be even	-
15:11	-	Reserved	-
10: 0	W	ISPHEIGHT : Target height	-

26.2.6 ISP difference size register (ISPDSIZE)

Address : C000 2010h

Bit	R/W	Description	Default Value
31: 27	-	Reserved	-
26:16	W	ISPDWIDTH : Difference of width $\text{ISPDWIDTH} = \text{source width} - \text{ISPWIDTH}$ In CCIR656 mode, source width must be greater or equal to horizontal active video width	-
15:11	-	Reserved	-
10: 0	W	ISPHEIGHT : Difference of height $\text{ISPHEIGHT} = \text{source height} - \text{ISPHEIGHT}$ In CCIR656 mode, source height must be greater or equal to vertical active video width	-

26.2.7 ISP size sub 1 register (ISPSIZE_1)

Address : C000 2014h

Bit	R/W	Description	Default Value
31: 27	-	Reserved	-
26:16	W	ISPWIDTH_1 : It must be ISPWIDTH – 1	-
15:11	-	Reserved	-
10: 0	W	ISPHEIGHT_1 : It must be ISPHEIGHT – 1	-

26.2.8 ISP base address register (ISPBASEADDR)

Address : C000 2018h

Bit	R/W	Description	Default Value
31: 3	W	ISPBASEADDR : target base address In Block addressing mode ISPBASEADDR[31:21] = segment; ISPBASEADDR[20:11] = y-coordinate[9:0] (in segment) ISPBASEADDR[10: 3] = x-coordinate[9:2] (in segment);	-
2: 0	-	Reserved	-

26.2.9 ISP stride register (ISPSTRIDE)

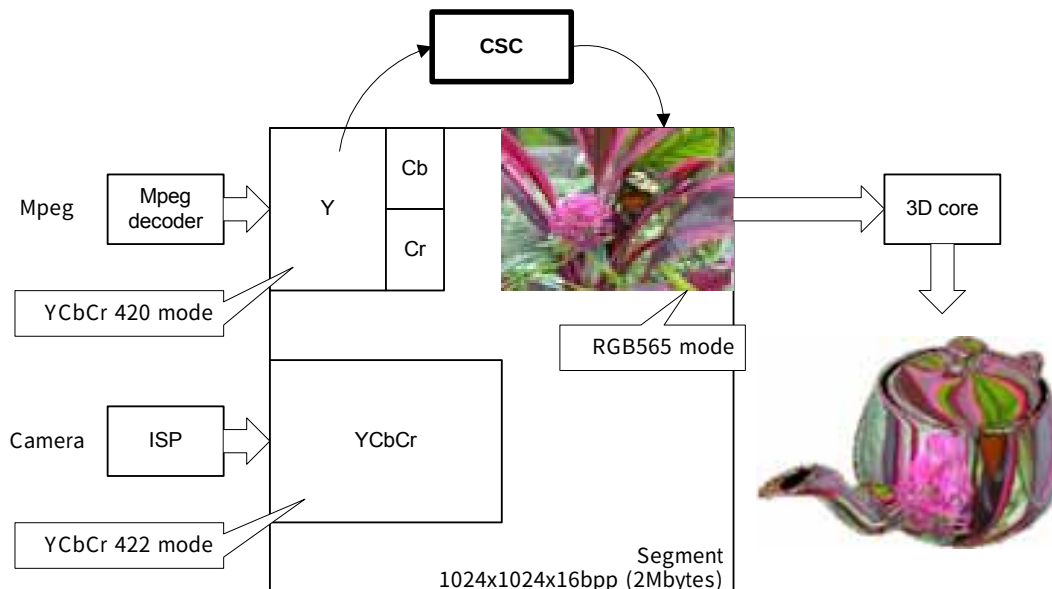
Address : C000 201Ch

Bit	R/W	Description	Default Value
31: 3	W	ISPSTRIDE : target stride ISPSTRIDE[31:3] = byte_per_line / 8 ; In Block addressing mode, byte_per_line must be 2048	-
2: 0	-	Reserved	-

27. Color Space Converter(CSC)

27.1 Overview

Color space converter (CSC) module converts YCbCr output from Mpeg decoder or Camera(ISP) to usable format for using as texture in 3D Core.



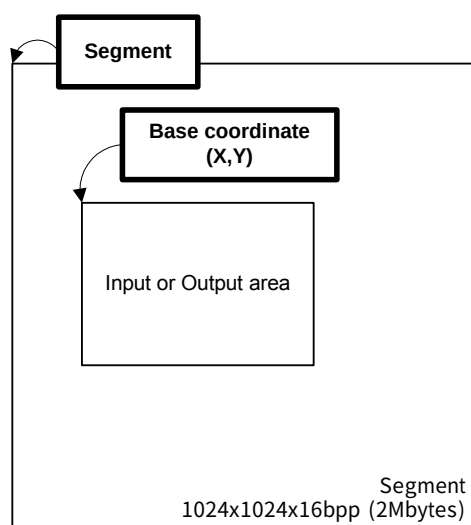
< Color space conversion >

27.1.1 memory structure

27.1.1.1 input/output data

Maximum convertible size is 1024x1024 pixel because CSC using Block addressing and operate by Segment. (refer to memory structure of 3D Core for Segment/block addressing)

All Input data and output data are assigned Segment and Base-coordinate.

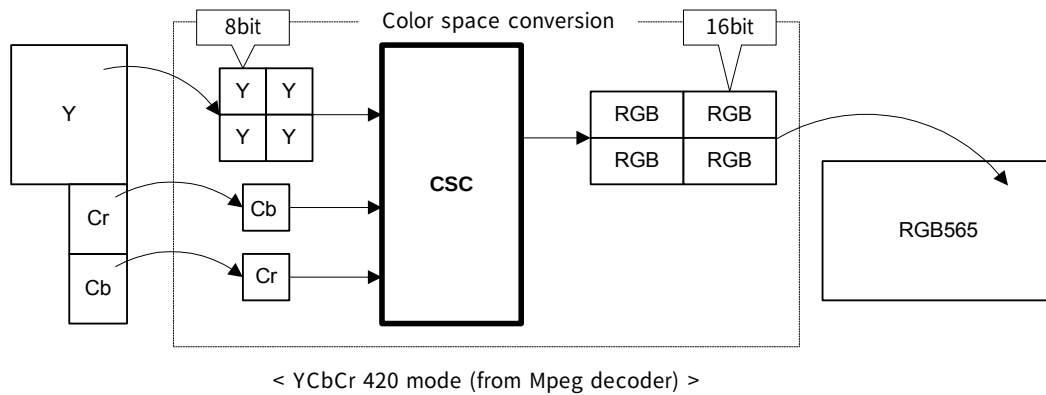


< input/output data addressing >

27.1.1.2 Input from Mpeg4 decoder

CSC are designed for input directly from Mpeg4 decoder. Y,Cb,Cr channel can be assigned Segment/Base-coordinate individually. Mpeg4 decoder output is 420mode which has split Y,Cb,Cr.

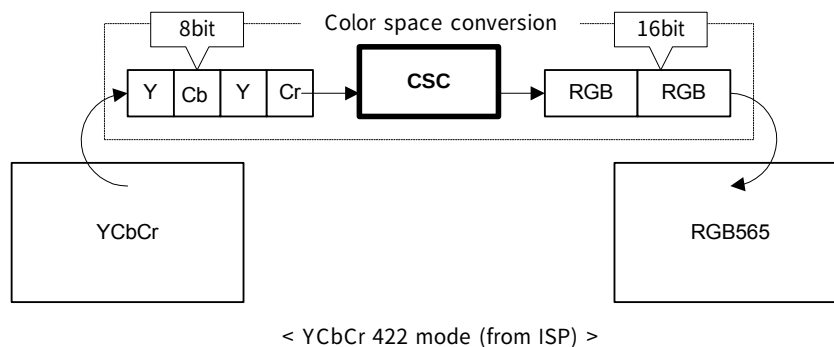
block addressing mode is display mode (refer to memory structure of 3D Core for block addressing mode)



27.1.1.3 Input from Image signal processor(ISP)

ISP module is store image into memory from Camera. CSC can convert input image from Camera into usable texture because it can use ISP output as input directly. ISP output is 422mode where Y, Cb, Cr coexist. It use Segment/Base-coordinate of Y channel only.

block addressing mode is display mode (refer to Memory structure of 3D core for block addressing mode)



27.1.1.4 Output to texture

3D core can use CSC output as texture. It fills rectangle area with color which is converted as RGB565 format when Segment/Base-coordinate of output position is assigned.

block addressing mode is texture mode (refer to Memory structure of 3D core for block addressing mode)

27.1.2 Transparency test

CSC replaces output color with certain value when Y, Cb, Cr has proscribed value. It is possible to make transparency effect to moving picture with this feature.

Bit	R/W	Description	Default Value
31: 26	-	Reserved	-
25:16	W	CSCHEIGHT : CSC height CSCHEIGHT = H – 1 H is height of result area in pixel. H must be multiple of 16	-
15:10	-	Reserved	-
9: 0	W	CSCWIDTH : CSC width in pixel CSCWIDTH = W – 1 W is width of result area in pixel. W must be multiple of 16	-

27.2.4 CSC Base Y channel register (CSCBASEY)

Address : C000 3208h

Bit	R/W	Description	Default Value
31: 21	W	CSCBASESEGMENTY : Y channel segment	-
20:11	W	CSCBASEYY : the y-coordinate of the upper-left corner of a Y channel in segment It must be multiple of 2^N ($2^N \geq$ height of result area in pixel)	-
10: 1	W-	CSCBASEYX : the x-coordinate of the upper-left corner of a Y channel in segment It must be multiple of 2^N ($2^N \geq$ width of result area in pixel)	-
0	-	Reserved	-

27.2.5 CSC Base Cb channel register (CSCBASCB)

Address : C000 320Ch

Bit	R/W	Description	Default Value
31: 21	W	CSCBASESEGMENTCB : CB channel segment In 422mode, this register is ignored.	-
20:11	W	CSCBASECBY : the y-coordinate of the upper-left corner of a CB channel in segment It must be multiple of 2^{N-1} ($2^N \geq$ height of result area in pixel) In 422mode, this register is ignored.	-
10: 1	W-	CSCBASECBX : the x-coordinate of the upper-left corner of a CB channel in segment It must be multiple of 2^{N-1} ($2^N \geq$ width of result area in pixel) In 422mode, this register is ignored.	-
0	-	Reserved	-

27.2.6 CSC Base Cr channel register (CSCBASECR)

Address : C000 3210h

Bit	R/W	Description	Default Value
31: 21	W	CSCBASESEGMENTCR : CR channel segment In 422mode, this register is ignored.	-
20:11	W	CSCBASECRY : the y-coordinate of the upper-left corner of a Cr channel in segment It must be multiple of 2^{N-1} ($2^N \geq$ height of result area in pixel) In 422mode, this register is ignored.	-
10: 1	W-	CSCBASECRX : the x-coordinate of the upper-left corner of a Cr channel in segment It must be multiple of 2^{N-1} ($2^N \geq$ width of result area in pixel) In 422mode, this register is ignored.	-
0	-	Reserved	-

27.2.7 CSC Base output register (CSCBASEOUT)

Address : C000 3214h

Bit	R/W	Description	Default Value
31: 21	W	CSCBASESEGMENTOUT : Output segment	-
20:11	W	CSCBASEYOUT : the y-coordinate of the upper-left corner of the output in segment It must be multiple of 2^N ($2^N \geq$ height of result area in pixel)	-
10: 1	W-	CSCBASEXOUT : the x-coordinate of the upper-left corner of the output in segment It must be multiple of 2^N ($2^N \geq$ width of result area in pixel)	-
0	-	Reserved	-

27.2.8 CSC Transparency control register (CSCTPCTRL)

Address : C000 3218h

Bit	R/W	Description	Default Value
31: 22	-	Reserved	-
21	W	CSCTPENBCR : Transparency test enable of Cr channel If Transparency test is true, pixel is replaced by transparency color(CSCTPCOLOR)	-
20	W	CSCTPENBCB : Transparency test enable of Cb channel	-
19	W	CSCTPENBY : Transparency test enable of Y channel	-
18	W	CSCTPINVCR : Inverse transparency range flag of Cr channel Transparency test is 0: CSCTPMINCR $\leq$ Cr $\leq$ CSCTPMAXCR 1: Cr < CSCTPMINCR or Cr > CSCTPMAXCR	-
17	W	CSCTPINVCB : Inverse transparency range flag of Cb channel Transparency test is 0: CSCTPMINCB $\leq$ Cb $\leq$ CSCTPMAXCB 1: Cb < CSCTPMINCB or Cb > CSCTPMAXCB	-
16		CSCTPINVY : Inverse transparency range flag of Y channel Transparency test is 0: CSCTPMINY $\leq$ Y $\leq$ CSCTPMAXY 1: Y < CSCTPMINY or Y > CSCTPMAXY	-
15:0		CSCTPCOLOR : Transparency color	-

27.2.9 CSC Transparency Y-range register (CSCTPY)

Address : C000 321Ch

Bit	R/W	Description	Default Value
31: 16	-	Reserved	-
15:8	W	CSCTPMAXY : Maximum value of Transparency range of Y channel	-
7: 0	W	CSCTPMINY : Minimum value of Transparency range of Y channel	-

27.2.10 CSC Transparency Cb-range register (CSCTPCB)

Address : C000 3220h

Bit	R/W	Description	Default Value
31: 16	-	Reserved	-
15:8	W	CSCTPMAXCB : Maximum value of Transparency range of Cb channel	-
7: 0	W	CSCTPMINCB : Minimum value of Transparency range of Cb channel	-

27.2.11 CSC Transparency Cr-range register (CSCTPCR)*Address : C000 3224h*

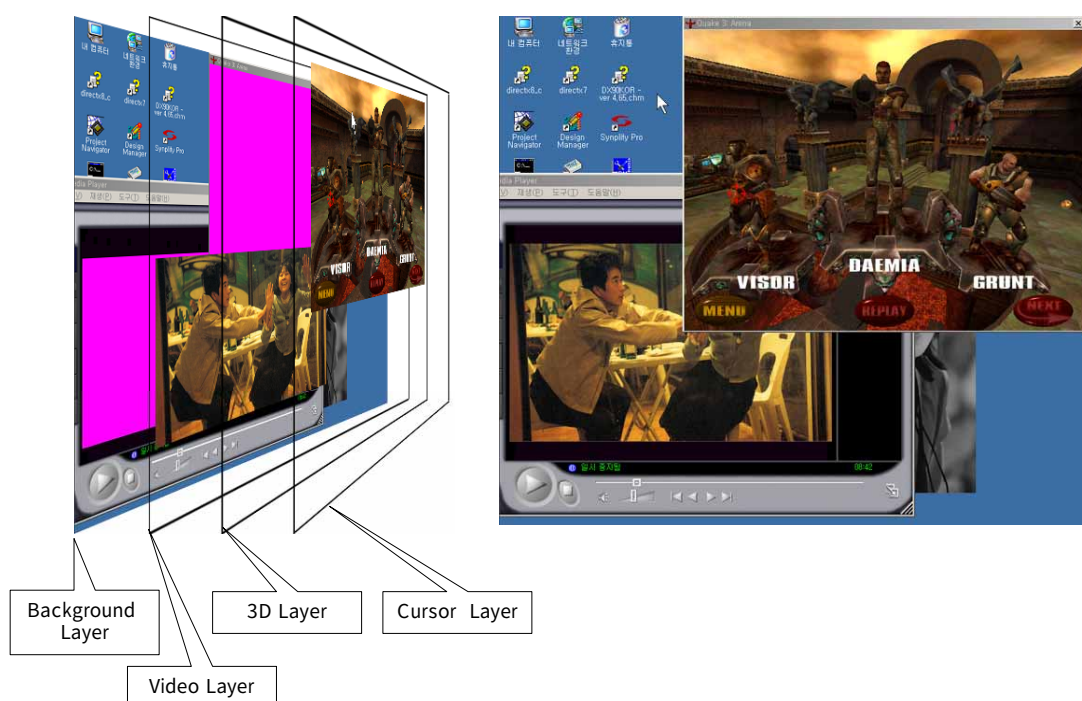
Bit	R/W	Description	Default Value
31: 16	-	Reserved	-
15:8	W	CSCTPMAXCR : Maximum value of Transparency range of Cr channel	-
7: 0	W	CSCTPMINCR : Minimum value of Transparency range of Cr channel	-

28. Multi Layer Controller (MLC)

28.1 Overview

VR3D can display 4 layers in one Screen at the same time.

Layer name	Pixel format	Addressing mode	Description
Background	RGB565	Linear/Block	Overlay function using ColorKey in 3D/ Video layer
Cursor	RGB565	Linear/Block	Configurable Cursor size Transparency Color / Invert Color
3D	RGB565	Block	Transparency Color Changing buffer address by 3D Core(page flipping)
Video	YCbCr 420 (MPEG) YCbCr 422 (Camera)	Linear/Block	Transparency Color Range(min~max) Scaling(Bilinear filtering)



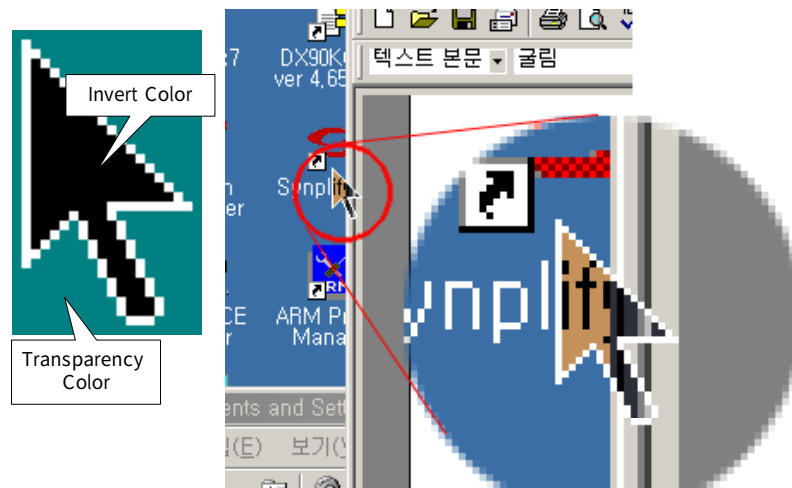
< Display layers >

28.1.1 Background layer

Background layer will be exposed unless other layers cover. Not like other layers, size and position of Background layer are not changeable. (Position is (0,0) and size is same as screen resolution). It can not be located above other layers but it can be exposed above 3D/Video layer when overlay function is used. When overlay is active, only part of 3D/Video layer, filled with certain color, will be exposed to screen.

28.1.2 Cursor layer

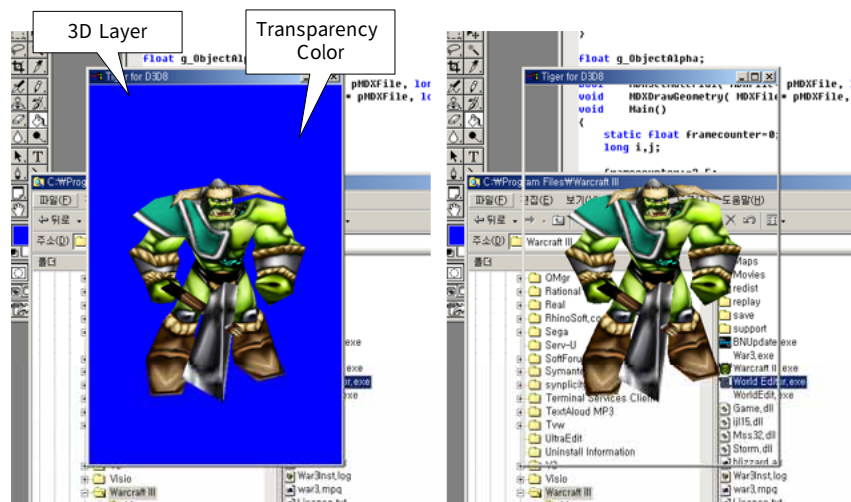
Cursor layer locates top of all layer. The size and position can be changeable. It also set to transparency and opposite color.



< Transparency color & Invert color >

28.1.3 3D layer

3D layer is for 3D frame buffer display. The size and position can be changed. Frame buffer context can not be scaled. Transparency option exist. All or part of 3D layer can be blocked by background layer if overlay function is active.

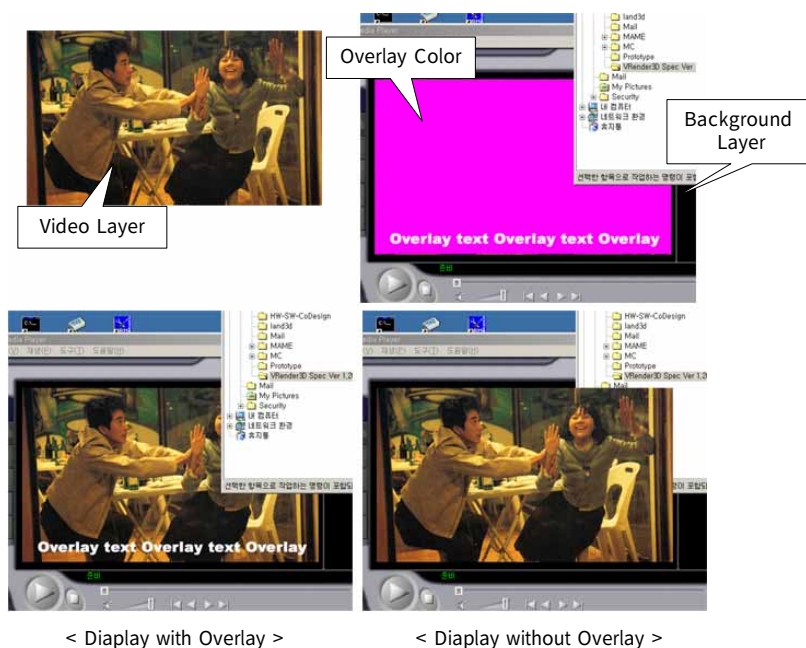


< 3D layer without Transparency >

< 3D layer with Transparency >

28.1.4 Video layer

Video layer display YCbCr pixel format. The size and position can be changeable. The contents can be scaled to fit window size. Transparency option exist. All or part of Video layer can be blocked by background layer if overlay function is active. Supported pixel formats are YCbCr422 mode and YCbCr420 mode. (refer to [Appendix](#))



Note : VR3D can not support filtered YUV layer in interlaced mode.

28.2 Multi Layer Controller Register

28.2.1 Register map (C000_1C00h ~ C000_1C63h)

Register Name	offset	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
MLCCCONTROL	00h																							MLCCCONTROL															
MLCCURPOS	04h					MLCCURY															MLCCURX																		
MLCCURSIZE	08h							MLCCURH															MLCCURW																
MLC3DPOS	0Ch					MLC3DY															MLC3DX																		
MLC3DSIZE	10h							MLC3DH															MLC3DW																
MLCVIDPOS	14h					MLCVIDY															MLCVIDX																		
MLCVIDSIZE	18h							MLCVIDH															MLCYVIDW																
MLCBKADDR	1Ch	MLCBKADDR																																					
MLCBKSTR	20h	MLCBKSTR																																					
MLCCURADDR	24h	MLCCURADDR																																					
MLCCURSTR	28h	MLCCURSTR																																					
MLCYADDR	2Ch	MLCYADDR																																					
MLCYSTR	30h	MLCYSTR																																					
MLCCBADDR	34h	MLCCBADDR																																					
MLCCBSTR	38h	MLCCBSTR																																					
MLCCRADDR	3Ch	MLCCRADDR																																					
MLCCRSTR	40h	MLCCRSTR																																					
MLCVIDSX	44h												MLCVIDSX																										
MLCVIDSY	48h												MLCVIDSY																										
MLCBKTP	4Ch																		MLCBKTPCOL																				
MLCCURTP	50h																		MLCCURTPCOL																				
MLCCURINV	54h																		MLCCURINVCOL																				
MLC3DTP	58h																		MLC3DTPCOL																				

MLCVIDTPMIN	5Ch																	MLCVIDRMIN	MLCVIDGMIN	MLCVIDBMIN
MLCVIDTPMAX	60h																	MLCVIDRMAX	MLCVIDGMAX	MLCVIDBMAX

28.2.2 MLC control register (MLCCONTROL)

Address : C000 1C00h

Bit	R/W	Description	Default Value
31: 11	-	Reserved	-
10	R	MLCREADIDLE : memory access flag 0: MLC access memory 1: MLC does not access memory	1
9	RW	MLCOUTMODE : output mode 0: RGB565 (dither ON) 1: RGB888 (dither OFF)	0
8	RW	MLCVIDQMODE : Video layer quantization mode 0: 220 quantization mode $R = Y + 1.375 * Cr$ $G = Y - 0.3359375 * Cb - 0.6953125 * Cr$ $B = Y + 1.734375 * Cb$ 1: 256 quantization mode $R = Y + 1.4020 * Cr$ $G = Y - 0.34414 * Cb - 0.71414 * Cr$ $B = Y + 1.7720 * Cb$	-
7	RW	MLCVIDFMT : Video layer format 0: 420mode Y,Cb,Cr 1: 422mode YcbCr	-
6	RW	MLCVIDFILTERENB : Video layer filter enable	-
5	RW	MLCPRIORITY : Video/3D layer priority 0: Video layer has priority over 3D layer 1: 3D layer has priority over Video layer	0
4	RW	MLCSHOWVIDEO : Show Video layer	0
3	RW	MLCSHOW3D : Show 3D layer	0
2	RW	MLCSHOWCUR : Show Cursor layer	0
1	RW	MLCSHOWBK : Show Background layer	1
0	RW	MLCON : MLC Enable 0 : Disable 1 : Enable	0

28.2.3 MLC cursor layer position register (MLCCURPOS)

Address : C000 1C04h

Bit	R/W	Description	Default Value
31: 28	-	Reserved	-
27:16	W	MLCCURY : the y-coordinate of the upper-left corner of a Cursor layer in screen	-
15:12	-	Reserved	-
11: 0	W	MLCCURX : the x-coordinate of the upper-left corner of a Cursor layer in screen	-

28.2.4 MLC cursor layer size register (MLCCURSIZE)

Address : C000 1C08h

Bit	R/W	Description	Default Value
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Bit	R/W	Description	Default Value
31: 26	-	Reserved	-
25:16	W	MLCCURH : Cursor layer height	-
15:10	-	Reserved	-
9: 0	W	MLCCURW : Cursor layer width	-

28.2.5 MLC 3D layer position register (MLC3DPOS)

Address : C000 1C0Ch

Bit	R/W	Description	Default Value
31: 28	-	Reserved	-
27:16	W	MLC3DY : the y-coordinate of the upper-left corner of a 3D layer in screen	-
15:12	-	Reserved	-
11: 0	W	MLC3DX : the x-coordinate of the upper-left corner of a 3D layer in screen	-

28.2.6 MLC 3D layer size register (MLC3DSIZE)

Address : C000 1C10h

Bit	R/W	Description	Default Value
31: 26	-	Reserved	-
25:16	W	MLCCURH : 3D layer height	-
15:10	-	Reserved	-
9: 0	W	MLCCURW : 3D layer width	-

28.2.7 MLC video layer position register (MLCVIDPOS)

Address : C000 1C14h

Bit	R/W	Description	Default Value
31: 28	-	Reserved	-
27:16	W	MLCVIDY : the y-coordinate of the upper-left corner of a Video layer in screen	-
15:12	-	Reserved	-
11: 0	W	MLCVIDX : the x-coordinate of the upper-left corner of a Video layer in screen	-

28.2.8 MLC video layer size register (MLCVIDSIZE)

Address : C000 1C18h

Bit	R/W	Description	Default Value
31: 26	-	Reserved	-
25:16	W	MLCVIDH : Video layer height	-
15:10	-	Reserved	-
9: 0	W	MLCVIDW : Video layer width	-

28.2.9 MLC background layer address register (MLCBKADDR)

Address : C000 1C1Ch

Bit	R/W	Description	Default Value
31: 1	W	MLCBKADDR : Background layer base address In Block addressing mode MLCBKADDR[31:21] = segment; MLCBKADDR [20:11] = y-coordinate[9:0] (in segment) MLCBKADDR [10: 1] = x-coordinate[9:0] (in segment);	-
0	W	MLCBKADDRMODE : Background layer addressing mode 0: linear addressing 1: block addressing (block addressing mode is display mode)	-

28.2.10 MLC background layer stride register (MLCBKSTR)

Address : C000 1C20h

Bit	R/W	Description	Default Value
31: 1	W	MLCBKSTR : Background layer stride MLCBKSTR[31:1] = byte_per_line / 2 ; In Block addressing mode, byte_per_line must be 2048	-
0	-	Reserved	-

28.2.11 MLC cursor layer address register (MLCCURADDR)

Address : C000 1C24h

Bit	R/W	Description	Default Value
31: 1	W	MLCCURADDR : Cursor layer base address In Block addressing mode MLCCURADDR[31:21] = segment; MLCCURADDR [20:11] = y-coordinate[9:0] (in segment) MLCCURADDR [10: 1] = x-coordinate[9:0] (in segment);	-
0	W	MLCCURADDRMODE : Cursor layer addressing mode 0: linear addressing 1: block addressing (block addressing mode is display mode)	-

28.2.12 MLC cursor layer stride register (MLCCURSTR)

Address : C000 1C28h

Bit	R/W	Description	Default Value
31: 1	W	MLCCURSTR : Cursor layer stride MLCCURSTR[31:1] = byte_per_line / 2 ; In Block addressing mode, byte_per_line must be 2048	-
0	-	Reserved	-

28.2.13 MLC video layer channel-Y address register (MLCYADDR)

Address : C000 1C2Ch

Bit	R/W	Description	Default Value
31: 3	W	MLCYADDR : Video layer channel-Y base address In Block addressing mode MLCYADDR[31:21] = segment; MLCYADDR [20:11] = y-coordinate[9:0] (in segment) MLCYADDR [10: 3] = x-coordinate[9:2] (in segment);	-

Bit	R/W	Description	Default Value
2: 1	-	Reserved	-
0	W	MLCVIDADDRMODE : Video layer addressing mode 0: linear addressing 1: block addressing (block addressing mode is display mode)	-

28.2.14 MLC video layer channel-Y stride register (MLCYSTR)

Address : C000 1C30h

Bit	R/W	Description	Default Value
31: 3	W	MLCYSTR : Video layer channel-Y stride MLCYSTR[31:3] = byte_per_line / 8 ; In Block addressing mode, byte_per_line must be 2048	-
2: 0	-	Reserved	-

28.2.15 MLC video layer channel-Cb address register (MLCCBADDR)

Address : C000 1C34h

Bit	R/W	Description	Default Value
31: 3	W	MLCCBADDR : Video layer channel-Cb base address In Block addressing mode MLCCBADDR[31:21] = segment; MLCCBADDR [20:11] = y-coordinate[9:0] (in segment) MLCCBADDR [10: 3] = x-coordinate[9:2] (in segment); In 422mode, this register is ignored.	-
2: 0	-	Reserved	-

28.2.16 MLC video layer channel-Cb stride register (MLCCBSTR)

Address : C000 1C38h

Bit	R/W	Description	Default Value
31: 3	W	MLCCBSTR : Video layer channel-Cb stride MLCCBSTR[31:3] = byte_per_line / 8 ; In Block addressing mode, byte_per_line must be 2048 In 422mode, this register is ignored.	-
2: 0	-	Reserved	-

28.2.17 MLC video layer channel-Cr address register (MLCCRADDR)

Address : C000 1C3Ch

Bit	R/W	Description	Default Value
31: 3	W	MLCCRADDR : Video layer channel-Cr base address In Block addressing mode MLCCRADDR[31:21] = segment; MLCCRADDR [20:11] = y-coordinate[9:0] (in segment) MLCCRADDR [10: 3] = x-coordinate[9:2] (in segment); In 422mode, this register is ignored.	-
2: 0	-	Reserved	-

28.2.18 MLC video layer channel-Cr stride register (MLCCRSTR)

Address : C000 1C40h

Bit	R/W	Description	Default Value
31: 3	W	MLCCRSTR : Video layer channel-Cr stride MLCCRSTR[31:3] = byte_per_line / 8 ; In Block addressing mode, byte_per_line must be 2048 In 422mode, this register is ignored.	-
2: 0	-	Reserved	-

28.2.19 MLC video layer scale-x register (MLCVIDSX)

Address : C000 1C44h

Bit	R/W	Description	Default Value
31:22	-	Reserved	-
21: 0	W	MLCVIDSX : Video layer scale-x $MLCVIDSX = ((SourceWidth-1) << 11) / (DisplayWidth-1)$	-

28.2.20 MLC video layer scale-y register (MLCVIDSY)

Address : C000 1C48h

Bit	R/W	Description	Default Value
31:22	-	Reserved	-
21: 0	W	MLCVIDSY : Video layer scale-y $MLCVIDSY = ((SourceHeight-1) << 11) / (DisplayHeight-1)$	-

28.2.21 MLC background layer transparency register (MLCBKTP)

Address : C000 1C4Ch

Bit	R/W	Description	Default Value
31: 18	-	Reserved	-
17	W	MLCBKOVVIDEO : Video layer overlay enable 0 : All rectangle area of Video layer will be displayed. 1 : Part of rectangle area, background with transparency color, of Video layer will be displayed	-
16	W-	MLCBKOV3D : 3D layer overlay enable 0 : All rectangle area of 3D layer will be displayed. 1 : Part of rectangle area, background with transparency color, of 3D layer will be displayed	-
15: 0	W	MLCBKTPCOL : Background transparency color (RGB565)	-

28.2.22 MLC cursor layer transparency register (MLCCURTP)

Address : C000 1C50h

Bit	R/W	Description	Default Value
31: 17	-	Reserved	-
16	W-	MLCCURTPENB : Cursor transparency enable	-
15: 0	W	MLCCURTPCOL : Cursor transparency color (RGB565)	-

28.2.23 MLC cursor layer invert register (MLCCURINV)

Address : C000 1C54h

Bit	R/W	Description	Default Value
31: 17	-	Reserved	-
16	W-	MLCCURINVENB : Cursor invert enable	-
15: 0	W	MLCCURINVCOL : Cursor invert color (RGB565)	-

28.2.24 MLC 3D layer transparency register (MLC3DTP)

Address : C000 1C58h

Bit	R/W	Description	Default Value
31: 17	-	Reserved	-
16	W-	MLC3DTPENB : 3D transparency enable	-
15: 0	W	MLC3DTPCOL : 3D transparency color (RGB565)	-

28.2.25 MLC video layer transparency min register (MLCVIDTPMIN)

Address : C000 1C5Ch

Bit	R/W	Description	Default Value
31: 25	-	Reserved	-
24	W	MLCVIDTPENB : Video transparency enable Transparency test is $MLCVIDMIN \leq \text{Video(}RGB888\text{)} \leq MLCVIDMAX$	-
23: 16	W	MLCVIDTPRMIN : Minimum value of Video transparency range of R channel	-
15: 8	W	MLCVIDTPGMIN : Minimum value of Video transparency range of G channel	-
7: 0	W	MLCVIDTPBMIN : Minimum value of Video transparency range of B channel	-

28.2.26 MLC video layer transparency max register (MLCVIDTPMAX)

Address : C000 1C60h

Bit	R/W	Description	Default Value
31: 24	-	Reserved	-
23: 16	W	MLCVIDTPRMAX : Maximum value of Video transparency range of R channel	-
15: 8	W	MLCVIDTPGMAX : Maximum value of Video transparency range of G channel	-
7: 0	W	MLCVIDTPBMAX : Maximum value of Video transparency range of B channel	-

29. Display Controller

29.1 Overview

The Display Controller (DPC) creates signals and data to connect to external display devices. Such external devices are TFT LCD, Video Encoder, DAC, etc. The connection to external devices requires various kinds of Synchronized Signals and Data Formats. The DPC can be set to a variety of Modes according to the types of external display devices. In addition, this block creates signals to be used for the Display FIFO in the Multi Layer Controller (MLC)

Feature

- Support for Display size of up to 1024x768 pixels
- Programmable HSYNC/VSYNC generation
- Supports Composite Sync
- Programmable polarity for DE, HSYNC, VSYNC
- Programmable Pixel Clock Phase
- Supports RGB, Multiplexed RGB(MRGB)

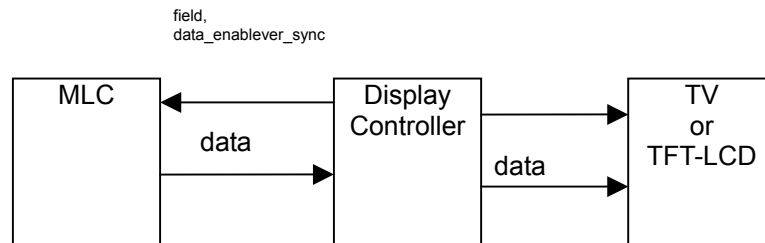


Figure 29-1 Display Controller

29.2 Operation

29.2.1 Display Clock Control

The DPC has two Clock Sources. The First Clock emerges from the PLL and the Second Clock is entered from the outside directly. In addition, the last clock is used for the DPC, synchronized with the ISP.

Once a clock source is determined, two clocks (CLK1 and CLK2) are created. The two created clocks are set differently according to the operation mode.

For RGB, one pixel per clock is transferred. Therefore, CLK1 and CLK2 have to be set as the same frequency. However, the MRGB transfer one pixel of data per two clocks. So, CLK2 has to be twice as fast as CLK1 in these cases

To change the clock source of CLK2, Set Display Clock Divisor Register (**DISPDIVCTRL**) of Clock Controller.

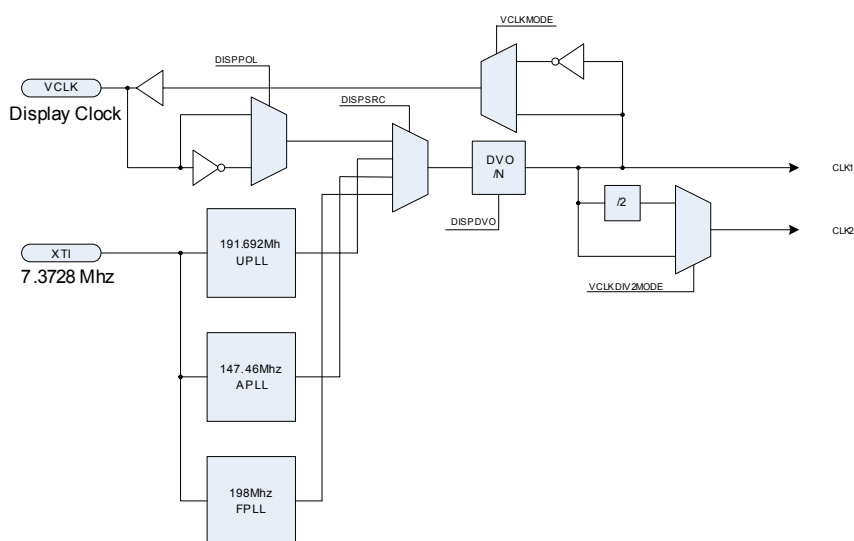


Figure 29-2, Clock Generation

Display Clock Divisor Register (**DISPDIVCTRL**) of Clock Controller.

Bit	R/W	Description	Default Value
15 : 14	R/W	DISPSRC : Set Clock Source for Display Controller 00 : External Video Clock PAD 01 : FPLL 10 : UPLL 11 : APLL	00
13 : 8	R/W	DISPDVO : Set division value of Display Clock (N-1)division	3Fh
7	R/W	DISPPOL : Set Polarity of External Video Clock. 0 : Inverted 1 : Normal	1
6	R/W	VCLKDIV2MODE : Set Clock VCLKDIV2. 0 : VCLKDIV2= VCLK 1 : VCLKDIV2= VCLK/2	0
5	R/W	VCLKMODE : Set Polarity of Display Clock. 0 : Normal 1 : Inverted	0
4 : 0	R	Reserved	0

29.2.2 Horizontal Sync(HSYNC) Control

The HSYNC is determined by four parameters (T2,T3,T7, and T8) as shown in Figure 29.3

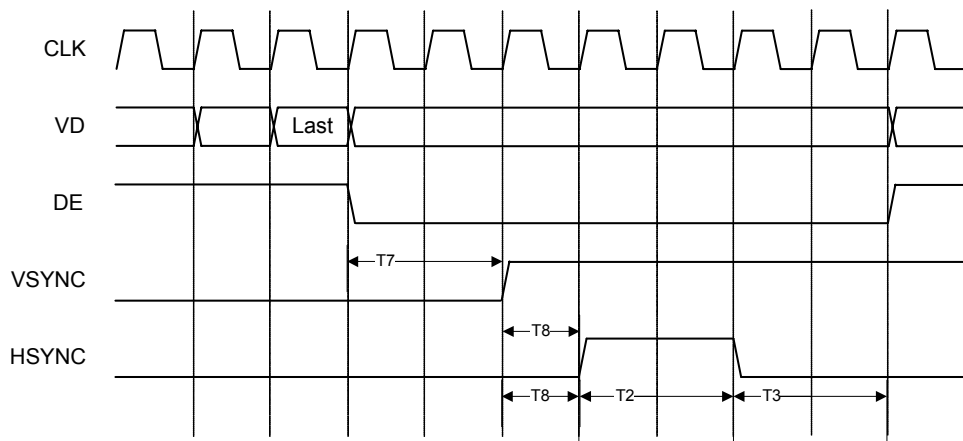


Figure 29-3 HSYNC Timing Diagram

T7 and T8 indicate the Front Porch of Horizontal Sync(FPHS). Especially, T7 shows the turning point of VSYNC between HSYNC and DE. T2 is the HSYNC width and T3 is the Back Porch of Horizontal Sync(BFHS). The following tables from 14-2 to 14-5 list the registers to be used for the parameter setting for T2, T3, T7 and T8

The HSYNC Pin can be set to several kinds of Modes according to the types of external display devices. The HSYNC PAD Configuration is determined by the HSPADCFG setting in Table 14-2. If the HSPADCFG is set to '0', the HSYNC Pin is used as a horizontal sync signal

29.2.3 Vertical Sync(VSYNC) Control

The VSYNC is determined by T9, T10, and T11 parameters. T10 indicates the Front Porch of Vertical Sync(FPVS). T11 points out the Vertical Sync Width. T9 is the Vertical Blank Line Width. In other words, the Back Porch of vertical Sync(BFVS) can be designated using T9, T10 and T11. The timing diagram for the VSYNC is as shown in Figure 29-4. The following tables from 14-5 to 14-6 list the registers to set T9, T10 and T11.

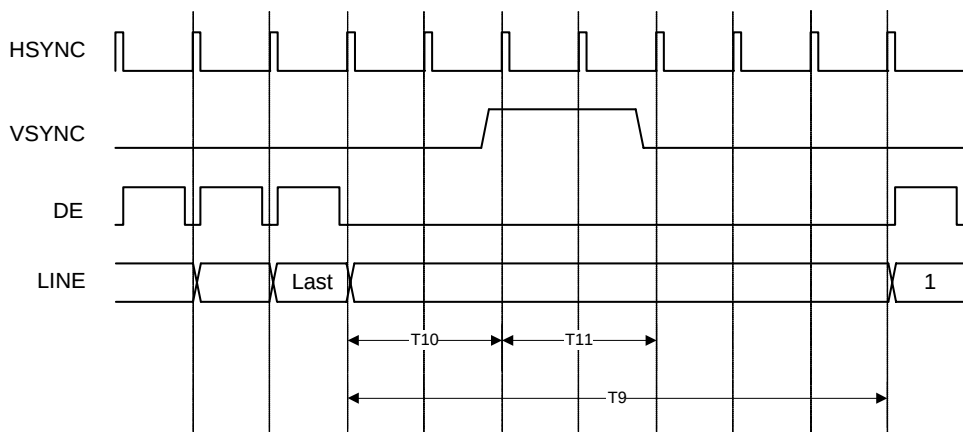


Figure 29-4 VSYNC Timing Diagram

29.2.4 Data Enable (DE) Control

According to the types of external display devices, the DE PAD can be set as Data Enable(DE), Start of Horizontal Pixel(STH), or Data Start Pulse. In addition, it can be used as the Composite Sync(CSYNC) output.

The VR3D has three Modes by the setting of the DEPADCFG and the DEMOD.

29.2.4.1 Data Enable Mode (DEPADCFG=0, DEMOD=0)

This Mode is activated only when Pixel data is outgoing. In addition, a phase can be shifted from 3 pixels to -3 pixels at the base of the effective pixel. The phase is determined by the DEWID Register value in Table 14-5. The timing diagram on the DE Mode operation is as shown in Figure 29-5

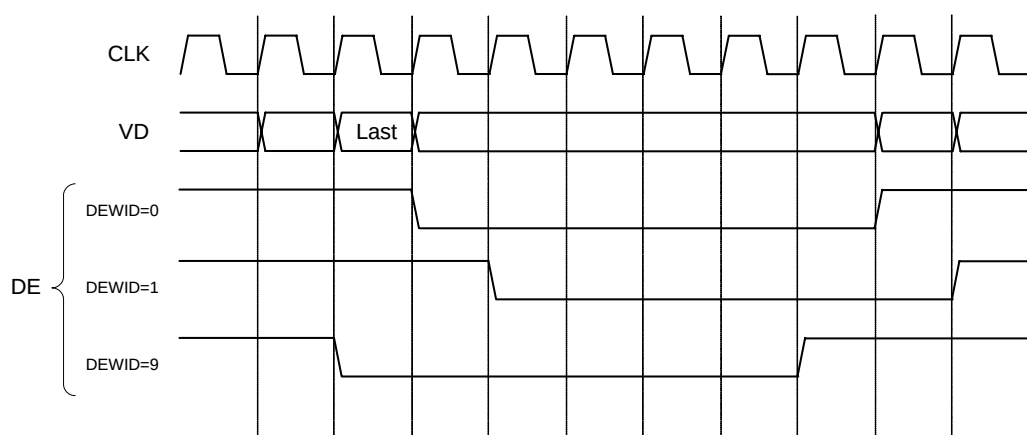


Figure 29-5 DE timing diagram in the Data Enable Mode

29.2.4.2 STH Mode (DEPADCFG=1, DEMOD=0)

In STH Mode, the DE PAD generates a pulse on the verge of the first pixel data transmission. The timing diagram on the DE Mode operation is as shown in Figure 14-6.

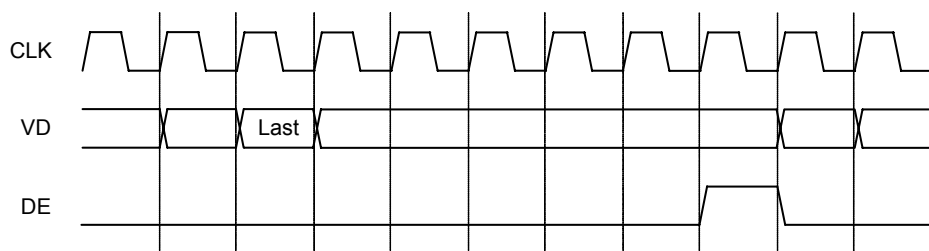


Figure 29-6. DE Timing Diagram in the STH Mode

29.2.4.3 PULSE Mode (DEPADCFG=0, DEMOD=1)

In PULSE mode, a pulse is generated at the same time as the first pixel data transmission. The pulse width is determined by the DEWID Register value in Table 14-5. In other words, the DEWID is used to adjust the DE phase in the DE Mode or the Pulse width in the PULSE Mode. The adjustable maximum pulse width is 16 Pixel Clocks..

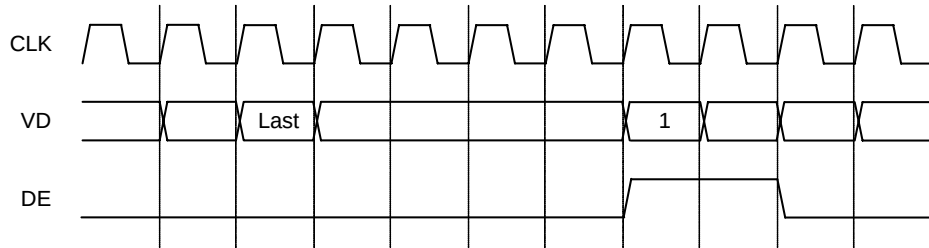


Figure 29-7. DE Timing Diagram in the PULSE Mode

29.2.4.4 When DEPADCFG=2, the CSYNC comes out through DE pad.

29.2.5 Active Display Size

The Display Size can be adjusted up to 1024 X 1024 in maximum. However, the real Display Size is determined by the system memory bandwidth used. Display Sizes are designated by T12 and T13.

In Interlace Mode, the Display Size has to be designated to half of the Active Display Size in the vertical direction since the Interlace Mode sends even fields and odd fields respectively

29.2.6 Video Data Bus Configuration

To transmit Data to external display devices, there are a total of 30 bits allocated. Out of these bits, 24 bits are allocated in the VD and the remaining are used with control signals. The 24-bit Data Bus is set to various Modes according to external devices.

Table 14-12 shows the VD setting in accordance with the types of Data.

DOF	0	1	2	3
RGB	4:4:4	5:6:5	6:6:6	8:8:8
MRGB	5:5:5	5:6:5	8:8:8a	8:8:8b

29.2.6.1 VD Configuration in RGB mode.

VD	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
4:4:4													R3	R2	R1	R0	G3	G2	G1	G0	B3	B2	B1	B0
5:6:5									R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B4	B3	B2	B1	B0
6:6:6							R5	R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B5	B4	B3	B2	B1	B0
8:8:8	R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0

29.2.6.2 VD Configuration in MRGB mode.

Out of 24 bits allocated in the VD, 12 bits are used in the MRGB Mode. Instead, 2 Clocks per pixel are required.

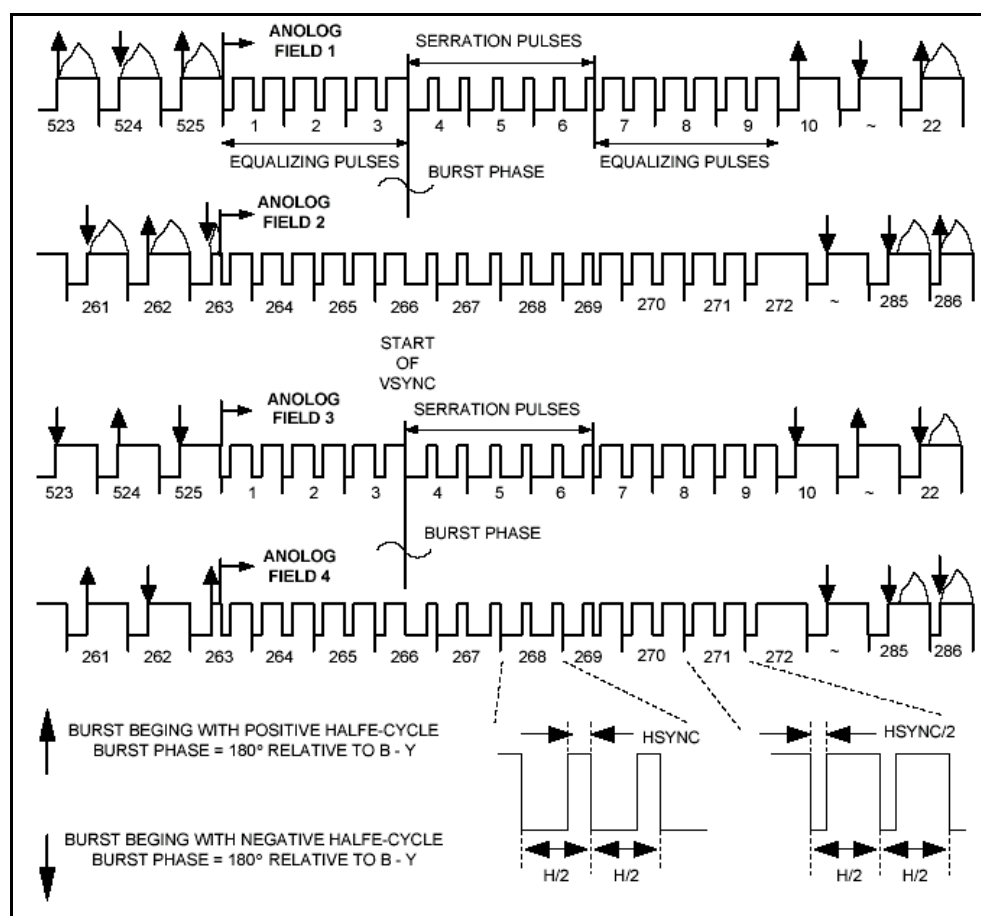
DOF	VD	11	10	9	8	7	6	5	4	3	2	1	0
5:5:5	1st					G5	G4	G3	R7	R6	R5	R4	R3
	2nd						B7	B6	B5	B4	B3	G7	G6
5:6:5	1st					G4	G3	G2	R7	R6	R5	R4	R3
	2nd					B7	B6	B5	B4	B3	G7	G6	G5
8:8:8a	1st	G3	G2	G1	G0	R7	R6	R5	R4	R3	R2	R1	R0
	2nd	B7	B6	B5	B4	B3	B2	B1	B0	G7	G6	G5	G4
8:8:8b	1st	G4	G3	G2	R7	R6	R5	R4	R3	G0	R2	R1	R0
	2nd	B7	B6	B5	B4	B3	G7	G6	G5	B2	B1	B0	G1

29.2.7 Composite Sync (CSYNC)

To make composite sync(CSYNC), the CSYNC of Table 14-13 has to be set to '1'. As shown in Figure 14-11. the EQUALIZING PULSE and the SERRATION PULSE have to be created in the Vertical blank area.

The timings about the EQUALIZING PULSE and the SERRATION PULSE are as shown in Figure 14-11.

As shown in the figure above, the pulse width is determined by the HSYNC width (HSWID), The pulse position in the middle of the line is determined by MIDSYNCPPOS. This value indicates the position of valid data area.



The CSYNC can be output through the HSYNC pin or DE pin. The HSYNC pin is determined by the HSPADCFG value and the DE

pin is determined by DEPADCFG value. The

29.2.8 Interrupt

When the HSYNC and the VSYNC are produced in the DPC, an interrupt can be initiated. In addition, each interrupt can be set to trigger at the first or the second edge of Sync.

This setting can be performed using the HSINTEEDGE/VSINTEEDGE in Table 14-31. Figure 14-19 shows when an interrupt initiated.

Once an interrupt is initiated, the VSINTFLAG and the HSINTFLAG in Table 14-31 are changed to '1'. The CPU can check the generation of the interrupt by reading the Flag. If the CPU completes the handling about the pertinent interrupt, the pertinent Flag has to be cleared. To clear the Flag, write '1', to the pertinent bit.

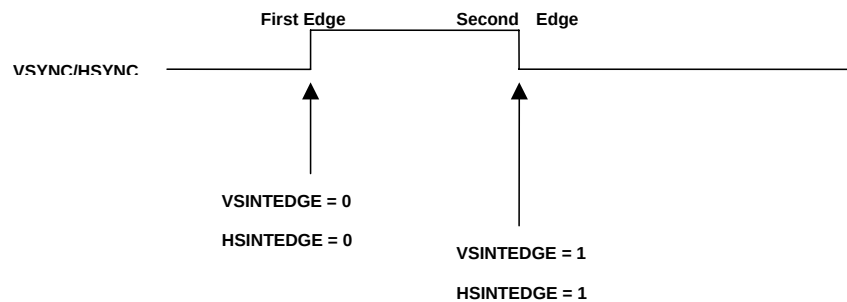


Figure 29-xx. HSYNC/VSYNC interrupt

29.3 Display Controller Register

29.3.1 Display control register (DISPCONTROL)

Address : C000 1800h

Bit	R/W	Description	Default Value															
15	R	SLVHV : Slave Sync Mode 0: HSYNC & FIELD, 1: HSYNC & VSYNC	0															
14	RW	SLAVE : Slave mode 0: Master mode 1: Slave mode	0															
13	RW	CSYNC : COMP SYNC mode	-															
12	RW	PAL : NTSC/PAL mode 0: NTSC 1: PAL	-															
11:9	RW	Reserved	-															
8	R/W	TVMODE : TV Mode enable 0: LCD 1:TV	0															
7:6	RW	DOT : Data Output Type 0: RGB 1: Multiplexed RGB(MRGB) 2: Reserved. 3: Reserved	0															
5	RW	INTERLACE : TV Interlaced mode 0: Progressive mode 1: Interlaced mode	0															
4:3	R	Reserved	0															
2:1	RW	Data Output Format: <table><tr><td>DOF</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>RGB</td><td>4:4:4</td><td>5:6:5</td><td>6:6:6</td><td>8:8:8</td></tr><tr><td>MRGB</td><td>5:5:5</td><td>5:6:5</td><td>888a</td><td>888b</td></tr></table>	DOF	0	1	2	3	RGB	4:4:4	5:6:5	6:6:6	8:8:8	MRGB	5:5:5	5:6:5	888a	888b	0
DOF	0	1	2	3														
RGB	4:4:4	5:6:5	6:6:6	8:8:8														
MRGB	5:5:5	5:6:5	888a	888b														
0	RW	ENB : Display Enable 0 : Disable 1 : Enable	0															

29.3.2 FPI Control register (MLCCURPOS)

Address : C000 1802h

Bit	R/W	Description	Default Value
15:13	RW	CHPAHSE : Select the phase for CLKH	000b
12	RW	CHMSKBNK : Mask pixel clock during blank area	0
11:3	R	Reserved	-
2	RW	PADSYNC : Sync Enable for ALL PAD Output 0 : Bypass 1: 1 clock delayed	0
1	RW	HMSKBNK : Mask HSYNC during the vertical blank	0
0	R	Reserved	-

29.3.3 FPI Polarity Control register (DISPFPIPOL1)

Address : C000 1804h

Bit	R/W	Description	Default Value
15:3	-	Reserved	-
2	RW	DEPOL : Polarity control for DE	0

Bit	R/W	Description	Default Value
1	RW	HSPOL : Polarity control for HSYNC	0
0	RW	VSPOL : Polarity control for VSYNC	0

29.3.4 FPI Polarity Control register (DISPFPIPOL2)

Address : C000 1806h

Bit	R/W	Description	Default Value
15:0	R	Reserved	-

29.3.5 FPI Polarity Control register (DISPFPIPOL3)

Address : C000 1808h

Bit	R/W	Description	Default Value
15:0	R	Reserved	-

29.3.6 FPI Enable Control register (DISPFPIATV1)

Address : C000 180Ah

Bit	R/W	Description	Default Value
15:3	R	Reserved	-
2	RW	DEATV : Enable control for DE	0
1	RW	HSATV : Enable control for HSYNC	0
0	RW	VSATV : Enable control for VSYNC	0

29.3.7 FPI Enable Control register (DISPFPIATV2)

Address : C000 180Ch

Bit	R/W	Description	Default Value
15:0	RW	VD1ATV : Enable control for VD[15:0]	0

29.3.8 FPI Enable Control register (DISPFPIATV3)

Address : C000 180Eh

Bit	R/W	Description	Default Value
15:8	R	Reserved	-
7:0	RW	VD2ATV : Enable control for VD[23:16]	0

29.3.9 Active Width register (DISPXMAX)

Address : C000 1816h

Bit	R/W	Description	Default Value
15:12	R	Reserved	-
11:0	RW	T12(XMAX) : X width	719

29.3.10 Active Height register (DISPYMAX)

Address : C000 1818h

Bit	R/W	Description	Default Value
15:12	R	Reserved	-
11:0	RW	T13(YMAX) : Y width	479

29.3.11 HSYNC Width register (DISPHSWIDTH)

Address : C000 181Ah

Bit	R/W	Description	Default Value
15:11	R	Reserved	-
10	RW	SFTDIR : Shift direction control for pixel data 0: Right 1: Left	1
9:8	RW	HSPADCFG : HSYNC PAD Configuration 00: HSYNC pin outputs horizontal sync signal (HSYNC) 01: HSYNC pin outputs the latch pulse (LP) : Reserved 10: HSYNC pin outputs the Data Enable(DE) 11: HSYNC pin outputs the Composite Sync (CSYNC)	00b
7:0	RW	T2(HSWID) : HSYNC Width	0

29.3.12 HSYNC Start register (DISPHSSTR)

Address : C000 181Ch

Bit	R/W	Description	Default Value
15:8	R	Reserved	-
7:0	RW	T8(HSSTR) : HSYNC Start Position	0

29.3.13 HSYNC End register (DISPHSEND)

Address : C000 181Eh

Bit	R/W	Description	Default Value
15:8	R	Reserved	-
7:0	RW	T7(HSEND) : HSYNC End Position	0

29.3.14 VSYNC Width register (DISPVSYNC)

Address : C000 1820h

Bit	R/W	Description	Default Value
15:8	RW	T10(VSSTR) : VSYNC Start Position	0
7	RW	VSFLDEN : Vertical Sync Field Enable	0
6	RW	VSFLDPOL : Vertical Sync Field Polarity	0
5:4	RW	VSPADCFG : VSYNC PAD Configuration 00: VSYNC pin outputs vertical sync signal (VSYNC) 01: VSYNC pin outputs the start of first line(STV) : Reserved 10: VSYNC pin outputs the field .	00b
3:0	RW	T11(VSWID) : VSYNC Width	0

29.3.15 VSYNC End register (DISPVSEND)

Address : C000 1822h

Bit	R/W	Description	Default Value
15	RW	Reserved	0
14	RW	Reserved	0
13	RW	Reserved	0
12	RW	Reserved	0
11	RW	SCANDIR : Scan direction control for line data (0: down, 1:Up)	1
10	RW	POLSTRVL : Toggle start level of POL per frame.	0
9	RW	SMPT274M : If its value is '1' in interlaced mode, the periods needed for the first field and the second field vertical sync are different. The second field period is 1H longer than the first period.	0
8:0	RW	T9(VSEND) : VSYNC End Position	0

29.3.16 TV Blank End register (DISPTVBNK)

Address : C000 1824h

Bit	R/W	Description	Default Value
15:14	R	Reserved	-
13:7	RW	TVBNKSTR : Blank Start position (Valid when TV mode is enabled)	0
6:0	RW	TVBNKEND : Blank End position (valid when TV mode is enabled)	0

29.3.17 DE Control register (DISPDE)

Address : C000 1826h

Bit	R/W	Description	Default Value
15:14	RW	DECFG : DE PAD Configuration, When DOF==3 00: DE pin outputs the Data enable (DE) 01: DE pin outputs the start of first pixel(STH) : Reserved 10: DE pin outputs the Composite Sync(CSYNC) 11: Reserved	0
13	RW	DEENB : Enable DE during blank area	0
12	RW	DEMODO : DE Mode 0: Enable Mode 1: Pulse Mode	0
11:4	RW	T3(DESTR) : DE Start position	0
3:0	RW	T14(IDEVID) : STH Pulse Width (When DEMOD==1 and DESEL=3)(T14) DE Delay (When DEMOD==0 and DESEL==3) 4'bxx00 : 0 delay 4'b0x01 : +1 delay 4'b0x10 : +2 delay 4'b0x11 : +3 delay 4'b1x01 : -1 delay 4'b1x10 : -2 delay 4'b1x11 : -3 delay	0

29.3.18 Reserved register (Reserved)

Address : C000 1828h~1834h (Reserved)

29.3.19 CSYNC Control register (DISPMIDSYNC)

Address : C000 1836h

Bit	R/W	Description	Default Value
15:12	R	Reserved	0
11:0	RW	MIDSYNCPOS : MID Sync Start Position	0

29.3.20 CSYNC Control register (DISPCSYNC)

Address : C000 1838h

Bit	R/W	Description	Default Value
15:12	R	Reserved	0
11:0	RW	CSYNCPOS : CSync Start Position	0

29.3.21 Reserved register (Reserved)

Address : C000 183Ah~1844h (Reserved)

29.3.22 Interrupt Control register (DISPINTR)

Address : C000 1846h

Bit	R/W	Description	Default Value
15:10	R	Reserved	0
9	RW	VSPLSSEL : Output selection for ver_sync	0
8:6	R	Reserved	-
5	RW	VSINTEN : VSYNC interrupt Enable	0
4	RW	HSINTEN : HSYNC interrupt Enable	0
3	RW	VSINTEDGE : Interrupt position for VSYNC (0: first edge, 1: second edge)	0
2	RW	HSINTEDGE : Interrupt position for HSYNC (0: first edge, 1: second edge)	0
1	R	VSINTFLAG: VSYNC Interrupt Status	0
0	R	HSINTFLAG: HSYNC Interrupt Status	0

30. ELECTRICAL DATA

30.1 ABSOLUTE MAXIMUM RATINGS

Parameter	Symbol	Rating		Unit
DC Supply Voltage	VDDi	1.8V Vdd	2.7	V
	VDDIO	3.3V Vdd	3.8	
DC Input Voltage	Vin	3.3V Input Buffer	3.8	
		3.3V Interface / 5V Tolerant input buffer	6.5	
DC output Voltage	VOOUT	3.3V Output buffer	3.8	
DC Input (Latch-up) Current	Iin	± 200		mA
Storage Temperature	Tstg	-65 to 150		°C

30.2 RECOMMENDED OPERATION CONDITIONS

Parameter	Symbol	Rating		Unit
DC Supply Voltage	VDDi	1.8V Vdd	1.65~1.95	V
		3.3V Vdd	3.0~3.6	
DC Supply Voltage for I/O Block	VDDi	1.8V Vdd	1.65~1.95	
		3.3V Vdd	3.0~3.6	
DC Supply Voltage for Analog Core	VDD	3.3V Vdd	3.3 $\pm$ 5%	
DC Input Voltage	VIN	3.3V Input buffer	3.3 $\pm$ 0.3	
		3.3V Interface / 5V Tolerant input buffer	3.8 ~5.25	
DC Output Voltage	VOOUT	3.3V Output buffer	3.3 $\pm$ 0.3	
Operating Temperature	TOPR	Commercial	0 to 70	°C

30.3 D.C. ELECTRICAL CHARACTERISTICS

CHARACTERISTIC

These DC Parameters guarantee the I/O cell characteristic at the static state only, not at the dynamic state.

30.3.1 Normal I/O – 1.8v

Vdd = 1.65 ~ 1.9v, Ta = 25°C

Characteristics	Symbol	Test Condition		Min	Typ	max	unit
Input High Current	Iih	Vin = Vdd	Normal	-10	-	10	uA
Input Low Current	Iil	Vin = Vss	Normal	-10	-	10	uA
Input High Voltage	Vih	CMOS		1.27	-	-	V
Input Low Voltage	Vil	CMOS		-	-	0.57	V
Output High Voltage	Voh	0mA Buffer, Ioh= - 0mA		1.2	-	-	V
Output High Voltage	Voh	3mA Buffer, Ioh= - 3mA		1.2	-	-	V
Output High Voltage	Voh	20mA Buffer, Ioh= - 20mA		1.2	-	-	V
Output Low Voltage	Vol	0mA Buffer, Ioh= 0mA		-	-	0.45	V
Output Low Voltage	Vol	3mA Buffer, Ioh= 3mA		-	-	0.45	V
Output Low Voltage	Vol	20mA Buffer, Ioh= 20mA		-	-	0.45	V

Table 32-3-1 Normal 1.8v I/O PAD DC Electrical Characteristics

30.3.2 Normal I/O – 3.3v

Vdd = 3.0 ~ 3.6v, Ta = 25°C

Characteristics	Symbol	Test Condition		Min	Typ	max	unit
Input High Current	Iih	Vin = Vdd	Normal	-10	-	10	uA
Input Low Current	Iil	Vin = Vss Pull-up	Normal Up	-10 -60	-	10 - 10	uA
Input High Voltage	Vih Vt+	CMOS Schmitt S (CMOS)		2.0 -	- -	- 2.0	V
Input Low Voltage	Vil Vt-	CMOS Schmitt S (CMOS)		- 0.8	- -	0.8 -	V
Output High Voltage	Voh	8mA Buffer, Ioh= - 8mA		2.4	-	-	V
Output High Voltage	Voh	12mA Buffer, Ioh= - 12mA		2.4	-	-	V
Output High Voltage	Voh	16mA Buffer, Ioh= - 16mA		2.4	-	-	V
Output Low Voltage	Vol	8mA Buffer, Ioh= 8mA		-	-	0.4	V
Output Low Voltage	Vol	12mA Buffer, Ioh= 12mA		-	-	0.4	V
Output Low Voltage	Vol	16mA Buffer, Ioh= 16mA		-	-	0.4	V
Tri-State Output Leakage Current	Ioz	Vout = Vdd or Vss		-10	-	10	uA

Note : Schmitt Trigger test condition : Vdd = 3.0 to 3.6 v, Ta = 25°C

Table 32-3-2 Normal 1.8v I/O PAD DC Electrical Characteristics

30.3.3 DDR Normal I/O

Characteristics	Symbol	Test Condition	Min	Typ	max	unit
Input High Voltage	Vih	CMOS	1.43	-	-	V
Input Low Voltage	Vil	CMOS	-	-	1.07	V
Output High Voltage	Voh	26mA Buffer, Ioh= -26mA	2.138	-	-	V
Output Low Voltage	Vol	26mA Buffer, Iol= 26mA	-	-	0.114	V

Comment : psbstl, psosstl, Vref=1.25V

Table 32-3-3 DDR Normal I/O PAD DC Electrical Characteristics

30.3.4 DDR Differential I/O

Characteristics	Symbol	Test Condition	Min	Typ	max	unit
Output High Voltage	Voh	26mA Buffer, Ioh= -26mA	2.138	-	-	V
Output Low Voltage	Vol	26mA Buffer, Iol= 26mA	-	-	0.114	V

Comment : psosstldiff, Vref=1.25V

Table 32-3-4 DDR Differential I/O PAD DC Electrical Characteristics

30.3.5 Special I/O

Characteristics	Symbol	Test Condition	Min	Typ	max	unit
Input High Voltage	Vt+	SCHITT (CMOS)	2.0	-	-	V
Input Low Voltage	Vt-	SCHITT (CMOS)	-	-	0.8	V
Output High Voltage	Voh	8mA Buffer, Ioh= -8mA	2.4	-	-	V
Output Low Voltage	Vol	8mA Buffer, Iol= 8mA	-	-	0.4	V

Comment : phbsu100ct8sm, phbsu100ct12sm, Controllable Pull-Up

Table 32-3-5 Special I/O PAD DC Electrical Characteristics

30.3.6 USB I/O

Characteristics	Symbol	Test Condition	Min	Typ	max	Unit
Input High Voltage	Vih		2.5	-	-	V
Input Low Voltage	Vil		-	-	0.8	V
Input High Current	Ih	Vin= 3.3V	-10		10	uA
Input Low Current	Il	Vin= 0.0V	-10		10	uA
Output High Voltage	Voh	15KΩ to GND	2.8	-	3.6	V
Output Low Voltage	Vol	1.5KΩ to 3.6v	-	-	0.3	V

Table 32-3-6 USB PAD DC Electrical Characteristics

30.3.7 Normal Operation

Parameter	Value	Unit	Condition
Operation Voltage	1.8/2.5/3.3	V	
Max. Operating Frequency (FCLK)	200	MHz	CPU Clock
Max. Operating Frequency (BCLK)	100	MHz	Bus Clock
Max. Operating Frequency (PCLK)	50	MHz	Internal Peripheral Clock
Typical normal mode power (VDDi Only)	390	mW	Full 3D Engine Operation ARM 200Mhz MPEG Core Operation
Idle mode power (VDDi Only)	160	mW	CPU Stop, Other Logic running.
Stop mode power (VDDi Only)	0.5	mW	All Clock Stop

Table 32-3-7 VRender3D Power Supply Voltage and Current